Asymmetric Labor Income Risk and Stock Market Participation: Evidence from the US*

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Job Market Paper (JMP)

September 2026

Abstract

I document that upside earnings opportunities predict stock market participation among initial nonholders, beyond mean earnings growth, total variance and measured household resources. I measure earnings distributions in historical Current Population Survey records and link them to subsequent portfolio transitions in the Survey of Income and Program Participation. A one-standard-deviation increase in upside composition predicts 1.6 percentage points more entry into stocks and mutual funds, nearly one fifth of the sample entry rate. The association extends to entry followed by ownership at two subsequent annual observations. The result concerns asymmetry in earnings itself, rather than its covariance with stock returns. A standard participation-cost model shows how favorable realizations can finance entry even at unchanged mean and variance. In an estimated life-cycle model, measured earnings dynamics strengthen the untargeted composition-entry relationship relative to pooled dynamics, mainly through differences in resources available to invest. Controlled experiments show why immediate entry and later access need not move together: the anticipated earnings distribution changes saving and the resources available for future participation.

Keywords: labor income risk; household portfolios; tail composition; stock market participation; measurement error
JEL codes: C18, D14, G11, J31

*I thank Lorenz Kueng, Alberto Plazzi, Luise Eisfeld, Fabrizio Colella, Darrell Duffie, Moritz Kuhn, Petra Todd, Harald Uhlig, Sylvain Catherine, Jessica Wachter, and seminar participants for helpful comments. All errors are my own.
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1 Introduction

I document that upside earnings opportunities predict the start of stock market participation. Among workers who initially hold no stocks or mutual funds, a more upside-oriented earnings distribution predicts more entry at a given mean earnings growth, total variance and measured household resources. A one-standard-deviation increase in upside composition predicts 1.6 percentage points more entry, nearly one fifth of the sample's 8.82% annual entry rate. The relationship also extends to entry followed by ownership at later annual observations.

The economic question is how earnings uncertainty affects becoming an investor when it includes opportunities for gains as well as risks of losses. Earnings finance household investment, while uncertainty about those earnings can discourage financial risk taking. Workers facing the same earnings variance may therefore have different prospects for accumulating the resources needed to enter, depending on which side of the earnings distribution accounts for that variation.

The central empirical distinction is between the two sides of earnings variation. I divide the variance of age-adjusted earnings growth into upper and lower partial variances, measuring variation above and below mean growth. Their sum is total variance; their difference is upside composition. Upside composition positively predicts entry when variance and its square root are included in the regression. Both variance terms are individually significant, whereas a linear variance-only specification yields an imprecise coefficient. In a linear specification entering the upper and lower partial variances jointly, their coefficients have opposite signs. The equivalent composition coefficient is 1.11 percentage points per standard deviation, compared with 1.60 when the variance relationship is allowed to curve. The direction survives this simpler specification, although its magnitude and precision depend on the functional form.

Standard saving and costly-participation models give this distinction an economic interpretation. A two-period consumption and saving problem yields a resource threshold for investing: a favorable earnings realization can make paying the fee worthwhile. Mean and variance do not determine how often earnings cover a household's shortfall to that threshold. An analytical example therefore generates different entry rates at unchanged mean and variance without group-specific preferences or costs. In a repeated version, more entry can also produce more subsequent ownership, even if entrants are no more likely to survive conditional on entry. The aggregate sign depends on households' positions relative to the threshold; the example does not imply that every increase in upside composition raises participation.

The empirical comparison uses earnings histories measured before, and separately from, portfolio decisions. I construct earnings distributions from the Current Population Survey (CPS), using histories with origins in 2013–2017 and destinations through 2018. I assign moments from 58 industry–education–age cells to workers in the Survey of Income and Program Participation (SIPP), then follow portfolios from 2019–2023 baselines to the next year. The entry sample contains 27,419 initial-nonholder transitions. Separating the surveys prevents a worker's own realized earnings change from mechanically determining the assigned distribution. The estimates compare historical

earnings environments; they do not require treating those environments as measured individual beliefs.

Measurement matters because adverse earnings outcomes include leaving wage work. I use arc growth, which remains defined when positive wages fall to zero, and remove industry–education-specific age profiles following Catherine, Sodini, and Zhang (2024). Including zero-wage destinations substantially improves the reliability of measured composition and its agreement with independently constructed PSID earnings distributions. Cells with more zero-wage destinations have less upside composition and greater total variance, helping explain why composition and variance are negatively correlated across workers' earnings environments. The participation association remains after controlling for that frequency and measured cyclical exposure. Partial variances also give the comparison a transparent interpretation: composition changes which side of mean growth accounts for variation while total variance is held fixed. Conventional skewness captures closely related asymmetry. The advantage here is a more reliably measured representation in these survey cells and an exact variance decomposition, rather than an empirically distinct source of asymmetry.

The main competing explanation is sorting into earnings environments. The baseline conditions on resources, demographics and industry, education, age-band and year effects. The entry association remains with joint industry–education effects, controls for cyclical exposure, and stricter portfolio-reporting screens. Adding baseline earnings and realized earnings changes, including separate gains and losses, does not attenuate it on a common sample. In the PSID, reported risk tolerance rises with variance but not detectably with composition. These comparisons address observable differences in resources and risk exposure, while leaving room for unobserved sorting. The paper establishes a conditional association across earnings environments, not a causal response to an isolated change in earnings risk.

The subsequent ownership evidence makes the entry result more economically informative. Greater wage upside composition predicts entry followed by ownership at both subsequent annual observations in the smaller cohort with longer follow-up. Entry followed by exit also rises at the shorter horizon. Both outcomes are measured among initial nonholders: more entry can increase the number who later stay and the number who later leave, without changing entrants' likelihood of staying. The estimates of that conditional likelihood are imprecise. These exploratory comparisons use selected complete-followup cohorts and measure ownership at annual interviews. Among initial holders, the main wage measure predicts lower retention but no clear average change in risky shares; the split by holding history is imprecise. The strongest evidence concerns becoming an investor, rather than a general increase in portfolio risk taking.

The entry pattern also appears outside the SIPP and before the pandemic. Among younger households, pre-pandemic PSID transitions show positive entry associations with the same opposing tail signs as the SIPP. Ownership associations extend to the 2019 Survey of Consumer Finances and the earlier SIPP panel. The main sample uses recent SIPP releases following the Census Bureau's revision of asset processing, including stock and mutual-fund ownership imputation. The PSID

supplies an independent comparison under a different survey design and in earlier years. Section 4.7 brings together this evidence and the comparison across SIPP survey vintages.

The PSID also connects earnings opportunities to other household decisions. Among initial stockholders, upside composition predicts greater participation in IRA or private-annuity contributions, with a significantly stronger association than among nonholders. Furnishings and household equipment also have a positive association among initial holders in the primary transformed specification, whereas broad expenditure, selected nondurables and services, and food show no clear association. These comparisons motivate distinguishing financial participation and occasional resource commitments from average spending. They do not establish that the responses differ statistically across expenditure categories. The two-period analytical model and access experiments in the life-cycle model show how a discrete participation margin can move substantially with modest changes in average consumption. The estimated model has one consumption good; the furnishings evidence is complementary to its mechanism.

I then use a life-cycle model to test the empirical relevance of this mechanism when households anticipate future earnings and repeatedly choose consumption, saving and portfolios. The 58 observed earnings cells inform 30 industry–education career processes, with adjustments before and after age 45. These measured laws affect both realized resources and the future earnings prospects used to choose optimal saving and investment rules. Preferences, participation costs and entry opportunities are common across groups. A nonholder can receive a fresh opportunity each year; holders retain access and decide whether the recurring payment is worthwhile. Five financial parameters are estimated from eight entry, holder and wealth outcomes, excluding all earnings-risk coefficients. Observed wages, wealth and portfolios initialize the model, so the exercise evaluates conditional decisions rather than fitting an unconditional wealth distribution.

At common financial parameters and observed starting positions, the composition coefficient is 0.241 percentage point with cell dynamics, -0.005 with pooled dynamics, and 1.816 in the matched data. Pooling removes the cell adjustments while retaining non-Gaussian education-specific processes and resolves household choices without refitting financial parameters. Cell dynamics recover the positive composition association and opposing tail signs in the selected specification, although the magnitude remains much smaller than in the data. The comparison tests what measured earnings heterogeneity adds; it does not establish that standard portfolio models require cells to generate the mechanism. Indeed, the permanent-access alternative produces a positive composition association even under pooled laws.

Crossing the earnings laws with rules previously optimized under either law attributes most of the additional composition gradient to differences in resources and earnings states reached before the entry decision. Rule adjustment partly offsets that increment. This compares measured environments that differ in growth and employment risk as well as asymmetry. A separate controlled experiment reallocates dollar earnings variance between favorable and adverse outcomes while preserving expected disposable earnings and total dollar variance. In this illustration, more upside reduces desired saving at given resources while initially making stockholding slightly more attractive.

A five-year change eventually lowers cumulative entry; a change lasting throughout working life lowers wealth more, but its cumulative-entry response fades toward zero as the investment incentive continues. The two exercises therefore do not imply that expectations are always secondary. Together they show how familiar resource and saving mechanisms can produce different short- and longer-run participation responses, with independent future stock returns. The controlled experiment illustrates this interaction rather than matching the empirical coefficient.

Related literature. The main contribution is to establish the economic importance of upside earnings opportunities for stock market participation. The allocation of earnings variation between favorable and adverse outcomes predicts economically important differences in entry beyond mean growth, variance and measured resources. This finding connects three questions: which features of earnings risk matter for portfolios, how households become investors, and what portfolio models should explain beyond average holdings. The starting point is the role of nontradable income, borrowing constraints and background risk in portfolio demand (Viceira, 2001; Cocco, Gomes, and Maenhout, 2005; Kimball, 1993; Gollier and Pratt, 1996). Empirical work relates subjective risk, earnings histories, employer shocks and human-capital exposure to portfolios (Guiso, Jappelli, and Terlizzese, 1996; Angerer and Lam, 2009; Betermier et al., 2012; Fagereng, Guiso, and Pistaferri, 2018; d'Astous and Shore, 2024). More recent evidence links wrongful-discharge protections to participation (Ahn, Jo, and Lam, 2026) and downside income risk to portfolio skewness (Zhang, 2018). I bring favorable and adverse variation into the same prospective entry comparison, holding mean growth and variance fixed when assessing composition. This reveals an entry pattern that treating earnings risk only as dispersion does not recover.

The distinction from cyclical tail risk concerns the earnings distribution being measured. Catherine (2022) and Shen (2024) model countercyclical earnings risk and portfolio choice. Catherine, Sodini, and Zhang (2024) show that Swedish workers with greater left-tail income exposure when equity markets perform poorly hold less equity, examining both participation and risky shares. I study asymmetry in the earnings distribution itself: whether favorable or adverse variation predicts becoming an investor, conditional on mean growth, total variance and measured resources. My composition measure is constructed from earnings histories without using stock returns; it is not a co-moment between earnings risk and market returns. Controls for cyclical exposure leave its entry association intact. The model then studies a resource-and-participation mechanism that operates with independent future stock returns, complementing the joint-downturn mechanism.

Participation costs make entry distinct from the allocation of an existing portfolio (Vissing-Jorgensen, 2002; Gomes and Michaelides, 2005; Fagereng, Gottlieb, and Guiso, 2017). Gálvez (2023) finds larger participation responses to favorable persistent earnings shocks among low-income households; wealth changes and lottery windfalls also reveal differences between entry and participants' allocations (Brunnermeier and Nagel, 2008; Briggs et al., 2021, 2023). Work on participation dynamics relates income risk and income changes to entry and exit (Bonaparte, Korniotis, and Kumar, 2020; Bonaparte et al., 2025), while Galaasen and Raja (2024) document

short ownership spells and the role of poor initial investment experiences. I compare entire earnings environments measured outside the portfolio sample and distinguish entry followed by ownership from survival conditional on entry. This connects the distribution of earnings opportunities to subsequent market access without interpreting every additional entrant as a lasting investor.

Gálvez and Paz-Pardo (2026) provide the closest quantitative benchmark. Their model combines nonlinear, non-Gaussian earnings dynamics, housing and participation costs to explain stockholding and risky shares with lower estimated risk aversion. I document a composition association with subsequent entry and entry followed by ownership, then apply the empirical specification to conditional life-cycle predictions without targeting these coefficients. The model's role is to provide a familiar theoretical foundation and assess it against this distributional entry evidence. Measured earnings environments add a positive entry gradient in the selected specification, chiefly through resources, while the magnitude gap and controlled saving experiment delimit the quantitative interpretation. The model omits housing choice and distinguishes intermittent entry opportunities from permanent access. The contribution is empirical discipline on established mechanisms beyond their fit to average financial outcomes.

The measurement draws on evidence that earnings risk is non-Gaussian and changes over the life cycle (Guvenen, Ozkan, and Song, 2014; Guvenen et al., 2021; Arellano, Blundell, and Bonhomme, 2017), and that household and government insurance alter the risk borne by individuals (Busch et al., 2022). The decomposition parallels the distinction between upside and downside semivariances in financial markets (Barndorff-Nielsen, Kinnebrock, and Shephard, 2010; Segal, Shaliastovich, and Yaron, 2015). Conditional return asymmetry also matters for international portfolio allocation (Ghysels, Plazzi, and Valkanov, 2016); related work studies firm-level conditional skewness (Dew-Becker, 2024) and labor-market event risk (Schmidt, 2025). I use partial variances to make a constant-variance comparison of entry and assess its measurement across public surveys. Subjective-expectations research offers a complementary route (Wang, 2023; Caplin et al., 2026; Arellano et al., 2026; Giglio et al., 2021; Drerup, Wibral, and Zimpelmann, 2023). I examine perceived risk separately in the supplementary Survey of Consumer Expectations exercise, keeping the distinction between historical earnings distributions and personal beliefs explicit.

Section 2 develops the participation mechanism. Section 3 describes the data and empirical design. Section 4 presents entry, sustained ownership and external comparisons. Section 5 examines retirement-saving participation, expenditure, portfolio-change accounting and subjective risk. Section 6 estimates the conditional life-cycle model and evaluates its entry predictions. Section 7 concludes.

2 Earnings Composition and the Participation Margin

A familiar participation cost gives earnings composition an economic role. A favorable realization can make investing worthwhile by reducing the fraction of resources absorbed by the cost. The probability of reaching that point depends on the earnings distribution, not just its mean and

variance. I first show this mechanism in a two-period consumption and saving problem, then isolate its implications for entry and subsequent ownership. The mechanism uses standard household-portfolio ingredients (Vissing-Jorgensen, 2002; Fagereng, Gottlieb, and Guiso, 2017); it requires neither different preferences across earnings groups nor a particular number of groups. The earnings cells introduced below supply measured environments in which to evaluate it.

2.1 A two-period consumption and investment decision

An earnings innovation Y is realized before the household chooses consumption and its portfolio. Current resources are $m = w + Y > 0$, where w is predetermined wealth. The household can save safely or pay a participation fee $\kappa > 0$ to hold stocks. The safe net return is zero; stock returns are independent of earnings and offer a positive optimized expected log-return gain b. There is no borrowing, short selling or terminal nontradable income. With logarithmic utility and discount factor $\beta > 0$, the value conditional on participation $d \in \{0, 1\}$ is

$$V_d(m) = \max_{0<c<m-\kappa d} \{\log c + \beta[\log(m - \kappa d - c) + bd]\}. \tag{1}$$

The household optimally divides resources remaining after the fee between consumption and saving. Its consumption and participation choices are

$$c_d(m) = \frac{m - \kappa d}{1 + \beta}, \qquad d^*(m) = \mathbf{1}\{m \geq m^*\}, \qquad m^* = \frac{\kappa}{1 - \exp[-\beta b/(1 + \beta)]}. \tag{2}$$

Participation is infeasible when $m \leq \kappa$; indifference at m^* is resolved in favor of participating. Appendix A.6 gives the derivation.

The household invests when earnings cover its shortfall to the threshold: $\Pr(d^* = 1) = \Pr(Y \geq m^* - w)$. Mean and variance do not determine that probability. Two distributions with the same mean and variance can therefore produce different entry rates in the same decision problem. No group-specific fee or preference is needed. The sign depends on which earnings realizations cross the threshold, so higher upside composition does not universally increase entry. The life-cycle model in Section 6 adds future earnings uncertainty and repeated saving decisions to determine how this resource channel interacts with expectations and accumulation.

2.2 Repeated participation at a resource threshold

To isolate entry and subsequent ownership, consider a repeated version with saving and initial wealth fixed. A household begins with financial wealth w and observes an earnings innovation Y, giving resources $w_1 = w + Y > 0$. It receives safe nontradable income $h > 0$ at the end of the period. Earnings are realized before portfolio choice. Utility is logarithmic; the safe net return is zero; and the independent risky excess return is $\bar{r} \in (0, 1)$ with probability $p \in (1/2, 1)$ and $-\bar{r}$ otherwise. There is no borrowing or short selling. Holding stocks costs $\kappa > 0$ each period. Wealth w does not

accumulate, and innovations are independent across periods. This benchmark holds accumulation fixed to isolate the participation decision.

Let $b > 0$ denote the optimized log-utility gain from risky investment before the cost is deducted. Appendix A.5 derives b and the allocation rule. When the allocation at the participation boundary is interior, the household participates if

$$w_1 \geq w^* = \frac{\kappa}{1 - e^{-b}} - h, \tag{3}$$

with indifference resolved in favor of participating. The fee absorbs a smaller fraction of resources after a favorable realization. Unlike a one-time access cost (Gomes and Michaelides, 2005), a recurring cost makes current holders revisit the participation decision (Fagereng, Gottlieb, and Guiso, 2017).

The participation probability of a household with wealth w is therefore

$$\pi(w; F_Y) = \Pr(Y \geq w^* - w) = 1 - F_Y\big((w^* - w)^-\big). \tag{4}$$

The relevant feature of the distribution is its probability mass at or above the household's resource shortfall. Independent innovations make π both the household's entry and retention probability. Aggregate rates differ because holders disproportionately have high π: in a stationary population, entry is $\mathbb{E}[\pi(1-\pi)]/\mathbb{E}[1-\pi]$, whereas retention is $\mathbb{E}[\pi^2]/\mathbb{E}[\pi]$.

2.3 An upside rotation at fixed mean and variance

Consider two earnings laws, with $d > 0$:

$$Y_0 = \begin{cases} -d, & \text{with probability } 1/2, \\ d, & \text{with probability } 1/2, \end{cases} \qquad Y_1 = \begin{cases} -d/2, & \text{with probability } 4/5, \\ 2d, & \text{with probability } 1/5. \end{cases}$$

Both have mean zero and variance d^2. The second replaces a large possible loss with a smaller, more frequent loss and a larger, less frequent gain. Upper-minus-lower partial variance rises from zero to $3d^2/5$.

For example, set $d = \$1,000$. A household \$1,500 short of either participation threshold enters with probability zero under Y_0 and 20% under Y_1: only the \$2,000 gain covers its shortfall. A household just \$500 short instead goes from 50% to 20%, because the smaller frequent gain under Y_0 was already sufficient. The aggregate entry response therefore depends on the distribution of resources relative to participation thresholds. The next proposition specifies a population for which the upside rotation raises entry and sustained ownership.

There are $N_m > 0$ marginal households with wealth $w_m > d$ and $w^* - w_m \in (d, 2d]$, and $N_e > 0$ established households with $w_e \geq w^* + d$. Only the largest favorable realization carries a marginal

household across the threshold; established households participate in every state. These are wealth types, not definitions based on observed tenure.

Proposition 1 (Participation responses to an upside rotation)**.** *Suppose optimal positions are interior whenever participation is chosen. Comparing stationary populations under Y_0 and Y_1:*

(i) marginal households' participation probability rises from 0 to $1/5$, while established households' probability stays at 1. Entry among baseline nonholders rises from 0 to $1/5$;

(ii) retention among baseline holders falls from 1 to $(N_e + N_m/25)/(N_e + N_m/5)$. All exits are by marginal households, and $4/5$ of exiters entered in the preceding period.

Under Y_0, only established households hold stocks. Under Y_1, the holder pool also includes marginal households that drew $2d$. Each remains only if the favorable state recurs, with probability $1/5$. Entry rises while aggregate retention falls because the holder population changes, not because every incumbent becomes more likely to exit. This comparison is between stationary populations in different earnings environments. It does not rank survival among entrants across the two laws: the symmetric law produces no entrants in this example.

The same rotation increases entry followed by sustained ownership. Let D_t indicate participation, with $t = 0$ the initial date. For $k \geq 1$, independence conditional on household type gives the population probability

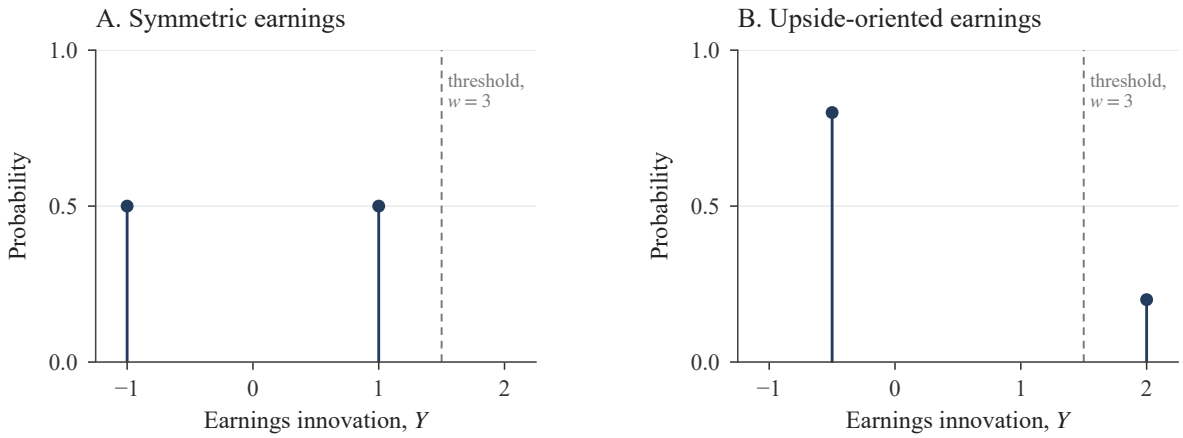
$$\Pr(D_1 = \cdots = D_{k+1} = 1 \mid D_0 = 0) = \frac{\mathbb{E}[(1-\pi)\pi^{k+1}]}{\mathbb{E}[1-\pi]}. \tag{5}$$

For the marginal households, the upside rotation raises this probability from zero to $(1/5)^{k+1}$: 4% for one follow-up period and 0.8% for two. Both sustained entry and entry followed by an interruption become possible. Dividing this joint probability by the entry probability gives survival among entrants, a different object. With persistent earnings or endogenous saving, resources in subsequent periods need not follow the independent, fixed-wealth benchmark; Appendix U illustrates how these features change duration.

Figure 1 illustrates the distributions and the resource threshold. The example in Appendix A.5 gives entry rates of 0% and 20%, and retention rates of 100% and 86.7%. Its threshold placement holds for an open interval of costs.

Empirical implications. The model links upside-oriented earnings to entry when favorable realizations bring the participation threshold within reach. This is not a monotone wealth effect: households far below the threshold never enter, those far above always participate, and the displayed rotation lowers participation for some intermediate wealth levels. Entry and sustained entry are therefore the central comparisons across earnings environments. Transitory gains can also generate short spells. Longer holding histories increasingly select established households, but observed tenure is only a proxy for wealth type; conditional on a known entry, the example's exit hazard is constant. Survival, earnings reversals and baseline resources help assess this particular transitory-income mechanism.

Figure 1. Earnings composition and the participation margin at fixed mean and variance



Notes: Panels A and B show earnings-innovation distributions with mean zero and variance one. The dashed line at $Y = 1.5$ is the innovation that carries a household with wealth $w = 3$ to the resource threshold $w^* = 4.5$. Neither realization of the symmetric law crosses the threshold; the upside law crosses it with probability 20%. The same household remains invested only if the favorable state recurs. Earnings are realized before allocation and saving is fixed. These are exact distributions from the illustration in Appendix A.5, not an estimated income process or a calibration to SIPP. Figure A.1 reports the corresponding participation and allocation rules.

The proposition establishes these signs for a specified rotation, not for every increase in upper partial variance. The rotation holds dollar-innovation moments fixed, whereas equal variance of adjusted log or arc growth need not preserve dollar-income variance. Earnings are realized before portfolio choice, so participation responds to realized resources. The proposition makes no prediction about unresolved background risk or an age cutoff. Appendix A.5 gives the allocation results and distinguishes these channels. Whether participation costs quantitatively account for the evidence is a separate question; windfall evidence and participation dynamics identify limitations of standard cost models (Briggs et al., 2021; Galaasen and Raja, 2024).

2.4 Financial participation and current consumption

A sizable participation response can coexist with a small average consumption response. Access to an investment opportunity is a discrete choice, whereas consumption is chosen continuously. Returning to the two-period problem in equations (1)–(2), consider two earnings environments with the same mean resources. Optimal consumption implies

$$\Delta \mathbb{E}[c] = -\frac{\kappa}{1+\beta} \Delta \Pr(d^* = 1). \tag{6}$$

An upside rotation that brings households across the access threshold can therefore change participation appreciably while changing mean current consumption by only the consumption-financed portion of the additional fees. Equal variance is not needed for this identity; the specified equal-mean, equal-variance rotation above supplies an example when its support straddles the new threshold. Appendix A.6 gives the derivation.

The identity concerns mean current consumption, not its variance or downside risk. Although the two earnings environments have equal mean resources, they can expose households to different consumption outcomes. Saving and future earnings also affect that exposure in a life-cycle problem. Section 5.1 therefore examines financial participation and observed expenditure together, while Section 6 allows consumption, safe saving and stockholding to adjust jointly over time. An IRA or private-annuity contribution is another discrete financial decision, although it need not increase equity exposure. The example allows a small consumption response but predicts neither exactly unchanged spending nor increased total saving, and leaves the allocation within retirement accounts unspecified.

The threshold logic also motivates examining occasional durable purchases. Transaction costs or indivisibilities can create an inaction region in which a household postpones a purchase (Eberly, 1994). When a favorable earnings realization carries resources into the purchase region, the timing of expenditure can change without a proportional increase in every spending category. This is an interpretation of the furnishings evidence in Section 5.1, rather than a durable-choice result derived from the present model. Its sign depends on the earnings distribution and the location of the purchase threshold, and it does not imply unchanged nondurable spending.

3 Data and Empirical Design

3.1 Earnings histories and stock market participation

I combine earnings histories from the Current Population Survey (CPS) with subsequent portfolio observations from the Survey of Income and Program Participation (SIPP). The CPS measures historical earnings distributions for different groups of workers; the SIPP records portfolio transitions among workers assigned to those groups. Using separate surveys prevents a worker's own earnings realization from mechanically determining the assigned risk measure.

The main CPS sample contains 88,862 validated annual earnings histories, including 5,490 with zero destination wages. Workers are aged 25–65 and have positive wages at the origin. Origins run from 2013 through 2017 and destinations through 2018, before the SIPP portfolio window. I use the public Census ASEC amounts at both endpoints and retain the redesigned 2014 questionnaire subsample. The wage measure includes transitions out of wage earnings, whether through nonemployment or a move to another income source; a zero wage is not a separately identified dismissal. Total labor earnings sum wages, net business earnings, and net farm earnings. The original positive-support comparison uses 83,372 histories with positive wages and total earnings at both endpoints. Appendix I.9 documents the linkage and treatment of signed earnings.

I group workers by harmonized industry and education. Within each of the 30 supported industry–education groups, the age bands 25–44 and 45–65 yield 58 supported earnings cells. The group determines the age profile of earnings growth; the finer cell determines the distribution around that profile. Appendix C reports the crosswalk and support rules.

The SIPP reports directly held stocks and mutual funds, other financial assets, and retirement balances. Let R_{it} denote stock-and-mutual-fund balances, F_{it} financial assets excluding retirement balances, and $s_{it} = R_{it}/F_{it}$ the nonretirement risky share. The public records do not reveal the asset composition of retirement accounts, and mutual funds need not consist entirely of equities. I aggregate eligible monthly records to annual observations and link consecutive years. The observations come from the 2020–2025 surveys: portfolio baselines refer to 2019–2023 and outcomes to 2020–2024. These releases follow the revised asset processing introduced with the 2018 SIPP, including model-based imputation of stock and mutual-fund ownership (U.S. Census Bureau, 2020). Section 4.7 compares the results with the earlier SIPP panel and with pre-pandemic PSID portfolios.

The entry sample contains 27,419 transitions beginning without stock or mutual-fund holdings. The prospective allocation sample requires positive broad and nonretirement financial denominators and usable ratios in $[0, 1]$ at both dates. It contains 31,569 transitions: 24,235 initial nonholders and 7,334 initial holders, including 2,165 who subsequently exit. Missing financial assets are not coded as zero shares. Table 1 describes this sample by initial ownership.

Baseline ownership is 23 percent in the analysis sample, compared with 19 percent of 2019 Survey of Consumer Finances (SCF) households with a head aged 25–64. The corresponding rates for college graduates are 34 percent in both surveys; for those without a degree they are 13 and 10 percent (Appendix Table S.23). The surveys differ in their population and sample selection. The SIPP unit is a worker transition involving household-reported assets. Changes in balances can reflect transactions, valuation, or household balance-sheet changes; they are not direct observations of stock purchases and sales. Appendix C.3 discusses the reporting unit and denominator support.

The PSID extends the analysis to spending and contributions to retirement-saving vehicles. I link the same reference person across adjacent biennial waves in 2005–2019, retaining 33,353 spending transitions among 9,183 people aged 25–65 at baseline. Official annualized expenditure components distinguish food, selected nondurable and service spending, and furnishings. Contributions are any reported money put into a private IRA or annuity, not the equity share in those accounts or a total saving rate. The regressions use baseline family weights and control for the earlier outcome, family income, liquid assets, homeownership, family size and demographics, with separate coefficients by initial non-IRA stock ownership. Section 5.1 explains the timing and Appendix V documents the measures and all specifications.

3.2 Age-adjusted earnings risk

The main measure uses nominal arc wage growth,

$$g_{i,t+1} = \frac{2(y_{i,t+1} - y_{it})}{y_{i,t+1} + y_{it}},$$

which equals -2 when destination wages are zero. Following the earnings-profile adjustment of Catherine, Sodini, and Zhang (2024), I remove a cubic in origin age separately within each industry–

Table 1. Prospective portfolio sample by initial ownership

	All		Nonholders		Holders	
	Mean	SD	Mean	SD	Mean	SD
Age (years)	43.74	11.27	43.56	11.31	44.34	11.09
Female (%)	47.47	49.94	49.82	50.00	39.71	48.93
College graduate (%)	49.99	50.00	43.12	49.52	72.65	44.57
Nonretirement financial assets ($1,000)	78.70	317.76	27.67	107.26	247.14	599.47
Retirement balances ($1,000)	129.46	411.21	83.29	310.00	281.89	616.26
Stocks and mutual funds ($1,000)	33.96	222.05	0.00	0.00	146.06	442.40
Baseline risky share (%)	10.76	25.12	0.00	0.00	46.27	32.73
Subsequent risky share (%)	11.18	25.77	4.27	16.86	33.99	35.22
Subsequent participation (%)	23.52	42.41	9.60	29.46	69.49	46.05
Person-year pairs	31,569		24,235		7,334	

Notes: Weighted means and standard deviations in the retained prospective sample, using outcome-year SIPP person weights. Ownership is measured at the baseline in 2019–2023; subsequent outcomes are in 2020–2024. Other characteristics and balances refer to the baseline. Dollar balances are nominal. The risky share is stocks and mutual funds divided by financial assets excluding retirement balances. Both dates have positive usable financial denominators and shares in $[0, 1]$; future zero holdings remain. The table describes selected consecutive person-year pairs, not a cross-sectional population of all households.

education group:

$$g_{i,t+1} = \theta_{0g} + \theta_{1g} a_{it} + \theta_{2g} a_{it}^2 + \theta_{3g} a_{it}^3 + \epsilon_{i,t+1}. \tag{7}$$

The regression uses CPS person weights. The residual is earnings growth net of the group's age profile. I apply the same procedure to total earnings and, for the original positive-support comparisons, to nominal arc growth and real log growth. Real log growth removes annual CPI-U inflation. The adjustment follows CSZ's method, but the object differs: historical pretax earnings composition, rather than the comovement of after-tax disposable income with stock returns.

For each industry–education–age cell c, let μ_c be mean residual growth. The lower and upper centered second partial moments are

$$V_c^- = \mathbb{E}_c[(\epsilon - \mu_c)^2 \mathbf{1}\{\epsilon < \mu_c\}], \qquad V_c^+ = \mathbb{E}_c[(\epsilon - \mu_c)^2 \mathbf{1}\{\epsilon > \mu_c\}]. \tag{8}$$

Total variance is $V_c = V_c^- + V_c^+$ and upside composition is $C_c^{\text{up}} = V_c^+ - V_c^-$. At fixed variance, higher composition reallocates variation from below to above mean growth. A specification using only total variance cannot distinguish these distributions. At fixed variance, an increase in composition combines a larger upper partial variance with a smaller lower partial variance. Section 4.2 compares the entry association of composition with specifications based on total variance.

Zero-inclusive wages lead the assigned-risk analysis because they measure cell composition more reliably and reproduce the cross-cell ranking in the PSID. I use the same wage measure for every portfolio outcome and report the positive-support measures for comparison. Section 4.10 reports the evidence behind this choice and when it was made.

3.3 Participation and allocation regressions

Stock market participation means reported ownership of stocks or mutual funds outside retirement accounts. Let H_{it} indicate that ownership. Entry is starting participation: $H_{it} = 1$ conditional on $H_{i,t-1} = 0$. I estimate

$$H_{it} = \beta_E \widetilde{C}^{\text{up}}_{c(i,t-1)} + Z'_{i,t-1}\gamma_E + u_{it}, \qquad H_{i,t-1} = 0. \tag{9}$$

A tilde denotes standardization using the construction's fixed source scale across CPS cells. For zero-inclusive arc growth and positive-support real log growth, I use the mean and standard deviation of the pre-adjustment cell moments, weighted by Kish effective cell size capped at 1,000. Positive-support nominal arc growth retains the fixed scales of the original positive-support design. The scales are unchanged across outcomes and initial ownership states. Consequently, the reported coefficient compares a one-standard-deviation increase in composition using this fixed scale, not a standard deviation of the variation remaining after controls.

Z includes variance, standard deviation, mean growth and its square, household controls, and outcome-year and baseline industry, education, and age-band effects. Including both variance and standard deviation permits curvature in dispersion. Household controls are age and its square, sex, log gross assets, transformed first-listed-job earnings, and an indicator for any eligible month with nonpositive first-listed-job earnings.[1] All main SIPP regressions use outcome-year person weights.

For allocation, every risk and control coefficient may differ by initial ownership:

$$\begin{aligned} s_{it} = {} & (1 - H_{i,t-1})\big(\beta_0 \widetilde{C}^{\text{up}}_{c(i,t-1)} + Z'_{i,t-1}\gamma_0\big) \\ & + H_{i,t-1}\big(\beta_1 \widetilde{C}^{\text{up}}_{c(i,t-1)} + Z'_{i,t-1}\gamma_1 + \phi s_{i,t-1}\big) + u_{it}. \end{aligned} \tag{10}$$

The common scaling permits a direct test of $\beta_1 - \beta_0$. Holders remain in their initial group if they exit, so the regression does not condition on continued participation. With the initial share controlled, the holder coefficient is also the composition coefficient in a regression of share changes. Replacing s_{it} with H_{it} yields entry and retention coefficients on the same sample.

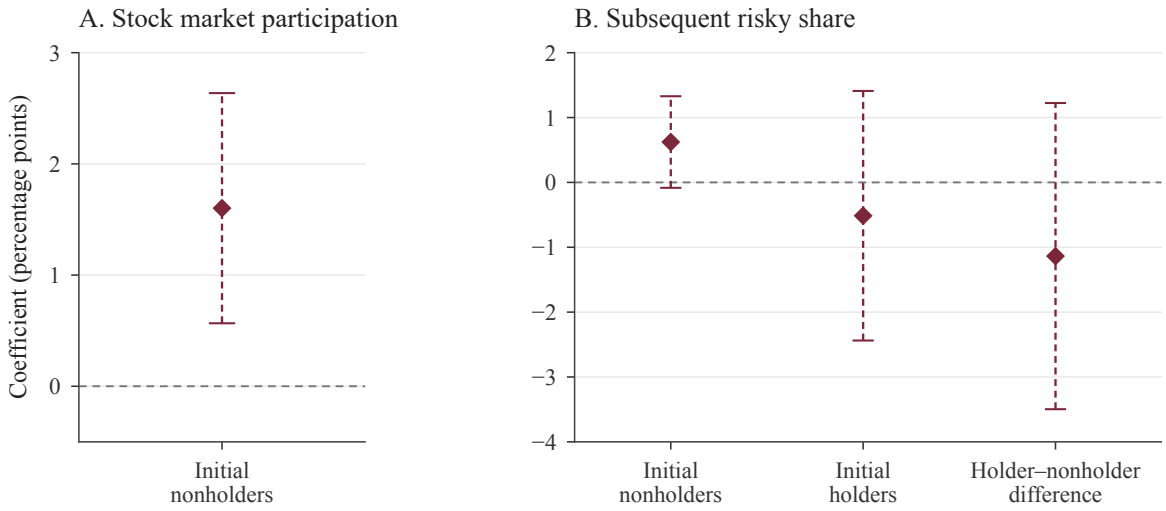
Main standard errors cluster by the 30 industry–education groups, with CR1 correction and a t_{29} reference distribution; restricted samples use their actual cluster count. Stars report unadjusted two-sided tests. The Online Appendix reports the full testing families, family-adjusted inference, finer-cell clustering, and remeasurement of the CPS moments. The analysis is exploratory. Predating the main portfolio window separates assigned earnings risk from contemporaneous own realizations, but does not make industry or education assignment exogenous. The coefficients are conditional associations; Section 4.8 examines the sorting explanations most relevant to them.

[1] For mean monthly job-1 earnings $\bar{y}^{(1)}$, the income control is $\log(\max\{\bar{y}^{(1)} + 1, 1\})$. Job 1 is the first-listed job, not necessarily the largest. The nonpositive-earnings indicator is not a separately observed unemployment spell. Appendix L gives the field definitions.

4 Earnings Composition and Stock Market Participation

Workers assigned greater upside composition are more likely to enter stocks and mutual funds, including entry followed by sustained ownership. I first report the magnitude and compare composition with total variance, then examine differences across households, portfolio outcomes, and surveys. The main estimates use zero-inclusive wage growth; positive-support measures provide comparisons.

Figure 2. Earnings opportunities and portfolio choice by initial ownership



Notes: Panel A reports starting stock market participation among initial nonholders. Panel B reports subsequent risky share by initial ownership and the directly estimated holder–nonholder difference. Points report composition coefficients per fixed pre-adjustment source standard deviation, holding total variance and the baseline controls fixed. The measure is the zero-inclusive wage construction of Section 3.2. Outcomes are dated 2020–2024. Entry has 27,419 initial-nonholder transitions; subsequent-share estimates use 24,235 initial nonholders and 7,334 initial holders, including later exits. Bars are 95% CR1/t_{29} intervals using 30 industry–education clusters, conditional on estimated CPS moments. Tables 2 and 5 report the specifications; Figure J.1 in Appendix J shows the original construction.

4.1 Starting stock market participation

At a given total earnings variance, a greater share of variation above mean growth predicts more entry. In Table 2, a one-standard-deviation increase in upside composition predicts 1.601 percentage points more entry across 27,419 baseline nonholder transitions. This is nearly a fifth of the 8.82 percent annual entry rate.

The positive association appears under all four earnings constructions. Zero-inclusive total earnings give 1.042 points; the original positive-support constructions give 0.875 point with real log growth and 0.695 with nominal arc growth. Observations, controls, and inference are unchanged across columns. Wild-cluster bootstrap inference supports the wage and positive-support entry results, with weaker evidence for zero-inclusive total earnings (Appendix Table S.20). This comparison addresses the modest number of industry–education groups.

Table 2. Earnings composition and starting stock market participation

	(1)	(2)	(3)	(4)
	Zero-inclusive wages	Zero-inclusive total earnings	Original real log	Original nominal arc
Earnings	Wages	Total	Total	Total
Growth	Arc, zeros kept	Arc, zeros kept	Real log, positive	Arc, positive
Earnings origin years	2013–2017	2013–2017	2013–2017	2013–2017
Upside composition	1.601^{***}	1.042^{**}	0.875^{***}	0.695^{***}
	(0.506)	(0.405)	(0.279)	(0.248)
Conditional-SD magnitude (pp)	0.590	0.506	0.573	0.534
Split-half reliability of composition	0.73	0.62	0.28	0.24
Group-specific cubic age adjustment	Yes	Yes	Yes	Yes
Risk and baseline controls	Yes	Yes	Yes	Yes
Industry, education, age-band, year FE	Yes	Yes	Yes	Yes
Nonholder transitions	27,419	27,419	27,419	27,419
Observed entry rate (%)	8.82	8.82	8.82	8.82
Assignment cells / inference groups	58 / 30	58 / 30	58 / 30	58 / 30

Notes: Weighted linear probability models of next-year stock-or-mutual-fund entry among baseline nonholders, using outcome-year SIPP person weights; baselines are 2019–2023 and outcomes 2020–2024. All four constructions use CPS earnings origins in 2013–2017 and destinations in 2014–2018, before every portfolio observation. Columns (1)–(2) use nominal arc growth of wages and of total earnings that keeps transitions to zero earnings (a zero destination contributes -2); columns (3)–(4) use the original construction on positive endpoints (Section 3.2). Survey-weighted group-specific origin-age cubics are removed before constructing centered moments in 58 industry–education–age cells; composition is upper minus lower partial variance. Coefficients are percentage points per fixed pre-adjustment source SD of each construction; the conditional-SD row uses residual composition after controls. Variance, SD, mean growth and its square are controlled; Section 3.3 specifies household controls and FE. Split-half reliability is the Spearman–Brown reliability of the cell composition across the 58 cells (Appendix S). Stars use unadjusted two-sided tests with CR1 standard errors clustered by 30 industry–education groups and t_{29}: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Inference conditions on the CPS moments; the complete construction comparison, including wages pooled over 2013–2021, is in Appendix S. Wild-cluster bootstrap p-values for the composition coefficients (null imposed, 1,999 Rademacher draws over the 30 groups) are 0.039, 0.080, 0.022 and 0.019 for columns (1)–(4) (Appendix Table S.20).

The magnitudes depend on the increment in composition being considered. The reported coefficients use its unconditional standard deviation across CPS cells. Scaling instead by the standard deviation remaining after the other regressors gives 0.590 point for wages, 0.506 for total earnings, and 0.573 and 0.534 for the positive-support real-log and nominal-arc constructions. For comparison, in the prospective regression using the positive-support real-log construction, a standard deviation of log financial assets predicts 3.2 points more entry and a standard deviation of log income 1.0 point, against 0.97 point for composition (Appendix Table S.27).

4.2 Composition beyond total variance

A linear specification containing total variance but omitting composition and the standard deviation gives an entry coefficient of -0.457 percentage point per source standard deviation (SE 0.584). Entering composition instead gives 1.091 points (SE 0.532). In the full specification, which includes composition, variance, and the standard deviation, the composition coefficient is 1.601 points (SE

0.506). Table S.18, Panel A, reports these comparisons. Upside composition thus predicts entry conditional on earnings dispersion.

Dispersion also has a nonlinear relationship with entry. In the full wage regression, the coefficients on V and $\sqrt{V}$ are -15.380 (SE 3.505) and 15.762 (SE 3.456) percentage points per respective source standard deviation, both with $p < 0.001$. Together they describe a concave variance–entry relationship. The imprecise coefficient in the linear variance-only specification therefore does not establish that dispersion is unrelated to entry. The composition coefficient measures the association with a more upside-oriented earnings distribution at a given level of dispersion.

Table 3 asks whether the positive composition association depends on allowing curvature in variance. Column (1) retains the baseline specification. Column (2) drops only $\sqrt{V}$, leaving composition and total variance with the same sample, weights, controls and fixed effects. Composition predicts 1.111 percentage points more participation among initial nonholders (SE 0.526, $p = 0.044$), while the variance coefficient is 0.053 point (SE 0.548). The positive association therefore remains in the simpler linear specification, with a smaller magnitude and less precision.

Column (3) writes that linear specification directly in the upper and lower partial variances, without composition, total variance or its square root as additional regressors. An increase of 0.01 in upper partial variance, holding lower partial variance fixed, predicts 0.331 percentage point more participation (SE 0.167, $p = 0.056$); the corresponding lower-partial-variance coefficient is -0.324 point (SE 0.154, $p = 0.044$). The common increment is measured in squared units of arc growth. Columns (2) and (3) are equivalent linear specifications with identical fitted values, not independent evidence. These coefficients come from the linear comparison, rather than local derivatives of the nonlinear baseline. Their signs support the distinction between favorable and adverse earnings variation, while the change from 1.60 to 1.11 points shows why the magnitude should not be treated as invariant to the variance specification.

Composition and conventional skewness capture closely related asymmetry. Their partial correlation is 0.91 after controlling for wage dispersion, against an unconditional correlation of 0.05; split-half reliability is 0.73 for composition and 0.39 for skewness. Keeping the headline variance and SD controls, replacing composition with skewness gives 0.91 point of entry. Entering both gives a composition coefficient of 2.58 points, with both slopes imprecise (Appendix Table S.22). The 1.32-point composition estimate with skewness omits the variance and SD controls (Table S.21, Panel A). Partial variances are more reliably measured and sum exactly to total variance. The joint regression does not establish that composition contains information beyond skewness. Appendix Lemma 3 connects the partial variances to bounds on tail probabilities.

4.3 Economic drivers of earnings asymmetry

Leaving wage work helps account for the cross-cell pattern in earnings risk. The share of CPS histories ending at zero wages correlates at -0.855 with upside composition and 0.924 with total variance. Cells with more zero destinations therefore tend to combine greater variance with more variation below the mean. This is a descriptive feature of the measured distribution: zero annual wages can

Table 3. Earnings composition, variance, and direct tail coefficients

	(1) Curved variance	(2) Linear variance	(3) Direct tails same fit as (2)
Upside composition C (per source SD)	1.601^{***}	1.111^{**}	
	(0.506)	(0.526)	
Total variance V (per source SD)	-15.380^{***}	0.053	
	(3.505)	(0.548)	
Standard deviation $\sqrt{V}$ (per source SD)	15.762^{***}		
	(3.456)		
Upper partial variance V^+ (per 0.01 growth-squared unit)			0.331^{*}
			(0.167)
Lower partial variance V^- (per 0.01 growth-squared unit)			-0.324^{**}
			(0.154)
Mean growth and its square	Yes	Yes	Yes
Baseline controls and fixed effects	Yes	Yes	Yes
Initial-nonholder transitions	27,419	27,419	27,419
Industry–education clusters	30	30	30
Annual entry rate (%)	8.82	8.82	8.82

Notes: Weighted linear probability models of next-year stock-or-mutual-fund participation among initial nonparticipants; coefficients are percentage points. The SIPP observations, outcome-year person weights, household controls and industry, education, age-band and year effects are identical across columns, with baselines in 2019–2023 and outcomes in 2020–2024. Risk measures use zero-inclusive, age-adjusted CPS wage growth from 2013–2017 origins. Column (1) is the main specification. Column (2) omits only $\sqrt{V}$. Column (3) reports the coefficients on V^+ and V^- of that same linear specification: because $C = V^+ - V^-$ and $V = V^+ + V^-$, columns (2) and (3) have identical fitted values. Thus column (3) is neither an additional regression nor a local derivative of column (1). Composition and variance in columns (1)–(2) use their fixed source SDs, 0.033913 and 0.135425; the tail rows use the common raw increment 0.01, rather than tail-specific SDs. Standard errors in parentheses use industry–education CR1 clustering and t_{29} inference, conditional on the CPS moments. Stars denote unadjusted two-sided tests against zero: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. The linear comparison was added in response to the functional-form question; it does not replace the main specification.

reflect nonemployment or a move to another income source and do not separately identify dismissal. Appendix S.1 illustrates the pattern with two worker groups that have almost the same total variance but different allocations between the upper and lower sides.

The participation association is not absorbed by the frequency of zero wages. Adding the cell's zero-destination share to the full design leaves a composition coefficient of 2.420 percentage points (SE 0.911); the zero-share coefficient is imprecise at 2.630 points (SE 2.030). The zero-share-only coefficient is also imprecise, at -1.163 points (Table S.18, Panel B). Zero-wage transitions help describe the variation in earnings environments, but their frequency alone does not account for the positive composition association. This comparison differs from the dismissal-risk quasi-experiment of Ahn, Jo, and Lam (2026).

Measured cyclical exposure does not absorb the relationship either. Adding the cell's mean wage growth in 2020 gives a composition coefficient of 1.783 points (SE 0.528). Controlling separately for historical PSID exposures of mean growth, lower partial variance, or composition to changes in national unemployment gives 1.612, 1.701 and 1.662 points; including all three gives 2.104 points

(SE 0.452). The exposures use biennial PSID cell moments over 1999–2019. These comparisons retain the headline sample, controls and weights (Table S.21, Panel B). They distinguish the association from these measured cyclical exposures without identifying a causal contribution of particular labor-market events.

The evidence is less decisive about persistent versus transitory earnings changes. An exploratory PSID variance-decomposition proxy initially gives a larger composition association in cells with a greater transitory share, but the interaction becomes smaller and imprecise after excluding very large log-wage changes. Appendix S.1 reports the nominal, real and large-change-screen comparisons together. These results do not establish that promotions, persistent job losses or transitory shocks drive a particular share of the participation association. The life-cycle model separately examines how realized earnings and the anticipated earnings distribution can affect resources, saving and participation.

4.4 Entry and sustained market access

The entry association extends to ownership at later annual observations. Table 4 divides entry into two joint outcomes: entry followed by sustained ownership and entry followed by an interruption. Within each panel, all outcomes use the same initial nonholders, weights and controls. Both outcomes are measured among all eligible initial nonholders, so differences do not arise from selecting different groups of entrants.

In Panel A's 9,236 initial-nonholder transitions, a one-standard-deviation increase in upside composition predicts 2.086 percentage points more entry. Of this, 0.978 point is entry followed by ownership at the next annual observation, and 1.108 points is entry followed by exit. The sustained component is suggestive, and the difference of -0.131 point between the two components is imprecise. The data do not establish that the association is concentrated in short spells.

Panel B follows 3,254 initial-nonholder transitions through two post-entry observations. Composition predicts 1.894 percentage points more entry with ownership at both dates. The remaining entry coefficient is 1.421 points, including any interruption or exit followed by reentry. This smaller, earlier complete-followup cohort extends the finding to successive annual dates; its larger coefficient does not establish an increasing association with duration. Annual observations also cannot establish uninterrupted ownership between interviews.

More sustained entry need not mean higher survival among entrants. The conditional-survival estimates in Table 4 are imprecise at both horizons and establish neither a survival advantage nor a penalty. As equation (5) illustrates, more households can enter and remain invested without a detectable change in entrants' survival rate.

Follow-up selection limits these exploratory comparisons. Missing ownership is not coded as exit, and follow-up must meet the worker-sample eligibility rules. Composition predicts eligible-followup availability by 1.820 percentage points, so complete cases need not represent all initial nonholders. Requiring reported ownership and balances at all three dates lowers the entry estimate to 0.921 point; both entry and sustained entry are then imprecisely estimated. This is a different restriction from

Table 4. Earnings opportunities and sustained market access

	(1) Entry	(2) Sustained entry	(3) Interrupted entry
A. One annual follow-up			
Upside composition	2.086***	0.978*	1.108**
	(0.697)	(0.501)	(0.512)
Outcome mean (percent)	9.26	4.74	4.53
Observations	9,236	9,236	9,236
Sustained minus interrupted		-0.131	
		(0.735)	
Conditional-survival gradient		-0.959	
		(4.036)	
B. Two annual follow-ups			
Upside composition	3.315**	1.894**	1.421
	(1.430)	(0.786)	(1.053)
Outcome mean (percent)	10.17	3.53	6.63
Observations	3,254	3,254	3,254
Sustained minus interrupted		0.473	
		(1.187)	
Conditional-survival gradient		7.296	
		(5.674)	
Controls and fixed effects	Yes	Yes	Yes
Industry–education clusters	30	30	30

Notes: Initial nonholders with ownership observed at entry and each required annual follow-up. Panel A covers entry years 2020–2023; Panel B covers 2020–2022 and a smaller complete-follow-up cohort. Sustained ownership requires holdings at every follow-up; interrupted ownership includes any observed interruption after entry. Nonentrants have zero for all three outcomes. Coefficients are percentage points per one-standard-deviation increase in predetermined zero-inclusive wage composition, holding variance fixed. All models retain the controls, fixed effects and outcome-year survey weights of equation (9). Columns (2) and (3) add to column (1); the first comparison subtracts their coefficients. The survival gradient is the local derivative of fitted sustained entry divided by fitted entry at the weighted mean design, not a finite one-standard-deviation change. Both comparisons use full joint covariance, including fitted-level uncertainty for the derivative. Standard errors in parentheses use CR1 clustering by 30 industry–education groups, conditional on the assigned CPS moments. Stars use unadjusted two-sided t_{29} tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Follow-up availability is related to composition; missing follow-up is not coded as exit. The panels do not identify a duration effect. Appendix U reports the complete exploratory family.

the headline's two-date reporting checks. Excluding sequences with a first follow-up in 2022 retains positive entry and no established survival penalty. Appendix U reports the complete construction, reporting and resource comparisons. Composition predicts entry followed by sustained ownership; its association with survival conditional on entry remains imprecise.

4.5 Participation costs and household resources

Participation costs make the resource distribution relevant for entry. In the model, favorable earnings realizations allow households near a resource threshold to enter. Financially wealthy nonholders provide a comparison: a fixed participation cost is small relative to their resources.

The positive-support real-log and nominal-arc constructions give larger entry coefficients among less wealthy nonholders. Between the 25th and 75th percentiles of financial assets, the real-log coefficient declines from 1.97 to 1.27 percentage points and the arc-growth coefficient from 0.68 to 0.18 point. The wealth interactions are imprecise, so the estimates are consistent with a participation threshold but also allow similar associations across wealth levels (Appendix Figure S.1 and Tables S.7–S.8).

Education gives a similar ordering. Among workers without a college degree, whose entry rate is 5.4 percent, the wage-composition coefficient is 1.98 points. Among college graduates, whose entry rate is 13.9 percent, the estimate is imprecise at 0.13 point. The wage difference of -1.85 points is also imprecise; the positive-support real-log difference of -1.34 points is only marginally significant (Table S.18, Panel C). The larger estimate among workers without a degree is consistent with participation costs mattering more at low resources, although education also differs with other household characteristics.

Age gives no uniform ordering within the SIPP. In the positive-support interaction design, real log growth gives a larger entry coefficient among younger households, but nominal arc growth gives a larger coefficient among older households. Separate wage regressions give positive coefficients in both bands, 2.65 points at ages 25–44 and 2.18 at ages 45–65. The estimated age pattern therefore depends on the earnings measure. Section 4.7 compares these age-band results with the PSID.

Observed earnings progression does not absorb the entry association. On 23,261 initial nonholder transitions with annual personal earnings at both dates, adding baseline earnings and the realized change moves the wage-composition coefficient from 1.518 to 1.574 percentage points. Separate gain and loss slopes give 1.569 points. The samples are identical and retain zero and negative destination earnings. The corresponding coefficient changes are 0.056 and 0.051 point (Appendix Table T.1). Since realized earnings may themselves reflect the earnings environment, these conditional comparisons do not identify a direct risk effect or households' perceptions.

Entry accompanied by a large earnings gain is one part of this relationship. Defining a gain as at least 25 percent growth in real annual personal earnings, it accounts for 0.546 point of the matched sample's 1.518-point composition coefficient; other entry accounts for 0.972 point, and their difference is imprecise (Appendix Table T.2). These joint outcomes do not identify mediation: gain-entry can rise because entry rises. In the model, gains and composition are complementary explanations, since composition changes the probability of crossing a resource threshold at fixed mean and variance.

The income-reversal explanation for short spells remains unconfirmed. Among 802 first observed entrants with complete follow-up, mean real earnings rise after entry for both exiters and retainers. Exiters' change exceeds retainers' by \$3,001, but the difference is imprecise (Appendix T). Among

entrants with a real earnings gain of at least 25 percent, retaining that income elevation predicts an imprecise -0.006 percentage point difference in retention. The composition interaction with pre-entry nonretirement financial assets is also imprecise (Appendix U). The earnings profiles do not establish income reversal among entrants who exit. They concern a selected group of entrants, whereas the sustained-entry estimates compare all eligible initial nonholders.

4.6 Initial holders: allocation and exit

Does greater entry accompany greater investment by existing holders? Table 5 allows all risk coefficients, controls, and fixed effects to differ between 24,235 initial nonholders and 7,334 initial holders. Baseline share also enters the holder equation.

Table 5. Earnings composition and subsequent portfolio allocation

Composition	(1) Zero-inclusive wages	(2) Original real log
Outcome: next-year nonretirement risky share		
Upside composition: baseline nonholders	0.622^{*}	0.360^{**}
	(0.345)	(0.142)
Upside composition: baseline holders	-0.514	-1.621^{**}
	(0.941)	(0.715)
Difference: holder minus nonholder	-1.136	-1.980^{**}
	(1.154)	(0.737)
Outcome: next-year ownership		
Nonholder entry	1.646^{**}	0.969^{***}
	(0.599)	(0.302)
Holder retention	-3.720^{**}	-2.650^{***}
	(1.708)	(0.952)
Difference: holder minus nonholder	-5.365^{**}	-3.619^{***}
	(2.098)	(1.070)
Baseline-nonholder observations	24,235	24,235
Baseline-holder observations	7,334	7,334
All risk and other controls $\times$ baseline status	Yes	Yes
Baseline share in holder block	Yes	Yes

Notes: Weighted OLS on the prospective sample of 31,569 consecutive person-year pairs (baselines 2019–2023, outcomes 2020–2024) with positive financial denominators and valid ratios in $[0, 1]$ at both dates; genuine future zeros, including exits, remain. The share is stock-and-mutual-fund balances divided by financial assets net of retirement balances. Every risk coefficient, household control and fixed effect is interacted with baseline ownership, and the baseline share enters the holder block. Coefficients are percentage points per fixed source SD of upside composition under each construction (Table 2); the difference row tests the holder-minus-nonholder slope directly using the saved covariance. Weights are destination-year person weights. Parentheses give CR1 standard errors clustered by 30 industry–education groups (t_{29}, conditional on the CPS moments): $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. The original-construction table with 58-cell sensitivity rows is Table R.1 in Appendix R.

The participation associations differ by initial ownership. Wage composition predicts 1.646 percentage points more entry and 3.720 points less holder retention, with a directly estimated

retention-minus-entry contrast of -5.365 points. The positive-support real-log construction gives entry of 0.969 point, retention of -2.650 points, and a contrast of -3.619 points. The wage retention association is not distinguishable from zero under the wild-cluster bootstrap, whereas the positive-support real-log association remains significant. Greater entry therefore coexists with lower estimated retention among initial holders.

The wage measure does not establish a decline in holders' risky shares. It predicts a 0.622-point higher subsequent share among nonholders and a 0.514-point lower share among holders; the latter and their difference are imprecise. The estimates are too imprecise to rule out a moderate decline. Positive-support real log growth gives 0.360 and -1.621 points, with a difference of -1.980 points. Positive-support nominal arc growth also predicts a higher nonholder share, but not a precise holder response. These regressions retain future zeros, so even the positive-support real-log construction's negative holder coefficient combines exit, reallocation, and valuation changes rather than identifying sales by continuing investors.

Exiters differ substantially from continuing holders. Their median risky balance is $12,000 rather than $25,000, and their median risky share is 0.375 rather than 0.456. Among those with prior history, 49.6 percent of exiters entered in the previous year, compared with 22.8 percent of stayers (Table S.13).

Appendix S examines how the portfolio associations differ between recent entrants and established holders. The recency comparisons depend on the earnings construction and are inconclusive for zero-inclusive wages. Established holders' positive-support real-log share estimate is too imprecise to rule out a modest decline.

Short spells are common. The exit hazard is 46.6 percent for last year's entrants, 33.0 percent in the second year, and 15.9 percent after three or more years (Table S.19), an ordering also documented by Galaasen and Raja (2024). Even the strictest reporting sample has 41.4 percent of entrants leave within a year. The composition coefficients for entry followed by exit and for persistent entry are not precisely distinguishable (Table S.17). Neither this comparison nor initial-holder retention establishes lower survival among upside-related entrants; Table 4 shows that wage composition also predicts sustained entry.

Most exits do not coincide with the acquisition of another large risky asset: 2,042 of the 2,165 exits involve neither business entry nor home purchase (Table S.11). Their small starting balances and infrequent business entry or home purchase suggest that many exits close small stock positions.

4.7 External evidence: the PSID and pre-pandemic surveys

Pre-pandemic PSID portfolios provide an independent test of whether the entry pattern is specific to the SIPP or to the years after COVID-19. The PSID records previous-calendar-year labor income and biennial direct, non-IRA stock ownership (Panel Study of Income Dynamics, 2023). Its 1999–2019 reference-person sample provides 45,840 two-year transitions at ages 25–65. The CPS earnings histories used for the ranking overlap or postdate many of these earlier portfolio transitions. The clearest

agreement with the SIPP is among younger households. The initial design preceded estimation; the age-band and window comparisons followed the pooled results and are exploratory.

Table 6. Entry and stock ownership by age band in two surveys

	(1) Zero-inclusive wages 2013–2017	(2) Zero-inclusive total earnings 2013–2017	(3) Original construction positive-support real log
A. Entry, percentage points per source SD of composition			
SIPP, ages 25–44 (annual, 2019–2024)	2.648^{***}	2.946^{***}	2.331^{***}
	(0.526)	(0.350)	(0.483)
PSID, ages 25–44 (two-year, 2009–2017 baselines)	3.564^{***}	2.134^{***}	1.308^{***}
	(0.728)	(0.448)	(0.377)
SIPP, ages 45–65	2.177^{***}	2.963^{***}	0.424
	(0.772)	(0.880)	(0.530)
PSID, ages 45–65 (two-year, 2009–2017 baselines)	0.442	-0.024	-0.036
	(1.302)	(1.077)	(0.634)
B. Stock market ownership at the baseline, percentage points per source SD			
SIPP, ages 25–44	3.109^{*}	2.791^{**}	1.803
	(1.710)	(1.024)	(1.110)
PSID, ages 25–44	6.317^{***}	4.893^{***}	2.026^{***}
	(0.809)	(0.615)	(0.515)
SIPP, ages 45–65	3.083	-1.388	0.112
	(2.425)	(2.258)	(1.293)
PSID, ages 45–65	1.801	-0.047	-0.238
	(2.163)	(1.624)	(0.829)

Notes: SIPP rows refit the entry design of Table 2 (Panel A) and a baseline-ownership regression with the same controls (Panel B) within age bands; ownership refers to stocks and mutual funds. PSID rows assign the CPS moments to reference persons aged 25–65 in the public PSID family files and fit the prospective design to two-year transitions in imputed direct (non-IRA) stock ownership (Panel A) and baseline ownership on the same moments with household controls, wave and industry effects (Panel B, baselines 1999–2017). Percentage points per fixed source SD of composition; CR1 standard errors clustered by the 29 industry–education groups represented in each age band appear in parentheses. Stars use unadjusted two-sided t_{28} tests conditional on the CPS moments: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. The age-band comparison was specified after a pooled null result and is exploratory; Table S.32 reports the measure agreement, the pooled and recent-window transitions and retention by band.

Among households aged 25–44, upside composition predicts entry in both surveys. Wage coefficients are 2.65 percentage points in annual SIPP transitions and 3.56 points in PSID transitions with 2009–2017 baselines. Original real log growth gives 2.33 and 1.31 points; zero-inclusive total earnings give 2.95 and 2.13. Restricting the PSID to 2013–2017 baselines retains positive entry associations of 2.90 points for wages and 1.50 for original real log growth (Table S.25). Young-household entry thus has a positive composition association in both surveys, including before the pandemic.

Ownership levels extend the comparison to other pre-pandemic sources. Among households under 45, wage composition predicts 3.11 points more ownership in the SIPP and 6.32 in the PSID, against rates of 23 and 18 percent. In the 2019 SCF, it predicts 5.0 points more ownership of any equity, including retirement accounts, and 3.1 points more direct stock-or-stock-fund ownership

among working heads aged 25–64; the direct-ownership association is marginally significant. In SIPP 2014 reference years 2013–2016, ownership among workers under 45 is positive in all four constructions, including 1.80 points with wages and 1.43 with original real log growth (Table S.24).

The PSID also reproduces the cross-cell ranking of zero-inclusive wage composition. Of 50,013 candidate PSID growth pairs, 43,805 contribute to pooled wage moments in the 58 matched cells. Their composition correlates at 0.33 with CPS composition, or 0.47 after correction for the two split-half reliabilities. The recent window gives raw and corrected correlations of 0.33 and 0.53. Original real-log composition is not reproduced: correlations are -0.19 pooled and -0.32 in the recent window (Table S.32). Including zero wages thus changes which earnings environments are ranked as having greater upside composition.

The comparison is strongest for younger households' entry and ownership. At ages 45–65, ownership estimates are imprecise and PSID entry estimates are small, whereas SIPP wage and total-earnings entry coefficients are positive, at 2.18 and 2.96 points. The all-age PSID entry coefficient is an imprecise -0.18 point, against 1.60 in the main SIPP sample. Holder transitions also differ: young-holder retention falls with composition in the SIPP but rises in the PSID, which does not reproduce recent-entrant exit. Within-person PSID ownership is positive but marginally significant, at 0.98 point per standard deviation; the sibling estimate is an imprecise -0.57 point (Tables S.32 and S.33). The age comparisons are exploratory and do not establish a sharp age gradient in SIPP wage entry alone.

Comparing annual entry across SIPP vintages requires attention to how asset ownership is recorded. The recent sample follows the Census Bureau's introduction of model-based imputation for stock and mutual-fund ownership in the 2018 SIPP (U.S. Census Bureau, 2020). These processing changes are a reason to examine survey comparability, rather than attribute every difference between panels to household behavior. They do not by themselves establish that measurement explains the earlier result. In the SIPP 2014 panel, the primary holdout test, specified before estimation for wages at ages 25–44, gives an imprecise -0.50 point, against 2.65 points in the later sample; their 3.14-point difference is significant (Table S.25).

Common annual ownership, income, wealth and record-aggregation definitions give an imprecise -0.15 point in the earlier SIPP and 2.73 points in the later one, with a significant difference of 2.88 points. Three symmetric comparisons—a liquid-asset control, ownership answers or positive balances, and December worker snapshots—also retain significant differences (Table T.4). Reweighting toward a common overlapping population balances earnings, gross and liquid assets, demographics and earnings-group representation. Coefficients become -0.03 point in the earlier panel and 2.60 points in the later panel. The 2.62-point gap is marginally significant, while the change in the gap is imprecise (Table T.5). The earlier SIPP entry result therefore remains different after harmonizing observed definitions and characteristics.

The PSID is especially useful for interpreting this comparison: it supplies positive young-household entry associations in overlapping pre-pandemic years. The SCF and earlier SIPP extend the ownership

evidence. Together, these sources show that the relationship between earnings composition and market access reaches beyond the main SIPP sample and its pandemic-era portfolio window.

4.8 Stability and occupational sorting

Common industry-year developments and persistent occupational differences are two alternatives to the earnings-opportunity interpretation. Adding industry-by-year and education-by-year effects leaves the wage entry coefficient at 1.663 percentage points; joint industry–education effects give 1.865 points (Table 7). The latter uses within-group age-band differences, controlling for the common age-band effect. Age-specific sorting into industries remains possible.

Portfolio reporting is a separate concern. Restricting both dates to reported ownership and balance components gives a wage entry coefficient of 1.699 points; additionally requiring self-reported interviews and stable household and residence records gives 2.557 points. Positive-support real-log estimates are 0.699 and 1.008 points. Holder retention becomes smaller and imprecise in both screens; wage holder-share coefficients are -0.480 and -2.275 points, the latter marginally significant. Since these restrictions also select households, their sensitivity does not establish that full-sample exits are erroneous.

Previous switching matters more for exit than entry. In the positive-support real-log construction, controlling for a prior switch on identical observations reduces the exit association by 0.814 point, with little change in entry. Every outcome-year deletion retains positive entry and negative retention, and early- versus late-period differences are imprecise (Appendix R).

Observed risk tolerance does not absorb the PSID composition association. Among reference persons who answered the 1996 hypothetical income gambles, tolerance rises with assigned earnings variance, but its composition coefficient is small and imprecise at 0.05. Adding the tolerance categories moves the composition coefficients by less than 0.1 point (Table S.23). The estimate still allows economically relevant sorting. The within-person ownership estimate is positive but only marginally significant, at 0.98 point per standard deviation, while sibling estimates are imprecise.

A common preference for greater earnings dispersion does not by itself explain the positive composition association conditional on dispersion or its persistence within industry–education groups. Preferences for upside-oriented earnings, financial sophistication, and age-specific sorting nevertheless remain possible. Earnings cells are not randomly assigned.

4.9 Construction comparisons

Entry is more consistent across constructions than the holder-share association. In the positive-support family, entry remains positive after the group-specific age adjustment for both earnings sources and growth transformations (Appendix Figure J.1). The real-log holder-share contrast remains negative with a broader financial denominator, alternative weights, a $1,000 initial-asset floor, and every group and year deletion. Arc holder-share estimates remain imprecise (Appendices J and Q).

Table 7. Stability of the entry and initial-holder associations

	(1) Nonholder entry	(2) Holder retention	(3) Initial-holder share	(4) N
A. Zero-inclusive wage composition (headline construction)				
Baseline prospective sample	1.646^{**}	-3.720^{**}	-0.514	31,569
	(0.599)	(1.708)	(0.941)	
Group-by-year effects	1.663^{**}	-4.098^{**}	-0.854	31,569
	(0.619)	(1.765)	(0.946)	
Joint industry–education effects	1.865^{**}	-2.362	-0.095	31,569
	(0.875)	(1.658)	(0.960)	
Reported components	1.699^{***}	-2.054	-0.480	24,114
	(0.449)	(1.752)	(1.232)	
Reported and stable households	2.557^{***}	-1.457	-2.275^{*}	13,939
	(0.555)	(2.021)	(1.298)	
B. Original real log composition				
Baseline prospective sample	0.969^{***}	-2.650^{***}	-1.621^{**}	31,569
	(0.302)	(0.952)	(0.715)	
Group-by-year effects	0.988^{***}	-2.707^{**}	-1.599^{**}	31,569
	(0.303)	(1.016)	(0.740)	
Joint industry–education effects	0.955^{*}	-2.713^{***}	-1.345^{*}	31,569
	(0.469)	(0.758)	(0.779)	
Reported components	0.699^{**}	-0.776	-0.788	24,114
	(0.303)	(1.129)	(0.932)	
Reported and stable households	1.008^{**}	-0.618	-0.593	13,939
	(0.375)	(1.115)	(0.887)	
Industry–education clusters	30	30	30	

Notes: Columns (1)–(2) reestimate the next-year ownership regression and column (3) the initial-holder block of the next-year share regression of Table 5; the first row of each panel reproduces the published estimates. Row 2 adds industry-by-year and education-by-year effects by initial ownership on the unchanged sample; row 3 adds joint industry–education effects, so that composition is identified only from the difference between the two age bands within each industry–education group (Table R.6 reports the complete family). Rows 4–5 keep transitions whose stock and mutual-fund ownership and balance components are reported rather than allocated at both dates, and additionally interviews coded as self-reported with stable household and residence records (Appendix R); initial holders number 7,334, 4,081 and 2,710. Panel A replays the saved designs with the zero-inclusive wage moments; Panel B is the original construction. Holder retention is next-year ownership among initial holders; exit is its negative with the same standard error. The initial-holder share includes exits. Coefficients are percentage points per fixed source SD of upside composition, with the original controls and destination weights; parentheses are industry–education CR1 standard errors ($G = 30$, t_{29}, conditional on the CPS moments). Stars use unadjusted two-sided tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$; family-adjusted q-values for the screened tests are in Appendix R.

A richer prediction using earnings ranks attenuates the original entry estimates and leaves them imprecise, as are the paired changes from the raw estimates (Appendix N). This adjustment removes more predictable progression than the CSZ age profile. It limits a pure-innovation interpretation without establishing that promotions explain the result.

The different coefficients by initial ownership also matter for the regression specification. A pooled Tobit imposes a common latent equation on zero holdings and positive shares, while the prospective specifications allow these associations to differ. Appendix J retains the standing-share and pooled Tobit comparisons.

4.10 Measurement reliability

Including zero earnings substantially improves the measurement of composition. Split-half reliability across CPS cells is 0.73 for wages and 0.62 for total earnings, against 0.28 and 0.24 for the original real-log and arc measures (Table S.28). Extending wage histories through origin year 2021 raises reliability to 0.85 but overlaps the portfolio window, so it is not the headline measure. The choice to lead with wages followed the reliability and external-survey comparisons; it did not precede estimation. The appendix retains the full family, including the initially primary total-earnings construction.

Including zero earnings also changes the ranking of cells. Wage composition correlates only 0.22 with original real-log composition and 0.35 with original arc composition, since it also describes transitions out of wage earnings. Composition and variance are more collinear: their correlation is -0.81 for pre-portfolio wages and -0.86 for pooled wages. Higher raw reliability therefore need not deliver smaller regression standard errors. The pooled-wage entry estimate is 1.23 points and imprecise (Table S.29).

Sampling uncertainty in the CPS affects the estimated entry association. Resampling CPS households while holding the SIPP fixed gives positive coefficients in 99.4 percent of 499 wage draws and 97.6 percent of total-earnings draws (Table S.30). Their 2.5th to 97.5th percentiles are 0.13 to 1.51 and 0.03 to 1.11 points. These are descriptive remeasurement ranges, not confidence intervals: the reconstructed coefficients are centered below the original estimates. Joint CPS–SIPP remeasurement of the positive-support family spans zero, and prospective allocation has not been jointly remeasured. Conventional tests remain conditional on the estimated cell moments.

Simulation and extrapolation (SIMEX) illustrates the sensitivity of the magnitude. Adding noise reduces the mean coefficient; linear and quadratic extrapolations give wage estimates of 1.96 and 2.82 points (Table S.31). This suggests attenuation, but depends on the extrapolant and a measurement-error model for jointly estimated moments. Neither is a corrected estimate or a lower bound. I therefore report the uncorrected estimate based on the pre-portfolio wage construction.

5 Saving Participation, Expenditure, and Portfolio Choice

The entry finding raises a broader question about household responses to earnings opportunities: do they accompany higher current spending, greater use of saving vehicles, or changes in existing portfolios? I first compare retirement-contribution participation and expenditure in the PSID, distinguishing furnishings and equipment from more regular spending. The SIPP then separates changes in risky balances from changes in other financial assets, while the SCE relates earnings beliefs to retirement portfolios. These comparisons connect the entry evidence to the joint consumption, saving and participation choices in Section 6.

5.1 Consumption and retirement-saving participation

Earnings risk matters for portfolios through the resources available to finance consumption and investment. Catherine, Sodini, and Zhang (2024) make this connection explicit by constructing lifetime consumption exposure from persistent disposable-income risk and human capital's share of total resources. Their consumption-risk measure is implied by earnings and wealth rather than observed expenditure. The distinction matters here as well: the persistence of earnings shocks, other household income and accumulated assets determine how individual wage risk translates into household consumption risk.

The PSID permits a complementary comparison using observed expenditure and retirement contributions. I ask whether the earnings composition associated with financial participation is also accompanied by a broad expansion in current spending. Table 8 uses the headline wage measure, conditions on the earlier outcome and baseline resources, and allows the full specification to differ between initial stockholders and nonholders. These regressions compare expenditure levels across earnings environments; they do not estimate the risk of consumption growth.

Greater upside composition predicts IRA or private-annuity contribution participation among initial stockholders. A one-standard-deviation increase corresponds to about 6.4 percentage points more participation, or 17% of their mean contribution rate. The nonholder estimate is close to zero and imprecise, and the directly tested difference supports a stronger association among holders. This outcome measures use of retirement-saving vehicles, without observing how their contributions are invested.

The tail signs agree with those for entry. Among initial holders, higher upper partial variance predicts more contribution participation and higher lower partial variance predicts less, each holding the other tail fixed. The dispersion component is small and imprecise. Opposing tail associations therefore also appear in a financial participation decision among households already holding stocks.

Broad expenditure has a small and imprecisely estimated association. The baseline-weight-share average in column (2) is roughly $450 annually against mean expenditure of about $57,000. The selected nondurable and service basket and food spending likewise show no clear association, as do the inverse-hyperbolic-sine specifications. Broad expenditure includes financing payments and is not a pure measure of consumption services; the narrower baskets help assess whether the result depends on those components. Taken together, the estimates allow modest spending changes but provide no clear evidence of a general spending expansion.

Furnishings and household equipment. Furnishings expenditure provides evidence on a different spending margin. Among initial stockholders, greater upside composition is positively associated with spending on furnishings and household equipment in the primary transformed specification. Table 9 presents this finding alongside selected nondurables and services and food. The directly tested holder–nonholder difference is positive. The holder association remains similar when realized family income is added, and its upper- and lower-tail derivatives have opposing signs. Furnishings also have

Table 8. Earnings opportunities, retirement contributions, and expenditure

	(1) IRA/annuity contribution	(2) Broad expenditure	(3) Selected nondurables and services	(4) Food
Outcome units	Percentage points	2019 dollars	2019 dollars	2019 dollars
Initial nonholders	0.215	504	124	53
	(0.331)	(449)	(273)	(91)
Initial holders	6.356***	225	508	-23
	(2.191)	(1,729)	(937)	(303)
Holder minus nonholder	6.141**	-279	384	-77
	(2.325)	(1,730)	(950)	(335)
Baseline-weight-share average	1.365***	452	196	39
	(0.407)	(519)	(296)	(83)
Nonholder outcome mean	10.10	51,004	29,885	8,841
Holder outcome mean	37.65	82,694	45,954	11,613
Overall outcome mean	15.26	56,966	32,909	9,362
Nonholder observations	28,337	28,569	28,569	28,569
Holder observations	4,719	4,784	4,784	4,784
Total observations	33,056	33,353	33,353	33,353
Distinct reference persons	9,148	9,183	9,183	9,183
Industry–education clusters	30	30	30	30

Notes: Coefficients per one-standard-deviation increase in wage upside composition, holding variance fixed. PSID baseline waves 2005–2017 are linked to 2007–2019. Initial ownership refers to non-IRA stocks. The fully interacted models include the earlier outcome, baseline resources and demographics, other earnings moments, and industry and wave fixed effects. Baseline family weights also determine the average of the two ownership-state slopes. Column (1) records any private IRA/annuity contribution, not its amount or asset allocation; means are percentages. Its reporting window starts in January of the destination year minus two and overlaps the baseline interview year. Columns (2)–(4) are declared dollar-level checks retaining the transformed earlier outcome. Spending is annualized but mixes current and previous-calendar-year recall periods; the overlapping baskets do not add. Appendix V defines their coverage and reports the primary asinh specifications. Standard errors in parentheses use CR1 clustering by 30 industry–education groups, conditional on the saved CPS moments. Stars use unadjusted two-sided t_{29} tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. The CPS window overlaps or postdates some observations; these are descriptive external associations.

a positive dispersion association, so composition is an additional dimension of their relationship with earnings risk.

The spending margin is economically relevant because durable purchases involve an outlay that can be postponed. Models of infrequent durable adjustment link purchase decisions to income growth, uncertainty and the household's position relative to an adjustment threshold (Eberly, 1994). The resource-threshold argument in Section 2.4 suggests a related interpretation: an earnings environment that changes the probability of reaching a resource threshold can matter for occasional purchases as well as financial participation. The positive furnishings association and imprecise nondurable estimates are consistent with this distinction, without establishing a difference in their economic responses.

The finding concerns a narrow category of durable expenditure, not the stock of durables or its service flow. Transformed spending combines whether a purchase occurs with its amount; these

Table 9. Earnings composition, furnishings and other expenditure

	(1) Furnishings and equipment	(2) Selected nondurables and services	(3) Food
Asinh dollar scale	\$1,000	\$10,000	\$10,000
Panel A. Wage composition, holding variance fixed			
Initial nonholders	0.007	0.001	0.002
	(0.013)	(0.008)	(0.005)
Initial holders	0.114***	0.012	0.009
	(0.041)	(0.020)	(0.017)
Holder minus nonholder	0.107**	0.011	0.006
	(0.046)	(0.024)	(0.019)
Baseline-weight-share average	0.027**	0.003	0.004
	(0.012)	(0.006)	(0.004)
Panel B. Composition checks among initial holders			
Add realized family income (asinh)	0.107**	0.008	0.006
	(0.039)	(0.019)	(0.016)
Positive-support real-log exposure (asinh)	0.027	0.007	0.003
	(0.017)	(0.010)	(0.007)
Wage exposure: outcome in 2019 dollars	188	508	-23
	(266)	(937)	(303)
Panel C. Wage dispersion among initial holders			
Dispersion, holding composition fixed	0.021***	0.003	0.001
	(0.006)	(0.004)	(0.003)
Holder outcome mean (2019 dollars)	2,433	45,954	11,613
Overall outcome mean (2019 dollars)	1,368	32,909	9,362
33,353 observations (28,569 nonholders; 4,784 holders)			
9,183 reference persons; 30 industry–education clusters			

Notes: Controls, weights, timing and inference follow Table 8. Composition is per construction-specific source SD at fixed variance; dispersion raises both partial variances equally at fixed composition using the wage-composition SD. Outcomes are $\operatorname{asinh}(x/s)$ at the displayed scales, except the dollar row, which retains the transformed earlier outcome. Transformed coefficients are not exact percentage changes or directly comparable across baskets. Furnishings measure narrow durable expenditure; the other baskets overlap. Stars use unadjusted t_{29} tests ($^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$). Furnishings holder/average BH q-values are 0.058/0.092 in the twelve ownership-state/six average composition tests; neither family covers dispersion or the ownership difference. No direct cross-outcome difference test was estimated. Appendix V reports the full family.

margins are not separately estimated. The dollar-level furnishings estimate and the composition estimate using the original positive-support real-log earnings measure are imprecise. Table 9 reports both checks together with the primary estimates. Appendix V retains the complete outcome family, including imprecise net non-IRA stock acquisitions.

The joint pattern is consistent with financial participation and occasional purchases responding without a broad expansion in average spending. In the analytical example, equal mean resources leave mean consumption affected only by the consumption-financed portion of additional participation fees (equation 6). The life-cycle model in Section 6 lets consumption and saving adjust to the entire future earnings process. Its access experiments generate large ownership differences alongside

modest consumption differences (Figure 9). Those experiments change access restrictions rather than earnings composition. They illustrate the distinction between participation and aggregate consumption; the model does not separately choose durable purchases, nondurable spending or IRA contributions.

The comparison uses 33,353 spending transitions in 2005–2019 and 33,056 contribution observations. CPS moments are assigned by earlier PSID industry, education and age, but overlap or postdate some historical observations. Contribution reports run from January of the earlier interview year through the next interview, overlapping the baseline year's beginning. These estimates therefore describe associations across earnings environments rather than responses to new information. The original positive-support real-log construction does not reproduce contribution participation. The evidence concerns financial participation and expenditure. It does not establish consumption insurance, changes in total saving, taxable-to-retirement substitution, or equity exposure within retirement accounts.

5.2 Risky balances and the financial-asset denominator

A risky share can fall because risky holdings decline or other financial assets grow. The original real-log construction's negative holder-share coefficient permits an accounting comparison of these explanations. For nonretirement financial assets F and risky balances R, define $\bar{F} = (F_t + F_{t-1})/2$ and $\bar{s} = (s_t + s_{t-1})/2$. Then

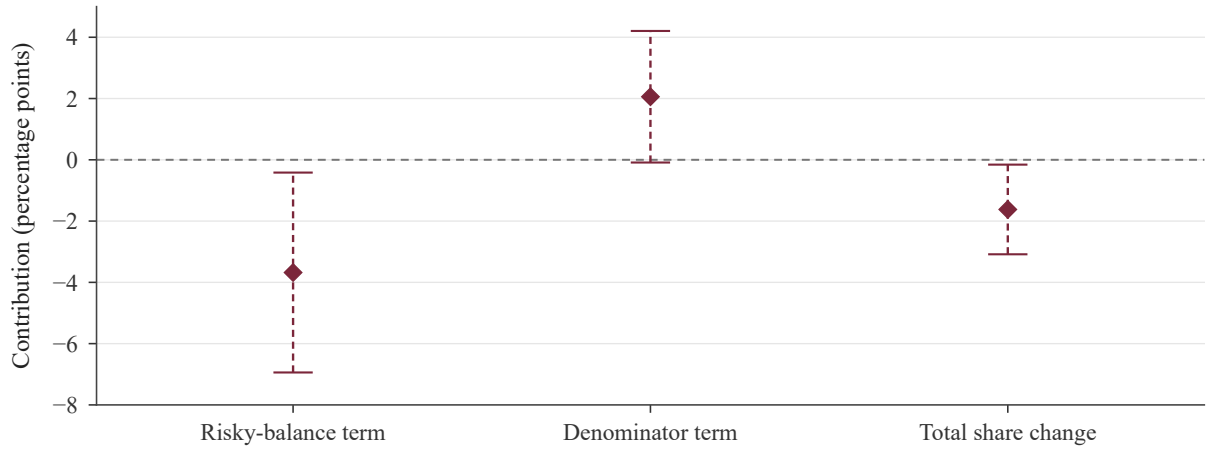
$$\Delta s = \frac{\Delta R}{\bar{F}} - \bar{s}\frac{\Delta F}{\bar{F}}. \tag{11}$$

The identity separates risky-balance and financial-asset components using midpoint weights. It accommodates entry and exit when both financial denominators are positive. On the same sample, weights, and regression design, the component coefficients sum to the share-change coefficient.

Lower risky balances account for the negative holder-share association (Figure 3). A one-standard-deviation increase in upside composition gives an estimated risky-balance contribution of -3.68 percentage points, partly offset by an imprecise $+2.06$-point denominator contribution. Their sum is the -1.62-point share coefficient. The risky-balance contribution is -4.57 points for higher upper partial variance and an imprecise $+2.79$ points for higher lower partial variance, each holding the other fixed (Table S.9). Table K.1 reports both original constructions and ownership states.

The regressions retain the prospective-allocation controls. They show which balance-sheet component accounts for the share association, but cannot separate trades from returns or exits from changes among continuing holders. This result uses the positive-support real-log construction; the wage share coefficient is imprecise. Section 4.6 separates the participation margins; Appendix K retains the accounting family, normalized-balance comparisons and alternative inference.

Figure 3. Accounting for the lower risky share of initial holders



Notes: The midpoint identity in equation (11) separates share changes into a risky-balance and a financial-asset-denominator contribution. Points report coefficients per fixed source standard deviation of age-adjusted real-log earnings composition (the original construction). The three regressions use the same 31,569 transitions and fully interacted design; plotted coefficients refer to 7,334 initial holders, retaining later exits. Point coefficients add exactly; the total uses its own estimated standard error. Bars are 95% CR1/t_{29} intervals using 30 industry–education clusters, conditional on CPS moments. Balance changes include valuation and are not measured net trades. Table K.1 reports both earnings constructions and initial states.

5.3 Elicited earnings beliefs and retirement portfolios

Respondents' own earnings beliefs provide a different comparison. The SCE relates individual perceived distributions to defined-contribution retirement portfolios, distinguishing standing equity shares from reported reallocation (Table 10). A one-standard-deviation increase in perceived dispersion accompanies a 4.75-percentage-point lower equity share conditional on normalized orientation (column (4)); using the sum of semideviations as the scale gives 3.80 points (column (2)). Orientation coefficients for the standing share are imprecise.

For reallocation, a one-standard-deviation increase in centered upside-minus-downside tilt accompanies a 0.028-unit higher riskier-minus-safer score at fixed total semideviation scale (column (1)). The score is one for riskier assets, minus one for safer assets, and zero for no change, not a change in equity share. Normalized orientation is imprecise (column (3)), and the tilt result does not survive the broader family adjustment in Appendix F.

Beliefs precede the Household Finance interview but generally fall within its twelve-month retrospective action window. They need not precede the reported action, so the comparison does not identify an effect of beliefs on adjustment. Repeated-belief tests and annual SIPP tail-change estimates also do not establish that tail revisions move shares (Appendices F.5 and D.2). The clearest SCE pattern is the negative association between perceived dispersion and standing retirement equity shares.

Table 10. Perceived earnings risk, portfolio shares, and reported adjustment

	Semideviation tilt and scale		Normalized orientation and SD	
	(1) Reallocation	(2) DC share	(3) Reallocation	(4) DC share
Upside-minus-downside tilt	0.028^{**}	-0.576		
	(0.013)	(0.664)		
Sum of semideviations	-0.017	-3.805^{***}		
	(0.018)	(1.011)		
Normalized upside orientation			0.019	0.059
			(0.014)	(0.937)
Total standard deviation			-0.016	-4.754^{***}
			(0.019)	(0.970)
Shape coefficient: marginal p	0.036	0.390	0.173	0.950
Mean and demographics	Yes	Yes	Yes	Yes
State and module-year FE	Yes	Yes	Yes	Yes
Respondents	2,136	3,332	1,893	2,952
State clusters	50	50	50	50

Reallocation is the reported direction of a DC-account adjustment over the preceding twelve months: $+1$ for riskier assets, -1 for safer assets, and zero for no change. DC share is the standing equity allocation in percentage points. Beliefs are averaged from reports one to eleven months before the Household Finance module, using monthly sampling weights; these reports generally overlap the retrospective action window. All regressions use official module weights and control for expected earnings growth, age, education, and household income.

Each risk regressor uses its unweighted strict-pre respondent standard deviation. Columns (1)–(2) center lower and upper semideviations on the elicited mean and retain degenerate densities with zero tilt and scale. Columns (3)–(4) normalize the upper-minus-lower semivariance difference by total variance, requiring positive variance. Their samples and risk scales therefore differ. Standard errors are clustered by state. Stars retain unadjusted two-sided t_{49} tests of a zero coefficient: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Appendix F reports the complete original shape-testing families and the fourteen-variant support and outcome comparison, including family-adjusted inference.

5.4 Implications for household-finance models

At a given earnings variance, portfolio models should be able to explain why greater upside composition predicts entry and ownership at successive annual dates. Raising both partial variances produces a much smaller estimated entry response than changing their composition. This adds to evidence on non-Gaussian earnings dynamics and portfolio choice (Catherine, 2022; Shen, 2024; Gálvez and Paz-Pardo, 2026). Matching total variance and average risky shares alone need not reproduce the participation pattern.

The PSID evidence adds a complementary comparison: retirement-contribution participation and transformed furnishings expenditure are positively associated with earnings composition among initial holders, while average broad expenditure has no clear association. This pattern motivates attention to discrete financial decisions and infrequent purchases alongside ordinary spending. Section 2.4 shows why a discrete access decision can respond when average consumption changes little. In the life-cycle model, that distinction emerges when access restrictions change at fixed preferences and earnings laws. Its single consumption good does not deliver a separate prediction for the durable margin.

The participation-cost interpretation depends on the specified earnings distribution and resource threshold. Standard or risk-vulnerable preferences can make uninsurable risk reduce financial risk taking (Kimball, 1993; Gollier and Pratt, 1996), without signing upside composition (Appendix A.3). Section 2 specifies a fixed-mean, fixed-variance change that creates both sustained and interrupted entries. A recurring cost (Vissing-Jorgensen, 2002; Fagereng, Gottlieb, and Guiso, 2017) permits exit after a transitory gain reverses; greater entry itself need not lower entrant survival. The resource responses in Briggs et al. (2021, 2023) connect to this mechanism, although standard models overpredict the windfall entry response in Briggs et al. (2021).

Several comparisons leave the participation-cost interpretation unresolved. Original real-log results link the upper tail more strongly to exit among recent entrants, but wage estimates are imprecise and the PSID does not reproduce the exit pattern. Earnings profiles, gain persistence and resource interactions do not establish the income reversal, while joint entry and gain does not identify mediation. Nor do the data establish a common intensive-margin response among continuing investors. Direct work-horizon and prior-participation comparisons are inconclusive (Appendix Q); the analysis neither estimates household-specific human capital nor reproduces the twin design of Catherine, Sodini, and Zhang (2024). These results therefore need to be evaluated separately for entry, sustained entry, and incumbent retention.

Endogenous saving makes income dynamics and household resources central to this interpretation. The illustrative endogenous-saving benchmark, which holds mean and variance of earnings levels fixed, reduces precautionary saving under more upside-oriented earnings but does not reproduce the positive entry association and implies nearly universal young-household participation (Appendix T.6). Its refined solution gives one-year survival of 97.48% for upside-law entrants versus 96.08% under the downside law: carried wealth cushions later fluctuations (Appendix U). Section 6 therefore evaluates optimal saving and funded participation from observed SIPP financial positions, using earnings laws fitted to CPS and PSID moments, own employee contributions and retirement benefits, and calendar returns on inherited portfolios. I estimate preferences, participation costs and opportunities from entry, retention, portfolio shares and wealth changes, requiring positive wealth growth. The model then predicts entry-risk gradients on the same regression design, decomposes the contributions of earnings laws and decision rules, and evaluates sustained ownership on common model and data cohorts.

6 Earnings Risk and Portfolio Choice over the Life Cycle

Standard saving and costly-participation mechanisms provide a theoretical foundation for the empirical composition association. Section 2.1 shows why earnings distributions with the same mean and variance can put different numbers of households across an investment threshold. A life-cycle model makes that argument more demanding: households can consume an earnings gain, save it safely or invest it, and anticipated earnings affect these choices before income arrives. I estimate a model with recurring participation costs and intermittent opportunities to enter, then ask whether

measured earnings environments generate the entry association without targeting it. Cell-specific dynamics yield a composition gradient of 0.24 percentage point, compared with approximately zero under pooled dynamics and 1.82 points in the matched data. The comparison evaluates what measured earnings heterogeneity adds to this familiar household problem, with a substantial remaining magnitude gap.

6.1 The workhorse mechanism and the earnings inputs

A Gaussian earnings innovation is determined by its conditional mean and variance. At fixed values of those moments, it leaves no independent composition change between favorable and adverse outcomes. Non-Gaussian processes, including those in Catherine (2022) and Gálvez and Paz-Pardo (2026), already allow distributional shape to affect portfolios. The theoretical ingredient is therefore an earnings distribution whose shape can vary beyond its first two moments; 58 distinct cells are not necessary for the mechanism. The cells supply measured heterogeneity with which to evaluate its relevance for entry.

Costless, continuously available investment also removes much of the distinction between choosing a risky share and becoming a stockholder. With a positive equity premium and independent earnings and stock returns, households with positive investable savings generally benefit from a small risky position. More cautious households can reduce that position without leaving the market. A recurring cost makes participation depend on whether its gain justifies the resources used to invest. Intermittent entry opportunities further distinguish an attractive investment from an investment the household can implement today. The experiments below remove these restrictions separately at fixed preferences and earnings laws.

Pooling earnings laws asks what measured cell differences add to common career dynamics and observed starting positions. The pooled benchmark retains non-Gaussian education-specific career dynamics, group wage levels and observed initial portfolios, but sets the cell adjustments to zero. I resolve the saving and portfolio rules under those pooled future laws at the same financial parameters. Workers retain their different wealth, wages and filtered earnings states, so predicted entry can still be associated with the original CPS composition measure. This fixed-parameter comparison is not a separately estimated pooled model, and it does not establish that workhorse models cannot generate the empirical signs. The permanent-access alternative below, for example, already has a positive composition gradient under pooled laws.

6.2 Households, participation costs and opportunities

Workers choose consumption, financial saving and a risky share through age 83, retire at 65, and consume terminal resources at 84. Preferences are Epstein–Zin, with discount factor β, risk aversion γ and elasticity of intertemporal substitution $\psi = 0.5$. Let M_t denote cash on hand, A_t end-of-period financial assets, α_t the risky share and D_t ownership. If the worker participates, a recurring cost $\kappa_R \xi_t$

is paid from current resources:

$$\begin{aligned} c_t + A_t + D_t\kappa_R\xi_t &= M_t, \\ M_{t+1} &= A_t\{R_{f,t+1} + \alpha_t(R^s_{t+1} - R_{f,t+1})\} + Y_{t+1}, \\ c_t > 0, \quad A_t \geq 0, \quad D_t \in \{0,1\}, \quad 0 \leq \alpha_t \leq D_t, \quad \alpha_t A_t &\geq \varepsilon D_t. \end{aligned} \tag{12}$$

The minimum funded position is $\varepsilon = \$1$. Financial assets are gross of liabilities and exclude retirement accounts. The empirical participation margin is separately reported stocks and mutual funds outside retirement accounts; it is not total equity exposure through pensions and retirement accounts. Working resources incorporate taxes, own employee contributions and income outside wage work. Retirement resources follow PSID reference persons' own Social Security and pension income by education and age.

After income and portfolio returns are realized, an opportunity indicator O_t is observed before the investment decision. Its probability is

$$\Pr(O_t = 1 \mid D_{t-1}) = q(D_{t-1}) = D_{t-1} + (1 - D_{t-1})q, \qquad D_t \leq O_t. \tag{13}$$

A nonholder receives a new opportunity with probability q each year. A current holder has access but may choose to exit; an exit exposes the worker to the opportunity restriction in the next period. A closed opportunity requires safe saving and consumes no money. When access is open, the worker observes the recurring-cost offer and decides whether to participate. The multiplier is iid lognormal, with $E\xi_t = 1$ and log standard deviation σ_κ.

The opportunity restriction represents occasions on which a nonholder can act on an investment decision. It can accommodate intermittent consideration or an opportunity to arrange participation, but the financial moments do not identify a particular attention, information or trust mechanism. There is no additional monetary entry charge. The recurring fee instead represents resources used while investing and is paid only upon funded holding. Both q and the cost distribution are common across earnings cells and independent of current earnings shocks. Differences in the estimated tail gradients therefore do not arise from assigning more opportunities to high-upside cells.

Let $V_t^0(m, z)$ be the optimized value of safe saving at cash m and earnings state z, and $V_t^1(m-k, z)$ the optimized value of funded holding after payment k. The worker accepts an open opportunity when

$$\Delta_t(m, z, k) = V_t^1(m - k, z) - V_t^0(m, z) > 0. \tag{14}$$

Both values allow future saving, consumption and portfolio decisions to adjust. Opportunities enter the entire lifetime problem, including the value of maintaining access. They are not applied as an entry-rate adjustment after solving an unrestricted model. Appendix X gives the recursion, which integrates open and closed opportunities in the Epstein–Zin risk certainty equivalent.

6.3 Earnings dynamics, asymmetry and the cell inputs

An earnings law specifies the amounts and probabilities of future wages and wage-work interruptions. While employed, a worker in industry–education group g and education type h earns

$$\begin{aligned} \log W_{gt} &= f_h(t) + \ell_g + B_g(t) + \eta_t + \epsilon_{gt}, \\ \eta_{t+1} &= \rho\eta_t + \zeta_{t+1} + J^+ I^+_{t+1} - J^-_b I^-_{t+1} - m_{gb}, \qquad \zeta_{t+1} \sim N(0, s_0^2). \end{aligned} \tag{15}$$

Here $f_h(t)$ is the education-specific age profile, ℓ_g a group wage level and $B_g(t)$ a group growth adjustment. The persistence parameter ρ carries earnings states forward. Ordinary innovations and occasional promotions or losses change η_t, with the latter indicated by the mutually exclusive I^+ and I^-. Their probabilities depend on age and the current earnings state; b indexes ten-year age bands. The compensating drift m_{gb} centers each band's stationary log-state distribution at zero. Temporary earnings ϵ_{gt} start from a two-component mixture with cell-adjusted node probabilities. A separate transition governs departure from and return to wage work. Unequal promotion and loss probabilities or sizes, the temporary-shock mixture, and the possibility of losing wage income all make the earnings distribution non-Gaussian. Survey reporting error is added to observed wages but does not enter household resources.

Two base processes, for college and noncollege workers, are calibrated to CPS one-year and PSID two-, four- and six-year earnings moments. The 58 observed industry–education–age cells then inform 30 industry–education career processes. Each has younger and older adjustments, corresponding to empirical ages 25–44 and 45–65; model workers switch adjustments at 45 and retire at 65. Two unsupported age bands borrow their group's supported-band adjustment. High-school and some-college groups share the noncollege base but have distinct adjustments. Thus the cells add measured heterogeneity to familiar earnings dynamics; their number is not a numerical setting of the household solver.

Each cell has four adjustments: the wage-work departure hazard, the growth profile $B_g(t)$, the scale of temporary earnings and reporting variation, and an asymmetry tilt a_c. The tilt shifts probability toward favorable temporary shocks and promotions, and away from adverse shocks and losses, when $a_c > 0$. In the persistent component, promotion and loss probabilities are multiplied by e^{a_c} and e^{-a_c} before a common normalization, state dependence and probability caps. The same tilt acts on temporary-shock probabilities. Appendix W.2 gives the exact formulas and their normalizations. The four adjustments are chosen jointly to fit each cell's zero-wage share, mean growth, upper partial variance and lower partial variance after empirical-Bayes shrinkage. There is no one-to-one mapping from the empirical composition measure to a_c: departures, growth and scale also generate composition differences. The data discipline the cell earnings moments more tightly than the allocation of their asymmetry between persistent and temporary shocks.

The law enters choices through two channels. Realized wages, net of taxes and employee contributions, supply current cash; workers outside wage employment receive the model's nonwage resources. These cash flows and the law's transition probabilities enter the expectation over future

resources in the household's continuation values. Solving that problem produces a rule mapping current cash, earnings state, age, ownership and access into consumption, saving and entry. Thus realized income changes what a household can invest today, while anticipated earnings change what it chooses to consume or save at the same current resources. Preferences, costs and entry opportunities remain common across groups. Future stock returns are independent of earnings shocks, distinguishing this mechanism from the joint stock–earnings downside risk emphasized by Catherine (2022).

6.4 Financial estimation with fixed earnings laws

Earnings moments, including upper and lower partial variances, are targets of the earnings calibration. The coefficients relating earnings risk to stock market participation are not targets. I fix the calibrated earnings laws before estimating the financial parameters and evaluating those portfolio coefficients.

Following the conditional approach of Briggs et al. (2021), I initialize the model from observed wages, wealth and portfolios. The financial sample contains 21,188 SIPP worker-year pairs aged 25–63. Wages filter latent earnings states. Calendar-year market and safe-return proxies apply to inherited portfolios in observed transitions, while decision rules use stationary future return expectations. Predictions condition within each observed origin on subsequent wage work and positive financial assets before aggregation with survey weights. Initial wealth, ownership and dated returns are inputs, not distributions fitted by the model.

I estimate $(\gamma, \beta, \kappa_R, \sigma_\kappa, q)$ from eight financial outcomes. Three entry rates by initial wealth discipline how participation changes with resources. Two retention rates and the next-year share of all initial holders, including exits at zero, discipline the holder margin. Mean financial-wealth changes for initial holders and nonholders discipline accumulation. These moments jointly inform preferences, recurring costs and opportunities; none is assumed to identify one parameter in isolation.

Table 11. Opportunity access: parameters

Parameter	Value
Risk aversion γ	8.801
Annual discount factor β	0.811
Mean recurring offer (2018 dollars)	524
Log-cost standard deviation	0.944
Opportunity probability q	0.176
Elasticity of intertemporal substitution (fixed)	0.50

Notes: Five parameters are estimated from the eight financial moments in Table 12. The elasticity of intertemporal substitution is fixed. The recurring offer has a mean-one lognormal multiplier; the reported mean is not the selected payment. There is no additional entry payment. The access probability is a model restriction, not a measured attention or trust variable.

The criterion is $Q_E + 0.1Q_H + Q_W$, with covariance-weighted entry, holder and wealth blocks. Both wealth changes must remain positive. The search places an exploratory upper bound of \$1,000 on the *mean* recurring offer, not on every lognormal draw. The estimated mean is about \$524 and

the annual opportunity probability is 17.6%; no parameter bound binds. Appendix Y reports the bounds and complete bounded search, including budget-limited paths. Parameters were fixed before the earnings-risk, duration and counterfactual comparisons. These are outcomes outside the fitting criterion within exploratory model development, rather than a claim of untouched external validation.

Table 12. Opportunity access: targeted financial outcomes

Financial outcome	Data	Model
Entry: lower wealth tercile (%)	2.93 (0.26)	3.18
Entry: middle wealth tercile (%)	5.84 (0.36)	5.51
Entry: upper wealth tercile (%)	10.36 (0.48)	10.44
Retention: lower initial risky-share bin (%)	66.19 (1.43)	67.75
Retention: upper initial risky-share bin (%)	76.31 (1.23)	74.34
Next risky share: all initial holders (%)	33.38 (0.84)	34.40
Annual financial wealth growth: initial nonholders (dollars)	3,939 (778)	9,941
Annual financial wealth growth: initial holders (dollars)	1,946 (5,177)	511

Notes: All eight outcomes are targets. Rates and portfolio shares are percentages; financial-wealth changes are annual real 2018 dollars and include portfolio gains. Data standard errors, in parentheses, use 299 household-lineage multiplier draws. Initial holders who exit have zero next-year shares. The conditional sample contains 17,606 initial nonholders and 3,582 holders. Observed initial states are inputs.

Table 12 shows that the model reproduces the steep increase in entry across wealth groups and broadly matches retention and the next-year holder share. Both wealth changes are positive, including under the finer numerical grid. The main remaining financial discrepancy is accumulation: nonholder wealth rises by roughly $10,000 against $4,000 in the data, while holder growth is about $500. These changes include capital gains. Positive holder wealth growth therefore does not imply positive saving net of returns.

The mean offer is also different from the payment made by a selected participant. Entrants pay about $235 and retained holders about $359 on average, compared with the potential mean of $524. The entrant calculation covers 99.994% of initial-nonholder survey weight; one origin lacks simulation support for a selected-payment mean. Appendix Table Y.2 makes this distinction explicit. The monetary cost has a direct budget interpretation, while the opportunity restriction governs whether the participation choice is available.

6.5 Entry gradients under cell and pooled earnings laws

I apply the empirical specification to predicted entry for the same 17,606 initial nonholders. In both calculations, predicted entry is related to the same original CPS risk measures. The cell and pooled calculations retain identical financial parameters, observed initial positions, cell-filtered earnings

posteriors, calendar returns, projection controls and survey weights. The pooled calculation removes future cell adjustments and resolves the optimal rules; it does not refit financial parameters. Table 13 reports the data and both model projections, together with their difference.

Table 13. Opportunity access: earnings-risk gradients

	Data	Data SE	Cell laws	Pooled laws	Difference
Upside composition	1.816^{***}	0.480	0.241	-0.005	0.247
Upper-tail derivative	1.967^{***}	0.513	0.115	-0.116	0.230
Lower-tail derivative	-1.665^{***}	0.470	-0.368	-0.105	-0.263
Dispersion at mean volatility	0.151	0.104	-0.127	-0.110	-0.016

Notes: All columns use the same 17,606 initial nonholders, weights, controls and source standardization. Units are percentage points per source standard deviation. Data use industry–education CR1 standard errors and two-sided t_{29} tests: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Model columns are deterministic conditional projections without sampling intervals or stars. No earnings-risk coefficient enters estimation. Tail rows are local partial-variance derivatives; dispersion is evaluated at common mean volatility. The difference is cell minus pooled laws at common parameters, not a causal share explained.

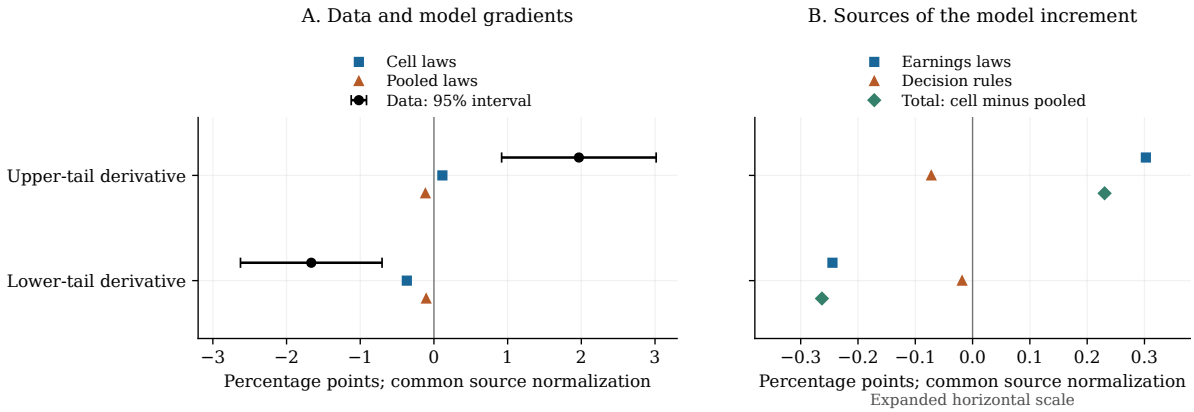


Figure 4. Earnings laws and the opposing entry gradients

Notes: Panel A applies the same projection to observed entry and model probabilities for 17,606 initial nonholders. Cell and pooled calculations share financial parameters, initial states, dated returns, controls and survey weights; only future earnings laws differ. Upper- and lower-tail derivatives use the same composition-standard-deviation normalization, with dispersion evaluated at common weighted mean volatility, as in Table 13. Data intervals are CR1/t_{29} 95% intervals; deterministic model projections have no sampling intervals. Panel B reports the signed earnings-law and decision-rule contributions to the cell-minus-pooled gradients, averaging the two change orders in Table 14. Its horizontal scale is expanded relative to Panel A. Each counterfactual conditions on its own future wage work and positive assets. The accounting is not causal mediation or an isolated change in perceived risk.

The composition coefficient is -0.005 percentage point under pooled laws and 0.241 under cell laws, compared with 1.816 points in the common data sample. The cell-minus-pooled difference, calculated before rounding, is 0.247 point and remains approximately 0.25 on the finer grid. Measured earnings heterogeneity recovers the positive association in this fitted specification, but most of the empirical magnitude remains unexplained. The 1.816-point common-sample benchmark is distinct

from the 1.601-point full empirical-sample finding. Neither coefficient is used to choose financial parameters, and the ratio of model to data is not a causal share explained.

Figure 4, Panel A, provides the corresponding tail comparison. Pooled laws predict negative entry gradients for both tails. Cell-specific laws reverse the upper-tail sign and strengthen the negative lower-tail gradient. The cell-minus-pooled upper- and lower-tail gradients are approximately 0.23 and -0.26 percentage point, with little net dispersion gradient. This shows what the full measured earnings environment adds in the selected opportunity model. Because cell adjustments change growth, dispersion and employment risk as well as asymmetry, it does not isolate the contribution of an asymmetry parameter.

6.6 Earnings laws and household decision rules

Table 14. Opportunity access: earnings laws and decision rules

Projection	Earnings laws	Decision rules	Total
Upside composition	0.274	-0.027	0.247
Upper-tail derivative	0.302	-0.072	0.230
Lower-tail derivative	-0.245	-0.018	-0.263
Dispersion at mean volatility	0.029	-0.045	-0.016

Notes: The four calculations cross cell and pooled earnings laws with their complete value functions and saving/portfolio rules. Contributions average the two orders of changing laws and rules. Each calculation conditions within origin on its own future wage-work and positive-asset events. Units are percentage points per source standard deviation. The total equals the cell-minus-pooled projection. This accounting is not causal mediation or a pure perception effect.

An earnings law and a decision rule are different objects. The law gives the probabilities of future earnings and employment states; the rule maps the household's current cash, earnings state, age, ownership and access offer into consumption, saving and stockholding. I solve one rule using cell-specific future laws and another using pooled future laws. Table 14 then crosses the law generating next-year earnings with the previously solved rule applied after those earnings arrive. For example, cell-law earnings and employment outcomes can be fed into the pooled-law saving and portfolio plan. That plan continues to evaluate future choices using pooled earnings dynamics; it is held fixed rather than reoptimized under the cell law. The crossed cases are imposed accounting comparisons, while the cell/cell and pooled/pooled cases each use one earnings law throughout.

Averaging the two change orders assigns 0.274 percentage point of the composition gradient to earnings laws and -0.027 to decision rules, totaling 0.247 point. The first component changes the resources and earnings states households reach; the second changes their response to those states because future earnings prospects differ. Panel B of Figure 4 shows both contributions. Rule adjustment dampens the positive upper-tail increment and reinforces the negative lower-tail increment. Earnings outcomes and resources dominate the net composition difference, although both plans already anticipate future income. The accounting also changes which future states satisfy the work and positive-asset conditions; it is not causal mediation or an isolated change in beliefs.

This accounting compares two measured earnings environments, not realized income against an absence of expectations. Both decision rules anticipate future earnings, and the changed-rule component measures how choices differ at a given state when those prospects change. Its small contribution here does not imply that expectations are generally unimportant. The next experiment transfers variation between upper and lower dollar tails at fixed dollar means and variances; it shows how a saving response can reverse the entry effect over time.

6.7 Composition, saving and the duration of earnings opportunities

I shift an equal amount of dollar earnings variance toward favorable or adverse outcomes. Within each age, earnings group and persistent destination state, the upside law adds to the upper partial variance exactly what it removes from the lower partial variance; the downside law reverses that transfer. Outcome probabilities and each tail's conditional mean remain fixed. Expected disposable earnings, total dollar variance and expected discounted disposable resources are therefore unchanged. This is a small mechanism illustration, not a shock calibrated to an empirical standard deviation or to the entry coefficient. Arc-growth means and variances are not held fixed. Financial parameters remain fixed and earnings-risk coefficients remain untargeted.

The duration comparison separates a lasting change in desired saving from an entry response that depends on when the earnings change ends. I apply the same annual perturbation for one year (age 26), five years (ages 26–30), or every remaining working year (ages 26–64). Retirement resources are unchanged from age 65, and households know each schedule. Five years is an illustrative window, not an estimate of persistence. Each experiment starts from the same observed age-25 portfolios, and the first new decision follows age-26 earnings. Consequently, the one-year change leaves no altered future earnings at the first new decision; its effects arise through realized resources. This timing does not measure a saving response at an earlier announcement date. Appendix Y.8 gives the construction and complete decomposition.

Figure 5 shows that the later entry reversal is duration-dependent. Under the five-year upside change, a small initial increase gives way to a cumulative-entry decline of 0.0138 percentage point after ten years. With upside lasting throughout working life, the initial increase is 0.0175 point but the year-ten response is near zero: -0.0042 point, with a paired simulation standard error of 0.0035 point. This is not a robust negative response. Persistent downside remains below baseline over the displayed decade, whereas five-year downside eventually raises cumulative entry. The one-year entry responses are near zero and their signs are sensitive to the numerical grid.

Wealth responds more strongly when the earnings change lasts longer (Figure 6). At year ten, persistent upside reduces mean financial wealth by about \$192, compared with \$13 under five years; persistent downside raises it by \$262, compared with \$22. The matched-state comparison in Appendix Figure Y.7 holds current cash and earnings states fixed. Under persistent upside, saving in the no-stock alternative falls by about \$69 in year one and \$41 in year ten. Under the five-year change, the same-state saving response is zero from year five, once the final affected earnings have arrived. Higher cumulative consumption accounts for \$163 of the persistent upside wealth decline



Figure 5. Duration of earnings composition and cumulative first entry

Notes: Differences from baseline for 395 initial nonholders drawn from 452 observed age-25 origins in 29 earnings groups. Both directions use mirrored transfers between upper and lower dollar partial variances, preserving conditional expected disposable earnings and dollar variance. The three durations change earnings at ages 26, 26–30 or 26–64; retirement resources are unchanged. Curves use perturbed earnings and their optimized rules and display ages 26–35. Cumulative first entry is the fraction of initial nonholders who have ever entered, regardless of subsequent exit. It is neither current ownership nor an annual hazard. Differences are percentage points, using the 201-node calculation and common draws. Paths condition on survival without future employment or positive-asset selection.



Figure 6. Duration of earnings composition and financial wealth

Notes: Mean financial-wealth differences in real 2018 dollars for the same 395 initial nonholders, earnings changes and full optimized paths as Figure 5. All initial nonholders remain in the wealth average, whether or not they subsequently enter. The persistent change covers ages 26–64, although the figure displays only ages 26–35. These magnitudes illustrate the model's mechanism and are not calibrated to an empirical composition change.

by year ten, lower investment gains for $28, and participation fees for less than one dollar (Appendix Figure Y.11). This supports lower buffer saving, rather than participation payments, as the principal source of the wealth loss.

A larger wealth decline need not produce a larger decline in cumulative entry. Wealth measures dollars accumulated by the whole initial cohort; first entry records whether a household has crossed the investment threshold at least once. A household can consume more and carry fewer assets forward while still finding a small stock position worthwhile. The persistent upside law continues to raise the same-state probability of taking a stock position among baseline current nonholders, even

while it lowers desired saving (Appendix Figure Y.8). In the five-year case, that direct encouragement disappears after the final affected earnings arrive, but the lower assets accumulated earlier remain. Under persistent upside, ongoing encouragement and resource erosion work in opposite directions. The same-state comparison includes possible re-entry and does not by itself isolate a first-entry mechanism.

The distinction between entry flows and cumulative entry also matters. Under persistent upside, annual first-entry differences turn negative from year three onward (Appendix Figure Y.6). These later shortfalls largely erase the extra entries in the first two years. Earlier entrants remain counted in cumulative entry even after a later exit, so the cumulative difference returns toward zero without becoming clearly negative. The resource–rule accounting in Appendix Figures Y.9 and Y.10 lets these entry and accumulation histories evolve rather than holding cash fixed.

These comparisons do not imply a universal entry sign. The no-stock saving comparison removes the current participation payment but retains future investment opportunities; it therefore does not uniquely identify pure prudence. The path decomposition changes saving and investment together and is not saving-only mediation. The lesson is that anticipated earnings composition changes both the desire to invest and the resources carried forward: the lower-saving and wealth response persists with a lasting change, while a clear reversal in cumulative entry depends on duration.

6.8 Earnings laws over working life

Figure 7 follows the same workers under cell-specific and pooled future earnings laws. Cell laws generate lower ownership early in working life and higher ownership later: the differences are about -4.1 percentage points at age 30 and $+6.2$ at age 60. Financial wealth follows a similar crossover. The earnings environment therefore changes the timing of accumulation and participation, rather than shifting ownership uniformly at every age. Together with the four-corner accounting, these paths illustrate how different earnings histories translate into different participation histories. The paths alone do not separate the contribution of accumulated resources from changes in decision rules.

6.9 Entry decisions at common resources

The earnings environment can also change the incentive to enter before cash balances evolve. Figure 8 holds current cash, age and earnings state fixed and compares the entry rules. The upper panels show participation probabilities, and the lower panels make their differences visible. The differences change sign across cash positions and earnings groups: cell laws do not imply a uniform increase in entry at a given resource level. This distinguishes a change in the willingness to invest from a change in the resources available to invest. The distinction is consistent with the accounting result that earnings and resource paths contribute more than decision rules to the composition gradient.

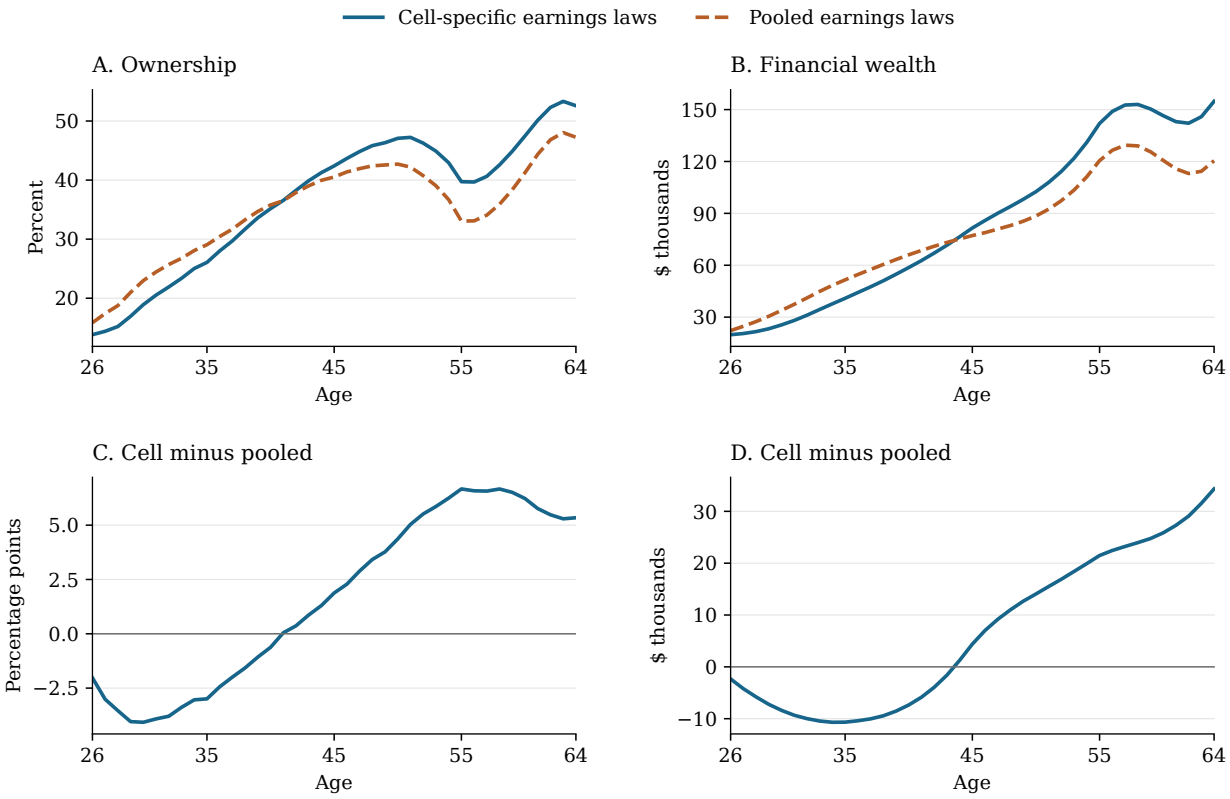


Figure 7. Earnings laws, accumulation and participation over working life

Notes: The same 135 observed age-25 origins in the two cells with the largest initial survey-weight totals: college graduates in education and health, and in professional and business services. The cohort includes 112 initial nonholders and 23 holders. Financial parameters, observed initial portfolios and earnings-state posteriors are common inputs. The first new decision is at age 26. Paths condition on survival, without future employment or asset screening. Panels A and B report ownership and mean financial wealth; Panels C and D show cell minus pooled differences. Ownership is distinct from first entry. Pooling changes the full future earnings laws, including means and persistence. Ages 26–64 are shown; the saved paths extend through age 84. Wealth is in thousands of real 2018 dollars. These conditional cohort projections are not fitted population age profiles.

6.10 Why the participation restrictions matter

Figure 9 removes the opportunity restriction, the recurring fee, and then both. Each economy reoptimizes saving and investment at the same preferences and earnings laws. Opening access raises participation, but recurring costs still make nonholding attractive for some households. Removing the recurring fee while retaining intermittent opportunities instead makes holding attractive to almost everyone who has obtained access. With open, costless investment, ownership is virtually universal in the displayed working-age cohort.

At age 45, removing both restrictions raises ownership from about 42% to nearly 100%, while mean annual consumption rises by only about 1.3%. Opening opportunities alone raises ownership to about 66% with a consumption increase of about 0.4%. Households can save safely when access is closed, so nonparticipation need not imply that resources are consumed instead. Financial wealth

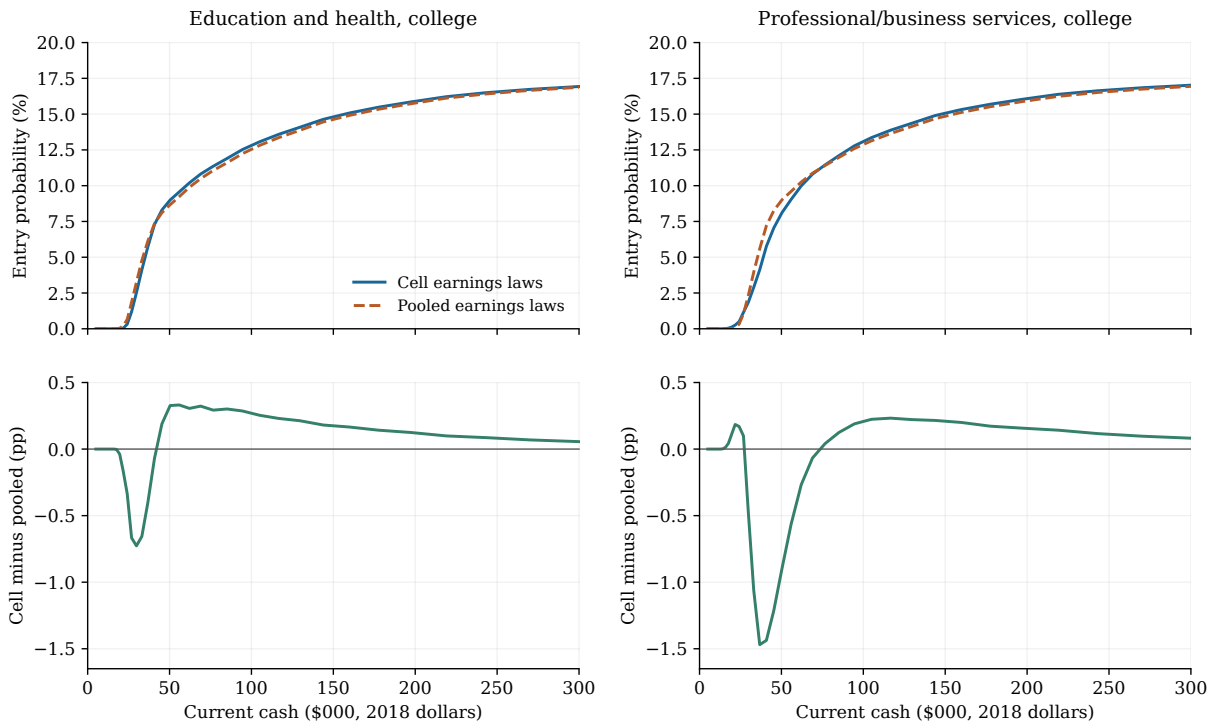


Figure 8. Entry under different earnings laws at the same current resources

Notes: Initially nonholding college workers aged 45 in education and health, and in professional and business services, at a common employed persistent earnings state (index 8). Each comparison fixes current cash, financial parameters, the opportunity probability and the recurring-cost distribution. The upper panels report entry probabilities under cell and pooled future earnings laws; the lower panels report their difference in percentage points. Cash is in thousands of real 2018 dollars. Probabilities integrate opportunities and cost offers without the future-work and positive-asset conditioning used in the observed-transition comparison. All future cell adjustments change jointly; this is not isolated upside news. Figure Y.12 reports the comparison across ages and earnings states, and Appendix Y reports numerical accuracy.

also changes much less than participation. Fitting accumulation or conditional risky shares alone therefore leaves an important part of the investment decision unresolved.

This comparison connects the model to Section 5.1: a large difference in financial participation can coexist with a modest difference in current consumption when both choices are optimized. It provides a quantitative example of the distinction in equation 6, with endogenous saving and future earnings. The access experiments hold earnings laws fixed, whereas the PSID regressions compare earnings environments, so their magnitudes are not a matched consumption test. Consumption outcomes were not targets in financial estimation, and retirement contributions enter working resources rather than being modeled as a separate participation choice. The experiment does not estimate a new earnings-tail regression for each restriction.

6.11 Accumulation, subsequent ownership and alternative access

Figure 10 follows observed age-25 origins forward with the estimated parameters. Ownership rises from about 15% at age 30 to 32% at 45 and 37% at 60 as savings and repeated opportunities support

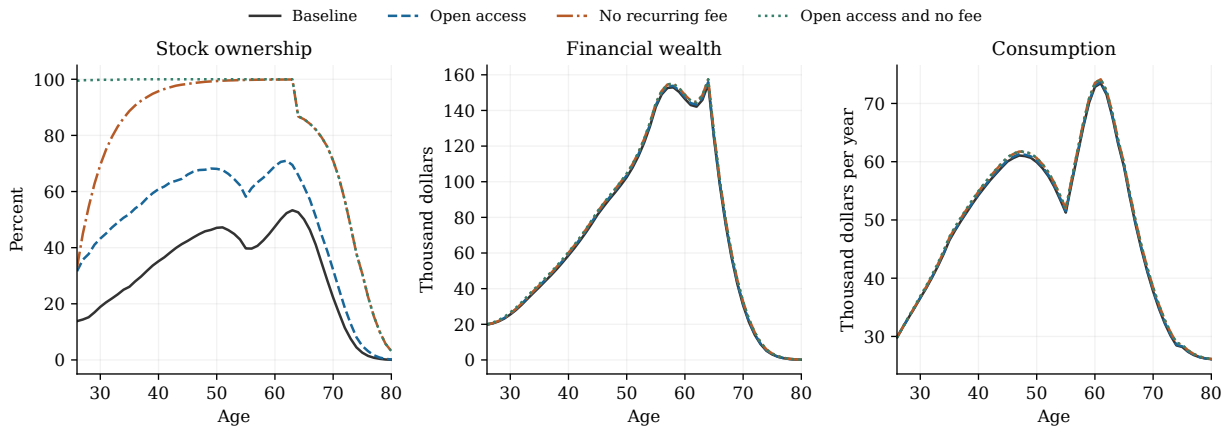


Figure 9. Entry opportunities and recurring participation costs

Notes: Fixed-parameter counterfactuals for the two age-25 cells with the largest initial survey-weight totals: college graduates in education and health, and in professional and business services. Observed initial portfolios and earnings-state distributions are common inputs. Paths condition on survival but impose no future employment or asset screen. Open access sets $q = 1$; no recurring fee sets $\kappa_R = 0$; both restrictions are removed in the final comparison. Other preferences and earnings laws remain fixed. These are conditional cohort projections, not reestimated specifications or population policy effects. Displayed paths stop at age 80; terminal liquidation at 84 is reported in the full experiment archive.

entry. Risky shares among holders need not move with ownership. The figure separates the extensive margin of access from the allocation decision conditional on holding.

The long-run exercise also shows the limits of the financial calibration. Wealth and participation fall sharply after retirement, with no bequest motive and low effective patience under the specified resource and terminal conventions. The projected life cycle is not a claim to match population retirement portfolios. In the observed-origin comparisons, sustained-entry levels are about 4.2% and 3.3% at two and three annual follow-ups, above 3.48% and 2.27% in the corresponding data cohorts. Their empirical composition gradients are imprecise.

Participation history provides an additional distinction. Among 408 nonholders with quality-screened reported prior ownership, the opportunity model predicts re-entry of about 10%, compared with 27% in the data. It also generates far fewer small entrant accounts than observed. These untargeted outcomes indicate where a common opportunity probability and the investment problem remain restrictive (Appendix Table Y.3).

Appendix Z replaces recurring opportunities with permanent access types known to households and estimates the same five dimensions from the same financial targets. Its financial fit is close and its longer sustained-ownership level fits better. Under permanent access, pooled laws already produce a positive composition coefficient of 0.211 percentage point and opposing tail signs; cell-specific laws raise the composition coefficient to 0.382 point (Appendix Table Z.3). However, exclusion imposes a fixed ceiling on eventual participation, and its treatment of reported prior holders overpredicts their re-entry. The opportunity specification allows changing resources to translate into entry over time without declaring initially inactive workers permanently outside the market. Both specifications leave a substantial gap between the observed and predicted earnings-risk magnitudes. Their comparison

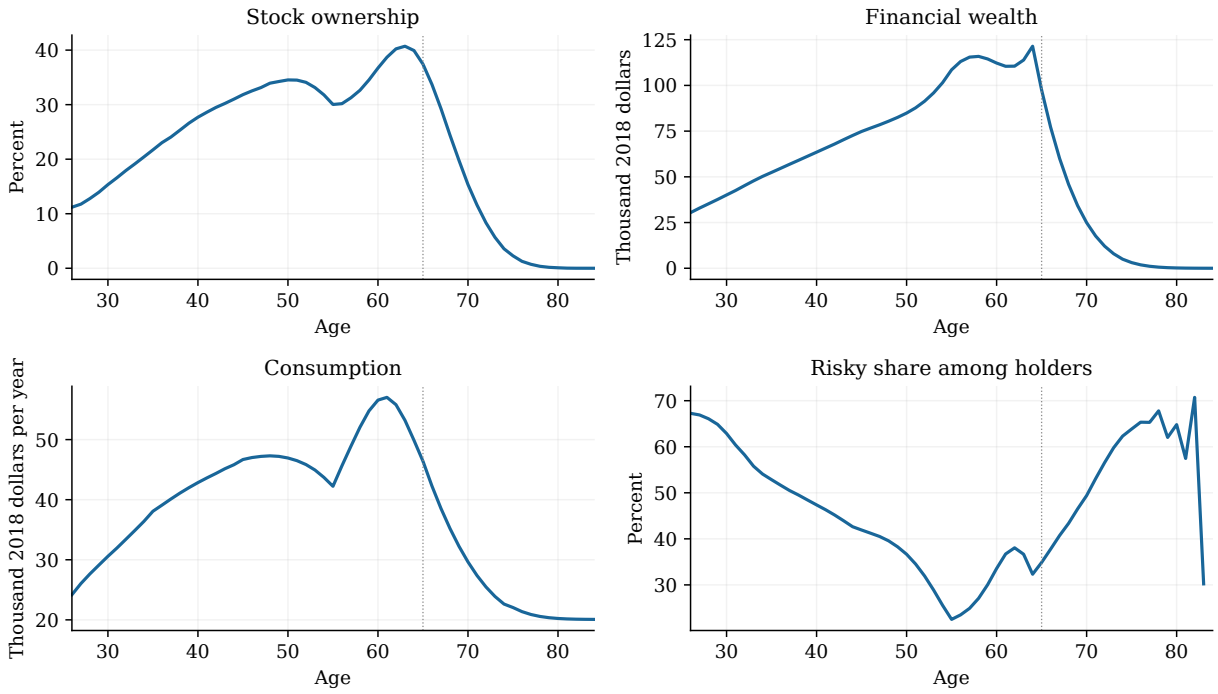


Figure 10. Participation, accumulation and portfolio allocation over the life cycle

Notes: Projections start from 452 observed age-25 origins in 29 earnings groups, with survey-weighted initial wealth, ownership, shares and earnings-state posteriors as inputs. The first new decision is at age 26. Paths condition on survival without future employment or positive-asset selection. Curves use the accepted 141-node calculation; the vertical line marks retirement at 65. Figure Y.14 compares numerical grids. These are conditional cohort projections, not a fitted unconditional distribution or an age–period–cohort comparison. There is no bequest motive and remaining resources are consumed at 84.

separates the economic role of access from the stronger claim that one particular access mechanism has been identified.

7 Conclusion

Upside earnings opportunities predict the start of stock market participation. At a given mean earnings growth, total variance and measured household resources, workers in more upside-oriented earnings environments are more likely to begin holding stocks or mutual funds. A one-standard-deviation increase predicts 1.6 percentage points more entry, nearly one fifth of the sample's annual entry rate. Upside composition predicts entry beyond the nonlinear relationship captured by variance and its square root. Its positive association also remains in the simpler linear partial-variance specification, at 1.11 percentage points per standard deviation.

The main comparison relates asymmetry in the earnings distribution itself to entry, rather than measuring how that asymmetry covaries with stock returns. Earnings distributions are measured before and separately from portfolio decisions. It relates historical earnings environments to participation after accounting for observed differences in resources and risk exposure. Transitions to zero wages help explain why more downside-oriented cells also have greater variance, but controlling for their frequency and measured cyclical exposure does not absorb the composition association. The finding also extends beyond the opening of a stock position: greater upside composition predicts entry followed by ownership at later annual observations among initial nonholders in the complete-followup cohorts. Among younger households, pre-pandemic PSID transitions show the same positive composition association and opposing tail signs. Ownership associations also appear in the SCF and the earlier SIPP panel. Together, these findings connect earnings composition to stock market participation across survey settings and periods.

The model grounds this evidence in familiar saving and costly-participation mechanisms. In the two-period example, earnings composition matters because mean and variance do not determine the probability of covering a household's shortfall to the investment threshold. The 58 observed earnings cells supply empirical inputs to 30 career processes; they are not required for the theoretical mechanism. In the estimated life-cycle model, measured cell dynamics raise the untargeted composition gradient from approximately zero under pooled dynamics to 0.24 percentage point, against 1.82 in the matched data. This fixed-parameter comparison evaluates measured earnings heterogeneity, not the ability of every standard model to fit the association. Differences in realized resources dominate the additional gradient, while rules optimized for the different future earnings prospects modestly offset it. The mechanism operates with future stock returns independent of earnings shocks.

A controlled composition experiment makes the saving margin explicit. The experiment shifts an equal amount of variation between earnings outcomes above and below their mean, holding expected disposable earnings and total variance fixed. Here, variance is calculated from earnings amounts rather than from earnings growth rates and is measured in squared dollars. In this illustration, more upside lowers desired saving at given resources while initially making stockholding slightly more attractive. When the change lasts five years, the incentive ends while lower accumulated resources remain, and cumulative entry eventually falls. When the change lasts throughout working life, wealth falls more but the ongoing incentive offsets resource erosion sufficiently that the cumulative-entry

response fades toward zero. The small response is a mechanism illustration, not a quantitative explanation of the empirical coefficient. The model therefore distinguishes realized earnings that finance entry from future earnings risk that changes saving and the resources available for later entry. This interaction can separate the immediate participation response from subsequent market access.

The empirical and model results place earnings opportunities alongside downside risk in explaining who becomes an investor. Workers with similar measured mean earnings growth, variance and current resources can have different entry patterns, depending on how earnings variation is divided between favorable and adverse outcomes. These entry relationships provide an additional test of the model beyond its fit to the targeted financial outcomes. How earnings vary, as well as how much they vary, helps explain who becomes an investor.

Data Availability. The CPS ASEC extract uses licensed IPUMS CPS microdata (Flood et al., 2025). SIPP (U.S. Census Bureau, 2025), PSID (Panel Study of Income Dynamics, 2023), SCE (Federal Reserve Bank of New York, 2026), and SCF (Board of Governors of the Federal Reserve System, 2019) are public. The replication package supplies download links, raw-data build scripts, frozen analysis inputs, and source-to-paper validation; redistribution of the IPUMS extract is subject to its license.

Disclosures. The author reports no research funding or conflicts of interest.

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Online Appendix

for

Asymmetric Labor Income Risk and Stock Market Participation: Evidence from the US

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A Earnings-Tail Measures and Analytical Results

Let q_p denote the survey-weighted pth quantile of earnings growth g, and let μ and σ denote its weighted mean and standard deviation. The signed quantile-orientation measures in Table A.1 are defined so that positive values indicate a longer or heavier lower tail relative to the upper tail. The unsigned distances L and U measure the lengths of the two tails separately. The variance-composition comparisons instead report the upside-positive contrast $C^{\text{up}} = V^{+} - V^{-}$; conventional third-moment skewness retains its usual sign.

Table A.1. Distributional estimators

Estimator	Definition	Interpretation
Upper tail	$U = q_{.90} - q_{.50}$	Magnitude above the median
Lower tail	$L = q_{.50} - q_{.10}$	Magnitude below the median
90–10 spread	$S = L + U = q_{.90} - q_{.10}$	Interquantile scale
Tail difference	$D = L - U$	Unscaled downside-minus-upside length
Kelley shape	$(L - U)/(L + U)$	Tail difference normalized by the 90–10 spread
Bowley shape	$(2q_{.50} - q_{.25} - q_{.75})/(q_{.75} - q_{.25})$	Central-quantile analogue of Kelley shape
Moment skewness	$-\mathbb{E}[((g - \mu)/\sigma)^3]$	Conventional skewness with sign reversed
Tail probability difference	$\Pr(g \leq \mu - \sigma) - \Pr(g \geq \mu + \sigma)$	Downside-minus-upside mass beyond one SD
Conditional tail balance	Tail probability difference divided by its probability sum	Direction conditional on being in either tail
L-skewness	$-\lambda_3/\lambda_2$ from probability-weighted L-moments (Hosking, 1990)	Robust full-distribution orientation
Incremental estimator	Residual from equation (M.5), applied to the named estimator	Cell ranking incremental to zero share, location, and scale

Weighted quantiles sort observations by g, cumulate survey weights, and select the first value whose cumulative weight share meets the requested probability. L-moments are approximated on a probability grid from 0.005 through 0.995. Weighted variance divides by the sum of weights because the target is the weighted empirical distribution in the cell.

For Kelley shape, $K = 1$ is the limiting case in which the 90–10 span lies entirely below the median, while $K = -1$ is the limiting upside case. A point mass at $g = -2$ can move the lower quantile to the support bound. This changes the unconditional distribution's shape; it is not treated as an outlier correction.

A.1 Destination zeros and the reliability of tail measures

Lemma 1 (A point mass below the positive-earnings support)**.** *Let F have support strictly above a, finite mean μ_F, and variance σ_F^2. For $p \in [0,1)$, let $G_p = p\delta_a + (1-p)F$. Using generalized-inverse quantiles, for $\tau \in (0,1)$,*

$$q_\tau(G_p) = \begin{cases} a, & \tau \le p, \\ F^{-1}\left((\tau - p)/(1-p)\right), & \tau > p, \end{cases}$$

while

$$\mu(p) = pa + (1-p)\mu_F, \qquad \mathrm{Var}(G_p) = (1-p)\sigma_F^2 + p(1-p)(\mu_F - a)^2.$$

Proof of Lemma 1. Because a lies below the support of F, the mixture distribution is zero below a, jumps to p at a, and equals $p + (1-p)F(x)$ above a. The generalized inverse is therefore a for $\tau \le p$. For $\tau > p$, solving $p + (1-p)F(x) \ge \tau$ gives the stated transformed quantile of F.

The mean follows by iterated expectations. Conditioning on the mixture component and applying the law of total variance gives

$$\mathrm{Var}(G_p) = (1-p)\sigma_F^2 + p(1-p)(\mu_F - a)^2.$$

Changing p therefore changes the quantile map and lower moments while leaving the conditional distribution F untouched. The formula imposes no global monotonicity on Kelley shape without additional restrictions on F. □

Lemma 2 (Reliability of a composite statistic)**.** *Suppose the two independent half-sample estimates of a group statistic satisfy*

$$\widehat{K}_c^{(h)} = a' Z_c + s_c + e_c^{(h)}, \qquad h \in \{1,2\},$$

where Z_c is a stable vector of lower distributional features, s_c is stable incremental tail variation, and the half-sample errors are mutually independent and independent of (Z_c, s_c). If their error variances are equal to σ_e^2, then

$$\mathrm{Corr}\left(\widehat{K}_c^{(1)}, \widehat{K}_c^{(2)}\right) = \frac{\mathrm{Var}(a' Z_c + s_c)}{\mathrm{Var}(a' Z_c + s_c) + \sigma_e^2}.$$

Reliability can therefore converge to one as sampling error vanishes even when $\mathrm{Var}(s_c) = 0$.

Proof of Lemma 2. Independence implies that the covariance between the two half-sample estimates is $\mathrm{Var}(a' Z_c + s_c)$. Each half-sample variance is $\mathrm{Var}(a' Z_c + s_c) + \sigma_e^2$. Dividing covariance by the common variance gives the displayed correlation. Setting $\mathrm{Var}(s_c) = 0$ and taking σ_e^2 to zero establishes the final statement. □

A.2 Properties of arc growth

For nonnegative earnings y_0 and y_1, with $y_0 > 0$, arc growth is defined at $y_1 = 0$ and equals -2. Writing $y_1 = y_0(1 + x)$,

$$g^{\text{arc}} = \frac{2x}{2 + x} = x + O(x^2), \qquad \log(y_1/y_0) = x + O(x^2).$$

Arc growth is invariant to the reporting unit, unlike $\log(y + 1)$ growth, and retains transitions from positive to zero earnings, unlike positive-to-positive log growth. Its bounded support and mass at the lower bound are therefore part of the economic distribution being measured.

A.3 A qualitative orientation benchmark

This benchmark shows how distributional asymmetry can matter for portfolio choice while clarifying why it does not sign the empirical quantile-tail coefficients. Let x denote terminal financial wealth before labor income and define

$$y = 1 + \sigma_y \eta, \qquad \mathbb{E}\eta = 0, \quad \mathbb{E}\eta^2 = 1, \quad m_3 = \mathbb{E}\eta^3,$$

with $x + y > 0$ almost surely. For CRRA utility, let

$$V(x; m_3) = \mathbb{E}u(x + y), \qquad A_V(x; m_3) = -\frac{V''(x; m_3)}{V'(x; m_3)}.$$

Within a smooth local family of shock laws that holds the first two moments fixed while varying m_3, suppose consumption remains bounded away from zero locally and derivatives of the higher-moment remainder are regular. A third-order expansion around $\sigma_y = 0$ gives

$$\frac{\partial A_V(x; m_3)}{\partial m_3} = -\frac{1}{2}\gamma(\gamma + 1)(\gamma + 2)\sigma_y^3(x + 1)^{-4} + O(\sigma_y^4).$$

The leading term is negative; under the stated regularity conditions, the derivative is therefore negative for sufficiently small σ_y. Thus, locally and within this approximation, a higher third central moment lowers derived absolute risk aversion.

For an interior unconstrained risky position in the local mean–variance approximation, write terminal financial wealth as $x = wR_f + w\alpha(R - R_f)$, with the risky return independent of labor income. A second-order expansion around wR_f gives the approximate terminal-wealth certainty-equivalent gain

$$G(w; m_3) \simeq \frac{(\mathbb{E}R - R_f)^2}{2A_V(wR_f; m_3)\,\text{Var}(R)}.$$

The preceding comparative static raises this approximate gain locally. If a fixed cost creates a regular participation boundary, households sufficiently close to that boundary can change participation. This

is not a global threshold theorem: portfolio constraints, a discrete wealth grid, and the cost itself can create flat regions and jumps.

The calculation does not sign the effect of the empirical lower or upper quantile distance. Holding mean and variance fixed, widening $q_{0.50} - q_{0.10}$ need not lower m_3, and widening $q_{0.90} - q_{0.50}$ need not raise it; either transformation can move the opposite tail and higher moments. Nor does it sign a general change in upside semivariance composition, which weights signed squared deviations rather than cubes. The calculation is a qualitative third-moment mechanism, not a prediction for the regression coefficients or their magnitudes.

An exact example separates the two objects. Consider the ordered support

$$x = (-1/2, 1/2, -1, 1, -3/2, 3/2).$$

The first law assigns probability $100/600$ to each point. The second assigns the probability vector $(124, 76, 79, 121, 106, 94)/600$. Both have mean zero and variance $7/6$; both assign total probability $1/3$ to each absolute magnitude, so every even moment agrees. Yet moving from the first law to the second increases $\mathbb{E}[x|x|]$ from zero to $1/200$ and reduces $\mathbb{E}[x^3]$ from zero to $-3/400$. The standardized third moment also falls because variance is unchanged. Thus even fixing the complete absolute-shock distribution does not make semivariance composition and third-moment skewness interchangeable.

A mirror comparison holds the distribution of absolute shocks fixed. Let F be a centered, unit-variance law formed from CPS arc-growth residuals and let F^- be the law of the negative of every realization. The two laws have the same distribution of absolute shocks and therefore identical even moments, while their odd moments have opposite signs. In the reported numerical one-period CRRA grid, replacing F with F^- changes the optimal risky position in some states. This establishes that matching every even moment need not make portfolio choice invariant. The exercise is not fitted to the SIPP entry rate and is not used to validate the empirical signs, magnitudes, or operative mechanism.

A.4 Entry value at fixed earnings moments

This section derives the comparison used in Section 2. The comparison applies standard participation costs to a specified change in the earnings distribution; its parameters are not estimated. Let $w > \kappa \geq 0$, let the distribution of risky excess returns r be fixed and independent of the earnings state z, and define

$$c_0(z) = w + \bar{y} + \lambda z, \qquad c_1(z, r; a) = c_0(z) - \kappa + ar.$$

Assume $\lambda \geq 0$, a fixed compact support, $u' > 0$, $u'' < 0$, and consumption bounded away from zero for every feasible $a \in [0, w - \kappa]$ and every distribution considered locally. Oriented earnings comparisons take $\lambda > 0$. The derivatives below require the corresponding smoothness and domination conditions. Income is affine in the earnings state by assumption. Observed arc growth need not correspond to an innovation in consumption resources.

The value of access. For a given position, define the earnings-state utility difference

$$k_a(z) = \mathbb{E}_r[u(c_1(z, r; a)) - u(c_0(z))] = \mathbb{E}_r\left[\int_0^{ar-\kappa} u'(c_0(z) + t)\, dt\right]. \tag{A.1}$$

The inner integral is signed when $ar - \kappa < 0$. Thus favorable and adverse financial payoffs receive the marginal-utility weights of the earnings states in which they occur. The participation gain is $\Gamma(F; w, \kappa) = \max_a \int k_a(z)\, dF(z)$.

Consider a valid local probability path $F_\theta = F_0 + \theta\nu$, where F_0 is centered and the signed measure ν satisfies

$$\int d\nu(z) = \int z\, d\nu(z) = \int z^2\, d\nu(z) = 0. \tag{A.2}$$

Probability mass, mean, and variance are then fixed along the path. At a unique maximizing position a^*, the envelope theorem and equation (A.2) imply

$$\begin{aligned} \partial_\theta \Gamma(F_\theta; w, \kappa)|_{\theta=0} &= \int k_{a^*}(z)\, d\nu(z) \\ &= \int \left[k_{a^*}(z) - k_{a^*}(0) - k'_{a^*}(0)z - \tfrac{1}{2}k''_{a^*}(0)z^2\right] d\nu(z). \end{aligned} \tag{A.3}$$

The equality is exact and uses no truncation of expected utility. If k_a is three times continuously differentiable on the convex hull of the support and zero, the bracket equals

$$\frac{z^3}{2}\int_0^1 (1-t)^2 k'''_{a^*}(tz)\, dt, \qquad k'''_a(z) = \lambda^3\{\mathbb{E}_r[u'''(c_1(z, r; a))] - u'''(c_0(z))\}.$$

Matching the first two moments therefore removes only the quadratic part of the state-value comparison. It need not remove the remainder. When $\lambda = 0$, consumption does not depend on the earnings state, so the earnings law drops out entirely.

The derivative of upside composition along the same centered path is $\int z|z|\, d\nu(z)$. This is a different functional from equation (A.3). Its sign alone does not determine whether the gain from entry rises or falls. The local third-moment example and the exact partial-moment counterexample in Appendix A.3 illustrate why a fully specified distributional comparison is necessary.

The positive position and the participation boundary. If the optimal position is interior, differentiating its first-order condition gives

$$\partial_\theta a^*|_0 = -\frac{\int \mathbb{E}_r[r\, u'(c_1(z, r; a^*))]\, d\nu(z)}{\mathbb{E}_{F_0, r}[r^2 u''(c_1(z, r; a^*))]}. \tag{A.4}$$

The denominator is strictly negative when risky returns are nondegenerate. The numerator uses a different earnings-state function from the entry-value derivative. At fixed w and κ, the share $a^*/(w - \kappa)$ scales this dollar derivative by $1/(w - \kappa)$. Binding position constraints require one-sided

comparisons instead. An observed balance-sheet share additionally reflects its denominator, valuation changes, and selection into holding.

Suppose a regular monetary-cost boundary $\kappa^*(\theta)$ satisfies $\Gamma(F_\theta; w, \kappa^*(\theta)) = 0$, with an interior maximizing position and positive expected marginal utility. The envelope theorem yields $\partial_\kappa \Gamma = -\mathbb{E}[u'(c_1)]$, so

$$\left.\frac{d\kappa^*}{d\theta}\right|_0 = \frac{\partial_\theta \Gamma|_0}{\mathbb{E}[u'(c_1)]}. \tag{A.5}$$

For nonholders with otherwise identical states and a smooth cost distribution held fixed along the path, the entry-probability derivative multiplies this expression by the cost density at κ^*. A low density near the boundary can produce a small entry response even if the gain changes. Conversely, observing entry does not identify the cost density or the underlying change in the value of financial risk. The empirical assignment may also sort households with different costs and preferences into different earnings groups.

Allowing accumulated financial resources to change adds $\partial_w \Gamma \, dw/d\theta$ to the fixed-resource derivative, even before other state variables adjust. The dynamic portfolio–saving comparisons of Catherine, Lu, and Paron (2026) also distinguish these responses. A conditional portfolio estimate alone does not determine how wealth accumulates.

The joint distribution with returns. Drop independence while holding both marginal distributions fixed. Conditional on having paid the entry cost, the marginal value of a risky dollar at zero is

$$\begin{aligned} M &= \mathbb{E}[r\, u'(w - \kappa + \bar{y} + \lambda z)] \\ &= \mathbb{E}[r]\mathbb{E}[u'(w - \kappa + \bar{y} + \lambda z)] + \mathrm{Cov}(r, u'(w - \kappa + \bar{y} + \lambda z)). \end{aligned} \tag{A.6}$$

Changing the coupling of earnings and returns changes the covariance term without changing earnings variance, either partial moment, or unconditional skewness. This is distinct from changing the earnings marginal in equation (A.3). Moreover, $M > 0$ is an incentive for a positive position after paying the cost, not a sufficient condition for paying it. The cyclical-tail exposure in Catherine, Sodini, and Zhang (2024) concerns the joint distribution, whereas the historical composition regressors in this paper describe the earnings marginal.

The replication code checks the fixed-moment identities exactly and verifies the envelope, position, and regular cost-boundary derivatives against reoptimized finite-distribution comparisons. A separate same-marginal exercise checks equation (A.6). These calculations verify the mathematical identities and derivatives; they estimate no parameters or empirical relationships.

A.5 Participation and allocation under a fixed-moment tail rotation

This subsection derives the portfolio rules and proves Proposition 1 in Section 2. The household observes a realized transitory earnings innovation before choosing its portfolio. Financial resources are $w_1 = w + Y$, remaining nontradable income is the safe payoff $h > 0$, utility is logarithmic, and

independent risky excess returns equal $\bar{r}$ with probability $p \in (1/2, 1)$ and $-\bar{r}$ otherwise. Investors cannot borrow or short stocks. Participation costs κ in every period in which the household holds stocks. All quantities are in arbitrary monetary units. Each earnings law governs both the baseline and the following period, so the proposition compares stationary populations across environments.

Optimal allocation and participation. For a participating household with net resources $\tilde{w}_1 = w_1 - \kappa$, the portfolio problem is

$$V_H(w_1) = \max_{0 \le a \le w_1 - \kappa} \{p \log(w_1 - \kappa + h + \bar{r}a) + (1-p)\log(w_1 - \kappa + h - \bar{r}a)\}. \tag{A.7}$$

For an interior solution, strict concavity and the first-order condition give

$$a^*(w_1) = \eta(\tilde{w}_1 + h), \qquad \alpha^*(w_1) = \frac{a^*(w_1)}{\tilde{w}_1} = \eta\left(1 + \frac{h}{\tilde{w}_1}\right), \qquad \eta = \frac{2p-1}{\bar{r}}. \tag{A.8}$$

The positions considered satisfy $0 < a^* < \tilde{w}_1$. Safe human capital acts like an implicit bond holding (Viceira, 2001; Cocco, Gomes, and Maenhout, 2005): the investor holds the fraction η of total resources in stocks, which is a larger fraction of financial wealth. Hence

$$\frac{\partial a^*}{\partial Y} = \eta > 0, \qquad \frac{\partial \alpha^*}{\partial Y} = -\frac{\eta h}{\tilde{w}_1^2} < 0. \tag{A.9}$$

There is no income target or satiation: a transitory inflow raises risky dollars less than proportionately to financial resources. If $h = 0$, the share is the constant η; increasing remaining safe income instead raises the share at given financial wealth, $\partial\alpha^*/\partial h = \eta/\tilde{w}_1 > 0$. The distinction between current resources and remaining income matters for this allocation comparison.

The value of participating is $\log(w_1 + h - \kappa) + b$, where

$$b = p\log(1 + \eta\bar{r}) + (1-p)\log(1 - \eta\bar{r}) > 0,$$

and the value of staying out is $\log(w_1 + h)$. Participation is worthwhile if

$$\log(w_1 + h - \kappa) + b \ge \log(w_1 + h),$$

which yields the threshold w^* of equation (3). The gain from access increases with realized resources:

$$G(w_1) = b + \log\left(1 - \frac{\kappa}{w_1 + h}\right), \qquad G'(w_1) = \frac{\kappa}{(w_1 + h - \kappa)(w_1 + h)} > 0. \tag{A.10}$$

A sufficient interiority condition at the threshold is $0 < \eta < 1$ and $\kappa > h(e^b - 1)/(1 - \eta)$; interiority then holds above the threshold. Below it, the unconstrained optimum is an upper bound on the constrained participation value. Because the cost recurs, the same comparison is made in every period by holders and nonholders alike. With a one-time access cost instead, initial holders have

already paid and do not face a fresh fee in this repeated-static environment. Dynamic models with a one-time cost can still generate exit; the result here depends on the repeated-static setting.

Proof of Proposition 1. The moment identities follow from the two probability laws. The symmetric law has upper and lower partial variances $d^2/2$ each; the upside-oriented law has $4d^2/5$ above zero and $d^2/5$ below. For a marginal household, $w^* - w_m \in (d, 2d]$ means that neither realization of Y_0 reaches the threshold and that only $Y_1 = 2d$ does. For an established household, $w_e - d \geq w^*$, so every realization of either law reaches it. Hence the participation probabilities are $\pi_m = 0$ under Y_0 and $1/5$ under Y_1, and $\pi_e = 1$ under both laws, which is (i): under Y_0 the baseline nonholders are all the marginal households and none enters; under Y_1 they are the marginal households that drew $-d/2$, and each enters with probability $1/5$.

For (ii), the baseline holders under Y_0 are the N_e established households, all of which remain. Under Y_1 they are the N_e established households and the $N_m/5$ marginal households that drew $2d$; the former remain with probability one and the latter with probability $1/5$, so retention is $(N_e + N_m/25)/(N_e + N_m/5) < 1$ and every exit is by a marginal holder. A marginal holder at the baseline was a nonholder in the previous period with probability $4/5$. Independence makes this fraction the same among marginal holders who subsequently exit.

□

Allocation corollary. Under the conditions of Proposition 1, suppose additionally that $h < 4(w_m + 2d - \kappa)$. The expected next-period risky share rises among baseline nonholders and falls among baseline holders. Established households' expected risky dollars are unchanged, while their expected share falls by a term of order $\eta h d^3/w_e^4$.

Proof. Write $\alpha_m = \eta\big(1 + h/(w_m + 2d - \kappa)\big)$ for the share a marginal household chooses when it participates. Under Y_1 the expected next-period share of the baseline holders is

$$\frac{N_e\,\mathbb{E}_1[\alpha^*(w_e + Y)] + (N_m/5)\,(\alpha_m/5)}{N_e + N_m/5},$$

against $\mathbb{E}_0[\alpha^*(w_e+Y)]$ under Y_0. The condition $h < 4(w_m+2d-\kappa)$ gives $\alpha_m/5 < \eta \leq \mathbb{E}_j[\alpha^*(w_e+Y)]$ for either law, and the established households' own expectation falls (below), so the weighted average falls. The baseline nonholders' expected next-period share rises from zero to $\alpha_m/5 > 0$. For established households, with $\bar{w} = w_e - \kappa > d$,

$$\mathbb{E}\left[\frac{1}{\bar{w} + Y_0}\right] = \frac{\bar{w}}{\bar{w}^2 - d^2}, \qquad \mathbb{E}\left[\frac{1}{\bar{w} + Y_1}\right] = \frac{\bar{w} + 3d/2}{\bar{w}^2 + 3\bar{w}d/2 - d^2},$$

so that, writing w for $\bar{w}$,

$$\mathbb{E}[\alpha^*(w_e + Y_1)] - \mathbb{E}[\alpha^*(w_e + Y_0)] = -\frac{3\eta h d^3/2}{(w^2 - d^2)(w^2 + 3wd/2 - d^2)} < 0. \tag{A.11}$$

Both denominators are positive for $w > d$, and the expression is of order $\eta h d^3/w^4$. Expected risky dollars are $\eta(\bar{w} + \mathbb{E}Y_j + h) = \eta(\bar{w} + h)$ under either law. Finally, the required threshold placement follows from the open cost interval

$$(w + d + h)(1 - e^{-b}) < \kappa < (w + 2d + h)(1 - e^{-b}),$$

evaluated at $w = w_m$. The numerical illustration below shows that this interval, the interiority conditions and the share condition can hold simultaneously. □

Participation tenure and the wealth types. The model's established type has enough wealth to participate in every earnings state. Observing a household hold stocks in consecutive years does not identify that type. Under the upside-oriented law, let $\pi_m = 1/5$. Among households that held stocks in each of the preceding $k \geq 1$ periods, the next-period exit rate is

$$\frac{N_m \pi_m^k (1 - \pi_m)}{N_e + N_m \pi_m^k}.$$

This rate declines with k because longer histories increasingly select established households, but it remains positive at every finite k. Recent entrants are all marginal in the example and leave with probability $1 - \pi_m$ at every subsequent spell age. The declining aggregate exit rate reflects selection across observed histories; each marginal household has a constant exit hazard. The empirical recent-entry indicator therefore tests the concentration of exit, not an exact zero response among households observed holding for two years. Independence also implies $\Pr(D_{t+k} = 1 \mid D_t = 0, w) = \pi(w)$ for every $k \geq 1$. Biennial observations miss intervening spells but do not, in this example, attenuate endpoint entry probabilities.

Unresolved earnings risk. The recurring-cost example realizes the earnings innovation before allocation. For comparison, let predetermined wealth satisfy $w > \kappa$, let future earnings be $\bar{y} + \lambda z$ with $z \sim F$ and $\lambda \geq 0$, and let utility be increasing and strictly concave. When earnings risk remains unresolved at the portfolio decision, the gain from access is

$$\Gamma(F; w, \kappa) = \max_{0 \leq a \leq w - \kappa} \mathbb{E}\left[u(w - \kappa + \bar{y} + \lambda z + ar) - u(w + \bar{y} + \lambda z)\right], \tag{A.12}$$

with positive consumption. An interior allocation satisfies

$$\mathbb{E}[r\, u'(w - \kappa + \bar{y} + \lambda z + a^* r)] = 0. \tag{A.13}$$

Adverse earnings states receive higher marginal-utility weights, while participation depends on the gain from access. Under risk-vulnerable preferences, independent background risk can reduce financial risk-taking (Kimball, 1993; Gollier and Pratt, 1996). Neither this observation nor an increase in upside composition alone signs the access-value comparison at fixed mean and variance.

Appendix A.4 derives the distributional derivatives, and Appendix A.3 distinguishes composition from third-moment skewness. These are separate mechanisms from realized earnings crossing a participation threshold.

Partial moments and threshold crossing.

Lemma 3. *Let y have mean μ, upper partial variance $V^+ = \mathbb{E}[((y-\mu)^+)^2]$ and lower partial variance $V^- = \mathbb{E}[((y-\mu)^-)^2]$, with $V = V^+ + V^-$ and $C = V^+ - V^-$. For any $a, b > 0$,*

$$\Pr(y - \mu \geq a) \leq \frac{V^+}{a^2} = \frac{V + C}{2a^2}, \qquad \Pr(y - \mu \leq -b) \leq \frac{V^-}{b^2} = \frac{V - C}{2b^2},$$

with equality in the first bound for a two-point distribution that places mass p at $\mu + a$ and mass $1 - p$ at $\mu - pa/(1-p)$, and in the second for its mirror image. The standardized skewness $\mathbb{E}[(y-\mu)^3]/V^{3/2}$ is unchanged when $y - \mu$ is multiplied by any $\lambda > 0$, whereas for bounded y the probability $\Pr(\lambda(y-\mu) \geq a)$ is zero once λ is small enough.

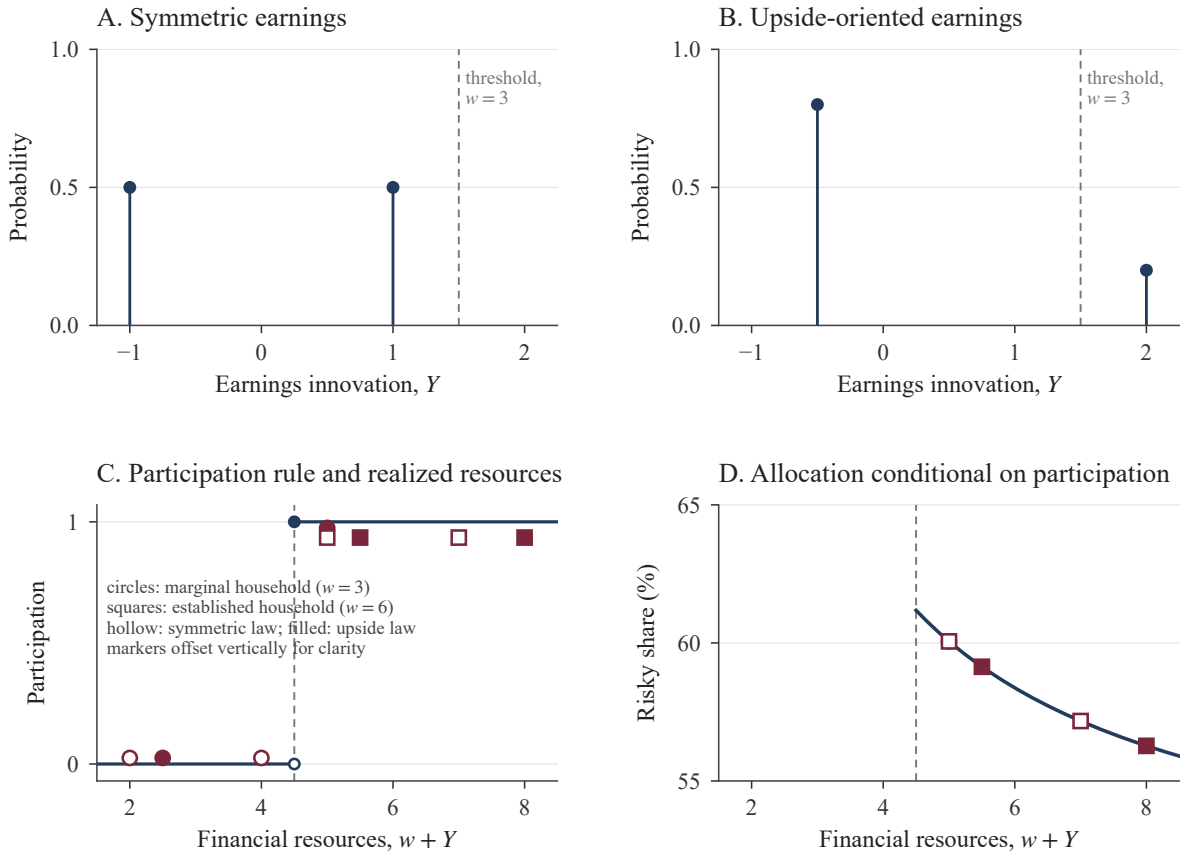
Proof. Markov's inequality applied to $((y-\mu)^+)^2$ gives $\Pr(y-\mu \geq a) = \Pr(((y-\mu)^+)^2 \geq a^2) \leq V^+/a^2$, and the same argument applied to $((y-\mu)^-)^2$ gives the second bound; the identities $V^\pm = (V \pm C)/2$ follow from the definitions. For the two-point distribution, $V^+ = pa^2$ and $\Pr(y - \mu \geq a) = p$. Scale invariance of the standardized third moment follows from homogeneity of degree three in the numerator and degree three in the denominator; if the support of $y - \mu$ lies in $[-L, U]$, then $\lambda < a/U$ gives $\Pr(\lambda(y - \mu) \geq a) = 0$. □

For a household whose expected resources lie a below the threshold, entry requires $y - \mu \geq a$. For a household whose expected resources lie b above it, exit requires a draw below $-b$. The respective partial variances supply the bounds above; at fixed variance, composition raises the first bound and lowers the second. Bounds do not order the probabilities themselves. The skewness comparison concerns changing the scale of a law: it does not establish that partial variances contain more information than skewness once variance is held fixed.

A feasible illustration. Set $w_m = 3$, $w_e = 6$, $d = 1$, $h = 1$, $p = 0.55$, and $\bar{r} = 0.20$, with equal numbers of marginal and established households. Then $\eta = 0.5$ and $b = 0.00500837$. The cost $\kappa = 5.5(1 - e^{-b}) = 0.02747715$ places the threshold at 4.5, so a marginal household participates only after the realization $Y = 2$ and an established household participates after every realization. All optimal positions satisfy the borrowing and short-sale constraints, and $h < 4(w_m + 2d - \kappa)$ holds.

The proposition concerns this particular tail rotation and population and an open set of costs and feasible portfolio parameters. It does not order arbitrary distributions by their upper-minus-lower partial variance. The sign of the aggregate entry response depends on the wealth distribution relative to the threshold. The lower expected holder share does not require lower positions among continuing investors: exits change the holder pool, while established households' expected risky dollars are identical across laws. Their share difference depends on remaining human capital h and

Figure A.1. Participation and allocation policies in the illustrative model



Notes: The distributions in Panels A and B are those of Figure 1. Panels C and D show the participation and conditional allocation rules in realized resources $w + Y$. Circles represent marginal households ($w = 3$) and squares established households ($w = 6$); hollow markers indicate the symmetric law and filled markers the upside-oriented law. Panel C offsets coincident markers vertically for visibility; the participation choices are zero or one. The threshold is 4.5. Earnings are realized before choice, saving is fixed, and the recurring fee is paid in every period of participation. Parameters and exact outcomes appear in Table A.2. These are illustrative policies, not estimates.

the dollar-shock scale d relative to wealth. It vanishes as h approaches zero, provided positions remain interior; changing h also changes the participation threshold, so there is no general work-horizon prediction. An increase in persistent future earnings raises the share in equation (A.8) at given financial resources. Finally, the illustration preserves moments of dollar innovations, whereas the empirical comparison conditions on moments of adjusted real-log or arc earnings growth. The model relates earnings distributions to participation, but it does not estimate the conversion from measured earnings growth to dollar innovations.

Table A.2. Illustrative participation and allocation outcomes under equal earnings mean and variance

	Symmetric law	Upside-oriented law
Mean earnings innovation	0	0
Variance of earnings innovation	1	1
Upper-minus-lower partial variance	0	0.6
Participation probability, marginal household (%)	0	20
Participation probability, established household (%)	100	100
Entry rate among baseline nonholders (%)	0	20
Retention rate among baseline holders (%)	100	86.7
Share of exits by holders who entered the period before (%)	–	80
Expected next-period risky share, baseline holders (%)	58.6	50.8
Expected next-period risky share, baseline nonholders (%)	0	12.0
Established households: expected risky share (%)	58.61	58.56
Established households: expected risky dollars	3.49	3.49

Notes: Exact outcomes from the illustrative model, not estimates or a calibration to SIPP. Marginal households hold financial wealth $w_m = 3$ and established households $w_e = 6$; both have safe remaining human income $h = 1$ and face the recurring participation cost $\kappa = 0.0275$, which places the threshold at 4.5. Earnings innovations are $(-1, 1)$ with equal probabilities under the symmetric law and $(-0.5, 2)$ with probabilities $(0.8, 0.2)$ under the upside-oriented law, independently over time and of returns. Baseline ownership is determined by the previous period's realization; the population has equal numbers of the two kinds of households. Next-period shares include zeros for households that do not participate and use resources net of the participation cost when they do.

A.6 Financial participation with endogenous consumption

This two-date illustration is separate from Proposition 1: saving is chosen and terminal nontradable income is set to zero. Current resources are $m > 0$, the discount factor is $\beta > 0$, and participation costs $\kappa > 0$. With safe gross return one and the independent risky excess return r used in Section 2, define

$$b = \max_{0 \le \alpha \le 1} \mathbb{E}[\log(1 + \alpha r)] > 0.$$

The risky gross payoff is strictly positive because $|r| = \bar{r} < 1$. The portfolio share is optimized after paying the fee; logarithmic utility makes b independent of the amount saved. A nonparticipant earns the safe return. Thus, conditional on $d \in \{0, 1\}$ and $m > \kappa d$, terminal consumption is $(m - \kappa d - c)(1 + \alpha r)$ for a participant and $m - c$ otherwise. Optimizing current consumption gives

$$c_d = \frac{m - \kappa d}{1 + \beta}, \qquad s_d = \frac{\beta(m - \kappa d)}{1 + \beta}.$$

Both are positive and satisfy $1/c_d = \beta/s_d$. Substitution yields

$$V_d(m) = (1 + \beta)\log(m - \kappa d) + \beta b d + K_\beta, \quad K_\beta = \beta \log \beta - (1 + \beta)\log(1 + \beta).$$

The value difference is increasing in m on the feasible domain. Setting it equal to zero gives

$$m^\dagger = \frac{\kappa}{1 - \exp[-\beta b/(1 + \beta)]} > \kappa.$$

With indifference resolved toward participation, $d^* = \mathbf{1}\{m \geq m^\dagger\}$. Hence

$$\mathbb{E}[c] = \frac{\mathbb{E}[m] - \kappa \Pr(d^* = 1)}{1 + \beta}.$$

Comparing laws with unchanged mean resources proves equation (6). The fixed-moment rotation can change the probability of crossing $m^\dagger$, just as it changes crossing of w^* in the fixed-saving example. Its direction depends on the location of the threshold, not on an arbitrary increase in upper partial variance.

The corresponding change in saved principal is

$$\Delta\mathbb{E}[s] = -\frac{\beta\kappa}{1 + \beta}\Delta \Pr(d^* = 1).$$

This example therefore does not predict higher aggregate saving when participation rises. Participation and the amount saved can respond differently. The consumption identity concerns current consumption after the earnings realization; terminal consumption also depends on portfolio returns. No tax treatment, contribution limit, account rollover, or retirement-account asset allocation is modeled. The PSID contribution indicator is a complementary participation outcome, not a direct observation of d^* or a quantitative test of this illustration.

B Reliability Design and Additional Results

B.1 Person-level splitting

Each repetition draws one half indicator for every unique person, attaches it to all records of that person, rebuilds weighted cell distributions independently in each half, re-estimates any projection separately within each half, and correlates the common eligible cells. Eligibility is frozen from the full sample; each half must also clear its fixed support floor. The same split assignment is used across paired construction comparisons.

The sequential reliability estimates use 30 deterministic person splits; comparisons across alternative estimators use 50. Splitting SIPP records rather than people would place some workers in both halves because a worker can contribute several monthly year-apart transitions. The matched construction comparison instead draws one eligible transition per person before imposing destination-positive restrictions.

Table B.1 compares positive-to-positive and zero-inclusive growth measures. Each comparison uses independent-person splits.

B.2 Support frontier

At an effective support floor of 400, early-versus-late Pearson correlations are 0.731 for the upper tail, 0.794 for the lower tail, 0.840 for the spread, 0.603 for raw Kelley, and 0.109 for mean/SD-residual

Table B.1. Positive-to-positive and zero-inclusive Kelley reliability

Data design	Growth construction	Cells	Raw reliability
CPS age-refined cells	Arc, zeros retained	60	0.735
CPS age-refined cells	Arc, positive-to-positive	60	0.018
CPS age-refined cells	Log, positive-to-positive	60	0.051
SIPP one transition/person	Arc, zeros retained	58	0.602
SIPP one transition/person	Log$(y+1)$, zeros retained	58	0.630
SIPP one transition/person	Arc, positive-to-positive	58	0.147
SIPP one transition/person	Log, positive-to-positive	58	0.120

Entries are Spearman–Brown transforms of mean independent-person half-sample correlations. The CPS construction uses the education mapping in Appendix C. The SIPP comparison selects one year-apart transition per person before construction-specific destination restrictions and uses identical split seeds in every row.

Table B.2. CPS reliability by effective sample support

Effective support floor	Common cells	90–10 spread	Raw Kelley	Mean/SD residual Kelley
100	63	0.922	0.728	0.379
200	60	0.929	0.746	0.334
400	58	0.921	0.741	0.368
800	51	0.937	0.791	0.297
1,200	34	0.951	0.839	0.344

Estimates use 50 global person splits in the broader CPS cell construction. The final column removes the mean and standard deviation but not the destination-zero share. Table O.2 uses the portfolio crosswalk and additionally removes the zero share; its residual measure is therefore a different conditional contrast.

Kelley. The corresponding rank correlations are 0.648, 0.785, 0.790, 0.622, and 0.225. These time-stability comparisons change both people and calendar periods. Split-sample reliability, by contrast, changes people while holding the period fixed.

B.3 Alternative centered estimators

Table B.3. Alternative measures of earnings-distribution asymmetry

Estimator	Raw rel.	Incr. rel.	Raw early/late	Incr. early/late	Min. 2018–24 BH q
Kelley	0.740	0.370	0.631	0.230	0.058
Bowley	0.768	-0.051	0.625	0.249	0.824
Moment skewness	0.794	0.233	0.733	0.290	0.244
L-skewness	0.685	0.178	0.597	0.050	0.381

Reliability is the Spearman–Brown transform of the mean global-person split correlation. Incremental signals remove the linear zero share, quadratic mean and standard deviation, and education and age components, with the projection re-estimated in every split. Early/late entries are Pearson correlations. The last column reports the smallest conditional false-discovery-rate q across four outcomes in the broader 2018–2024 sample: participation, the positive-financial-assets share, the all-worker share, and the holder share. No measure passes the 5% family-adjusted threshold. The Kelley minimum corresponds to the positive-assets share, whose plug-in first-stage-uncertainty p-value is 0.115. The main text reports the strict-post Kelley family; the alternative-estimator comparison was not rerun in that window. The complete estimator-by-outcome results are retained in the replication files.

The alternative measures do not strengthen the portfolio-level evidence. None of the twelve Bowley, moment-skewness, and L-skewness coefficients survives within-estimator family adjustment. The controlled-raw and residualized parameterizations yield the same p-values, as required by Frisch–Waugh–Lovell. The negative estimated incremental reliability for Bowley shape indicates that this design does not detect a stable component; it is not a negative population reliability.

B.4 Assignment, transition-selection, and cross-survey sensitivities

Table B.4 separates sensitivity to analyst-generated person splits and transition selections from formal sampling uncertainty. It also replaces the overlapping-month SIPP object with complete calendar-year job-1 earnings and directly compares CPS and SIPP cells over common origin years.

Table B.4. Reliability and cross-survey sensitivity ranges

Design	Raw Kelley	Incremental Kelley	Zero share
Panel A. Spearman–Brown reliability: center [P10, P90]			
CPS exact crosswalk, 30 person splits	0.740 [0.679, 0.802]	0.370 [0.109, 0.549]	0.940 [0.916, 0.956]
SIPP, 30 one-transition selections	0.566 [0.501, 0.630]	0.092 [−0.072, 0.239]	0.829 [0.814, 0.851]
SIPP, complete-year job-1 earnings	0.442 [0.213, 0.607]	0.128 [−0.203, 0.369]	0.774 [0.703, 0.859]
Panel B. Direct common-year CPS–SIPP Pearson correlation: mean [P10, P90]			
Pooled common-year cells	0.482 [0.440, 0.522]	0.053 [−0.025, 0.137]	0.619 [0.598, 0.633]

Notes: Brackets are analyst-generated sensitivity ranges, not confidence intervals. The CPS row varies the 30 frozen person-half assignments; its center applies the Spearman–Brown transform to the mean half-sample correlation. The second row reports the median and P10–P90 range across 30 deterministic choices of one monthly 12-month-apart transition per person, with 30 person splits per choice. The complete-year row sums twelve finite nonnegative decoded monthly job-1 earnings observations, uses December origin characteristics, selects one adjacent-year transition per person, and reports the range across 30 person splits; it contains 40,635 people in 57 supported cells. Panel B compares 58 common cells over the eight origin years shared by both surveys and varies the SIPP transition selection 30 times. CPS measures total annual wage-and-salary earnings, whereas SIPP measures job-1 earnings; the direct correlation is therefore a comparability sensitivity, not validation against an identical construct. Incremental Kelley is residualized on the zero share, quadratic mean and standard deviation, education, and age within each design.

Table B.5. Strict out-of-time prediction across SIPP transition selections

Later SIPP destination outcome	Early CPS signal	Median [P10, P90]	Positive / $q < 0.05$
Any monthly job-1 zero	Raw Kelley	0.297 [0.068, 0.450]	29 / 0
Any monthly job-1 zero	Incremental Kelley	0.154 [-0.013, 0.338]	26 / 0
Any monthly job-1 zero	Destination-zero share	0.965 [0.104, 1.662]	28 / 0
Employer-arrangement zero	Raw Kelley	0.200 [0.081, 0.328]	30 / 1
Self-employment-arrangement zero	Raw Kelley	0.059 [-0.045, 0.165]	22 / 0
Zero with any no-job spell	Raw Kelley	-0.006 [-0.043, 0.057]	14 / 0

Entries are percentage-point coefficients per one early-CPS signal standard deviation across 30 deterministic choices of one eligible 2018–2023 SIPP origin transition per person, selected without conditioning on destination state. Destination outcomes occur in 2019–2024, after the final destination year entering the CPS signal. Each cell-year regression uses 360 cell-years in 60 signal cells, official person weights, baseline earnings and demographics, destination-year and additive industry, education, and age-band fixed effects, and standard errors clustered by signal cell. The incremental specification also includes the CPS zero share and quadratic mean and dispersion. The final column counts positive coefficients and selections whose four-outcome Benjamini–Hochberg q-value is below 0.05. Brackets are transition-selection sensitivity ranges, not sampling confidence intervals. SIPP arrangement classes are mutually exclusive; the no-job-spell flag can coexist with an arrangement within a month.

C CPS-to-SIPP Crosswalk and Portfolio Construction

C.1 Exact public industry mapping

Table C.1. CPS IND1990 to SIPP broad-industry crosswalk

SIPP code	Sector	CPS IND1990 support
1	Agriculture	10–32
2	Mining	40–50
3	Construction	60
4	Manufacturing	100–392
5	Trade	500–691
6	Transportation and utilities	400–432
7	Information	440–472
8	Finance, insurance, and real estate	700–712
9	Professional and business services	721–760 and 861–893
10	Education and health	800–860
11	Leisure and hospitality	761–791
12	Other services	No clean IND1990 analogue; unmatched
13	Public administration	900–932
14	Military	940–960; excluded ex ante

Education is harmonized to high school or less, some college, and college; IPUMS CPS code 73 is a high-school diploma and belongs to the first group. Age is divided into 25–44 and 45–65. The 12 matched industries, three education groups, and two age bands give 72 candidate cells. Base support requires pooled CPS Kish support of at least 200 and at least 100 in each early and late period; 65 cells pass. The early-signal rule requires early CPS Kish support of 200; 60 cells pass.

The SIPP source contains 7,026,846 person-months for 263,280 people. Age, weight, and industry restrictions leave 2,495,083 worker-months for 109,401 people. Annualization produces 224,910 worker-years for 108,606 people; matching to supported pooled CPS cells leaves 212,635 worker-years for 104,093 people. The primary strict-post sample contains 90,110 worker-years for 52,115 people during 2019–2024 in 60 signal cells. All outcomes therefore follow the CPS destination years, which end in 2018. The broader 2018–2024 timing comparison contains 107,555 worker-years for 61,162 people in the same 60 cells.

C.2 Portfolio outcomes

Let M_{it} denote the observed stock-and-mutual-fund balance and F_{it} reported financial assets. The level outcomes are

$$\mathrm{PART}_{it} = \mathbf{1}\{M_{it} > 0\}, \tag{C.1}$$

$$\mathrm{SMF}_{it}^{+} = 100 M_{it}/F_{it} \quad \text{for } F_{it} > 0, \tag{C.2}$$

$$\mathrm{SMF}_{it}^{\mathrm{all}} = 100 M_{it}/F_{it} \quad \text{when } F_{it} > 0, \text{ and } 0 \text{ otherwise}, \tag{C.3}$$

$$\mathrm{SMF}_{it}^{\mathrm{holder}} = 100 M_{it}/F_{it} \quad \text{for } M_{it} > 0, F_{it} > 0. \tag{C.4}$$

The ratio is not total household equity exposure. Mutual funds need not be pure equity, retirement balances enter reported financial assets without their internal allocation being observed, and employer-stock compensation cannot be cleanly removed.

C.3 Portfolio sample and measurement

The person-year measure averages available monthly balance-sheet records within the calendar year. Table C.2 describes the strict-post sample before outcome-specific exclusions and reports the broader timing window for comparison. Asset-imputation flags are unavailable in this cleaned extract. The separate public-source audit in Appendix R uses component amount and root ownership flags for the prospective sample. The observed records do permit checks of missingness, denominator support, ratio bounds, and duplication.

Table C.2. SIPP portfolio sample and measurement diagnostics

Sample characteristic	Value	Definition
Strict-post primary frame	90,110	Worker-years on 52,115 people during 2019–2024 in 60 supported cells
Broader timing-sensitivity frame	107,555	Worker-years on 61,162 people during 2018–2024 in the same cells
Strict-post positive financial assets	93.40%	Observations entering the positive-assets share
Strict-post weighted participation	19.49%	Positive observed stock-and-mutual-fund balance
Strict-post all-worker share equal to zero	80.49%	Nonholders, including nonpositive financial-asset years
Strict-post fewer than 12 eligible monthly records	15.77%	Annual mean uses the available within-year records
Strict-post positive-assets ratio outside $[0, 100]$	0.00%	No mechanical ratio-tail deletion is required
Broader-frame person-years in multiworker household-years	61.09%	More than one eligible worker shares the household-year
Broader-frame multiworker household-years with same asset tuple	0.87%	Exact cross-worker duplication is rare

Values precede outcome-specific missing-value exclusions. The primary sample covers 2019–2024; the two household-record rows explicitly use the broader 2018–2024 window. The asset tuple comprises the annual stock-and-mutual-fund balance, financial assets, total assets, and net worth. Monthly asset values are constant within each retained person-year in the cleaned source. The participation rate uses official person-year weights; all other shares are unweighted sample diagnostics. The replication files retain the underlying definitions, weighted ratio quantiles, and sample counts.

The strict-post sample contains 90,110 worker-years in 61,658 household-years and 36,333 households, identified by the composite SIPP key SSUID plus SHHADID. Using SSUID alone would combine distinct current households in 811 sample-unit-years, affecting 1,651 household-years. Exactly 36,049 household-years contain one worker satisfying all sample and support restrictions. Table C.3 first clusters the full sample by household and signal cell, then restricts it to these single-supported-worker household-years. The comparison tests sensitivity to observing several eligible workers in the same household; it does not identify the household's financial decision maker.

Table C.3. Household decision-unit sensitivity, strict-post sample

	All supported workers		One supported worker	
	(1)	(2)	(3)	(4)
Kelley measure	Raw	Incremental	Raw	Incremental
Panel A. Participation				
Kelley downside shape	0.0038	-0.0092^{*}	0.0053	-0.0108^{*}
	(0.0073)	(0.0048)	(0.0086)	(0.0058)
BH q-value	0.879	0.080	0.538	0.091
Worker-years	90,110	90,110	36,049	36,049
Panel B. Share among workers with positive financial assets				
Kelley downside shape	0.0608	-0.2610^{***}	0.2704	-0.3123^{**}
	(0.1836)	(0.0976)	(0.2272)	(0.1501)
BH q-value	0.879	0.039	0.344	0.091
Worker-years	84,161	84,161	33,228	33,228
Panel C. Share among all workers				
Kelley downside shape	0.0800	-0.2341^{**}	0.2648	-0.2742^{*}
	(0.1833)	(0.0985)	(0.2189)	(0.1382)
BH q-value	0.879	0.042	0.344	0.091
Worker-years	90,110	90,110	36,049	36,049
Panel D. Share among current holders				
Kelley downside shape	−0.0967	−0.1525	0.9947	−0.2065
	(0.6303)	(0.4292)	(0.8708)	(0.8321)
BH q-value	0.879	0.724	0.344	0.805
Worker-years	17,578	17,578	6,497	6,497

Coefficients are per one early-CPS standard deviation of the raw or incremental Kelley measure specified in the column header. Participation is a probability; share outcomes are percentage points. All regressions use the strict-post 2019–2024 sample. Standard errors in parentheses cluster by the composite SIPP household key (SSUID plus SHHADID) and signal cell. Superscripts use unadjusted two-sided conditional tests: $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$. Benjamini–Hochberg q-values adjust the four outcomes separately within each column. Columns (3)–(4) retain household-years with exactly one worker in the supported analytic sample, not necessarily the household's sole earner or financial decision maker. The selected person's official weight is retained because no dedicated household weight is available in the cleaned source. These estimates do not propagate CPS first-stage uncertainty.

The all-worker share coefficient is -0.234 with household rather than person clustering (SE 0.099; $q = 0.042$), and -0.274 in the single-supported-worker sample (SE 0.138; $q = 0.091$). The negative point estimate therefore does not require multiple supported workers per household-year, although the smaller sample is less precise. Both comparisons condition on the estimated CPS measures and retain person weights; they are not household-representative estimates.

C.4 Quantile orientation across margins and earnings supports

Table C.4 reports the complete standing-outcome and transition families for raw and incremental Kelley orientation. Table C.5 varies the zero-endpoint construction for the entry comparison. Appendix N discusses both comparisons; they use the original quantile-based assignment, not the age-adjusted prospective allocation specification in Table R.1.

Table C.4. Assigned tail orientation, portfolio levels, and transitions

Panel A. Standing outcomes, 2019–2024

	(1) Participation	(2) Share: positive financial assets	(3) Share: all workers	(4) Share: current holders
Raw Kelley	0.380	0.061	0.080	−0.097
	(0.727)	(0.185)	(0.185)	(0.638)
Conditional BH q	0.880	0.880	0.880	0.880
Incremental Kelley	−0.915*	−0.261***	−0.234**	−0.153
	(0.478)	(0.096)	(0.097)	(0.424)
Conditional BH q	0.081	0.035	0.039	0.721
Observations	90,110	84,161	90,110	17,578
People	52,115	49,175	52,115	11,867
CPS signal cells	60	60	60	60

Panel B. Consecutive-year transitions; both observations in 2019–2024

	(5) Signed participation change	(6) Nonholder to positive holdings	(7) Holder to zero holdings
Raw Kelley	−0.335	−1.093**	−2.236
	(0.428)	(0.526)	(1.538)
Conditional BH q	0.437	0.126	0.227
Incremental Kelley	−0.731***	−1.047***	−0.813
	(0.260)	(0.272)	(1.152)
Conditional BH q	0.010	< 0.001	0.483
Observations	35,112	27,752	7,360
People	23,170	19,300	5,308
CPS signal cells	60	60	60

Notes: Coefficients and standard errors are percentage points per one cross-cell standard deviation. Raw and incremental estimates in each outcome column come from separate regressions. Both include household controls and year, industry, education, and age-band fixed effects; incremental specifications also include the CPS zero share and quadratic mean and dispersion. Panel B assigns lagged signals and controls, uses lagged-state risk sets, and additionally controls for lagged participation in column (5). Regressions use current-year SIPP weights. Parentheses report conditional standard errors clustered by person and CPS signal cell. Stars use unadjusted conditional two-sided tests: $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$. BH q-values adjust exactly the four standing outcomes or three transitions within each specification (Benjamini and Hochberg, 1995). All outcomes strictly postdate CPS destination earnings through 2018. Shares among all workers or workers with positive financial assets include nonholders and are not holder-only intensive-margin outcomes.

Table C.5. Tail coefficients under alternative zero-endpoint constructions

Early-CPS construction	(1) A. Baseline	(2) B. Exclude self-employed	(3) C. Zero: no weeks worked	(4) D. Interior $q_{0.10}$
Panel A. Lower and upper tails entered jointly				
Lower distance, L	-2.712^{***}	-0.907	-0.758	-1.215
	(0.646)	(0.887)	(0.895)	(1.352)
Upper distance, U	1.917^{**}	1.392	1.660^{**}	2.306^{**}
	(0.846)	(0.902)	(0.817)	(1.114)
Panel B. Total span and composition entered jointly				
Total span, $L+U$	2.721	1.557	1.845	4.466
	(1.850)	(1.640)	(1.372)	(2.793)
Downside composition, $L-U$	-4.456^{***}	-1.730^{**}	-1.767^{**}	-4.458^{**}
	(1.530)	(0.857)	(0.757)	(1.845)
Panel C. Incremental orientation				
Incremental Kelley	-1.047^{***}	-0.325	-0.432^{*}	-0.683^{**}
	(0.272)	(0.230)	(0.243)	(0.337)
Cross-cell S.D. of L	0.398	0.249	0.231	—
Lagged controls and fixed effects	Yes	Yes	Yes	Yes
Observations	27,752	27,752	27,752	23,858
CPS signal cells	60	60	60	49

Notes: The dependent variable is one-year entry into positive reported stock-and-mutual-fund holdings among lagged nonholders. Each panel–column combination is a separate regression. Coefficients are percentage points per one cross-cell standard deviation; parentheses report conditional standard errors clustered by SIPP person and lagged CPS cell. Columns (1)–(3) use the same 27,752 transitions. Column (2) excludes CPS transitions whose destination class of worker is self-employed (CLASSWKR 13 or 14); column (3) excludes destination wage zeros with positive weeks worked. Their early-CPS quantiles, zero share, mean, dispersion, and incremental Kelley are rebuilt and restandardized. Column (4) retains baseline measures and scaling but restricts the portfolio sample to the 49 cells with a 10th percentile above the arc-growth bound of -2. Thus its lower-distance S.D. is not reported as a newly standardized construction. All regressions use current SIPP weights and the same lagged controls and fixed effects as Table N.3. Incremental Kelley is the residual in equation (M.5); Panel C does not add total span. The S.D. of L is Kish-weighted and in arc-growth units. Stars use unadjusted conditional two-sided tests: $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$. These checks do not propagate CPS first-stage uncertainty.

C.5 Transition durability and positive-balance thresholds

Table C.6 tests whether the entry association persists beyond a single-year classification or a small positive balance. Sustained entry requires zero holdings at $t-1$ and positive holdings at both t and $t+1$. Threshold entry requires a nominal stock-and-mutual-fund balance to move from no more than \$1,000 or \$5,000 to above that amount. Each outcome is estimated in the primary sample and among workers whose assigned signal cell is unchanged.

Table C.6. Durability and positive-balance threshold checks

	Sustained entry		Cross \$1,000		Cross \$5,000	
	(1)	(2)	(3)	(4)	(5)	(6)
Sample	Primary	Same cell	Primary	Same cell	Primary	Same cell
Incremental Kelley	-0.697^{**}	-0.578^{**}	-1.104^{***}	-1.131^{***}	-0.453^{**}	-0.537^{**}
	(0.293)	(0.268)	(0.222)	(0.273)	(0.221)	(0.260)
Conditional p-value	0.0205	0.0349	< 0.0001	0.0001	0.0445	0.0436
BH q-value	0.041	0.045	< 0.001	< 0.001	0.045	0.045
Baseline controls and fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Worker transitions	9,331	7,328	28,686	22,242	29,823	23,132

Coefficients and standard errors in parentheses are percentage points per one early-CPS incremental-signal standard deviation. Specifications retain the lagged signal, lagged controls, current-year weight, fixed effects, and two-way person and lag-cell clustering of the primary transition design. Superscripts use unadjusted conditional two-sided tests: $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$. BH q-values adjust the complete six-coefficient family. These checks condition on the CPS first stage. Sustained entry requires positive holdings at both t and $t+1$ after zero holdings at $t-1$ and is observed through 2023. Threshold entry moves from a balance at or below the stated nominal amount to one above it; thresholds are not inflation-adjusted.

The entry association is not concentrated in one outcome year. Omitting each year in turn produces estimates from -1.203 to -0.881 percentage point, with a largest conditional p-value of 0.0031. Restricting to stable households gives -1.057 point (SE 0.275; three-outcome $q = 0.001$); selecting the eligible worker with the highest mean lagged job-1 earnings in each household transition gives -0.869 point (SE 0.360; $q = 0.056$). These estimates assess sensitivity to the observational unit, rather than testing a household risk-pooling mechanism.

Table C.7. Finite-cell inference for the entry tail decomposition

Component	Coef.	Two-way p	Wild p	Wild BH q
Lower distance, $q_{0.50} - q_{0.10}$	-2.712^{***}	< 0.001	0.0097	0.0194
Upper distance, $q_{0.90} - q_{0.50}$	1.917^{**}	0.0272	0.0865	0.0865
Downside composition, $L - U$	-4.456^{***}	0.0051	0.0327	—

Coefficients are percentage points per one early-CPS cross-cell standard deviation in the locked 27,752-transition entry risk set. The conventional column uses two-way person and signal-cell clustering. The wild column uses 99,999 null-imposed Rademacher wild-bootstrap draws with weights at the 60 CPS signal cells and retains the two-way covariance structure (Roodman et al., 2019). The BH family contains the lower and upper coefficients; signed composition is an exact rotation of the same two-dimensional column space and is reported separately. Superscripts use the conventional two-way-clustered tests: $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$. This check conditions on the frozen CPS tail estimates; Table N.3 separately reports CPS first-stage uncertainty.

Table C.8. Formal contrasts with other asset-entry margins

Comparison entry	Lower contrast	Lower BH q	Upper contrast	Upper BH q	Joint p / BH q	N
Bank deposits	-3.038	0.689	4.717	0.689	0.585 / 0.750	2,712
Bonds	-2.512^{***}	0.086	2.013^{*}	0.279	0.0035 / 0.017	26,913
Retirement assets	-3.033	0.324	3.566	0.324	0.110 / 0.274	12,312
Home equity	-0.485	0.821	0.310	0.838	0.920 / 0.920	13,137
Vehicle equity	-2.265	0.689	2.356	0.689	0.600 / 0.750	7,864
Bonds, one worker per lag household	-2.253^{**}	0.234	2.492^{**}	0.234	0.020 / 0.102	20,337

Each row restricts the locked worker-transition frame to records whose household reports neither stock-and-mutual-fund holdings nor the comparison asset in the lag year; N counts worker transitions. The dependent variable is stock-and-mutual-fund entry minus comparison-asset entry, so a coefficient is a formal cross-outcome slope difference on the same observations. Coefficients are percentage points per one early-CPS cross-cell standard deviation. Plug-in covariance adds the covariance of coefficients across CPS person-within-cell resamples to the locked two-way-clustered SIPP covariance. Stars use the unadjusted plug-in two-sided tests: $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$. Lower and upper BH q-values adjust one family of ten scalar contrasts; joint q-values adjust the five two-dimensional tests. The first five rows retain all eligible workers, cluster by person and signal cell, and use 200 CPS draws. Because the asset reports are household-level, the final row re-estimates the complete five-comparison family with 1,000 CPS draws after selecting the eligible worker with the highest mean lagged job-1 earnings in each lag-household transition and clusters by lag household and signal cell; only its bond row is displayed. The all-worker bond vector passes its joint family, but the one-worker version narrowly misses the 10% joint-family threshold and neither of its scalar contrasts survives the ten-coefficient family. These are exploratory diagnostics, not literal placebos: labor income tails can affect other assets for economic reasons, and neither a rejection nor a null contrast establishes a stock-and-mutual-fund mechanism.

Table C.9. Entry estimates under alternative weights, coverage, and controls

	(1) Cell–year	(2) Lag-year weights	(3) Unweighted	(4) Full-year records	(5) Quadratic zero share	(6) Cubic zero share
Incremental Kelley	-0.985^{***}	-0.975^{***}	-0.806^{***}	-0.862^{**}	-0.982^{***}	-0.901^{***}
	(0.297)	(0.272)	(0.242)	(0.345)	(0.273)	(0.298)
Conditional p-value	0.0016	0.0007	0.0015	0.0152	0.0007	0.0037
Observations	300	27,752	27,752	20,522	27,752	27,752

The dependent variable is entry into stock-and-mutual-fund holdings. Coefficients and standard errors in parentheses are percentage points per one early-CPS incremental-signal standard deviation. Column (1) uses 300 lag-cell-by-outcome-year observations and signal-cell-clustered inference with 59 reference degrees of freedom; all other columns use worker transitions and two-way person and lag-cell clustering. Column (4) requires twelve eligible monthly records in both years. Columns (5)–(6) add powers of the zero-incidence control to the outcome equation without redefining the first-stage signal. Superscripts use unadjusted two-sided conditional tests: $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$. These checks were conducted post hoc; their p-values are not multiplicity-adjusted and omit CPS first-stage uncertainty.

Nor does any single signal cell or represented industry account for the negative coefficient. Omitting each of the 60 cells produces estimates from -1.223 to -0.890 percentage point; the largest unadjusted conditional p-value is 0.0020. Omitting each of the 11 represented broad industries gives estimates from -1.316 to -0.700 point, with a largest p-value of 0.0155. These are influence summaries, not 60- or 11-test confirmatory families.

D Additional Evidence on the Entry Association

This section examines within-person variation, sample heterogeneity, the relation between assigned and perceived tails, and alternative assignment cells. Unless stated otherwise, regressions use the strict-post entry risk set, baseline controls and fixed effects, current SIPP weights, and two-way clustering by person and lagged CPS cell. These estimates condition on the CPS first-stage measures. Section N.7 discusses their implications for the main result.

D.1 Out-of-cell prediction

The in-sample tail decomposition does not establish improved forecasting for workers in other earnings-risk cells. I assess that question by leaving out each of the 60 signal cells in turn, estimating the SIPP coefficients on the other 59, and predicting entry in the omitted cell. The three comparisons use the same 27,752 transitions and contrast the separate-tail model with controls alone, total tail span, and incremental Kelley shape. The model family was fixed in the analysis script before its prediction results were read; this was not a preregistered design.

The separate-tail model reduces the Brier score by only 0.039 percent relative to total span ($p = 0.571$, $q = 0.695$). Its improvement over controls alone is 0.179 percent, but does not pass the 5% three-comparison adjustment ($q = 0.091$). Clipping predicted probabilities to $[0, 1]$ leaves the span comparison small and imprecise: the improvement is 0.025 percent ($p = 0.698$). Thus the

Table D.1. Out-of-cell predictive value of separate earnings tails

Comparison model	Brier score	Improvement (percent)	p-value	BH q
Controls only	0.077509	0.179	0.030	0.091
Total 90–10 span	0.077400	0.039	0.571	0.695
Incremental Kelley	0.077380	0.013	0.695	0.695
Separate lower and upper tails	0.077370	—	—	—

Each cell is held out once. Scores use current SIPP weights and the original linear-probability predictions in the 27,752-transition sample, which contains 19,300 people, 2,566 entries, and 60 signal cells. A lower Brier score is better. Improvement is the percentage reduction in the comparison model's score obtained by the separate-tail model. Two-sided tests use cell-clustered uncertainty for the difference in squared prediction errors; BH q-values adjust the complete three-comparison family. These comparisons condition on the CPS measures and do not address sorting, first-stage measurement error, perceptions, or causality. The total-span comparison is not a test against variance or third-moment skewness.

coefficient decomposition reveals conditional associations, but this exercise does not demonstrate an out-of-cell forecasting gain.

D.2 Annual changes in the earnings distribution

Annual changes in earnings tails may help explain portfolio adjustment. I first assess whether the public SIPP can measure these changes using the same estimator and averaging window at both dates.

The construction uses first-listed-job earnings in the 2018–2025 public files. Each annual transition matches the same person and calendar month one year apart, with positive origin earnings and nonnegative destination earnings; missing earnings are not coded as zero. One eligible month is selected per person and origin year before restricting the data to the fixed 58 industry–education–age cells. Origin characteristics determine the cell. The resulting 53,082 transitions yield distributions ending in 2018–2024 and adjacent distributional changes ending in 2019–2024. Weighted means, centered partial moments, and inverse-CDF quantiles are recomputed identically at both dates. These are changing cohorts of earnings histories, not observations of individual belief revisions; the first-listed slot need not identify the same employer across waves.

Table D.2 reports support and split-sample reproducibility. Of 348 possible cell-year changes, 78 have person-level Kish support of at least 100 at both dates. The 2021 and 2022 changes each have only six such cells. One hundred global household splits keep each full sample-unit lineage together across all years and cells. Requiring Kish support of at least 50 at all four half-sample endpoints leaves 61–75 paired cell-year changes in pooled comparisons and only 4–6 in each of 2021 and 2022. The latter cannot meet the declared eight-cell minimum for a within-year correlation.

The pooled median Pearson half-sample correlation is 0.270 for upper-distance changes and -0.038 for lower-distance changes. For the upside-minus-downside semivariance change it is 0.105. Conditioning on initial composition and on levels and changes in mean, variance, and zero incidence, with update-year fixed effects, reduces the median to 0.085. The incremental signed-quadratic composition change has a median correlation of 0.062 after also conditioning on linear composition

and parallel nonlinear variance changes. Each of these five measures has both positive and negative partition correlations; this refers to the full range, not the table's 10th–90th percentiles. These partition summaries are not confidence intervals. Limited support in 2021 and 2022 and weak incremental reproducibility prevent this construction from resolving the proposed share-adjustment mechanism. I therefore do not estimate portfolio responses with these measures; their weak reproducibility leaves the share-adjustment question unresolved.

Table D.2. Support and split-sample stability of annual SIPP earnings-tail changes

Panel A: Cell-update support

Update year	Full cells	Median paired	Paired range	All splits
2019	22	19	16–22	14
2020	22	19	16–22	14
2021	6	5	4–6	4
2022	6	5	4–6	4
2023	13	12	10–13	7
2024	9	7.5	6–9	5

Panel B: Pooled half-sample Pearson correlations

Measure	Median r	10th pct.	90th pct.	Negative/defined
Composition, ΔC	0.105	-0.043	0.241	18/100
Variance, ΔV	0.207	0.060	0.346	6/100
Mean growth, $\Delta\mu$	0.167	-0.010	0.293	11/100
Zero incidence, Δp_0	-0.020	-0.163	0.158	55/100
Upper span, ΔU	0.270	0.091	0.422	4/100
Lower span, ΔL	-0.038	-0.168	0.134	62/100
Quantile orientation, $\Delta(U-L)$	0.054	-0.108	0.178	32/100
Total span, $\Delta(U+L)$	0.108	-0.061	0.266	20/100
Nonlinear composition, $h(x)$	0.139	-0.042	0.280	19/100
Incremental composition, ΔC	0.085	-0.037	0.202	22/100
Incremental nonlinear composition, $h(x)$	0.062	-0.099	0.244	32/100

The source is public SIPP first-listed-job pay, with positive origin pay, nonnegative destination pay, origin age 25–65, and positive weights at both endpoints. Earnings distributions use origin person weights. One eligible same-person, same-month twelve-month transition is selected per person and origin year before applying the fixed 58-cell map. Seven distributions end in 2018–2024; each update compares adjacent distributions constructed with the same estimator. Missing pay is not zero. Changing source cohorts and job slots are not individual belief updates.

Full cells require person Kish effective size at least 100 at both distribution dates. Each of 100 globally balanced full-SSUID splits keeps entire lineage histories together. Paired cells additionally require Kish size at least 50 at all four half-by-date endpoints. All splits is the intersection across the 100 partitions. Pooled samples have 61–75 cell-updates (median 69); 48 pass in every partition. These partition percentiles are descriptive, not confidence intervals or portfolio-response tests.

$C = V^+ - V^-$, $V = V^+ + V^-$, μ is mean bounded earnings growth, and p_0 is the fraction with recorded zero destination pay. $U = q_{90} - q_{50}$ and $L = q_{50} - q_{10}$. Moments are re-centered and quantiles re-estimated in each half and date. Nonlinearity is $h(x) = x|x|$, where $x = \Delta C/\tau$ and τ is the fixed cross-cell SD of C in 2018. Incremental ΔC removes the complete span of initial C, V, μ, p_0, changes in V, μ, p_0, an intercept, and update-year indicators separately in each half. Incremental $h(x)$ additionally removes x and $h(\Delta V/\tau)$. Correlations use equal cell-update weights and retain negative values. No portfolio outcomes enter.

D.3 Within-person variation and sample heterogeneity

Table D.3. Within-person changes in the assigned tails

Design	Lower L	Upper U	ΔL	ΔU	Obs.	Persons
Table N.3 baseline (lagged cell)	-2.712	1.917			27,752	19,300
	(0.646)	(0.846)				
Lagged levels plus changes, full frame	-2.623	1.988	0.058	0.315	27,752	19,300
	(0.664)	(0.847)	(0.408)	(0.397)		
Movers only, changes given lagged levels	-1.669	5.347	-0.057	0.563	6,231	5,338
	(1.242)	(1.163)	(0.567)	(0.546)		
Current-cell assignment, full frame	-0.416	0.609			27,752	19,300
	(0.440)	(0.413)				
Current-cell assignment, movers only	-0.042	0.647			6,231	5,338
	(0.567)	(0.547)				
Person fixed effects, lagged cell	-0.578	4.266			14,241	5,789
	(1.303)	(1.987)				
Person fixed effects, current cell	-0.046	-1.171			14,241	5,789
	(0.964)	(0.752)				

Coefficients are percentage points per one cross-cell standard deviation of the named early-CPS component; standard errors in parentheses. A mover is a transition whose signal cell differs between the lagged and current observation (6,231 transitions, 22.5 percent); ΔL and ΔU are the current minus lagged standardized distances and equal zero for stayers. The full-frame change specification also includes a mover indicator, whose coefficient is 2.057 (SE 0.439). Person-fixed-effect rows use the 5,789 workers with at least two entry-risk transitions, 2,226 of whom change cell, and replace the year and lagged-cell fixed effects by person and year effects. Among movers the standard deviations of ΔL and ΔU are 1.21 and 1.27 and their correlation is 0.56.

Table D.4. Hold-out year and split-sample heterogeneity

Sample	Lower L	Upper U	Obs.	Entry rate
Panel A. Outcome years				
Locked frame, outcomes 2020–2024	-2.712 (0.646)	1.917 (0.846)	27,752	0.088
Hold-out, 2018 to 2019 transitions	-0.343 (1.079)	-0.462 (0.987)	6,033	0.066
Pooled, outcomes 2019–2024	-2.419 (0.593)	1.573 (0.738)	33,785	0.085
Panel B. Lagged bank-account balances, locked frame				
Below \$1,000	-0.526 (0.754)	1.981 (0.877)	9,759	0.045
\$1,000 to \$7,500	-1.012 (0.948)	1.565 (0.983)	8,903	0.077
Above \$7,500	-6.204 (1.681)	0.643 (1.782)	9,090	0.143
Panel C. Retirement account at $t-1$, locked frame				
None	-1.892 (0.610)	0.031 (0.699)	14,036	0.061
Owns	-3.108 (1.232)	3.566 (1.121)	13,716	0.115
Panel D. Age band, locked frame				
25–44	0.950 (1.249)	3.523 (1.785)	13,765	0.091
45–65	-2.885 (0.698)	3.593 (2.095)	13,987	0.084

Coefficients are percentage points per one cross-cell standard deviation; standard errors in parentheses. The hold-out row uses consecutive 2018 to 2019 worker-years in the same 60 supported cells with the same lagged assignment, controls, and clustering; the strict design excludes them because CPS destination earnings extend through 2018. Bank-balance terciles use weighted cutpoints from the locked frame and lagged annual means of the public SIPP monthly balance. Joint interaction tests that hold controls and fixed effects common across bank-balance terciles do not reject equal lower-tail or upper-tail coefficients within the locked frame ($p = 0.385$ and 0.890), within the 2018–2019 hold-out frame ($p = 0.661$ and 0.665), or within the pooled 2019–2024 outcome frame ($p = 0.373$ and 0.862), respectively. A continuous interaction with the inverse hyperbolic sine of the balance gives -0.154 (SE 0.080) for the lower and -0.089 (SE 0.097) for the upper tail. Adding the worker's own realized arc growth over the transition year does not move the upper coefficient in the bottom tercile (1.981 to 1.978).

Table D.5. Assigned historical CPS tails and individual SCE earnings expectations

Perceived outcome	(1) Lower distance	(2) Upper distance	(3) Fall $> 4\%$	(4) Rise $> 4\%$	(5) S.D.	(6) Job-loss probability
Assigned lower tail, L	0.122	0.094	0.0070	-0.0093	0.149	0.070
	(0.097)	(0.088)	(0.0051)	(0.0146)	(0.118)	(0.853)
Assigned upper tail, U	-0.137	-0.154	-0.0080	-0.0065	-0.176	-0.918
	(0.062)	(0.064)	(0.0031)	(0.0079)	(0.074)	(0.516)
Joint p-value	0.018	0.016	0.016	0.635	0.015	0.115
Controls and fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Respondent-months	75,669	75,669	75,863	75,863	75,863	68,813

Assigned tails use CPS origins through 2017. Of 306 candidate state-by-education-by-age cells, 144 meet the Kish support rule of 200; standardized cell measures match to 75,863 SCE responses from 10,861 respondents aged under 60 in 143 cells. All regressions include the assigned zero share, quadratic mean and dispersion, household-income and numeracy categories, and state, education, age-category, and year fixed effects, with SCE sampling weights and cell-clustered standard errors in parentheses. Perceived lower and upper distances are median-minus-25th and 75th-minus-median quantiles, respectively. Distances and standard deviation are percentage points of expected earnings growth; the fall and rise probabilities are shares. Job loss concerns the next twelve months. In outcome-standard-deviation units, the largest coefficient is -0.07 for the assigned upper tail in column (2).

Table D.6. Finer assignment cells: occupation groups

Cell scheme	Cells	Raw rel.	Incr. rel.	Lower L	Upper U	Incr. Kelley	Obs.
Industry, education, age (paper)	60	0.710	0.390	-2.712 (0.646)	1.917 (0.846)	-1.047 (0.272)	27,752
Industry, occupation, education	69	0.671	0.315	-0.666 (0.725)	0.316 (0.860)	-0.369 (0.227)	26,290
Industry, occupation, education, age	96	0.529	0.180	-1.218 (0.976)	0.552 (0.906)	-0.438 (0.286)	23,735
Occupation, education, age	18	0.817	0.344	-1.259 (0.531)	-1.738 (0.976)	-0.269 (0.134)	27,751

Occupation groups are professional and managerial (CPS OCC1990 3–199; SIPP 2018 codes 10–3655), sales, office, and administrative support (203–389; 4700–5940), and service, production, construction, and transportation (403–889; 3700–4655 and 6005–9760). SIPP workers receive their modal annual occupation group. Cells are early-CPS cells clearing Kish support of 200. Reliability is the Spearman–Brown transform of the mean correlation across 30 random global person splits generated inside the script, which is why the paper's scheme shows 0.710 and 0.390 rather than the 0.687 and 0.397 of the frozen deterministic splits in Table O.2. Coefficients are percentage points per one cross-cell standard deviation under the scheme's own standardization, with lagged assignment, the scheme's zero-share, mean, and dispersion controls, the paper's household controls and fixed effects, and clustering by person and lagged cell. Observations fall when a worker-year lacks a supported cell under the finer scheme.

E Portfolio Levels in the Broader Timing Window

Table E.1. Portfolio levels and alternative Kelley parameterizations, 2018–2024

Outcome	Raw	Raw + controls	Incremental	BH q
Participation	0.004 (0.007)	-0.022^{*} (0.012)	-0.008^{*} (0.005)	0.106
Share, positive financial assets	0.097 (0.162)	-0.552^{**} (0.219)	-0.209^{**} (0.083)	0.058
Share, all workers	0.115 (0.163)	-0.489^{**} (0.227)	-0.185^{**} (0.086)	0.071
Share, current holders	0.200 (0.590)	0.070 (0.993)	0.027 (0.376)	0.944

This table is the broader 2018–2024 timing-sensitivity window, not the strict-post primary sample. Conditional two-way-clustered standard errors are in parentheses. Superscripts denote unadjusted two-sided conditional significance: $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$; they omit BH adjustment and CPS first-stage uncertainty. Raw Kelley is standardized using early supported cells. "Raw plus controls" adds the linear CPS zero share and quadratic mean and dispersion; Incremental Kelley is the standardized residual from that projection. Those two columns are FWL-equivalent—their t statistics, p-values, and four-outcome q-values agree—and are not independent tests. Plug-in first-stage-uncertainty p-values are 0.182 for participation, 0.115 for the positive-assets share, and 0.168 for the all-worker share.

Raw Kelley coefficients are positive and imprecise in the broader 2018–2024 sample. Conditioning on the zero share, location, and dispersion reverses the sign for participation and the two broad share measures, but no person-year coefficient passes the 5% four-outcome adjustment. Adding CPS first-stage uncertainty further weakens inference. Aggregating to 420 signal-cell-years in 60 cells yields an incremental positive-assets-share coefficient of -0.224 (SE 0.085; four-outcome BH $q = 0.044$). The participation, all-worker-share, and holder-share coefficients are -0.007, -0.170, and 0.092, with q-values of 0.198, 0.125, and 0.816. The positive-assets result is thus specific to a conditional aggregation comparison in the broader timing window. It does not establish a holder-allocation response or overcome sorting, imperfect portfolio measurement, or the distinction between historical and perceived risk.

F Perceived Earnings Risk and Portfolio Adjustment

The SCE provides complementary evidence on perceived risk, which need not coincide with risk assigned from earnings histories. Guiso, Jappelli, and Terlizzese (1996) relates directly elicited income uncertainty to household portfolios; Wang (2023), Caplin et al. (2026), and Arellano et al. (2026) document the distinction between perceived and realized earnings risk. Related evidence connects elicited asset-return skewness to allocations conditional on other expected moments (Drerup, Wibral, and Zimpelmann, 2023) and investor beliefs to the direction of adjustment conditional on trading (Giglio et al., 2021). Dew-Becker (2024) separates forward-looking from realized skewness and uses relative upper and lower semivariance to distinguish orientation from scale. The comparisons below examine the centering and timing of the SCE fixed-threshold imbalance and its relationship with portfolio outcomes.

F.1 Fixed thresholds are not centered shape

Lemma 4 (Translation of a fixed-threshold imbalance)**.** *Let X have distribution function F, with density f positive at the relevant thresholds, and let $c > 0$. For the translated variable $X + m$, define*

$$A_c(m) = \Pr(X + m < -c) - \Pr(X + m > c) = F(-c - m) + F(c - m) - 1.$$

Then $A_c'(m) = -f(-c - m) - f(c - m) < 0$, although every translation-invariant centered shape measure is unchanged.

Proof. Continuity gives $\Pr(X+m < -c) = F(-c-m)$ and $\Pr(X+m > c) = 1-F(c-m)$. Subtraction yields the expression; differentiation yields the derivative. A translation-invariant centered-shape functional takes the same value for X and $X + m$. □

The public SCE density bins use fixed percentage-growth thresholds. The imbalance subtracts probability above $+4$ percent from probability below -4 percent. In the state-education-year reconstruction, this threshold imbalance is correlated -0.941 with the forecast mean. Additive cubics in the mean and standard deviation explain 89.4 percent of its variation, leaving 32.6 percent of its raw standard deviation. The residual correlates 0.018 with centered Bowley shape and 0.029 with centered moment skewness. Residualizing a fixed-threshold statistic therefore does not recover the same ranking as a centered shape estimator.

F.2 Lagged group beliefs and subsequent SIPP portfolio changes

I first aggregate SCE beliefs by state, education, and year and relate these lagged beliefs to subsequent SIPP portfolio changes. The matched sample contains 42,375 consecutive transitions for 26,005 people in 31 states and 366 state-education-outcome-year cells. Regressions include state-by-year, state-by-education, and education-by-year fixed effects, perceived mean and dispersion, and baseline

household controls; standard errors are clustered by state. The comparison establishes temporal ordering across surveys, but not a causal effect of beliefs.

Table F.1. Lagged SCE beliefs and subsequent SIPP active share changes

	Subsequent active share change		
	(1)	(2)	(3)
Fixed-threshold residual	0.1543		
	(0.2136)		
Centered Bowley downside shape		-0.3471^{*}	
		(0.1997)	
Centered moment downside skewness			-0.0121
			(0.1745)
Focal coefficient p-value	0.476	0.093	0.945
Mean, dispersion, and household controls	Yes	Yes	Yes
State–year, state–education, education–year FE	Yes	Yes	Yes
Worker transitions	42,375	42,375	42,375
State clusters	31	31	31

Beliefs in year $b-1$ are assigned to SIPP portfolio changes from b to $b+1$ among state–education stayers. Coefficients are percentage points per one standard deviation of the assigned belief measure. Regressions use baseline SIPP weights and control for baseline income, wealth, age, sex, risky share, and participation. Standard errors in parentheses are clustered by state. Superscripts denote unadjusted two-sided significance: $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$. Split reliability is 0.198 for the raw threshold, 0.147 for its leave-year-out residual, 0.068 for Bowley shape, and 0.121 for moment skewness. The fixed-threshold coefficient is imprecise, and the two centered measures do not provide consistent evidence of a subsequent share response.

F.3 Belief timing and reported retirement reallocation

The Household Finance module records the direction of defined-contribution retirement-account reallocation over the preceding twelve months: $+1$ for a move into riskier assets, -1 for a move into safer assets, and 0 for no change. Because the measures in this subsection are downside-positive, a positive coefficient associates more perceived downside with a riskier reported adjustment. Belief timing matters. Among respondents with a reallocation answer and a same-calendar-year belief report, 87.1 percent have at least one belief report after the module month. Moreover, no same-calendar-year report predates the start of the twelve-month action window. The preferred construction averages only reports one through eleven months before the module and uses the official module weight with forecast mean and standard deviation controls. These reports precede the interview, but usually overlap the period in which the reported action occurred.

Table F.2. Perceived earnings risk, belief timing, and retirement portfolios

	(1) Pre-module	(2) No realized shock	(3) Realized income fall	(4) Post-module	(5) Joint
Panel A. Fixed-threshold imbalance; outcome: reported reallocation					
Pre-module imbalance	0.067^{**}	0.013	0.240^{***}		0.109^{**}
	(0.031)	(0.051)	(0.083)		(0.049)
Post-module imbalance				-0.052	-0.091^{*}
				(0.041)	(0.047)
Observations	2,136	1,270	190	1,805	1,749
Panel B. Centered Bowley downside shape; outcome: reported reallocation					
Pre-module shape	-0.024^{*}	-0.037^{**}	0.015		-0.022
	(0.013)	(0.017)	(0.035)		(0.015)
Post-module shape				-0.010	0.001
				(0.015)	(0.017)
Observations	2,136	1,270	190	1,804	1,748
Forecast mean, dispersion, and demographics	Yes	Yes	Yes	Yes	Yes
Module-year and state fixed effects	Yes	Yes	Yes	Yes	Yes

Panel C. Perceived dispersion; outcome: standing DC equity share			
	(1)	(2)	(3)
Shape control	Fixed threshold	Bowley	Moment skewness
Perceived standard deviation	-3.633^{***}	-3.766^{***}	-4.734^{***}
	(1.002)	(1.006)	(0.961)
Forecast mean and demographics	Yes	Yes	Yes
Module-year and state fixed effects	Yes	Yes	Yes
Observations	3,332	3,332	2,952

Standard errors in parentheses are clustered by state. Superscripts denote unadjusted two-sided significance: $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$. All specifications use the official Household Finance quarterly weight; demographics comprise age, education, and household income. The comparisons are exploratory and are not family-adjusted. In Panels A and B, the outcome is $+1$ for a riskier reallocation, -1 for a safer reallocation, and zero for no change during the preceding twelve months. Pre-module beliefs are recorded in months -11 to -1; post-module beliefs, in months $+1$ to $+11$, cannot have caused the reported action. Column (5) includes both terms. Centered moment skewness on the same sample grid produces coefficients from -0.083 to $+0.026$; only the realized-fall subsample and joint post-module term have $p < 0.05$. Panel C uses pre-module beliefs and reports percentage points of standing DC equity share per one-standard-deviation increase in perceived earnings dispersion. These are descriptive associations with an accumulated portfolio stock.

Table F.2, Panel A, shows that the fixed-threshold association depends on both the sample and the belief window. The pre-module coefficient is 0.013 after excluding intervening realized shocks and 0.240 in the 190-person large-income-fall subsample. This exploratory contrast does not decompose the full-sample coefficient or establish a subgroup difference. In the joint specification, pre-module and post-module beliefs enter with opposite signs ($+0.109$ and -0.091). Averaging reports from both sides of the interview can therefore combine different associations. Nor can an income fall after a belief report explain that earlier belief mechanically: selection, common shocks, or overlap between the action and shock windows may account for the pattern. Together with Lemma 4, these results show why a fixed-threshold coefficient cannot be read as a prospective response to centered tail orientation.

Panel B gives a more conventional background-risk sign for pre-module Bowley shape, but moment skewness does not corroborate it. The dispersion comparison in Panel C is more stable: a one-standard-deviation increase in perceived earnings dispersion is associated with a 3.633–4.734 percentage-point lower standing DC equity share across the three shape controls. Thus the SCE contains a clear descriptive scale association, whereas the evidence on centered shape remains sensitive to its definition and timing.

F.4 Centered upper and lower semimoments

To measure centered tails directly, I reconstruct each respondent-month's ten-bin public density. Open tails are assigned ± 16 percentage points, the three reports whose integer bins sum to 99 or 101 are renormalized, and lower and upper semideviations are centered on the respondent's elicited mean. Denote these semideviations by d_i^- and d_i^+. The regression includes their sum and their difference, or tilt, $d_i^+ - d_i^-$. Holding the sum fixed, a positive tilt coefficient associates a shift in perceived tail magnitude from downside toward upside with a riskier reported reallocation. This gap is measured in semideviation units; it is not a scale-free orientation measure.

Table F.3. Perceived earnings-tail tilt and reported retirement reallocation

	Reported retirement reallocation					DC share
	(1)	(2)	(3)	(4)	(5)	(6)
	Primary	Equal months	No large change	Post-module	Normalized	Primary
Upside-minus-downside semideviation tilt	0.0276^{**}	0.0284^{**}	0.0435^{***}	0.0047		-0.576
	(0.0128)	(0.0127)	(0.0137)	(0.0178)		(0.664)
Normalized semivariance orientation					0.0189	
					(0.0137)	
Focal coefficient p-value	0.036	0.030	0.003	0.793	0.173	0.390
BH q-value	0.054	—	0.008	—	0.347	0.390
Scale, mean, and demographics	Yes	Yes	Yes	Yes	Yes	Yes
Module-year and state fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,136	2,136	1,270	1,805	1,893	3,332

The reallocation outcome equals $+1$ for a reported move into riskier assets, -1 for a move into safer assets, and zero for no change over the preceding twelve months. Column (6) measures standing DC equity share in percentage points. Except in column (2), repeated belief reports are averaged within respondent using the public monthly SCE sampling weight. The standardized tilt enters jointly with total semideviation scale, expected earnings growth, age, education, household income, and the displayed fixed effects. Column (5) instead divides the upper-minus-lower semivariance difference by total variance and controls for total standard deviation. Regressions use official Household Finance quarterly weights; standard errors in parentheses are clustered by state. Superscripts denote unadjusted two-sided significance: $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$.

The primary $q = 0.054$ adjusts the three displayed focal shape tests in its outcome family; expanding to fourteen support and outcome variants gives $q = 0.169$. The no-large-change restriction excludes a reported large income, health, or wealth change and a respondent or spouse retirement-contribution cut; it is a post hoc subgroup and its stronger coefficient is scale dependent. The post-module report cannot have caused the retrospective action, but the formal pre-minus-post contrast is also imprecise ($p = 0.191$). Beliefs recorded before the module usually fall inside the retrospective action window. The separate lower and upper semideviations are correlated 0.95 and are therefore not presented as independent discoveries. The result is exploratory and descriptive: it does not survive the full support family, normalized orientation is null, standing share is null, and adding the tilt changes five-fold cross-validated RMSE by only -0.065%.

The semideviation estimates have opposing tail signs. Entered jointly, lower semideviation has a coefficient of -0.107 (SE 0.047; $p = 0.026$) and upper semideviation a coefficient of $+0.087$ (SE 0.045; $p = 0.060$). Their correlation of 0.95 motivates reporting the gap and sum rather than treating the two estimates as independent findings. In the full sample, the tilt mainly predicts fewer reported moves into safer assets; the increase in moves into riskier assets is imprecise, and the association among active movers alone is also imprecise. The full testing family, normalized-orientation comparison, and standing-share result in Table F.3 limit the interpretation. These estimates neither establish that perceived risk dominates assigned histories nor supply an independent confirmation of the CPS–SIPP entry result.

F.5 Belief revisions and the size of updates

The level comparisons leave open whether larger changes in the perceived earnings distribution are more closely associated with portfolio adjustment. I examine this possibility using repeated SCE earnings densities. For each respondent, I select the earliest and latest valid reports one to eleven months before the Household Finance interview, requiring at least six months between reports. Both endpoint distributions must have positive variance. Together with valid baseline demographics and positive weights, these restrictions define 1,132 respondents before portfolio outcomes are selected. Valid outcome reports leave 539 respondents for reported reallocation and 827 for the standing DC equity share.

For each of three shape measures, let x_i be the change between endpoints divided by the measure's baseline standard deviation in this common endpoint cohort. The linear specification includes x_i and the baseline shape level, together with baseline and changes in scale and expected earnings growth. An amplitude specification adds $x_i|x_i|$ and the analogous signed quadratic change in scale. The conditional slope is then $\beta_x + 2\beta_a|x_i|$: the amplitude coefficient allows it to vary with the absolute size of the revision, without choosing a large-update cutoff. A positive amplitude coefficient need not increase the absolute response when the linear coefficient is negative. All specifications include baseline age, education, and household-income categories, state and module-year fixed effects, and official module weights. The endpoint and specification rules were fixed after examining earlier analyses, so the exercise is exploratory.

Table F.4 reports the complete comparison. Linear revisions in centered tilt and normalized orientation are imprecisely associated with both outcomes. Their amplitude coefficients for reported reallocation are positive, with marginal p-values of 0.093 and 0.058, respectively. The downside-positive fixed-threshold revision has a positive standing-share coefficient with an unrounded p-value just below 0.100. None of the twelve focal tests survives the family adjustment: the smallest BH q-value is 0.399. The table also retains the supplementary joint tests of the revision and amplitude coefficients.

These results do not establish a stronger response to large changes in perceived tail risk. The outcomes are reported adjustment and standing shares, not changes in shares. Although both belief reports precede the interview, they overlap the retrospective action window, and the date of any

adjustment is unknown. Conditioning on the initial belief does not supply a person fixed effect or eliminate common shocks. A causal revision effect and an attention mechanism remain unidentified.

Table F.4. Earnings-belief revisions and retirement portfolios

	Reported reallocation			Standing DC equity share		
	(1)	(2)	(3)	(4)	(5)	(6)
Shape measure	Tilt	Orient.	Threshold	Tilt	Orient.	Threshold
Panel A. Linear revisions						
Revision, x	-0.024	0.012	0.088	0.717	1.613	5.673^{*}
	(0.031)	(0.024)	(0.126)	(1.202)	(1.398)	(3.380)
Focal marginal p	0.435	0.633	0.487	0.554	0.255	0.100
Focal BH-twelve q	0.664	0.690	0.664	0.664	0.611	0.399
Panel B. Revisions and their amplitude						
Revision, x	-0.093^{*}	-0.046	0.132	0.250	0.064	3.523
	(0.052)	(0.041)	(0.114)	(2.413)	(2.454)	(4.074)
Amplitude, $x\|x\|$	0.032^{*}	0.025^{*}	-0.039	0.294	0.654	0.897
	(0.019)	(0.013)	(0.028)	(0.881)	(0.978)	(1.120)
Focal marginal p	0.093	0.058	0.174	0.740	0.507	0.427
Focal BH-twelve q	0.399	0.399	0.523	0.740	0.664	0.664
Joint p: x, $x\|x\|$	0.193	0.092	0.288	0.699	0.459	0.219
Respondents	539	539	539	827	827	827
State clusters	46	46	46	48	48	48

Reallocation is -1 for safer assets, zero for no change, and $+1$ for riskier assets over the preceding twelve months. Equity share is the standing DC allocation in percentage points, not its change. Columns (1)–(3) use the same respondents, as do columns (4)–(6). Each panel reports separate regressions.

The earliest and latest valid earnings densities lie one to eleven months before the module and six to ten months apart. Tilt is the upper-minus-lower semideviation gap; orientation is the upper-minus-lower semivariance difference divided by total variance. Both are upside-positive. The threshold measure is $\Pr(g < -4) - \Pr(g > 4)$ and is downside-positive. Revisions use each measure's baseline standard deviation in the common 1,132-person endpoint cohort. Positive variance at both endpoints is required for this common comparison.

All models control for baseline shape, baseline and changes in scale and expected growth, baseline age, education, and household income, and state and module-year fixed effects, using official module weights. Scale is the semideviation sum in tilt models and standard deviation otherwise. Panel B also includes the signed quadratic change in scale. State-clustered CR1 standard errors appear in parentheses. Stars use unrounded marginal two-sided t_{G-1} tests: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. The twelve-test BH family comprises six linear-revision tests in Panel A and six amplitude tests in Panel B. Joint tests in Panel B are supplementary $F_{2,G-1}$ tests. This exploratory comparison does not identify responses to belief revisions or test attention directly.

G Uncertainty in the Estimated CPS Risk Measures

Suppose the portfolio regression assigns $\widehat{K}_c = T(\widehat{G}_c)$. Conventional regression inference conditions on the observed $\widehat{K}_c$. Clustering by cell allows second-stage residuals to covary within a cell, but does not account for the sampling variation of the nonlinear first-stage statistic.

The CPS bootstrap resamples unique people within each frozen signal cell. If a draw repeats a person m times, weighted first moments use mw and Kish denominators use mw^2, not $(mw)^2$. Every draw rebuilds weighted mean, standard deviation, zero share, quantiles, support selection,

equation (M.5), standardization, SIPP assignment, and the outcome regression. Across 200 draws, all requested estimates succeed; early support ranges from 59 to 61 cells and pooled support from 63 to 66. The strict-post level coefficient draws are not perfectly centered on the corrected point estimates. For participation, their mean is -0.0062 versus a point estimate of -0.0092 (mean-minus-point 0.0029; 2.5th–97.5th percentiles $[-0.0145, 0.0013]$). For the all-worker share, the corresponding values are -0.150 versus -0.234 (mean-minus-point 0.084; percentiles $[-0.363, 0.045]$).

The transition bootstrap additionally reassigns both annual observations and rebuilds the risk sets in each draw. For nonholder-to-positive-holdings entry, the draw mean is -0.650 point versus a point estimate of -1.047 points, a mean-minus-point displacement of 0.397 point. The across-draw standard deviation is 0.258 point, 198 of 200 coefficients are negative, and the 2.5th–97.5th percentile range is $[-1.125, -0.138]$ points. For signed change the corresponding standard deviation is 0.250 point; for exit it is 0.845 point. These ranges are descriptive summaries of rebuilt-CPS coefficients with SIPP held fixed, not confidence intervals, probability statements, or bias corrections; their displacement further motivates treating the variance combination as a sensitivity.

Table G.1 reports the corresponding resampling distributions for all six coefficients in the entry parameterization comparison. Some distributions are materially displaced from the original estimates, especially those involving the upper tail or total span. The added variance should therefore be interpreted as a first-stage sensitivity calculation, not as a recentered estimate or a bootstrap confidence interval.

Table G.1. CPS resampling distributions for the entry tail specifications

	Original	CPS-resampled coefficients			
Regressor	estimate	Mean	S.D.	P2.5	P97.5
Panel A. Lower and upper distances					
Lower distance	−2.712	−2.138	0.562	−3.269	−1.137
Upper distance	1.917	0.707	0.619	−0.530	2.134
Panel B. Total span and signed composition					
Total span	2.721	0.209	1.331	−2.286	3.187
Lower-minus-upper composition	−4.456	−2.076	0.986	−4.486	0.089
Panel C. Total span and incremental Kelley					
Total span	−0.755	−1.410	0.770	−2.981	−0.110
Incremental Kelley	−0.870	−0.430	0.293	−1.035	0.128

Entries are percentage-point coefficients per one draw-specific cross-cell standard deviation. Each of 200 CPS person-within-cell resamples rebuilds the risk measures, projections, and standardization while holding the 27,752 SIPP entry transitions fixed. All 200 draws succeed. P2.5 and P97.5 are percentiles of the rebuilt-CPS coefficients, not confidence limits for joint CPS–SIPP inference. The table reports the complete three-parameterization comparison; no coefficient is recentered or bias-corrected.

Given the independence of CPS and SIPP, I combine the two sources of variance as follows:

$$\widehat{\mathrm{Var}}_{\text{plug-in}}(\widehat{\beta}) = \widehat{\mathrm{Var}}_{\mathrm{SIPP}}(\widehat{\beta} \mid \widehat{G}) + \widehat{\mathrm{Var}}_{\mathrm{CPS}}(\widehat{\beta}).$$

The first term is the conditional two-way-clustered SIPP variance evaluated at the observed signal, and the second is the across-draw variance induced by resampling CPS people and rerunning the stated first-stage pipeline. This is not a joint nonparametric bootstrap of CPS and SIPP. It propagates CPS signal-estimation uncertainty within the frozen cell schema, crosswalk, and observed SIPP sample; it does not cover crosswalk or functional-form choice, SIPP sample-reconstruction uncertainty, or a latent-shape correction. The resampler also does not model cross-cell covariance from a person appearing in multiple cells, but no CPS person does so in the frozen annual-link sample. Reported plug-in p-values use a two-sided standard-normal reference. The primary strict-post first-stage results use this calculation; Table G.2 reports the corresponding design sensitivity.

Table G.2. Strict-post power for the early incremental signal

	80% minimum detectable effect		Power at observed magnitude	
Outcome	Conditional	CPS plug-in	Conditional	CPS plug-in
Participation	1.34	1.79	0.481	0.298
All-worker asset share	0.273	0.409	0.672	0.361

Entries use the strict-post 2019–2024 sample and are per one standard deviation of the early CPS incremental signal. Both MDE columns are in percentage points. The first pair gives two-sided 5%-size, 80%-power minimum detectable effects under a normal approximation; the second gives power at the absolute magnitude of the observed coefficient. "Conditional" treats the estimated CPS signal as fixed. "CPS plug-in" uses the conditional variance plus the CPS first-stage bootstrap variance and remains a sensitivity rather than joint CPS–SIPP inference. The table deliberately reports no latent-signal correction: weak cross-survey transport and possible nonclassical error preclude an equivalence mapping from the assigned signal to latent orientation.

Table G.3. Reporting group-estimated earnings-risk measures

Layer	Required disclosure	Diagnostic reported here
Population and support	Define origin population, destination zeros, growth transformation, cell schema, weights, and effective support.	Positive-origin workers; zero-inclusive arc growth; exact crosswalk; Kish support.
Reproducibility	Split globally by person and report rank and time stability.	Person-split Spearman–Brown reliability and early-versus-late correlations.
Incremental content	State the conditional interpretation and report the sequential relation with zero share, mean, and dispersion.	Raw; after zero share; after zero share, mean, and dispersion.
Downstream inference	Preserve support, nonlinear estimation, projection, standardization, and assignment inside first-stage resampling.	Independent CPS person bootstrap combined with conditional SIPP variance.

H Related Applications of Estimated Risk Measures

Table H.1 places the measurement issues in a broader set of finance applications. Each study estimates a risk measure from histories or groups and uses it in an analysis of portfolios, consumption, or performance. Their data and identification strategies differ, and rich administrative histories can substantially reduce the constraints that arise in public survey cells. The common consideration is that construction and estimation of the risk measure affect the interpretation of the subsequent analysis.

Table H.1. Finance applications using estimated risk measures

Study	Estimated risk measure	Application	Relation to the present analysis
Ranish (2013)	Permanent variance, disaster exposure, and cyclical labor income risk estimated in PSID histories and mapped to SCF households	Household stockholding and leverage	Direct two-sample group assignment; bootstraps estimated risk, multiple imputation, and generated-regressor uncertainty
Angerer and Lam (2009)	Occupation-by-education permanent and transitory income risk	Household risky-asset participation and shares	Early public group-risk assignment; does not isolate a nonlinear tail-orientation contrast
Fagereng, Guiso, and Pistaferri (2018)	Firm-level shocks to individual wage income	Household risky shares	Administrative and instrumental-variables benchmark for the effect of wage risk rather than a public-cell measurement design
Chang et al. (2022)	Individual income-volatility breaks and firm-volatility instruments	Household portfolio choice	Causal second-moment benchmark; distinct from higher-moment reliability and transport
d'Astous and Shore (2024)	Permanent income volatility shifted by university-admission discontinuities	Household participation and risky shares	Quasi-experimental sorting benchmark; stronger identification but a different risk object
Shen (2024)	Regional earnings-risk skewness	PSID consumption and risky portfolio shares	Nonlinear group risk assigned to households; cell support and incremental shape are relevant interpretation gates
Catherine, Sodini, and Zhang (2024)	Cyclical skewness and unconditional third moments in Swedish registers	Household equity participation and shares	Closest empirical comparison; the unconditional skewness–share relation attenuates within twins
Catherine (2022)	Estimated non-Gaussian, state-contingent income process	Life-cycle portfolio model	Shows the economic object that a reduced-form group statistic seeks to proxy
Gálvez and Paz-Pardo (2026)	Nonlinear earnings process estimated from PSID histories	Structurally estimated consumption, housing, participation, and risky shares	Quantitative nonlinear-risk benchmark, not a test of public group-assigned composition
Akey, Robertson, and Simutin (2026)	Methodologically revised Fama–French factor vintages	Mutual-fund performance, stock alphas and betas, and corporate-finance conclusions	Finance analogue showing that construction revisions can change downstream inference

Notes: The table contrasts risk constructs, portfolio margins, and identification strategies with the present historical composition–entry comparison. It distinguishes substantive portfolio benchmarks from measurement and inference precedents.

I Variance Composition and Joint Inference

I.1 The original wage benchmark

This appendix preserves the original-linkage wage benchmark and the matched wage–total extension. These comparisons use different earnings links and inference from the validated comparison in Section N.1. Table I.1 reports the wage comparison using upside variance composition, $C_c^{\text{up}} = V_c^+ - V_c^-$. In the original 60 cells, its coefficient is 3.390 percentage points per cross-cell standard deviation. Almost 94% of the SIPP-weighted variation in that regressor is absorbed by the other regressors. A one-standard-deviation difference in the remaining composition is therefore associated with 0.799 point more entry, a more representative magnitude for the conditional comparison than the full cross-cell shift.

Restricting to the common 59 cells changes the coefficient little. Reconstructing the earnings distribution from positive destinations within those cells also leaves it positive: 0.972 point per construction-specific standard deviation, or 0.640 point per standard deviation of residual composition. The estimates are not measured in a common cross-cell unit, so their ratio is not an attenuation estimate. The positive point relation is therefore not confined to a construction retaining wage zeros.

The upside and downside descriptions are economically symmetric. At fixed variance, more upside composition necessarily means less downside composition. Reversing the regressor's orientation gives equal-magnitude negative coefficients, identical standard errors, and identical two-sided tests; it is not a second empirical confirmation. Nor do the estimates establish that upper semivariance independently increases entry with lower semivariance unchanged. Both separate partial-moment coefficients are negative in the original sample, and the upper one is imprecise. Their difference at fixed total variance is the interpretable composition contrast.

The uncertainty about this contrast is greater than its conditional significance suggests. Independent-person reliability of incremental variance composition is only 0.22 in the original cells and 0.26 in the positive-destination construction. In 2,000 draws, rebuilding the CPS moments moves the respective coefficient means from 3.390 to 1.798 and from 0.972 to 0.544 points. Joint resampling of both surveys produces 95% percentile intervals of $[-0.346, 3.932]$ and $[-0.232, 1.343]$, although basic intervals exclude zero. Section N.6 examines this disagreement. Changing the sign convention leaves this sensitivity of inference unchanged.

Explicit treatment of the 2014 survey design preserves the point pattern. Retaining only its redesigned origin subsample gives 0.835 point more entry per residual standard deviation of upside composition in the zero-inclusive comparison and 0.646 point more entry in the positive-destination comparison. Refitting the original moments on the same SIPP observations shows that these results are not generated by the small change in supported cells. The zero-inclusive comparisons retain local CPS-plus-two-way-SIPP evidence; the positive-destination comparison is borderline, and broader crossed-reweighting inference is inconclusive. Omitting 2014 entirely attenuates all three magnitudes while preserving their signs. Appendix Table I.6 reports every declared convention and testing family.

Table I.1. Variance composition and stock market participation

Earnings distribution CPS cell frame	(1) Zero-inclusive Baseline	(2) Zero-inclusive Common	(3) Positive-to-positive Common
Upside composition, C^{up}	3.390^{***}	3.265^{**}	0.972^{**}
	(1.243)	(1.229)	(0.483)
Per residual S.D. (percentage points)	0.799	0.767	0.640
Conditional two-way SIPP: S.E.	0.692	0.662	0.328
Local IF + two-way SIPP: BH q	0.015	0.015	0.049
Local IF + product SIPP: S.E.	1.589	1.585	0.591
Local IF + product SIPP: BH q	0.066	0.066	0.105
CPS-only remeasurement mean	1.798	1.776	0.544
Joint remeasurement S.D.	1.060	1.084	0.398
Joint percentile 95% interval	$[-0.346,\ 3.932]$	$[-0.351,\ 3.952]$	$[-0.232,\ 1.343]$
Total variance V and $\sqrt{V}$	Yes	Yes	Yes
Other controls and fixed effects	Yes	Yes	Yes
Destination-zero probability	Yes	Yes	Not applicable
Observations	27,752	27,545	27,545
People	19,300	19,157	19,157
CPS signal cells	60	59	59

Notes: Entry is a transition from zero to positive reported stock-and-mutual-fund holdings in the strictly later SIPP window. For arc growth g centered at its cell mean μ, $V^- = \mathbb{E}[\min(g-\mu,0)^2]$, $V^+ = \mathbb{E}[\max(g-\mu,0)^2]$, $V = V^- + V^+$, and $C^{\mathrm{up}} = V^+ - V^-$. At fixed V, higher C^{up} means more upside and less downside variance: one composition contrast, not two independent effects. Each column regresses entry on C^{up} and the scale block $V, \sqrt{V}$, quadratic mean growth, lagged individual controls, and year, industry, education, and age-band fixed effects, using SIPP weights. Coefficients and uncertainty are percentage points per original construction-specific cross-cell S.D.; the residual-S.D. row uses the variation remaining after the regression controls. Parentheses combine local CPS IF and person-and-cell-clustered SIPP uncertainty. Stars use the corresponding two-sided t tests with cells minus one degrees of freedom: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. The conditional S.E. row omits CPS uncertainty. Local IF propagates CPS moment-estimation noise through the first-order influence function, including estimation of the centering mean, and combines it separately with two-way clustered or crossed-product-bootstrap SIPP covariance. Each combined BH q row adjusts the same three composition tests, using a t reference with cells minus one degrees of freedom. The remeasurement means and joint intervals use 2,000 draws with fixed original scales; the joint engine also applies positive crossed person-and-cell SIPP weights. All three joint percentile intervals cross zero. These are exploratory assignment associations; displaced redraw means and nonlinear attenuation prevent treating the joint redraw S.D. as stronger inferential evidence.

I.2 Wages and total labor earnings: original links

The wage-based comparison leaves open whether its interpretation depends on excluding other labor earnings. To address this question, I recover wages, net business earnings, and net farm earnings from the original public Census ASEC files. Total labor earnings sum these three components. Wages and totals come from the same release at both endpoints; only the redesigned 2014 origin subsample is retained. I keep the original positive-wage-origin worker population and compare the two measures on identical CPS histories. Positive total origins and nonnegative total destinations are required in both constructions; a second comparison additionally requires positive wage and total destinations. The exercise therefore broadens measured earnings without adding a different initial worker population or treating business losses as zero.

Table I.2 reports the original-linkage benchmark: four models on a common 58-cell sample of 27,419 subsequent SIPP entry transitions. With zero destinations retained, upside variance composition remains positively associated with entry for wages and total earnings. The total-earnings coefficient is 2.203 percentage points per cross-cell standard deviation, or 0.684 point per residual standard deviation, compared with 0.789 point for wages. Both survive local CPS-plus-two-way-SIPP inference within the complete four-test family ($q = 0.037$). The association is thus not confined to a wage-only measure that omits business and farm income. Section N.1 separately rebuilds this matched comparison using validated earnings links; these local-inference tests are not reassigned to the rebuilt estimates.

The positive-destination comparisons also have positive point estimates: 0.647 point per residual standard deviation for wages and 0.655 for total earnings. Both have four-test $q = 0.052$, making them suggestive rather than 5% family rejections. Under the broader SIPP product covariance, all four comparisons have $q = 0.111$. Formal wage–total slope differences, expressed in common wage-composition units and retaining cross-model covariance from both surveys, are imprecise. These tests neither establish equality nor show that total earnings generate a stronger portfolio response. The extension addresses the scope of measured labor earnings; it does not resolve weak incremental reliability, endogenous assignment, or sensitivity to the covariance approximation. Appendix I.9 records the source linkage, signed-earnings restrictions, and complete inference design.

Allowing dependence among household members leaves the comparison virtually unchanged. Clustering the CPS moment influences by household and the SIPP regression scores by sample-unit lineage and assignment cell gives four-test $q = 0.038$ for both zero-inclusive constructions and $q = 0.052$ for both positive-destination constructions. This permits common earnings disturbances and portfolio reporting across different workers assigned to different risk cells. This accounts for household dependence; endogenous assignment and the nonlinear-resampling discrepancy remain unresolved. Appendix Table I.7 reports the comparison.

I.3 Second-moment construction and conditional support

The variance components in equation (M.1) use the same early-CPS linked earnings histories and original assignment cells as the quantile comparison. Each early CPS person contributes one linked annual pair. The 60-cell sample contains 106,404 pairs. I use survey-weighted cell means and partial moments, with $V = V^- + V^+$ and $C^{\text{up}} = V^+ - V^-$. Both identities hold before any standardization. Cell standard deviations used to scale regressors are weighted by Kish effective support capped at 1,000. Original scales remain fixed in all remeasurement draws. The underlying downside-oriented estimates are reflected exactly for upside reporting: means and coefficients change sign, interval endpoints change sign and order, and standard errors and two-sided tests remain unchanged.

Positive-to-positive histories retain positive origin earnings and require positive destination earnings. One of the original 60 cells has positive-destination Kish support below 200, leaving 59 cells. Both the common-cell unconditional and positive-destination regressions use exactly 27,545 SIPP entry-risk observations on 19,157 people. Only the CPS distribution and its associated controls

Table I.2. Total labor earnings and variance composition

	(1)	(2)	(3)	(4)
Earnings	Wages	Total	Wages	Total
Destinations	Zero-inclusive		Both positive	
Upside composition, C^{up}	3.272^{**}	2.203^{**}	0.982^{*}	1.036^{**}
	(1.235)	(0.908)	(0.495)	(0.508)
Per residual S.D. (percentage points)	0.789	0.684	0.647	0.655
Local IF + two-way SIPP: BH q	0.037	0.037	0.052	0.052
Local IF + product SIPP: S.E.	1.611	1.183	0.606	0.622
Local IF + product SIPP: BH q	0.111	0.111	0.111	0.111
Incremental composition reliability	0.197	0.276	0.239	0.249
Total minus wage slope	Zero-inclusive		Both positive	
Common-unit coefficient	-0.610		0.028	
	(0.702)		(0.122)	
Local IF + two-way SIPP: BH q	0.777		0.816	
Local IF + product SIPP: BH q	0.844		0.844	
Observations	27,419	27,419	27,419	27,419
People	19,081	19,081	19,081	19,081
CPS signal cells	58	58	58	58

Notes: The outcome is subsequent entry from no reported stock-and-mutual-fund holdings. Wages and total labor earnings (wages plus net business and farm earnings) come from the same Census ASEC release at both endpoints. The population retains the original positive-wage-origin workers. Both constructions require positive total origins and nonnegative total destinations; columns (3)–(4) additionally require both wage and total destinations positive. Negative total earnings are excluded and recorded, not recoded to zero. The 2014 origins retain only the redesigned questionnaire subsample; later years are unchanged. Exact record linkage passes, but 232 high-income 2014 wage values differ from the retained IPUMS version, so these are not hybrid or silently harmonized earnings. All four models use identical later SIPP observations in the four-way effective-support intersection. Each includes C^{up}, total variance and its square root, quadratic mean growth, its own zero probability when applicable, and the same household controls and fixed effects. Coefficients use construction-specific cross-cell standard deviations. Parentheses combine local CPS influence-function and two-way SIPP covariance; stars use two-sided t_{57} tests: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. The broader covariance substitutes 1,000 paired person-by-cell SIPP reweightings. Four composition tests and two wage–total differences have separate complete BH families within each covariance convention. Differences rescale the total-earnings slope to the corresponding wage-composition unit and retain cross-model CPS and SIPP covariance; different conditional risk controls prevent a causal decomposition. Reliability uses 200 paired person partitions, not a sampling confidence interval. All results are exploratory.

change. In the positive-destination specification the zero-share control is identically zero and omitted. The full- and positive-support coefficients use different original standard deviations and cannot be compared as a common-unit attenuation ratio.

The weighted SIPP projection removes all regressors other than standardized variance composition, including the variance and standard-deviation scale block. Its residual standard deviation is 0.236 original cross-cell units in the baseline and 0.658 in the positive-destination construction. Multiplying by the corresponding coefficient gives the residual-standard-deviation comparisons in Table I.1. These are normalizations of the original estimates, not additional regressions or new tests. Independently splitting CPS people into halves 200 times gives incremental-composition Spearman–Brown reliability of 0.220, 0.253, and 0.261 across the three constructions. The cell-level reliability projection includes quadratic mean growth, variance, standard deviation, zero incidence when applicable, and industry,

education, and age-band components. Split-to-split dispersion is a conditional partition sensitivity, not a sampling confidence interval.

I.4 Uncertainty in the assigned earnings distributions

Sampling error in the historical earnings distributions contributes to uncertainty in the portfolio estimates. The CPS resampling exercise reconstructs weighted quantiles, support, zero incidence, mean and dispersion, residualization, standardization, and assignment before re-estimating the SIPP regressions. Following the two-sample household-finance logic of Ranish (2013), this carries uncertainty in the measured earnings environment into the portfolio calculation. The reported plug-in variance adds the coefficient variance across CPS resamples to the conditional two-way-clustered variance. Table I.3 shows the full three-outcome transition family.

Entry retains the clearest association. For incremental Kelley, the CPS resampling standard deviation is 0.258 point and the plug-in standard error is 0.375 point, giving $p = 0.0052$ and three-outcome $q = 0.0157$. A t_{59} reference instead gives $p = 0.0070$ and $q = 0.0211$. Signed participation change has plug-in $q = 0.064$, while exit remains imprecise. The corresponding standing all-worker share and participation have plug-in p-values of 0.109 and 0.153. At 5% size and 80% power, the measured-signal minimum detectable differences are 0.409 point for the all-worker share and 1.79 points for participation (Appendix Table G.2). These are limits on detection, not evidence of zero portfolio responses.

The separate two-tail calculation keeps the 27,752 entry transitions and 60 assignment cells fixed and reconstructs the tail measures within each of 200 CPS draws. The pair is jointly distinguishable from zero under the plug-in calculation ($p = 0.0022$), although the upper coefficient alone has $p = 0.068$. In a distinct 99,999-draw null-imposed wild bootstrap at the 60 cells, the lower and upper two-component BH q-values are 0.019 and 0.087; signed composition has $p = 0.033$. The two exercises address different sources of uncertainty. Neither combines a CPS remeasurement draw with a resampled SIPP outcome process; the joint exercise below does so.

The CPS draw distributions are displaced from the original estimates. Their mean coefficients are -2.138, $+0.707$, and -2.076 points for the lower tail, upper tail, and signed composition, compared with the original estimates of -2.712, $+1.917$, and -4.456. The mean incremental-Kelley entry coefficient is likewise -0.650 rather than -1.047 point. The remeasurement distributions are not centered on the original estimates, and SIPP is held fixed. Their means, sign frequencies, and percentiles are therefore diagnostics, not bias-corrected estimates or confidence intervals.

Allowing both surveys to vary makes the inference sensitive to the treatment of this displacement. In 2,000 joint draws, the lower, upper, and signed-composition means are -1.906, 0.698, and -2.028 points. Their 95% percentile intervals all contain zero; for composition the interval is $[-5.755, 1.481]$. Basic intervals, which reflect the draw distribution around the original estimate, instead exclude zero for all three. The same disagreement appears for the smooth variance components, so a quantile at the zero-earnings boundary cannot be its only source. Appendix Table I.4 reports both interval constructions.

Table I.3. Portfolio transitions after rebuilding the CPS first stage

	(1) Signed participation change	(2) Zero to positive holdings	(3) Positive to zero holdings
Incremental Kelley orientation	-0.731^{**}	-1.047^{***}	-0.813
	(0.360)	(0.375)	(1.429)
Conditional S.E.	0.260	0.272	1.152
Conditional BH q	0.010	< 0.001	0.483
CPS coefficient S.D.	0.250	0.258	0.845
Plug-in p	0.043	0.005	0.569
Plug-in BH q	0.064	0.016	0.569
Lagged controls and fixed effects	Yes	Yes	Yes
Lagged participation control	Yes	No	No
Observations	35,112	27,752	7,360
People	23,170	19,300	5,308
CPS signal cells	60	60	60
CPS resamples	200	200	200

Notes: Each column is a separate regression using consecutive SIPP observations in 2019–2024, lagged signals and controls, and current-year SIPP weights. Entries are percentage points per one early-CPS cross-cell standard deviation. All columns include lagged zero share, quadratic mean and dispersion, household controls, and year, lagged industry, education, and age-band fixed effects. Parentheses report plug-in standard errors. The 200 deterministic CPS person-within-cell resamples rebuild weighted quantiles, Kish support selection, residualization, standardization, assignment to both SIPP observations, and the transition risk sets. Every draw produces all three estimates; supported cells range from 59 to 61. SIPP is held fixed; two-way clustering by person and lagged CPS cell supplies the conditional variance. Plug-in S.E.^{2} = conditional S.E.^{2}+ CPS coefficient S.D.^{2}, with a two-sided standard-normal reference. This is a sensitivity, not a joint nonparametric bootstrap or inference for an error-free latent signal. Conditional and plug-in BH q-values each adjust exactly the three displayed transitions. Stars use unadjusted plug-in tests: $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$.

Adding CPS remeasurement can also reduce the variance of the refitted coefficients: for the upper quantile distance, the joint resampling standard deviation is 1.088 points, compared with 1.500 when only SIPP varies. Noisy regressors alter the conditional comparison and can contract both the center and spread of the refitted coefficients. A smaller original-point p-value from the joint draws is consequently not evidence of greater precision.

For the smooth variance components, a local influence-function calculation provides a separate check without refitting at a noisier set of cell moments. It includes uncertainty in the estimated centering mean and propagates all moment covariances through the same regression. In the original 60 cells, the resulting CPS standard error is 1.033 points, almost twice the 0.527-point standard deviation of nonlinear CPS redraws. Adding conditional two-way-clustered SIPP variance gives a total standard error of 1.243 points and three-comparison $q = 0.015$. The positive-destination comparison has $q = 0.049$. Replacing the SIPP component with the person-by-cell product-reweighting covariance gives $q = 0.066$ and $q = 0.105$, respectively. These first-order covariance approximations leave the coefficient unchanged. Their disagreement with the nonlinear draws shows that uncertainty depends on the method used; neither corrects the coefficient for bias.

I.5 Joint resampling and interval construction

The joint exercise fixes the SIPP observations, original cell support, and coefficient units. Within each original CPS cell, it draws multinomial counts over the observed people and rebuilds the weighted distribution. Independently, each SIPP person and lagged assignment cell receives a mean-one exponential weight. Their product multiplies the original SIPP survey weight, and the weighted regression is re-estimated. Positive multipliers preserve the original design support. The CPS-only, SIPP-only, and joint versions use matched underlying random draws, with 2,000 draws per version. For the quantile model, span and composition are exact fixed linear rotations of the jointly estimated lower and upper coefficients; each draw preserves fitted values and the joint Wald comparison.

Crossed reweighting allows dependence along the person and assignment-cell dimensions (Owen and Eckles, 2012). Its general conservative rationale does not imply exact finite-sample coverage for these nonlinear generated-regressor estimates. CPS resampling is conditional on the matched weighted sample, not a reconstruction of Census sampling strata and primary sampling units. Draw-specific support is recorded but does not redefine the original comparison.

For an original coefficient $\widehat{b}$ and draw quantiles $q^*_{.025}, q^*_{.975}$, Table I.4 reports the percentile interval $[q^*_{.025}, q^*_{.975}]$ and the basic interval $[2\widehat{b} - q^*_{.975}, 2\widehat{b} - q^*_{.025}]$. It also reports the original-point normal-reference test using the draw standard deviation. The percentile construction inherits the displaced coefficient distribution; the basic construction reflects its displacement around the original estimate. Neither is designated as definitive when they disagree. The full replication output retains all engines, draw means, covariances, and 90% as well as 95% intervals.

I.6 Local propagation of moment uncertainty

The smooth partial moments admit a first-order check that does not refit the regression at a noisier distribution. Write $d = g - \mu$, $m_- = \mathbb{E}[\min(d, 0)]$, and $m_+ = \mathbb{E}[\max(d, 0)]$. The moment influence functions are

$$\psi_\mu(g) = d, \tag{I.1}$$

$$\psi_-(g) = \min(d, 0)^2 - V^- - 2m_-d, \tag{I.2}$$

$$\psi_+(g) = \max(d, 0)^2 - V^+ - 2m_+d, \tag{I.3}$$

$$\psi_0(g) = \mathbf{1}\{g = -2\} - p_0. \tag{I.4}$$

The last terms in the partial-moment expressions account for estimation of the centering mean. In a cell with weights w_i, let $h_i = w_i\psi(g_i)/\sum_j w_j$. The local covariance under the within-cell multinomial draw is $\sum_i h_i h_i' - (\sum_i h_i)(\sum_i h_i)'/n$. For the positive-destination distribution, excluded zero observations contribute zero and the denominator sums positive-destination weights. The regression Jacobian propagates the full covariance of these primitive moments, retaining the identities $V = V^- + V^+$ and $\sigma = \sqrt{V}$ and the original scales.

Table I.4. Joint remeasurement and the choice of interval

Coefficient row / Interval row	(1) Original	(2) Draw mean	(3) Draw S.D.	(4) Normal p
	Percentile 95%		Basic 95%	
Panel A. Quantile distances: baseline 60 cells				
Lower distance	-2.712	-1.906	1.323	0.040
	$[-4.462, 0.872]$		$[-6.297, -0.962]$	
Upper distance	1.917	0.698	1.088	0.078
	$[-1.377, 2.870]$		$[0.964, 5.211]$	
Total tail span	2.721	0.456	2.645	0.304
	$[-4.667, 5.882]$		$[-0.440, 10.109]$	
Tail composition	-4.456	-2.028	1.872	0.017
	$[-5.755, 1.481]$		$[-10.394, -3.157]$	
Panel B. Centered second partial moments: upside composition				
Zero-inclusive, 60 cells	3.390	1.831	1.060	0.001
	$[-0.346, 3.932]$		$[2.847, 7.126]$	
Zero-inclusive, 59 cells	3.265	1.816	1.084	0.003
	$[-0.351, 3.952]$		$[2.577, 6.881]$	
Positive-to-positive, 59 cells	0.972	0.569	0.398	0.014
	$[-0.232, 1.343]$		$[0.601, 2.177]$	

Notes: Each contrast uses 2,000 paired CPS and SIPP draws with the original cell assignment, outcome rows, and coefficient units fixed. CPS persons are sampled within cells; independent mean-one exponential person and lagged-cell multipliers are multiplied into SIPP weights. Values are percentage points per original cross-cell S.D. Panel A retains quantile composition $D = L - U$; Panel B reports upside variance composition $C^{\text{up}} = V^{+} - V^{-}$. Reversing either contrast would reverse estimates and interval endpoints without changing its two-sided tests; this is not independent evidence. The upper row gives the original estimate, redraw mean, redraw S.D., and the two-sided standard-normal reference $p = 2\Phi(-|\widehat{\beta}/\mathrm{SD}_b|)$. The lower row gives the percentile interval in columns (1)–(2) and the basic interval $[2\widehat{\beta} - Q_{0.975}, 2\widehat{\beta} - Q_{0.025}]$ in columns (3)–(4). The normal reference ignores mean displacement; basic intervals extrapolate it as a bias correction; percentile intervals retain it. All seven percentile intervals contain zero. These interval conventions are sensitivity diagnostics, not interchangeable confirmatory tests. The crossed-weight procedure has no finite-sample exact guarantee, and weighted quantiles have atoms. Panel B avoids quantile jumps but remains sensitive to noisy composition and nonlinear coefficient attenuation; adding CPS noise can reduce redraw variance. No interval convention is selected because it excludes zero. Panel A uses 27,752 transitions; Panel B uses 27,752, 27,545, and 27,545 transitions, respectively.

The resulting local CPS covariance is combined separately with the locked two-way SIPP covariance and with the SIPP-only crossed-reweighting covariance. Independence of the two surveys justifies adding their first-order contributions; it does not make either covariance approximation exact. The reported reference uses a t distribution with the number of original cells minus one degrees of freedom, followed by BH adjustment over the three composition comparisons within each covariance combination. Numerical perturbations of the actual weighted moment estimator and every cell-by-moment regression derivative verify the analytic calculation.

Local CPS standard deviations exceed the nonlinear CPS redraw standard deviations by factors of 1.96, 1.94, and 1.83 in the three constructions. The discrepancy persists for smooth moments and therefore cannot be attributed solely to quantile jumps. It indicates curvature in the downstream

regression when little composition variation remains after conditioning. These calculations do not correct measurement error, identify latent household risk, or supply a bias-corrected coefficient.

I.7 Separating earnings-zero states

The earnings-state exercise uses the same 59-cell common support for three comparisons. The first retains unconditional quantiles and the total zero share. The second replaces that share with separate probabilities of zero wage earnings with no weeks worked and zero wage earnings despite some weeks worked. The third uses positive-to-positive quantiles, positive-distribution mean and dispersion, and the same two unconditional zero-state probabilities. CPS annual weeks worked classify the states; neither is labeled an exogenous job-loss or business-loss shock.

All four displayed coefficients in each model—lower distance, upper distance, span, and signed composition—form one twelve-scalar family. The lower–upper vectors form a separate three-model joint family. The span–composition model is an exact reparameterization, not additional identifying evidence. Each of 500 CPS draws reconstructs the distribution and controls on the fixed common SIPP sample. Plug-in covariance adds the draw covariance to the conditional two-way SIPP covariance; scalar reference tests use t_{58} and joint tests use $F_{2,58}$.

The two unconditional vectors retain joint plug-in evidence, while no scalar coefficient survives its complete family adjustment. For positive-to-positive quantiles, incremental signed-composition reliability is -0.113 across 200 independent-person splits. Its near-zero portfolio coefficient is therefore not a persuasive test against economically relevant positive-earnings orientation. The positive upside-variance-composition coefficient in Table I.1 concerns a different, smoother feature of that distribution, whose reliability is higher but still limited. Neither construction resolves sorting into the assigned earnings environments.

I.8 The 2014 survey design

The 2014 ASEC randomly assigned households to traditional and redesigned income questionnaires. Both subsamples have independently calibrated population weights. Combining them without adjustment gives earnings origin year 2013 a 28.10% share of the original early-pair weight. Retaining only the redesigned subsample reduces that share to 14.81%, consistent with the official recommendation for income comparisons with 2015 onward.[2] I therefore examine this restriction using the original IPUMS wage amounts, without introducing the Census wage-version differences discussed below. Two separately declared sensitivities omit 2014 or halve all its weights. The latter produces a population-normalized average of the two questionnaire regimes; it does not harmonize their instruments or correct selection into longitudinal linkage.

Each convention reconstructs the three original variance-composition comparisons, retaining their original units and requiring effective support of 200 before estimating portfolio coefficients. The

[2] See IPUMS CPS, "2014 3/8 ASEC File Information." The redesign did not extend to the job-income questions, but its independently population-weighted samples still require an explicit pooling convention.

Table I.5. Earnings tails and the composition of destination zeros

	(1)	(2)	(3)
Earnings distribution	Zero-inclusive	Zero-inclusive	Positive-to-positive
Zero-state controls	Total zero rate	Separate states	Separate states
Panel A. Lower and upper distances			
Lower distance, L	-2.129	-2.331	1.536
	(1.097)	(1.173)	(0.898)
BH q (12-test family)	0.112	0.112	0.139
Upper distance, U	2.312	2.367	1.891
	(1.230)	(1.249)	(1.280)
BH q (12-test family)	0.112	0.112	0.193
Panel B. Exact span and composition rotation			
Tail span, $S = L + U$	3.996	4.002	3.193
	(2.936)	(3.000)	(1.571)
BH q (12-test family)	0.204	0.204	0.112
Downside composition, $D = L - U$	-4.941	-5.124	0.007
	(2.040)	(2.070)	(0.540)
BH q (12-test family)	0.112	0.112	0.990
Joint L, U BH q (three models)	0.019	0.019	0.140
CPS mean: upper	0.786	0.791	1.018
CPS mean: composition	-2.101	-2.122	-0.031
Lagged controls and fixed effects	Yes	Yes	Yes
Observations	27,545	27,545	27,545
CPS signal cells	59	59	59

Notes: All columns use the same 59 CPS cells and SIPP lagged-nonholder sample. Column (1) includes total destination-zero probability; columns (2)–(3) replace it with no-work/wage-zero and worked-without-wage probabilities, measured on the full positive-origin CPS frame using annual weeks worked. The latter is not an identified business-loss state. Column (3) estimates tails and mean/SD controls conditional on positive destination wages; no person is excluded from the SIPP risk set by that CPS condition. All columns retain quadratic growth mean/SD, lagged individual controls, and year, industry, education, and age-band fixed effects. Coefficients are percentage points per construction-specific cross-cell S.D.; scaling weights use full-frame Kish support in every column. Parentheses report plug-in S.E.s from conditional person-and-cell variance plus coefficient variance from 500 CPS redraws, with SIPP fixed. Stars, if any, use BH q over all 12 displayed scalar tests: $^{***}q < 0.01$, $^{**}q < 0.05$, $^{*}q < 0.10$. Marginal tests use a t_{58} reference; joint tests use $F_{2,58}$ and a separate three-model BH family. The rotations are equivalent regressions, not independent evidence. Positive-only composition reliability is 0.038 before controls and -0.113 after residualization across 200 person splits, failing the declared measurement gate. Its imprecise coefficient does not establish economic irrelevance. The zero controls do not address nonlinear boundary mechanics or endogenous assignment.

redesigned-only construction leaves 59 cells in the zero-inclusive comparison and 58 cells on common positive-destination support. An additional accounting refit uses the original moments on each identical SIPP sample, separating sample changes from changes in the assigned distribution. Table I.6 reports the complete nine-model exercise. The redesigned-only comparisons form a three-test family; the other two conventions form a six-test sensitivity family. All-nine adjustments are retained as well. Local CPS covariance includes estimation of the centering mean; the broader SIPP covariance uses 1,000 matched positive person-by-cell reweightings.

All nine upside-composition point estimates remain positive. The redesigned-only zero-inclusive comparisons retain local CPS-plus-two-way evidence, including within the full nine-test family,

Table I.6. The 2014 ASEC design and variance composition

Earnings frame	Original moments	C^{up}	Per residual S.D.	q: two-way	q: product	Cells
A. Redesigned 2014 subsample only						
With zeros, full	3.410	3.338^{***}	0.835	0.015	0.064	59
$N = 27,626$		(1.194)				
With zeros, common	3.276	3.192^{***}	0.791	0.015	0.064	58
$N = 27,419$		(1.196)				
Positive, common	0.967	0.923^{*}	0.646	0.053	0.114	58
$N = 27,419$		(0.467)				
B. Omit interview year 2014						
With zeros, full	3.390	2.293^{*}	0.545	0.101	0.193	60
$N = 27,752$		(1.342)				
With zeros, common	3.265	2.426^{*}	0.575	0.101	0.193	59
$N = 27,545$		(1.359)				
Positive, common	0.972	0.699	0.501	0.101	0.193	59
$N = 27,545$		(0.420)				
C. Half of every 2014 weight						
With zeros, full	3.390	3.493^{**}	0.773	0.042	0.145	60
$N = 27,752$		(1.380)				
With zeros, common	3.265	3.425^{**}	0.759	0.042	0.145	59
$N = 27,545$		(1.353)				
Positive, common	0.972	1.013^{**}	0.640	0.089	0.193	59
$N = 27,545$		(0.494)				

Notes: The outcome is stock-and-mutual-fund entry from nonholding. $C^{\text{up}} = V^{+} - V^{-}$ is upper minus lower centered semivariance. Models retain original IPUMS wages, $V, \sqrt{V}$, quadratic mean growth, lagged household controls, fixed effects, and SIPP weights. Positive/common uses positive CPS destinations and the same SIPP sample as zero-inclusive/common. Cells retain original support and Kish counts of at least 200. Coefficients are percentage points per original cross-cell S.D.; residual S.D.s remove all other regressors. Original-moment estimates refit the old assignment on identical rows and units; they are not additional tests. Parentheses combine local CPS influence-function uncertainty, including estimation of the centering mean, with person-and-cell-clustered SIPP covariance. Stars use unadjusted two-sided t references with cells minus one degrees of freedom: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. The product column instead combines local CPS covariance with 1,000 crossed person-by-cell SIPP reweightings. BH q-values adjust the three Panel A tests and, separately, the six Panel B–C tests, fixed before these exploratory estimates. With a single nine-test adjustment, two-way q-values are 0.032 for zero-inclusive A/C, 0.079 for positive A/C, and 0.101 for B; corresponding product q-values are 0.109, 0.171, and 0.193. Neither covariance is full Census survey-design inference or a causal test.

whereas the positive-destination comparison is borderline. None of the redesigned-only results survives a 5% family threshold under the broader product covariance. Omitting 2014 attenuates the associations and does not preserve 5% family significance. The sign is unchanged under these pooling conventions, while significance depends on the covariance estimator.

I.9 The same-source total-earnings comparison

The public Census files for interview years 2014–2019 preserve wages, net business income, net farm income, and their signed earnings total. I link these records to the retained IPUMS extract using the documented ASEC ordering method, checking record counts, person order, household

boundaries, identifiers, demographics, work measures, and weights.[3] All wage comparisons in 2015–2019 agree exactly. In 2014, all non-wage checks pass, but 232 high-income wage values differ from the retained IPUMS version. Each lies at a documented earnings-disclosure threshold; the particular source-version revisions remain unresolved. I preserve the failed exact-wage check, flag the record-linked release separately, and use Census wages and Census total earnings consistently. No IPUMS-wage/Census-business hybrid is constructed, and the discrepant records are not selectively discarded.

Before portfolio estimation, the design was amended to use the redesigned 2014 origin subsample and the same Census earnings source. These choices followed the source and sampling-design audit, not inspection of new portfolio coefficients. The restriction leaves 90,657 original wage-worker histories. Eleven have negative total origins and 61 have negative total destinations, with one overlap, leaving 90,586 nonnegative-destination histories with positive total origins. The matched positive-destination comparison contains 84,902 histories. Negative totals are retained in an exclusion ledger and never recoded to zero. Effective support of 200 in all four constructions leaves 58 common cells and fixes the SIPP sample before estimation. These populations exclude histories with negative total labor earnings and do not represent workers without positive origin wages. Business or farm losses offset by other earnings can remain in the sample.

The four regressions use identical SIPP observations and household controls but their own earnings-distribution mean, variance, standard deviation, composition, and zero rate when applicable. Original construction-specific scales remain fixed throughout inference. Across 200 paired person partitions, incremental-composition reliabilities are 0.197 and 0.276 for zero-inclusive wages and totals, and 0.239 and 0.249 for the corresponding positive-destination distributions. Partition ranges cross zero; the total-earnings measure is not a high-precision risk benchmark.

For the wage–total contrasts, local CPS covariance retains all four estimating-function contributions from the same person. The SIPP covariance similarly stacks the four person-level regression scores before clustering by person and assignment cell; its diagonal is independently verified against each conditional regression. A separate 1,000-draw product-reweighting covariance uses the same person and cell multipliers in all four models. Each total-earnings coefficient is rescaled into the corresponding wage-composition unit before differencing, and both covariance stages retain the cross-model terms. Four composition tests and two differences form separate complete families under each covariance convention. None of these comparisons is a causal decomposition or an equivalence test.

I.10 Dependence within households and sample units

Members of the same household can contribute earnings histories in different assignment cells, while several SIPP workers can share portfolio-reporting circumstances. I therefore aggregate the existing CPS coefficient influence functions by the validated longitudinal household identifier rather than

[3] IPUMS describes the procedure in "IPUMS FAQs: How can I merge my IPUMS CPS file with an NBER CPS file?" The replication retains each original Census file, its dictionary, source URL, checksums, and the complete row-level comparison.

treating all people as independent. The 89,951 retained CPS case-influence records map to 62,085 households. In SIPP, I replace person clustering with clustering by the validated original sample-unit lineage, retaining assignment-cell clustering and the corresponding lineage–cell intersection subtraction. The 27,419 records on 19,081 people belong to 13,849 lineages. A lineage can include different current addresses or spawned households; it is not labeled a current household. These changes retain all point estimates and allow covariance across members assigned to different cells.

Table I.7 compares the expanded covariance with the original two-way and product-reweighting results. The zero-inclusive and positive-destination four-test q-values change negligibly, and the wage–total differences remain imprecise. The complete outputs also expand each survey's dependence group separately. Neither identifier reconstructs the Census strata and primary sampling units, and the exercise does not supply a full survey-design correction. It remains a local influence-function sensitivity conditional on the retained sample, not a correction for noisy latent regressors or endogenous sorting.

Table I.7. Household dependence and the composition–entry comparison

	(1)	(2)	(3)	(4)
Earnings	Wages	Total	Wages	Total
Destinations	Zero-inclusive		Both positive	
Upside composition, C^{up}	3.272^{**}	2.203^{**}	0.982^{*}	1.036^{**}
	(1.244)	(0.911)	(0.495)	(0.508)
Household/lineage covariance: BH q	0.038	0.038	0.052	0.052
Original person covariance: S.E.	1.235	0.908	0.495	0.508
Original person covariance: BH q	0.037	0.037	0.052	0.052
Original product covariance: S.E.	1.611	1.183	0.606	0.622
Original product covariance: BH q	0.111	0.111	0.111	0.111
Observations	27,419	27,419	27,419	27,419
SIPP people	19,081	19,081	19,081	19,081
SIPP sample-unit lineages	13,849	13,849	13,849	13,849
CPS assignment cells	58	58	58	58

Notes: The dependent variable is entry into reported stock-and-mutual-fund holdings. Coefficients, construction-specific units, samples, survey weights, risk controls (variance, standard deviation, quadratic mean growth, and zero incidence when applicable), household controls, and fixed effects are unchanged from Table I.2. Parentheses add CPS local influence-function covariance clustered by validated household to SIPP covariance clustered by validated sample-unit lineage and assignment cell. The CPS mapping contains 62,085 households; a SIPP lineage can contain multiple current or spawned households. Stars use unadjusted two-sided t_{57} tests: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Each displayed q row adjusts all four models. Original person-based and product-reweighting covariances remain separate benchmarks. These are local dependence sensitivities, not complete Census strata/PSU survey-design inference. The companion output retains both intermediate covariance combinations and both wage–total contrasts.

I.11 Uncertainty in the economic increment

The residual-standard-deviation magnitude already reported in the paper is estimated from both surveys. Write r_C for composition after projecting out every other regressor, $B = \sum_i w_i r_{Ci}^2$, $W_0 = \sum_i w_i$, and $A = \sum_i w_i r_{Ci} H_i$. Then the original slope is $\widehat{\beta} = A/B$, whereas the reported magnitude is

$M = \widehat{\beta}\sqrt{B/W_0} = A/\sqrt{BW_0}$. The normalization can be uncertain and correlated with the estimated slope.

I differentiate this entire expression, including all generated CPS risk controls, centering means, SIPP weights, and nuisance projections. CPS case influences are aggregated by household; SIPP influences are clustered by sample-unit lineage and assignment cell. For a multiplicative perturbation of SIPP weight i, with full-regression residual e_i and $s = \sqrt{B/W_0}$, the influence is

$$h_{Mi} = s\,\frac{w_i r_{Ci} e_i}{B} + \frac{\widehat{\beta} s}{2}\left(\frac{w_i r_{Ci}^2}{B} - \frac{w_i}{W_0}\right).$$

The second term captures variation in the reported increment. The CPS derivative similarly includes the change in B induced by every generated regressor. Direct perturbations of primitive moments and individual weights verify both derivatives.

Table I.8 reports the resulting local intervals. The total-earnings magnitude of 0.684 point has a 95% t_{57} interval of $[0.134, 1.235]$. These supplementary tests must not be substituted for the original coefficient tests. Since $s > 0$, $M = 0$ and $\beta = 0$ are the same population null, but finite-sample delta-Wald tests change under nonlinear normalization. The original positive-destination coefficient family remains borderline, and the broader product-covariance comparison remains inconclusive. These intervals quantify uncertainty in the normalized magnitude; nonlinear remeasurement remains unresolved.

Table I.8. Uncertainty in the residual-composition magnitude

	(1)	(2)	(3)	(4)
Earnings	Wages	Total	Wages	Total
Destinations	Zero-inclusive		Both positive	
Magnitude per residual S.D. (pp)	0.789^{***}	0.684^{**}	0.647^{**}	0.655^{**}
	(0.292)	(0.275)	(0.302)	(0.293)
Full-normalization 95% interval	[0.205, 1.372]	[0.134, 1.235]	[0.043, 1.251]	[0.069, 1.241]
Supplementary magnitude-test BH q	0.031	0.031	0.036	0.036
Holding residual S.D. fixed: S.E.	0.300	0.283	0.326	0.321
Original slope-test BH q	0.038	0.038	0.052	0.052
Broader original slope-test BH q	0.111	0.111	0.111	0.111

Notes: The displayed object is the existing entry coefficient times the weighted standard deviation of composition after removing all other regressors, not a newly estimated portfolio model. All four constructions retain the same 27,419 observations and 58 cells. Parentheses propagate both the coefficient and this normalization, including CPS moment means and nuisance projections and SIPP weights. Covariance permits CPS household and SIPP sample-unit-lineage/assignment-cell dependence. Stars and supplementary q-values use unadjusted two-sided t_{57} magnitude tests and their complete four-test BH family: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. The coefficient and magnitude have the same population zero restriction, but finite-sample delta-Wald tests are not invariant to nonlinear normalization. These tests do not replace the original slope tests, whose household-expanded and broader product-covariance families are retained below. In particular, the positive-destination slope family remains borderline. Intervals are local delta-method approximations, not nonlinear bootstrap intervals or a correction for measurement error.

I.12 Observable progression and conventional skewness

Two outcome-free checks examine the economic content of the composition statistic on the original-linkage benchmark. First, predictable earnings progression can contribute to within-cell dispersion even after removing the cell mean. I predict arc growth out of fold using a fixed basis of origin-year wage-rank splines, exact age, sex, origin-year, industry, and education effects. Five folds keep all members of each CPS household together. Each of 200 separate household-disjoint reliability splits reconstructs ranks and refits all predictors within each half. The residuals are centered again within cells. Their zero control remains the probability of zero *raw earnings*, not the probability of a zero regression residual.

Panel A of Table I.9 shows positive held-out earnings-prediction gains in all four constructions. Adjusted composition remains incrementally reproducible, but only weakly so. The adjustment changes the positive-earnings rankings substantially. These residuals remove progression associated with observed covariates; they are not elicited forecasts or identified unanticipated or permanent shocks. Conditioning on a noisy origin wage can also absorb measurement-error mean reversion, and year effects remove common realized growth.

Second, a comparison beyond variance need not be a comparison beyond conventional skewness. Panel B holds the original mean, scale, zero, and assignment controls fixed and adds $m_3/V^{3/2}$. Conditional correlations with upside composition are high. Only zero-inclusive total earnings passes the fixed incremental-information gates under both the CPS support metric and the exact SIPP household-control-profiled metric. The wage-only and positive-destination constructions have negative incremental-composition reliability. Their low incremental reliability does not establish that conventional skewness is economically sufficient. The full outputs also report the unnormalized third moment.

Table I.9. Observable progression and incremental earnings shape

Earnings	(1) Wages	(2) Total	(3) Wages	(4) Total
Destinations	Zero-inclusive		Both positive	
Panel A. Removing observable progression				
Held-out growth MSE reduction (%)	8.1	10.6	23.0	22.5
Raw–adjusted composition correlation	0.945	0.923	0.595	0.598
Original incremental reliability	0.184	0.268	0.212	0.215
Adjusted incremental reliability	0.188	0.221	0.270	0.267
Panel B. Composition given conventional skewness				
Conditional correlation with skewness	0.901	0.904	0.970	0.967
Incremental reliability: CPS metric	-0.032	0.223	-0.115	-0.045
Incremental reliability: SIPP metric	-0.109	0.182	-0.136	-0.078
Passes both information gates	No	Yes	No	No

Notes: These are measurement diagnostics, not portfolio coefficients or significance tests. All constructions retain the same 58 cells. Panel A predicts arc growth in five household-disjoint folds using origin-year wage-rank splines, exact-age polynomials, sex, origin-year, industry, and education effects. Ranks and predictions are re-estimated independently in each of 200 household-disjoint reliability halves. Residual distributions are re-centered within cells; zero incidence remains the raw earnings-zero probability. Incremental reliability removes mean and squared mean, variance, standard deviation, zero incidence when applicable, and industry/education/age components using the declared CPS projection. Panel B additionally removes standardized third-moment skewness, $m_3/V^{3/2}$, using either fixed CPS support weights or the exact SIPP household-control-profiled metric without fitting portfolio outcomes. Its correlation row uses the latter metric and the lower-moment controls. Gates require at least ten residual degrees of freedom in the full sample and every half, one percent of metric-specific target variation remaining, finite results in all 200 splits, and positive Spearman–Brown reliability under both metrics. Negative reliability means no detected incremental signal, not negative population reliability. Partition ranges, retained in the outputs, are not sampling confidence intervals. Passing a gate does not establish a precise signal or a valid instrument.

I.13 Validated earnings links and observable progression

The progression comparison requires equal positive CPSIDV values at the two earnings endpoints. This identifier validates longitudinal demographic consistency using the underlying CPS rotation history.[4] Among the eligible original histories in the matched 58 cells, the restriction retains 88,862 of 89,881 zero-inclusive histories and 83,372 of 84,287 positive-destination histories, in 61,480 source households. All 58 cells retain the required support. The original quantile and variance comparisons, local covariance estimates, and skewness comparison are not rebuilt by this restriction and remain explicitly separate benchmarks.

I refit the same fixed earnings-prediction basis on the retained histories before estimating the eight portfolio models. All four constructions retain positive held-out prediction gains, but incremental-composition reliability remains weak: 0.212, 0.215, 0.259, and 0.263 for zero-inclusive wages and totals and positive-destination wages and totals. All 200-split sensitivity ranges cross zero; these partition ranges are not confidence intervals. The four progression-adjusted models and their four raw-growth references use identical 58-cell support and 27,419 SIPP transitions. Each pair

[4] See IPUMS CPS, "Validated Longitudinal Identifier." Equality at two annual endpoints is the operational restriction here. An unequal identifier is not proof that this particular annual pair links different people.

retains the original-linkage raw-composition standard deviation for coefficient units, fixed throughout inference. The adjusted model conditions on its own residual-growth mean, squared mean, variance, and standard deviation, plus the probability of zero raw earnings when applicable. Its residual-standard-deviation magnitude is recomputed after removing every other regressor. A paired slope difference compares descriptive specifications with different conditioning distributions; it is not a causal mediation decomposition.

Exactly 1,000 matched draws rebuild both stages. One mean-one exponential multiplier per CPS household is shared across people, cells, and constructions. Multipliers are generated on the original ordered frame of 62,062 households and applied to the 61,480 retained households; the original random streams therefore also pair the two linkage versions exactly. Each draw reconstructs the weighted origin-year wage ranks, the fixed predictor basis, all five held-out fits, and the residual distributions. Original household-fold membership and spline knots remain fixed. Origin-covariate ranks use all folds, so this is not an untouched new-sample forecast. Independently, a SIPP sample-unit-lineage multiplier and an assignment-cell multiplier multiply the original survey weight; household controls are reprofiled under these weights. The original portfolio observations and supported cells are never selected again within a draw. Both earnings measures use the same multipliers within each support comparison.

The outputs retain CPS-only, SIPP-only, and joint versions of all eight models and all four adjusted-minus-raw slope contrasts. Within each engine, adjusted coefficients, raw references, and contrasts form separate complete four-test BH families. Original-point t_{57} approximations use the draw standard deviation. Percentile and basic intervals are shown together, alongside draw means and displacement; the full outputs also retain 90% and 99% intervals and magnitude uncertainty. Neither the reference tests nor one favorable interval construction is treated as definitive when nonlinear remeasurement shifts the distribution. These are dependence sensitivities, not a full complex-survey bootstrap or a correction for endogenous assignment.

Table I.10 reports the joint intervals for all eight slopes and four paired contrasts. Adjusted slopes remain positive, but none survives its four-test family adjustment. Every adjusted percentile interval includes zero; basic slope intervals exclude zero for both zero-inclusive constructions, barely so for total earnings. For the raw references, every percentile interval includes zero and every basic interval excludes it, with draw-mean attenuation of roughly 42–46%. Joint raw-coefficient standard deviations are smaller than their SIPP-only counterparts even though both stages vary. Rebuilding noisy regressors can contract the coefficient distribution rather than supply additional precision.

All paired adjusted-minus-raw slope differences are negative at the original estimates but imprecise under the t, percentile, and basic 95% intervals. Their four-test q-values are 0.281. The adjusted residual-standard-deviation magnitudes are similarly inconclusive, with four-test q-values ranging from 0.445 to 0.498 and all 95% point-centered and percentile intervals containing zero. Only the zero-inclusive wage magnitude has a basic interval excluding zero; the corresponding total-earnings magnitude does not, despite its barely positive basic slope interval. Neither a statistically established attenuation nor a progression-adjusted population association follows from this exercise.

Table I.11 reports validated-minus-original-linkage slope differences using the same 1,000 draws. These paired differences retain covariance between the two source versions, rather than treating them as independent estimates. Every joint 95% point-centered, percentile, and basic interval includes zero. The saved results retain the corresponding own-residual-standard-deviation magnitude comparisons, with the same qualitative conclusion. Source validation therefore leaves the raw point pattern close to its predecessor but does not establish equivalence or resolve the uncertainty. The original-linkage progression results and all source-specific inference remain in the replication output.

Table I.10. Observable-progression adjustment: intervals and paired comparisons

	(1)	(2)	(3)	(4)
Earnings	Wages	Total	Wages	Total
Destinations	Zero-inclusive		Both positive	
Panel A. Adjusted composition slope				
Slope	1.793^{*}	1.259	0.380	0.394
	(1.041)	(0.981)	(0.499)	(0.520)
Separate four-test BH q	0.362	0.409	0.451	0.451
95% t_{57} interval	[-0.292, 3.877]	[-0.705, 3.222]	[-0.620, 1.379]	[-0.646, 1.435]
95% percentile interval	[-1.109, 3.025]	[-1.296, 2.508]	[-0.727, 1.256]	[-0.768, 1.321]
95% basic interval	[0.560, 4.694]	[0.009, 3.814]	[-0.497, 1.486]	[-0.532, 1.557]
Draw mean minus original point	-0.866	-0.659	-0.145	-0.148
Panel B. Raw composition slope				
Slope	3.242^{***}	2.260^{**}	0.974^{**}	1.029^{**}
	(1.067)	(0.858)	(0.403)	(0.412)
Separate four-test BH q	0.014	0.019	0.019	0.019
95% t_{57} interval	[1.105, 5.380]	[0.543, 3.978]	[0.166, 1.781]	[0.203, 1.855]
95% percentile interval	[-0.329, 3.851]	[-0.562, 2.972]	[-0.262, 1.357]	[-0.265, 1.420]
95% basic interval	[2.634, 6.814]	[1.549, 5.083]	[0.590, 2.210]	[0.638, 2.323]
Draw mean minus original point	-1.483	-1.013	-0.411	-0.444
Panel C. Adjusted-minus-raw slope contrast				
Paired slope difference	-1.450	-1.002	-0.594	-0.635
	(1.093)	(0.920)	(0.491)	(0.504)
Separate four-test BH q	0.281	0.281	0.281	0.281
95% t_{57} interval	[-3.639, 0.739]	[-2.844, 0.841]	[-1.577, 0.389]	[-1.643, 0.374]
95% percentile interval	[-3.132, 1.395]	[-2.443, 1.156]	[-1.298, 0.578]	[-1.353, 0.649]
95% basic interval	[-4.294, 0.232]	[-3.159, 0.439]	[-1.765, 0.111]	[-1.918, 0.084]
Draw mean minus original point	0.617	0.354	0.266	0.296

Notes: Entries use Table N.1's fixed original raw-composition slope units (percentage points). All panels use 27,419 SIPP observations, 19,081 people, 13,849 lineages, and 58 cells. Parentheses are S.D.s from 1,000 paired joint full-pipeline draws. CPS weights are generated on the original 62,062-household random-number frame, then restricted to 61,480 validated source households; SIPP weights are lineage-by-cell products. Original-point-centered t_{57} tests are approximate and not null-imposed bootstrap tests; stars use strict unrounded thresholds: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Each panel has a separate BH-four family. Percentile bounds are empirical 2.5th and 97.5th percentiles; basic bounds subtract their reversed order from twice the original point. All eight slope percentile intervals contain zero, whereas both zero-inclusive adjusted basic intervals exclude zero; no method is selected for its significance. Panel C retains paired cross-model covariance; these descriptive contrasts are not mediation effects. Appendix I.13 gives the procedure. Saved outputs retain all three source engines, 90/95/99% intervals, covariances, and own-S.D. magnitude inference. This 58-cell source repair does not rebuild original 60/59-cell estimates or original-link local inference.

Table I.11. Validated-link versus original-source portfolio associations

	(1)	(2)	(3)	(4)
Earnings	Wages	Total	Wages	Total
Destinations	Zero-inclusive		Both positive	
Panel A. Adjusted composition: validated minus original				
Paired slope difference	-0.112	0.002	-0.116	-0.116
	(0.174)	(0.163)	(0.088)	(0.093)
Separate four-test BH q	0.697	0.993	0.435	0.435
95% t_{57} interval	[-0.460, 0.236]	[-0.325, 0.328]	[-0.292, 0.060]	[-0.302, 0.070]
95% percentile interval	[-0.419, 0.279]	[-0.377, 0.261]	[-0.265, 0.101]	[-0.283, 0.110]
95% basic interval	[-0.503, 0.195]	[-0.257, 0.380]	[-0.334, 0.033]	[-0.341, 0.051]
Draw mean minus original point	0.061	-0.030	0.064	0.063
Panel B. Raw composition: validated minus original				
Paired slope difference	-0.029	0.057	-0.009	-0.007
	(0.180)	(0.135)	(0.065)	(0.068)
Separate four-test BH q	0.916	0.916	0.916	0.916
95% t_{57} interval	[-0.389, 0.331]	[-0.214, 0.328]	[-0.139, 0.122]	[-0.143, 0.129]
95% percentile interval	[-0.332, 0.409]	[-0.218, 0.305]	[-0.124, 0.142]	[-0.127, 0.155]
95% basic interval	[-0.468, 0.273]	[-0.191, 0.332]	[-0.159, 0.107]	[-0.169, 0.112]
Draw mean minus original point	0.048	-0.024	0.009	0.010

Notes: The outcome, 27,419 SIPP rows, household controls, control specification, 58 assignment cells, and construction-specific original raw-composition units are fixed across source definitions. Generated risk controls are rebuilt for each source definition. Entries are validated-minus-original slopes in percentage points. Validated histories require equal positive endpoint CPSIDV; rejection does not prove that an original link is false. Both definitions use identical 1,000 CPS and SIPP multiplier streams, including the original 62,062-household CPS frame before restriction. Parentheses retain the full paired covariance, not the sum of marginal variances. Stars use unadjusted, original-point-centered approximate two-sided t_{57} tests: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$, with separate raw/adjusted BH-four families. All three 95% interval methods are displayed; every joint source-difference interval includes zero. These are not equivalence tests or causal effects of linkage errors. Each version's own conditional-S.D. magnitude differences, all source engines, and complete 90/95/99% interval families are retained in the replication outputs. The adjusted positive-destination wage and total magnitude differences are -0.062 and -0.059 percentage points (joint BH $q = 0.507$ and 0.507). This repair concerns only the matched partial-moment models; it does not validate or replace the paper's original 60/59-cell quantile/variance models or original-link local-inference calculations.

I.14 Portfolio composition conditional on standardized skewness

Only zero-inclusive total earnings meets the incremental-information requirements for the exploratory portfolio comparison. I add standardized skewness, $S_c = m_{3,c}/V_c^{3/2}$, to the existing composition model on the unchanged 58 cells and 27,419 entry transitions. Composition and every original risk control retain their original scales; skewness uses its full-sample cross-cell standard deviation, fixed throughout inference. Composition conditional on skewness and skewness conditional on composition form a symmetric two-test family. Their joint zero restriction is a separate $F_{2,57}$ test. No alternative third-moment regression is selected after inspecting these results.

Local CPS uncertainty propagates the centering mean through both shape statistics. For $x = g - \mu$, the third-central-moment influence is $x^3 - m_3 - 3Vx$; the chain rule for S_c also includes the variance influence. Aggregating all primitive influences by CPS household preserves their cross-cell and cross-moment covariance. SIPP covariance clusters by sample-unit lineage and assignment cell, including the intersection subtraction. A separate sensitivity combines the same local CPS covariance

with 1,000 matched SIPP lineage-by-cell positive reweightings. Both conventions retain the full two-coefficient covariance and testing family.

Table I.12. Total-earnings composition conditional on conventional skewness

	(1)	(2)
Dependent variable	Stock market participation	
Two-survey covariance	Primary	Broader product
Upside composition, C^{up}	3.782^{*}	3.782
	(2.261)	(3.045)
Symmetric marginal-family BH q	0.200	0.439
Standardized skewness, $m_3/V^{3/2}$	-1.234	-1.234
	(1.705)	(2.315)
Symmetric marginal-family BH q	0.472	0.596
Joint $F(2, 57)$ statistic	3.471	1.996
Joint zero-restriction p-value	0.038	0.145
Risk-scale and zero-incidence controls	Yes	Yes
Household controls and fixed effects	Yes	Yes
Observations	27,419	27,419
SIPP people	19,081	19,081
SIPP sample-unit lineages	13,849	13,849
CPS households in influence ledger	62,085	62,085
CPS assignment cells	58	58

Notes: Both columns report the same single regression on the zero-inclusive total-labor-earnings frame of Table I.2. Composition and standardized skewness enter jointly, alongside variance, standard deviation, quadratic mean growth, zero probability, lagged household controls, and year, industry, education, and age-band fixed effects. Coefficients are percentage points per fixed full-CPS cross-cell standard deviation; skewness uses weights capped at a person Kish count of 1,000. Parentheses in column (1) combine CPS local influence-function covariance clustered globally by household with SIPP sample-unit-lineage/assignment-cell covariance, subtracting lineage–cell intersections. The CPS calculation propagates estimation of the centering mean and all moment covariances. Column (2) replaces the SIPP component with covariance from 1,000 lineage-by-cell product reweightings, retaining the same CPS component. Stars use each column's unadjusted two-sided t_{57} test of a zero coefficient and strict unrounded thresholds: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. The two marginal coefficients form one symmetric BH family within each column; the joint F test is separate. Only this earnings construction passed the prior outcome-free incremental-skewness information gates; all three other construction failures remain visible in Table I.9. This is an exploratory gate-selected comparison, not a correction for the project's complete search history. Conditional-only inference and all product draws remain in the companion output.

Neither individual coefficient survives its two-test adjustment. The primary joint test rejects at 5%, but the broader joint test does not. Thus this exercise does not establish an incremental composition association beyond conventional skewness. The other three constructions have insufficient incremental information for this comparison.

I.15 An early-SIPP total-earnings validation attempt

An additional measurement check uses annual total earnings from the four waves of the 2014 SIPP panel. It retains complete annual earnings records and adjacent within-panel transitions, with positive origins and nonnegative destinations, and compares total earnings with job-1 earnings on matched records. Negative monthly amounts enter the annual sums; negative annual destinations are recorded separately and excluded, not recoded to zero. The resulting sample contains 29,307 transitions on 15,785 people in the original 60 cells. Repeated transitions remain together in person splits, and effective support uses each person's summed weight. Only 18 cells meet the original effective-support threshold of 200. On this overlap, CPS wage variance correlates with SIPP total-earnings variance at 0.577, but upside variance composition correlates at only 0.162. Its raw split reliability is 0.336 across 200 person partitions. The declared conditional projection leaves too few residual degrees of freedom to assess incremental transport. I therefore do not use this exercise as independent validation or proceed to portfolio regressions with these measures. Historical SIPP total earnings also contain a documented business-profit collection and processing problem, preventing their treatment as an error-free benchmark.[5]

[5]The Census Bureau's March 24, 2022 user note, "Business Profits Data Collection and Processing Issue," describes the possible repeated reporting of annual profits across employment spells and its effects on total earnings. No assumed correction is applied here.

I.16 Own prior earnings: construction and complete comparisons

The own-history comparison conditions on a worker's earnings before the baseline portfolio observation. If destination year is d, nonholding is measured in $d-1$ and the earnings years are $d-3$ and $d-2$. The supported panel paths produce destinations in 2020, 2022, 2023, and 2024; no 2021 destination is supplied by a substitute path. The primary sample requires self or proxy interview status in all 24 prior monthly records. These are not 24 independent interviews or a guarantee of unimputed earnings. Complete histories retain 6,639 of the original 27,419 nonholder transitions, on 6,639 workers and 5,291 sample-unit lineages in 58 cells. Original destination survey weights are retained; they are not history-longitudinal weights.

I sum the twelve monthly total-person earnings recodes, `TPEARN`, within each prior calendar year. They include all jobs and signed business profits or losses, not just first-listed-job wages or household disposable income. Exact nominal sums determine zero and negative indicators. Native missing values do not become zeros; a structural zero is used only when the source record establishes a valid no-job state. Incomplete years are excluded by the fixed support rule, not annualized. Amounts can contain allocation, suppressed-component, disclosure, and business-profit processing error. The analysis applies the documented source restrictions, although these measurement errors may remain.

Annual CPI-U converts the two totals to 2022 dollars. For levels Y_0, Y_1, the primary block contains $\operatorname{asinh}(Y_0/10{,}000)$, $\operatorname{asinh}(Y_1/10{,}000)$, signed arc change $2(Y_1-Y_0)/(|Y_1|+|Y_0|)$, and separate zero and negative indicators at both dates. The arc is zero when both totals are zero and otherwise uses the stated denominator. It is bounded signed growth, not percentage growth or an identified surprise. The decline-kink sensitivity replaces the arc by $\max\{\operatorname{asinh}(Y_0/10{,}000) - \operatorname{asinh}(Y_1/10{,}000), 0\}$, retaining the other controls. Neither block estimates a personal tail distribution, earnings persistence, or a loss followed by recovery.

Two fixed source sensitivities accompany the primary sample. The panel-2022 processing sensitivity retains 1,275 primary histories, with earnings in 2021–2022, baseline holdings in 2023, and entry in 2024. Its post-fix processing label does not certify error-free income. The Type-Z-inclusive sample contains 7,406 domain-valid histories, including legal Type-Z months, and nests the primary sample. It changes interview coverage, not the accuracy of reported amounts. Both sensitivities retain 58 assignment cells. Sample membership and both history blocks were fixed before the new amounts and portfolio estimates were inspected; this is an exploratory follow-up, not an original preregistration.

For each of the three samples, baseline, signed-change, and decline-kink specifications retain all eight raw/adjusted wage/total and zero-inclusive/positive CPS constructions. The separate eight-model full-sample reference makes 80 models in total. Within every history comparison, observations, destination weights and all original noncomposition controls remain the same; the history block is added without replacing variance, standard deviation, mean, mean squared, applicable zero incidence, personal controls or fixed effects. Positive CPS destinations restrict the source earnings distribution, not the recipient history sample. The original composition units are held fixed, and each model's residual-standard-deviation magnitude is recomputed after profiling all its controls. The 48

augmented-minus-matched-baseline slope comparisons retain their joint cross-model covariance; their differences are not mediation shares.

Exactly 1,000 matched draws rebuild CPS ranks, prediction fits and moments while allowing dependence within CPS households and within SIPP lineages and assignment cells. CPS-only, SIPP-only and joint calculations retain the original random-weight frames before recipient subsetting. The joint calculation is the primary uncertainty summary. Stars use the declared approximate original-point two-sided t_{57} tests; BH8 adjusts all eight raw/adjusted constructions within each sample, control block and target. Paired differences have their own eight-test families. The original four-construction adjustments remain in the saved output for continuity. Point-centered, percentile and basic intervals are retained at 90%, 95% and 99%; no interval method is selected for favorable significance.

Tables I.13 and I.14 report every joint slope and paired 95% interval. All 80 slope percentile intervals include zero. All 48 paired changes include zero under each of the three 95% methods. The primary signed-change slope family has no BH8 rejection; its eight paired q-values are 0.994. Replacing the signed change by a decline kink gives the same qualitative limitation. In both the panel-2022 and Type-Z-inclusive sensitivities, progression-adjusted positive-destination slopes are negative but highly imprecise. Their imprecision precludes interpreting the negative signs as opposite effects. None of the eight primary signed-change comparisons satisfies the declared positive-reporting screen requiring a positive primary point, BH8 $q < 0.05$, and positive lower bounds under all three joint 95% methods. This screen is a reporting rule, not a separately calibrated hypothesis test. The history controls leave the primary raw point pattern nearly unchanged but do not establish equivalence, a stable population association, or a causal/perception mechanism.

Table I.13. Prior earnings and entry: complete slope intervals

Notes: All entries use 1,000 joint CPS-household/SIPP-lineage-by-cell reweighted draws and report percentage points per original raw-source CPS composition SD, unchanged across control blocks and within raw/adjusted pairs. Point-centered intervals use the original point, draw SD and t_{57} critical value. Percentile intervals use empirical 2.5th/97.5th percentiles; basic intervals reflect them about the original point. BH8 uses unadjusted two-sided point-centered reference probabilities within each sample/control-block slope family of eight raw/adjusted constructions; paired differences form separate families. These approximate positive-weight sensitivity tests do not certify complex-survey coverage. No interval is selected by its sign. Full 90/95/99 intervals, CPS-only/SIPP-only/joint engines, own-SD magnitudes and covariance arrays remain in the replication output.

All specifications retain baseline controls and their complete nuisance span. Both history blocks add two asinh annual total-earnings levels and four zero/negative indicators; the primary block adds signed arc change and the sensitivity block adds a decline kink. CPS positive-destination restrictions do not change recipient rows. Original destination weights are not history-longitudinal weights. Only positive-support destination years are listed. The full reference is not a matched no-history comparator. Processing and Type-Z sensitivities cannot substitute for the primary comparison; separate BH8 families do not authorize an omnibus claim. Neither histories nor paired differences identify beliefs, causal responses or mediation shares.

CPS earnings construction	Point-centered 95%	Percentile 95%	Basic 95%	BH8 q
Full-sample reference: 27,419 nonholder transitions; 19,081 people; 13,849 lineages; 58 cells. Destination years: 2020, 2021, 2022, 2023, 2024. Original baseline-nonholder sample; a selection reference, not a matched baseline.				
Full-sample reference; baseline controls; raw growth				
Wages, zero-inclusive	[1.105, 5.380]	[-0.329, 3.851]	[2.634, 6.814]	0.029
Total earnings, zero-inclusive	[0.543, 3.978]	[-0.562, 2.972]	[1.549, 5.083]	0.038
Wages, positive	[0.166, 1.781]	[-0.262, 1.357]	[0.590, 2.210]	0.038
Total earnings, positive	[0.203, 1.855]	[-0.265, 1.420]	[0.638, 2.323]	0.038
Full-sample reference; baseline controls; progression-adjusted growth				
Wages, zero-inclusive	[-0.292, 3.877]	[-1.109, 3.025]	[0.560, 4.694]	0.145
Total earnings, zero-inclusive	[-0.705, 3.222]	[-1.296, 2.508]	[0.009, 3.814]	0.273
Wages, positive	[-0.620, 1.379]	[-0.727, 1.256]	[-0.497, 1.486]	0.451
Total earnings, positive	[-0.646, 1.435]	[-0.768, 1.321]	[-0.532, 1.557]	0.451
Primary histories: 6,639 nonholder transitions; 6,639 people; 5,291 lineages; 58 cells. Destination years: 2020, 2022, 2023, 2024. Self/proxy interview status in all 24 prior monthly records; not a guarantee of unimputed earnings.				
Primary histories; baseline controls; raw growth				
Wages, zero-inclusive	[0.256, 8.282]	[-1.601, 6.562]	[1.976, 10.139]	0.216
Total earnings, zero-inclusive	[-0.691, 5.751]	[-1.645, 4.797]	[0.263, 6.705]	0.216
Wages, positive	[-0.309, 2.680]	[-0.788, 2.232]	[0.138, 3.159]	0.216
Total earnings, positive	[-0.372, 2.694]	[-0.864, 2.282]	[0.039, 3.185]	0.216
Primary histories; baseline controls; progression-adjusted growth				
Wages, zero-inclusive	[-0.826, 6.587]	[-1.855, 5.491]	[0.270, 7.616]	0.216
Total earnings, zero-inclusive	[-1.483, 5.344]	[-2.169, 4.467]	[-0.606, 6.030]	0.350
Wages, positive	[-1.675, 1.875]	[-1.578, 1.926]	[-1.725, 1.779]	0.910
Total earnings, positive	[-1.480, 2.199]	[-1.527, 2.123]	[-1.404, 2.245]	0.797
Primary histories; signed-change history controls; raw growth				
Wages, zero-inclusive	[0.271, 8.270]	[-1.641, 6.524]	[2.016, 10.181]	0.217
Total earnings, zero-inclusive	[-0.697, 5.746]	[-1.668, 4.821]	[0.228, 6.717]	0.217
Wages, positive	[-0.308, 2.677]	[-0.766, 2.216]	[0.153, 3.135]	0.217
Total earnings, positive	[-0.375, 2.693]	[-0.847, 2.255]	[0.063, 3.165]	0.217

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Continued

CPS earnings construction	Point-centered 95%	Percentile 95%	Basic 95%	BH8 q
Primary histories; signed-change history controls; progression-adjusted growth				
Wages, zero-inclusive	[-0.770, 6.647]	[-1.824, 5.524]	[0.354, 7.702]	0.217
Total earnings, zero-inclusive	[-1.374, 5.463]	[-2.096, 4.538]	[-0.450, 6.185]	0.315
Wages, positive	[-1.631, 1.923]	[-1.566, 1.981]	[-1.689, 1.859]	0.870
Total earnings, positive	[-1.442, 2.243]	[-1.490, 2.192]	[-1.391, 2.291]	0.760
Primary histories; decline-kink history controls; raw growth				
Wages, zero-inclusive	[0.265, 8.267]	[-1.602, 6.546]	[1.987, 10.134]	0.215
Total earnings, zero-inclusive	[-0.685, 5.754]	[-1.633, 4.818]	[0.251, 6.703]	0.215
Wages, positive	[-0.308, 2.674]	[-0.762, 2.221]	[0.146, 3.129]	0.215
Total earnings, positive	[-0.370, 2.692]	[-0.848, 2.244]	[0.078, 3.170]	0.215
Primary histories; decline-kink history controls; progression-adjusted growth				
Wages, zero-inclusive	[-0.808, 6.639]	[-1.885, 5.524]	[0.306, 7.715]	0.215
Total earnings, zero-inclusive	[-1.415, 5.442]	[-2.142, 4.529]	[-0.502, 6.169]	0.326
Wages, positive	[-1.655, 1.906]	[-1.572, 1.954]	[-1.702, 1.824]	0.888
Total earnings, positive	[-1.465, 2.224]	[-1.502, 2.154]	[-1.395, 2.261]	0.779
Panel-2022 processing sensitivity: 1,275 nonholder transitions; 1,275 people; 1,031 lineages; 58 cells. Destination years: 2024. Primary panel-2022 histories only; no processing partition is certified clean.				
Panel-2022 processing sensitivity; baseline controls; raw growth				
Wages, zero-inclusive	[-2.626, 13.754]	[-5.149, 11.349]	[-0.221, 16.276]	0.911
Total earnings, zero-inclusive	[-3.956, 8.976]	[-4.793, 8.192]	[-3.172, 9.813]	0.911
Wages, positive	[-2.177, 3.915]	[-2.362, 3.623]	[-1.885, 4.101]	0.911
Total earnings, positive	[-2.514, 3.722]	[-2.634, 3.570]	[-2.363, 3.841]	0.911
Panel-2022 processing sensitivity; baseline controls; progression-adjusted growth				
Wages, zero-inclusive	[-3.782, 10.921]	[-4.460, 9.771]	[-2.632, 11.599]	0.911
Total earnings, zero-inclusive	[-5.747, 7.620]	[-6.332, 7.248]	[-5.375, 8.205]	0.911
Wages, positive	[-3.687, 3.294]	[-3.312, 3.483]	[-3.876, 2.919]	0.911
Total earnings, positive	[-3.899, 3.332]	[-3.380, 3.737]	[-4.303, 2.814]	0.911
Panel-2022 processing sensitivity; signed-change history controls; raw growth				
Wages, zero-inclusive	[-2.467, 13.915]	[-4.738, 11.250]	[0.198, 16.185]	0.862
Total earnings, zero-inclusive	[-3.796, 9.212]	[-4.745, 8.202]	[-2.786, 10.160]	0.862
Wages, positive	[-2.078, 4.023]	[-2.257, 3.675]	[-1.730, 4.202]	0.862
Total earnings, positive	[-2.394, 3.856]	[-2.466, 3.692]	[-2.230, 3.928]	0.862
Panel-2022 processing sensitivity; signed-change history controls; progression-adjusted growth				
Wages, zero-inclusive	[-3.125, 11.469]	[-4.220, 10.070]	[-1.726, 12.564]	0.862
Total earnings, zero-inclusive	[-5.106, 8.159]	[-5.623, 7.521]	[-4.467, 8.676]	0.862
Wages, positive	[-3.592, 3.348]	[-3.242, 3.487]	[-3.731, 2.998]	0.944
Total earnings, positive	[-3.820, 3.367]	[-3.359, 3.650]	[-4.102, 2.906]	0.944
Panel-2022 processing sensitivity; decline-kink history controls; raw growth				
Wages, zero-inclusive	[-2.608, 13.765]	[-4.856, 11.346]	[-0.189, 16.014]	0.880
Total earnings, zero-inclusive	[-3.860, 9.132]	[-4.672, 8.261]	[-2.989, 9.944]	0.880
Wages, positive	[-2.105, 3.978]	[-2.252, 3.766]	[-1.893, 4.125]	0.880
Total earnings, positive	[-2.423, 3.812]	[-2.499, 3.549]	[-2.161, 3.887]	0.880
Panel-2022 processing sensitivity; decline-kink history controls; progression-adjusted growth				
Wages, zero-inclusive	[-3.226, 11.429]	[-4.428, 9.996]	[-1.792, 12.632]	0.880
Total earnings, zero-inclusive	[-5.172, 8.100]	[-5.953, 7.402]	[-4.474, 8.882]	0.880
Wages, positive	[-3.594, 3.344]	[-3.239, 3.532]	[-3.782, 2.989]	0.943
Total earnings, positive	[-3.822, 3.361]	[-3.364, 3.700]	[-4.160, 2.903]	0.943

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CPS earnings construction	Point-centered 95%	Percentile 95%	Basic 95%	BH8 q
Type-Z-inclusive histories: 7,406 nonholder transitions; 7,406 people; 5,672 lineages; 58 cells. Destination years: 2020, 2022, 2023, 2024. Domain-valid histories including legal Type-Z months; primary histories are nested within this sample.				
Type-Z-inclusive histories; baseline controls; raw growth				
Wages, zero-inclusive	[-0.970, 7.389]	[-2.333, 5.953]	[0.466, 8.752]	0.482
Total earnings, zero-inclusive	[-1.607, 5.108]	[-2.381, 4.363]	[-0.862, 5.881]	0.482
Wages, positive	[-0.695, 2.419]	[-1.088, 2.050]	[-0.325, 2.813]	0.482
Total earnings, positive	[-0.730, 2.456]	[-1.193, 2.075]	[-0.349, 2.919]	0.482
Type-Z-inclusive histories; baseline controls; progression-adjusted growth				
Wages, zero-inclusive	[-0.838, 6.731]	[-1.951, 5.528]	[0.364, 7.844]	0.482
Total earnings, zero-inclusive	[-2.002, 4.876]	[-2.648, 4.054]	[-1.179, 5.522]	0.541
Wages, positive	[-2.052, 1.545]	[-1.881, 1.725]	[-2.232, 1.374]	0.890
Total earnings, positive	[-1.879, 1.845]	[-1.842, 1.936]	[-1.969, 1.808]	0.986
Type-Z-inclusive histories; signed-change history controls; raw growth				
Wages, zero-inclusive	[-0.973, 7.380]	[-2.434, 5.884]	[0.523, 8.841]	0.487
Total earnings, zero-inclusive	[-1.621, 5.097]	[-2.419, 4.366]	[-0.891, 5.895]	0.487
Wages, positive	[-0.709, 2.402]	[-1.131, 2.038]	[-0.344, 2.825]	0.487
Total earnings, positive	[-0.742, 2.443]	[-1.222, 2.094]	[-0.393, 2.923]	0.487
Type-Z-inclusive histories; signed-change history controls; progression-adjusted growth				
Wages, zero-inclusive	[-0.821, 6.740]	[-1.965, 5.642]	[0.277, 7.884]	0.487
Total earnings, zero-inclusive	[-1.935, 4.915]	[-2.597, 3.974]	[-0.994, 5.577]	0.516
Wages, positive	[-2.033, 1.563]	[-1.904, 1.725]	[-2.195, 1.435]	0.908
Total earnings, positive	[-1.864, 1.857]	[-1.882, 1.924]	[-1.931, 1.875]	0.997
Type-Z-inclusive histories; decline-kink history controls; raw growth				
Wages, zero-inclusive	[-0.968, 7.357]	[-2.396, 5.785]	[0.605, 8.785]	0.481
Total earnings, zero-inclusive	[-1.602, 5.092]	[-2.361, 4.310]	[-0.820, 5.851]	0.481
Wages, positive	[-0.695, 2.404]	[-1.144, 1.996]	[-0.288, 2.852]	0.481
Total earnings, positive	[-0.728, 2.443]	[-1.197, 2.074]	[-0.359, 2.912]	0.481
Type-Z-inclusive histories; decline-kink history controls; progression-adjusted growth				
Wages, zero-inclusive	[-0.901, 6.662]	[-2.000, 5.512]	[0.248, 7.760]	0.481
Total earnings, zero-inclusive	[-1.974, 4.880]	[-2.617, 4.003]	[-1.097, 5.523]	0.533
Wages, positive	[-2.041, 1.557]	[-1.866, 1.710]	[-2.195, 1.381]	0.901
Total earnings, positive	[-1.872, 1.849]	[-1.874, 1.936]	[-1.958, 1.851]	0.990

Table I.14. Prior earnings and entry: matched slope changes

Notes: All entries use 1,000 joint CPS-household/SIPP-lineage-by-cell reweighted draws and report percentage points per original raw-source CPS composition SD, unchanged across control blocks and within raw/adjusted pairs. Point-centered intervals use the original point, draw SD and t_{57} critical value. Percentile intervals use empirical 2.5th/97.5th percentiles; basic intervals reflect them about the original point. BH8 uses unadjusted two-sided point-centered reference probabilities within each sample/control-block slope family of eight raw/adjusted constructions; paired differences form separate families. These approximate positive-weight sensitivity tests do not certify complex-survey coverage. No interval is selected by its sign. Full 90/95/99 intervals, CPS-only/SIPP-only/joint engines, own-SD magnitudes and covariance arrays remain in the replication output.

All specifications retain baseline controls and their complete nuisance span. Both history blocks add two asinh annual total-earnings levels and four zero/negative indicators; the primary block adds signed arc change and the sensitivity block adds a decline kink. CPS positive-destination restrictions do not change recipient rows. Original destination weights are not history-longitudinal weights. Only positive-support destination years are listed. The full reference is not a matched no-history comparator. Processing and Type-Z sensitivities cannot substitute for the primary comparison; separate BH8 families do not authorize an omnibus claim. Neither histories nor paired differences identify beliefs, causal responses or mediation shares.

CPS earnings construction	Point-centered 95%	Percentile 95%	Basic 95%	BH8 q
Primary histories: 6,639 nonholder transitions; 6,639 people; 5,291 lineages; 58 cells. Destination years: 2020, 2022, 2023, 2024. Self/proxy interview status in all 24 prior monthly records; not a guarantee of unimputed earnings.				
Primary histories; signed-change history controls minus matched baseline; raw growth				
Wages, zero-inclusive	[-0.292, 0.294]	[-0.296, 0.310]	[-0.308, 0.298]	0.994
Total earnings, zero-inclusive	[-0.242, 0.232]	[-0.248, 0.239]	[-0.249, 0.237]	0.994
Wages, positive	[-0.110, 0.108]	[-0.117, 0.107]	[-0.109, 0.115]	0.994
Total earnings, positive	[-0.115, 0.111]	[-0.118, 0.110]	[-0.114, 0.114]	0.994
Primary histories; signed-change history controls minus matched baseline; progression-adjusted growth				
Wages, zero-inclusive	[-0.229, 0.346]	[-0.236, 0.353]	[-0.236, 0.352]	0.994
Total earnings, zero-inclusive	[-0.146, 0.374]	[-0.172, 0.343]	[-0.115, 0.400]	0.994
Wages, positive	[-0.092, 0.184]	[-0.102, 0.178]	[-0.086, 0.194]	0.994
Total earnings, positive	[-0.102, 0.184]	[-0.104, 0.182]	[-0.100, 0.186]	0.994
Primary histories; decline-kink history controls minus matched baseline; raw growth				
Wages, zero-inclusive	[-0.297, 0.291]	[-0.298, 0.319]	[-0.324, 0.292]	0.997
Total earnings, zero-inclusive	[-0.235, 0.245]	[-0.244, 0.260]	[-0.250, 0.254]	0.997
Wages, positive	[-0.111, 0.107]	[-0.115, 0.111]	[-0.114, 0.112]	0.997
Total earnings, positive	[-0.114, 0.114]	[-0.112, 0.114]	[-0.114, 0.113]	0.997
Primary histories; decline-kink history controls minus matched baseline; progression-adjusted growth				
Wages, zero-inclusive	[-0.264, 0.333]	[-0.281, 0.351]	[-0.282, 0.351]	0.997
Total earnings, zero-inclusive	[-0.187, 0.354]	[-0.217, 0.340]	[-0.174, 0.384]	0.997
Wages, positive	[-0.117, 0.168]	[-0.122, 0.168]	[-0.117, 0.173]	0.997
Total earnings, positive	[-0.130, 0.170]	[-0.126, 0.178]	[-0.138, 0.166]	0.997
Panel-2022 processing sensitivity: 1,275 nonholder transitions; 1,275 people; 1,031 lineages; 58 cells. Destination years: 2024. Primary panel-2022 histories only; no processing partition is certified clean.				
Panel-2022 processing sensitivity; signed-change history controls minus matched baseline; raw growth				
Wages, zero-inclusive	[-0.961, 1.281]	[-0.836, 1.377]	[-1.056, 1.156]	0.827
Total earnings, zero-inclusive	[-0.726, 1.121]	[-0.733, 1.155]	[-0.760, 1.128]	0.827
Wages, positive	[-0.319, 0.526]	[-0.366, 0.490]	[-0.284, 0.572]	0.827
Total earnings, positive	[-0.316, 0.570]	[-0.369, 0.532]	[-0.278, 0.623]	0.827

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CPS earnings construction	Point-centered 95%	Percentile 95%	Basic 95%	BH8 q
Panel-2022 processing sensitivity; signed-change history controls minus matched baseline; progression-adjusted growth				
Wages, zero-inclusive	[-0.392, 1.597]	[-0.538, 1.489]	[-0.284, 1.743]	0.827
Total earnings, zero-inclusive	[-0.315, 1.496]	[-0.471, 1.282]	[-0.102, 1.651]	0.827
Wages, positive	[-0.424, 0.573]	[-0.388, 0.593]	[-0.443, 0.537]	0.827
Total earnings, positive	[-0.464, 0.578]	[-0.386, 0.610]	[-0.496, 0.500]	0.827
Panel-2022 processing sensitivity; decline-kink history controls minus matched baseline; raw growth				
Wages, zero-inclusive	[-1.042, 1.072]	[-0.874, 1.217]	[-1.188, 0.904]	0.978
Total earnings, zero-inclusive	[-0.768, 1.019]	[-0.739, 1.054]	[-0.803, 0.990]	0.952
Wages, positive	[-0.337, 0.472]	[-0.338, 0.473]	[-0.338, 0.472]	0.952
Total earnings, positive	[-0.338, 0.519]	[-0.356, 0.515]	[-0.334, 0.537]	0.952
Panel-2022 processing sensitivity; decline-kink history controls minus matched baseline; progression-adjusted growth				
Wages, zero-inclusive	[-0.458, 1.523]	[-0.626, 1.418]	[-0.353, 1.691]	0.952
Total earnings, zero-inclusive	[-0.352, 1.408]	[-0.523, 1.156]	[-0.101, 1.579]	0.952
Wages, positive	[-0.412, 0.555]	[-0.387, 0.565]	[-0.422, 0.530]	0.952
Total earnings, positive	[-0.449, 0.555]	[-0.421, 0.580]	[-0.474, 0.526]	0.952
Type-Z-inclusive histories: 7,406 nonholder transitions; 7,406 people; 5,672 lineages; 58 cells. Destination years: 2020, 2022, 2023, 2024. Domain-valid histories including legal Type-Z months; primary histories are nested within this sample.				
Type-Z-inclusive histories; signed-change history controls minus matched baseline; raw growth				
Wages, zero-inclusive	[-0.293, 0.280]	[-0.275, 0.287]	[-0.300, 0.263]	0.965
Total earnings, zero-inclusive	[-0.256, 0.231]	[-0.247, 0.248]	[-0.272, 0.222]	0.965
Wages, positive	[-0.124, 0.094]	[-0.114, 0.099]	[-0.129, 0.083]	0.965
Total earnings, positive	[-0.126, 0.100]	[-0.115, 0.101]	[-0.126, 0.089]	0.965
Type-Z-inclusive histories; signed-change history controls minus matched baseline; progression-adjusted growth				
Wages, zero-inclusive	[-0.262, 0.288]	[-0.282, 0.295]	[-0.269, 0.308]	0.965
Total earnings, zero-inclusive	[-0.202, 0.307]	[-0.214, 0.295]	[-0.189, 0.319]	0.965
Wages, positive	[-0.120, 0.158]	[-0.115, 0.167]	[-0.128, 0.154]	0.965
Total earnings, positive	[-0.130, 0.156]	[-0.128, 0.161]	[-0.135, 0.154]	0.965
Type-Z-inclusive histories; decline-kink history controls minus matched baseline; raw growth				
Wages, zero-inclusive	[-0.330, 0.300]	[-0.334, 0.313]	[-0.343, 0.304]	0.969
Total earnings, zero-inclusive	[-0.262, 0.252]	[-0.260, 0.266]	[-0.277, 0.250]	0.969
Wages, positive	[-0.124, 0.108]	[-0.117, 0.118]	[-0.134, 0.101]	0.969
Total earnings, positive	[-0.127, 0.115]	[-0.120, 0.118]	[-0.129, 0.108]	0.969
Type-Z-inclusive histories; decline-kink history controls minus matched baseline; progression-adjusted growth				
Wages, zero-inclusive	[-0.371, 0.239]	[-0.337, 0.276]	[-0.408, 0.204]	0.969
Total earnings, zero-inclusive	[-0.263, 0.294]	[-0.271, 0.286]	[-0.255, 0.302]	0.969
Wages, positive	[-0.136, 0.158]	[-0.134, 0.160]	[-0.137, 0.157]	0.969
Total earnings, positive	[-0.149, 0.160]	[-0.151, 0.161]	[-0.150, 0.162]	0.969

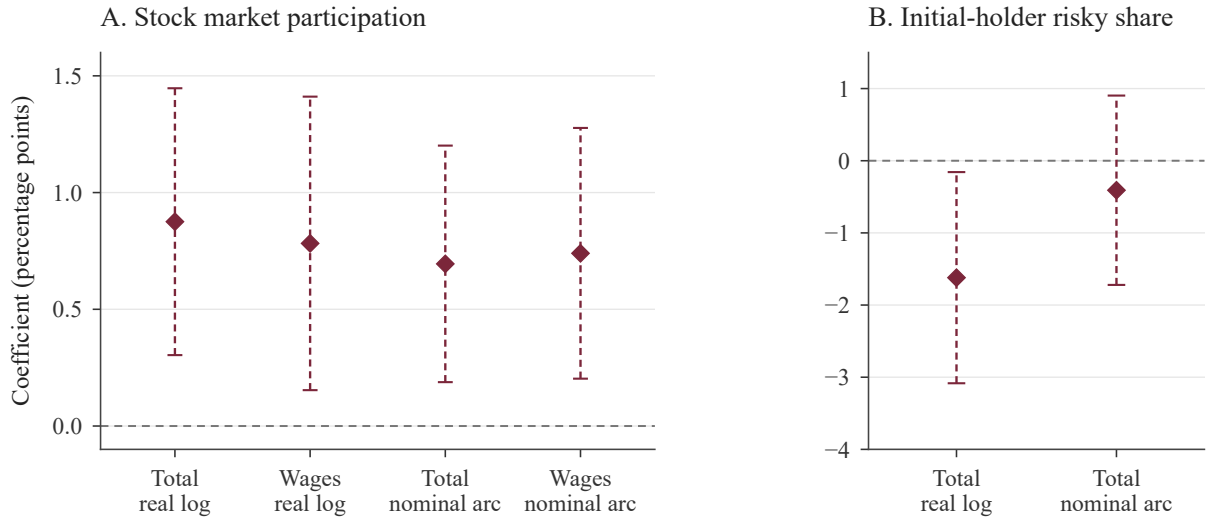
J Age-Adjusted Earnings: Complete Construction Comparisons

This appendix reports the complete families underlying the original positive-support age-adjusted earnings comparisons. That source contains 83,372 positive-endpoint histories with validated longitudinal identifiers, grouped into 58 industry–education–age cells within 30 industry–education groups; the current zero-inclusive wage construction retains 88,862 histories. Real log growth removes CPI-U inflation; nominal arc growth uses the same positive-support histories. Survey-weighted origin-age cubics are fitted separately by industry–education group, following the profile adjustment in Catherine, Sodini, and Zhang (2024). These are U.S. pretax earnings rather than CSZ's Swedish disposable income, and the historical composition measure is not their stock-return comovement measure.

J.1 Entry: source, transformation, and profile adjustment

Figure J.1 places the four adjusted entry estimates alongside the prospective initial-holder comparison. The positive entry association extends across the constructions; the nominal-arc holder interval includes zero. The tables below retain the complete raw–adjusted and uncertainty comparisons.

Figure J.1. Construction sensitivity of entry and subsequent holder allocation



Notes: Panel A compares four age-adjusted earnings constructions on identical validated CPS histories and 27,419 initial-nonholder transitions. Panel B uses 7,334 initial-holder transitions and retains later exits. Each coefficient is per its construction-specific fixed pre-adjustment source standard deviation. Real log versus nominal arc changes both inflation treatment and the growth transformation; arc growth is $2(y_1 - y_0)/(y_1 + y_0)$. Bars are 95% CR1/t_{29} intervals using 30 industry–education clusters, conditional on estimated CPS moments. The comparison does not itself test whether the real-log and arc coefficients differ.

Table J.1 retains all eight combinations of wage versus total earnings, real log versus nominal arc growth, and raw versus age-adjusted growth. Observations and source scaling are common within the raw–adjusted pairs. Conventional coefficients and standard errors use 30-group CR1 inference with t_{29} tests. The finer 58-cell results and 499-draw joint remeasurement results are reported separately.

The joint calculation uses a t_{57} reference, distinct from the 30-group inference. All four adjusted slopes have joint eight-test q-values of 0.073, and their percentile intervals include zero. Table J.2 reports all four paired adjustment contrasts using their own shared-draw covariance.

J.2 Standing shares and pooled allocation

Table J.3 reports all eight standing-holder regressions: two earnings sources, two growth transformations, and two financial denominators. It also reports the four direct broad-minus-nonretirement denominator contrasts. The current-holder sample is not the prospective initial-holder sample in the main paper. Age-adjusted real-log composition has a negative nonretirement holder association for both earnings sources; the corresponding arc estimates are imprecise.

Table J.4 retains all eight pooled Tobit specifications, with latent coefficients and average marginal effects distinguished and each using its own uncertainty calculation. The marginal effects are imprecise throughout. These estimates complement the prospective comparisons in the main text.

J.3 Relationship to the coarser CSZ benchmark

The 58-cell main construction pools historical residual earnings within age bands. A coarser annual-aggregation benchmark instead builds annual moments within coarser industry–education groups and averages them over time, more closely matching the annual aggregation in Catherine, Sodini, and Zhang (2024). All eight conditional composition tests in that 29-supported-group benchmark are imprecise (the smallest p-value is 0.370). The replication package contains the complete outputs. The results depend on aggregation and concern a different measure from CSZ's systematic risk. The matched 58-cell comparison holds the public-data assignment and earnings histories fixed while importing their age-profile adjustment.

Table J.1. Raw and age-adjusted earnings: complete matched entry comparison

Earnings Growth	(1) Total Raw	(2) Total Age adjusted	(3) Wages Raw	(4) Wages Age adjusted
A. Real log growth				
Upside composition	1.096^{**}	0.875^{***}	0.732	0.782^{**}
	(0.443)	(0.279)	(0.495)	(0.307)
58-cell sensitivity p	0.0112	0.0014	0.1142	0.0062
Conditional coefficient BH8 q	0.0223	0.0168	0.1498	0.0220
Conditional-SD magnitude (pp)	0.523	0.573	0.374	0.524
Joint-refit coefficient SE	0.555	0.421	0.548	0.413
Joint original-point p	0.0533	0.0421	0.1869	0.0635
Joint coefficient BH8 q	0.0726	0.0726	0.1869	0.0726
Joint percentile 95% interval	$[-0.522, 1.622]$	$[-0.353, 1.299]$	$[-0.614, 1.590]$	$[-0.348, 1.251]$
Joint point-centered 95% interval	$[-0.016, 2.208]$	$[0.032, 1.718]$	$[-0.365, 1.830]$	$[-0.045, 1.610]$
Mean coefficient across joint draws	0.575	0.486	0.435	0.445
Nonholder transitions / joint draws	27,419 / 499	27,419 / 499	27,419 / 499	27,419 / 499

Notes: Coefficients and intervals are entry percentage points per fixed pre-adjustment source SD. The conditional-SD magnitude instead uses residual composition after all other regressors. Stars use unadjusted two-sided zero-coefficient tests with CR1 standard errors clustered by 30 industry–education groups and t_{29}: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Fine-cell rows instead use 58-cell CR1 and t_{57}. All eight regressions across both panels use the same observations and controls as Table S.36. Raw versus age-adjusted refers only to the group-specific cubic age fit, not the richer earnings-rank adjustment. Joint draws rebuild CPS age fits and moments with household multipliers and independently reweight SIPP sample lineages and assignment cells. Joint original-point tests retain their saved t_{57} reference and use the coefficient SD across 499 draws; their point-centered intervals use that same reference. Percentile intervals use the draw quantiles. BH8 covers all eight raw/adjusted, earnings-definition and growth-transformation slopes within each inference convention. The methods are not interchangeable when the refit distribution is displaced. Every percentile interval includes zero. Saved joint magnitude tests, distinct from slope tests, give $q = 0.1038$ for all four adjusted conditional-SD magnitudes.

Table J.1. Raw and age-adjusted earnings: complete matched entry comparison

Earnings Growth	(1) Total Raw	(2) Total Age adjusted	(3) Wages Raw	(4) Wages Age adjusted
B. Nominal arc growth				
Upside composition	1.029^{***}	0.695^{***}	0.974^{**}	0.740^{***}
	(0.350)	(0.248)	(0.356)	(0.263)
58-cell sensitivity p	0.0031	0.0042	0.0053	0.0044
Conditional coefficient BH8 q	0.0168	0.0168	0.0168	0.0168
Conditional-SD magnitude (pp)	0.660	0.534	0.649	0.582
Joint-refit coefficient SE	0.413	0.359	0.406	0.349
Joint original-point p	0.0157	0.0579	0.0197	0.0382
Joint coefficient BH8 q	0.0726	0.0726	0.0726	0.0726
Joint percentile 95% interval	$[-0.162, 1.336]$	$[-0.263, 1.123]$	$[-0.144, 1.352]$	$[-0.166, 1.146]$
Joint point-centered 95% interval	$[0.201, 1.856]$	$[-0.024, 1.413]$	$[0.161, 1.786]$	$[0.042, 1.438]$
Mean coefficient across joint draws	0.596	0.452	0.574	0.471
Nonholder transitions / joint draws	27,419 / 499	27,419 / 499	27,419 / 499	27,419 / 499

Notes: Coefficients and intervals are entry percentage points per fixed pre-adjustment source SD. The conditional-SD magnitude instead uses residual composition after all other regressors. Stars use unadjusted two-sided zero-coefficient tests with CR1 standard errors clustered by 30 industry–education groups and t_{29}: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Fine-cell rows instead use 58-cell CR1 and t_{57}. All eight regressions across both panels use the same observations and controls as Table S.36. Raw versus age-adjusted refers only to the group-specific cubic age fit, not the richer earnings-rank adjustment. Joint draws rebuild CPS age fits and moments with household multipliers and independently reweight SIPP sample lineages and assignment cells. Joint original-point tests retain their saved t_{57} reference and use the coefficient SD across 499 draws; their point-centered intervals use that same reference. Percentile intervals use the draw quantiles. BH8 covers all eight raw/adjusted, earnings-definition and growth-transformation slopes within each inference convention. The methods are not interchangeable when the refit distribution is displaced. Every percentile interval includes zero. Saved joint magnitude tests, distinct from slope tests, give $q = 0.1038$ for all four adjusted conditional-SD magnitudes.

Table J.2. Matched changes from removing group-specific age profiles

Earnings Growth	(1) Total Real log	(2) Wages Real log	(3) Total Nominal arc	(4) Wages Nominal arc
Adjusted minus raw composition slope	-0.221	0.050	-0.334^{*}	-0.234
	(0.278)	(0.304)	(0.183)	(0.180)
58-cell sensitivity p	0.4349	0.8651	0.0674	0.1640
Conditional BH4 q	0.5787	0.8703	0.3138	0.4076
Joint original-point p	0.5655	0.8897	0.2412	0.3921
Joint BH4 q	0.7540	0.8897	0.7540	0.7540
Joint percentile 95% interval	$[-0.839, 0.690]$	$[-0.674, 0.717]$	$[-0.696, 0.478]$	$[-0.658, 0.475]$
Joint point-centered 95% interval	$[-0.985, 0.544]$	$[-0.668, 0.768]$	$[-0.900, 0.231]$	$[-0.777, 0.309]$

Notes: Stars use unadjusted two-sided zero-coefficient tests with CR1 standard errors clustered by 30 industry–education groups and t_{29}: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Fine-cell rows instead use 58-cell CR1 and t_{57}. All four slope differences use identical observations and common fixed source-SD units within each raw/adjusted pair. Their SEs retain covariance between the paired fits. Joint results use the same 499 matched draws and saved t_{57} original-point convention as Table J.1. BH4 covers all four paired differences within each convention. These are differences between descriptive specifications, not mediation effects. All joint paired intervals include zero.

Table J.3. Age-adjusted earnings composition and standing holder shares

Earnings Growth	(1) Total Real log	(2) Wages Real log	(3) Total Nominal arc	(4) Wages Nominal arc
A. Nonretirement financial denominator				
Composition	-1.317^{**}	-1.070^{**}	-0.349	-0.384
	(0.513)	(0.504)	(0.394)	(0.372)
58-cell sensitivity p	0.0066	0.0200	0.4461	0.3585
Full-family BH16 q (coefficient)	0.2372	0.2372	0.4712	0.4712
Conditional-SD magnitude (pp)	-0.762	-0.705	-0.223	-0.261
Worker-year observations	17,455	17,455	17,455	17,455
People	11,773	11,773	11,773	11,773
Assignment cells / inference groups	58 / 30	58 / 30	58 / 30	58 / 30
Risk controls, personal controls, and FE	Yes	Yes	Yes	Yes

Notes: Weighted OLS among current positive stock-and-mutual-fund holders; these are standing associations, not within-person reallocations. Panel C uses paired coefficient covariance. Both panels use identical valid 2019–2024 person-years. Shares divide stocks/mutual funds by financial assets net of retirement balances (A) or broad financial assets (B). Both denominators are positive and ratios in $[0, 1]$; invalid values are not recoded or clipped. Coefficients/AMEs are percentage points per fixed pre-adjustment source SD; the conditional-SD magnitude uses residual composition after the other controls. Stars use unadjusted two-sided zero-coefficient tests with CR1 standard errors clustered by 30 industry–education groups and t_{29}: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Fine-cell rows instead use 58-cell CR1 and t_{57}. Age adjustment, composition, risk controls and personal controls follow Table S.36, using current rather than lagged values and current-year person weights. The full family spans two earnings definitions, two growth measures, two denominators and two estimators. Coefficient and AME BH16 families are separate. Inference conditions on CPS moments; no first-stage bootstrap is supplied for these models.

Table J.3. Age-adjusted earnings composition and standing holder shares (continued)

Earnings Growth	(1) Total Real log	(2) Wages Real log	(3) Total Nominal arc	(4) Wages Nominal arc
B. Broad financial denominator				
Composition	-0.817^{*}	-0.826^{**}	-0.638	-0.600
	(0.465)	(0.393)	(0.438)	(0.398)
58-cell sensitivity p	0.0778	0.0400	0.1947	0.1808
Full-family BH16 q (coefficient)	0.3573	0.2372	0.4160	0.4160
Conditional-SD magnitude (pp)	-0.472	-0.544	-0.407	-0.407
C. Paired broad-minus-nonretirement slope				
Difference	0.500	0.244	-0.288	-0.216
	(0.349)	(0.369)	(0.260)	(0.242)
Paired-difference BH4 q	0.5075	0.5126	0.5075	0.5075
Worker-year observations	17,455	17,455	17,455	17,455
People	11,773	11,773	11,773	11,773
Assignment cells / inference groups	58 / 30	58 / 30	58 / 30	58 / 30
Risk controls, personal controls, and FE	Yes	Yes	Yes	Yes

Notes: Weighted OLS among current positive stock-and-mutual-fund holders; these are standing associations, not within-person reallocations. Panel C uses paired coefficient covariance. Both panels use identical valid 2019–2024 person-years. Shares divide stocks/mutual funds by financial assets net of retirement balances (A) or broad financial assets (B). Both denominators are positive and ratios in $[0, 1]$; invalid values are not recoded or clipped. Coefficients/AMEs are percentage points per fixed pre-adjustment source SD; the conditional-SD magnitude uses residual composition after the other controls. Stars use unadjusted two-sided zero-coefficient tests with CR1 standard errors clustered by 30 industry–education groups and t_{29}: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Fine-cell rows instead use 58-cell CR1 and t_{57}. Age adjustment, composition, risk controls and personal controls follow Table S.36, using current rather than lagged values and current-year person weights. The full family spans two earnings definitions, two growth measures, two denominators and two estimators. Coefficient and AME BH16 families are separate. Inference conditions on CPS moments; no first-stage bootstrap is supplied for these models.

Table J.4. Age-adjusted earnings composition and censored standing shares

Earnings Growth	(1) Total Real log	(2) Wages Real log	(3) Total Nominal arc	(4) Wages Nominal arc
A. Nonretirement financial denominator				
Composition: latent Tobit coefficient	-0.872	-0.934	0.643	0.412
	(0.835)	(0.810)	(0.710)	(0.776)
Observed-share average marginal effect	-0.187	-0.200	0.138	0.088
	(0.179)	(0.174)	(0.152)	(0.166)
58-cell sensitivity p (AME)	0.3767	0.2939	0.4269	0.6188
Full-family BH16 q (coefficient)	0.4712	0.4712	0.4712	0.6849
Full-family BH16 q (AME)	0.4712	0.4712	0.4712	0.6851
Conditional-SD magnitude (pp)	-0.120	-0.135	0.103	0.068
Worker-year observations	81,603	81,603	81,603	81,603
People	48,016	48,016	48,016	48,016
Assignment cells / inference groups	58 / 30	58 / 30	58 / 30	58 / 30
Risk controls, personal controls, and FE	Yes	Yes	Yes	Yes

Notes: Weighted Gaussian Tobit includes holders and genuine zero positions, with lower censoring at zero and no upper censor at one. Latent coefficients and observed-share AMEs are separate estimands; each is shown with its own SE and stars from its own zero test. AME SEs propagate the full coefficient/log-scale covariance. The AME is not a purely intensive-margin effect. Both panels use identical valid 2019–2024 person-years. Shares divide stocks/mutual funds by financial assets net of retirement balances (A) or broad financial assets (B). Both denominators are positive and ratios in $[0, 1]$; invalid values are not recoded or clipped. Coefficients/AMEs are percentage points per fixed pre-adjustment source SD; the conditional-SD magnitude uses residual composition after the other controls. Stars use unadjusted two-sided zero-coefficient tests with CR1 standard errors clustered by 30 industry–education groups and t_{29}: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Fine-cell rows instead use 58-cell CR1 and t_{57}. Age adjustment, composition, risk controls and personal controls follow Table S.36, using current rather than lagged values and current-year person weights. The full family spans two earnings definitions, two growth measures, two denominators and two estimators. Coefficient and AME BH16 families are separate. Inference conditions on CPS moments; no first-stage bootstrap is supplied for these models.

Table J.4. Age-adjusted earnings composition and censored standing shares (continued)

Earnings Growth	(1) Total Real log	(2) Wages Real log	(3) Total Nominal arc	(4) Wages Nominal arc
B. Broad financial denominator				
Composition: latent Tobit coefficient	-0.529	-0.657	0.198	0.051
	(0.537)	(0.529)	(0.490)	(0.551)
Observed-share average marginal effect	-0.107	-0.133	0.040	0.010
	(0.109)	(0.107)	(0.099)	(0.111)
58-cell sensitivity p (AME)	0.3838	0.2345	0.7068	0.9259
Full-family BH16 q (coefficient)	0.4712	0.4712	0.7347	0.9263
Full-family BH16 q (AME)	0.4712	0.4712	0.7348	0.9263
Conditional-SD magnitude (pp)	-0.068	-0.090	0.030	0.008
Worker-year observations	81,603	81,603	81,603	81,603
People	48,016	48,016	48,016	48,016
Assignment cells / inference groups	58 / 30	58 / 30	58 / 30	58 / 30
Risk controls, personal controls, and FE	Yes	Yes	Yes	Yes

Notes: Weighted Gaussian Tobit includes holders and genuine zero positions, with lower censoring at zero and no upper censor at one. Latent coefficients and observed-share AMEs are separate estimands; each is shown with its own SE and stars from its own zero test. AME SEs propagate the full coefficient/log-scale covariance. The AME is not a purely intensive-margin effect. Both panels use identical valid 2019–2024 person-years. Shares divide stocks/mutual funds by financial assets net of retirement balances (A) or broad financial assets (B). Both denominators are positive and ratios in $[0, 1]$; invalid values are not recoded or clipped. Coefficients/AMEs are percentage points per fixed pre-adjustment source SD; the conditional-SD magnitude uses residual composition after the other controls. Stars use unadjusted two-sided zero-coefficient tests with CR1 standard errors clustered by 30 industry–education groups and t_{29}: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Fine-cell rows instead use 58-cell CR1 and t_{57}. Age adjustment, composition, risk controls and personal controls follow Table S.36, using current rather than lagged values and current-year person weights. The full family spans two earnings definitions, two growth measures, two denominators and two estimators. Coefficient and AME BH16 families are separate. Inference conditions on CPS moments; no first-stage bootstrap is supplied for these models.

K Portfolio-Change Accounting and Balance Diagnostics

Table K.1 reports the share-change accounting for both initial ownership states and earnings constructions, complementing the real-log holder comparison in Figure 3.

The balance-sheet family fixes the same 31,569 prospective observations, earnings constructions, weights, and fully interacted control design as the main allocation comparison. Before fitting the diagnostics, the analysis plan fixed two constructions and five diagnostic or accounting outcomes: midpoint risky-balance contribution, midpoint denominator contribution, growth in financial assets normalized by initial assets, similarly normalized risky-balance growth, and future normalized risky balances conditional on future normalized financial assets. Together with one share-change identity replay per construction, this gives twelve fitted models. Controlling for baseline share makes the composition coefficients on future normalized risky balances and risky-balance growth identical. Two further calculations verify the conditional-outcome equivalence. No observations are clipped or winsorized in response to the results.

Equation (11) is an exact finite-change identity, not a first-order approximation. Both share-change components are bounded on the retained positive-denominator, valid-share sample. Because all regressions use identical designs, the estimated composition contributions add exactly to the original share slope, including each ownership contrast. Covariance between components is retained for the arithmetic checks. Table K.2 gives the complete accounting family, and Table K.3 reports the normalized-balance diagnostics. Conventional stars use 30-group CR1/t_{29} inference; 58-cell p-values are sensitivities.

Normalizing by initial financial assets avoids controlling for the same future denominator in every outcome but makes very small starting balances influential. Holder financial-asset growth reaches approximately 1,893 times initial assets; the corresponding nonholder maximum is approximately 464,061. Ranking observations by the absolute normalized risky-growth outcome, the largest 1% account for 66.44% of absolute weighted risky growth among holders and 99.08% among nonholders. These diagnostics must therefore be distinguished from the bounded midpoint accounting result.

Conditioning on future assets changes the estimand. Because future assets are endogenous and include risky assets, the resulting coefficient does not identify a direct effect or causal mediation. The real-log holder coefficient is positive and imprecise ($p = 0.508$); it does not establish that greater labor income reduces demand for equity income at given financial wealth. An income-target explanation would require an additional preference or saving channel. Unconditioned normalized risky-balance growth is also imprecise and sensitive to small initial denominators. The midpoint risky-balance component combines transactions, returns, and changes in reporting; it is not net purchases or sales.

Table K.4 reports all family corrections separately: twelve accounting-component tests, eighteen normalized-balance tests, and the combined thirty-test family, each within its clustering convention. The share replays are arithmetic checks, not additional hypotheses. The real-log holder numerator coefficient has a twelve-test q-value of 0.090 and a thirty-test q-value of 0.113. The marginal significance does not identify a separate mechanism.

Table K.1. Earnings-tail composition and the sources of portfolio-share changes

	Real log		Nominal arc	
	(1)	(2)	(3)	(4)
Baseline ownership	Nonholder	Holder	Nonholder	Holder
Risky-share change, Δs	0.360^{**}	-1.621^{**}	0.264^{**}	-0.409
	(0.142)	(0.715)	(0.128)	(0.642)
Risky-balance term, $\Delta R/\bar{F}$	0.589^{**}	-3.679^{**}	0.362	-1.244
	(0.226)	(1.595)	(0.227)	(1.339)
Asset term, $-\bar{s}\Delta F/\bar{F}$	-0.229^{**}	2.058^{*}	-0.099	0.835
	(0.101)	(1.050)	(0.111)	(0.847)
Observations in baseline-status block	24,235	7,334	24,235	7,334
Industry–education clusters	30	30	30	30
Fully interacted baseline controls	Yes	Yes	Yes	Yes

Notes: Composition slopes are percentage points per common fixed source SD. The 31,569 consecutive SIPP pairs and fully baseline-status-interacted controls are those in Table R.1; future zeros remain. With risky balances R, nonretirement financial assets F, $s = R/F$, and two-date means denoted by bars, $\Delta s = \Delta R/\bar{F} - \bar{s}\Delta F/\bar{F}$. Parentheses report 30-group CR1 standard errors. Stars use unadjusted two-sided t_{29} tests: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. These are conditional associations and an accounting decomposition, not trading flows or causal channels.

Table K.2. Portfolio-share accounting: full status contrasts and clustering sensitivity

	Real log			Nominal arc		
	(1)	(2)	(3)	(4)	(5)	(6)
Baseline ownership	Nonholder	Holder	H−N	Nonholder	Holder	H−N
Risky-share change, Δs	0.360^{**}	-1.621^{**}	-1.980^{**}	0.264^{**}	-0.409	-0.673
	(0.142)	(0.715)	(0.737)	(0.128)	(0.642)	(0.683)
p, 30 groups	0.0172	0.0311	0.0118	0.0490	0.5288	0.3325
p, 58 cells	0.0058	0.0150	0.0044	0.0178	0.4902	0.2887
Risky-balance term, $\Delta R/\bar{F}$	0.589^{**}	-3.679^{**}	-4.268^{**}	0.362	-1.244	-1.606
	(0.226)	(1.595)	(1.619)	(0.227)	(1.339)	(1.387)
p, 30 groups	0.0141	0.0284	0.0133	0.1214	0.3608	0.2562
p, 58 cells	0.0044	0.0207	0.0087	0.0566	0.3114	0.2092
Asset term, $-\bar{s}\Delta F/\bar{F}$	-0.229^{**}	2.058^{*}	2.288^{**}	-0.099	0.835	0.933
	(0.101)	(1.050)	(1.058)	(0.111)	(0.847)	(0.851)
p, 30 groups	0.0301	0.0595	0.0389	0.3813	0.3324	0.2820
p, 58 cells	0.0139	0.0567	0.0360	0.2743	0.2922	0.2421
Nonholder / holder observations	24,235 / 7,334					
Consecutive person-year pairs	31,569					

Notes: All entries are composition slopes in percentage points per common fixed pre-adjustment source SD. H−N is the directly estimated holder-minus-nonholder contrast, with shared cluster covariance. The rows decompose changes in stocks and mutual funds divided by nonretirement financial assets; their point estimates sum exactly. The share-change row reproduces the retained prospective-share result, because initial share is controlled for holders and is zero for nonholders. The sample retains all 31,569 valid consecutive pairs, including 2,165 holder-to-zero transitions. Each risk and baseline-control coefficient varies by initial ownership. Controls are composition, variance, mean growth, mean growth squared, SD, age and age squared, sex, transformed first-listed-job earnings, log gross assets, a nonpositive first-listed-job-earnings indicator, and outcome-year and baseline industry, education and age-band effects; initial share also enters the holder block. Future-year person weights are used. Parentheses and stars use primary 30-group CR1 inference and unadjusted two-sided t_{29} tests; the 58-cell sensitivity uses t_{57}. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Table K.4 reports the complete family adjustments. Inference conditions on estimated earnings moments, without a first-stage bootstrap. The accounting terms are not net trades or identified causal mechanisms.

Table K.3. Prospective balance diagnostics: full exploratory family

Baseline ownership	(1) Nonholder	(2) Holder	(3) H−N
Panel A: Real log			
Financial-asset growth, $\Delta F/F_0$	-2458.081	-252.542^{*}	2205.539
	(3005.350)	(134.976)	(3049.184)
p, 30 groups	0.4201	0.0715	0.4753
p, 58 cells	0.4099	0.1023	0.4633
Risky-balance growth, $\Delta R/F_0$	2507.971^{*}	-125.930	-2633.901^{*}
	(1356.996)	(108.983)	(1389.596)
p, 30 groups	0.0748	0.2573	0.0680
p, 58 cells	0.1075	0.3098	0.0962
Future risky balance, R_1/F_0	3762.382^{***}	24.813	-3737.569^{***}
	(1323.286)	(36.991)	(1327.363)
p, 30 groups	0.0081	0.5077	0.0087
p, 58 cells	0.0065	0.5070	0.0070
Future financial assets controlled		Final outcome only	
Nonholder / holder observations		24,235 / 7,334	
Consecutive person-year pairs		31,569	

Notes: Coefficients are nominal dollars per $100 of initial nonretirement financial assets, per common fixed source SD of composition. All three diagnostic outcomes, both transformations, and all direct status contrasts are retained. The sample, weighting, baseline controls, and inference are those in Table K.2. The final outcome additionally conditions on future assets F_1/F_0, separately by baseline ownership; this descriptive conditioning is not causal mediation, and future assets include risky balances. All balances and exits are untrimmed. Initial assets can be as low as $1 for nonholders and $7 for holders, producing very large normalized growth. Within baseline status and each normalized growth outcome, observations are ranked by the absolute unweighted outcome. The top 1% then accounts for 57.8% of the weighted absolute sum of holder asset growth and 66.4% of risky growth; corresponding nonholder shares are 91.2% and 99.1%. Parentheses and stars use 30-group CR1 and unadjusted two-sided t_{29} tests; the 58-cell p rows use t_{57}. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. The complete 18-test diagnostic and 30-test combined BH adjustments appear in Table K.4. Balance changes do not identify net purchases or sales.

Table K.3. Prospective balance diagnostics: full exploratory family (continued)

Baseline ownership	(1) Nonholder	(2) Holder	(3) H−N
Panel B: Nominal arc			
Financial-asset growth, $\Delta F/F_0$	-2107.772	-169.689^{*}	1938.083
	(1334.091)	(92.835)	(1359.102)
p, 30 groups	0.1250	0.0779	0.1645
p, 58 cells	0.2332	0.0863	0.2754
Risky-balance growth, $\Delta R/F_0$	803.290	-31.321	-834.610
	(772.808)	(58.424)	(769.597)
p, 30 groups	0.3072	0.5960	0.2871
p, 58 cells	0.4346	0.6253	0.4138
Future risky balance, R_1/F_0	1878.875^{**}	69.989^{*}	-1808.886^{**}
	(776.996)	(35.339)	(781.568)
p, 30 groups	0.0221	0.0572	0.0279
p, 58 cells	0.0285	0.0615	0.0352
Future financial assets controlled		Final outcome only	
Nonholder / holder observations		24,235 / 7,334	
Consecutive person-year pairs		31,569	

Notes: Coefficients are nominal dollars per $100 of initial nonretirement financial assets, per common fixed source SD of composition. All three diagnostic outcomes, both transformations, and all direct status contrasts are retained. The sample, weighting, baseline controls, and inference are those in Table K.2. The final outcome additionally conditions on future assets F_1/F_0, separately by baseline ownership; this descriptive conditioning is not causal mediation, and future assets include risky balances. All balances and exits are untrimmed. Initial assets can be as low as $1 for nonholders and $7 for holders, producing very large normalized growth. Within baseline status and each normalized growth outcome, observations are ranked by the absolute unweighted outcome. The top 1% then accounts for 57.8% of the weighted absolute sum of holder asset growth and 66.4% of risky growth; corresponding nonholder shares are 91.2% and 99.1%. Parentheses and stars use 30-group CR1 and unadjusted two-sided t_{29} tests; the 58-cell p rows use t_{57}. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. The complete 18-test diagnostic and 30-test combined BH adjustments appear in Table K.4. Balance changes do not identify net purchases or sales.

Table K.4. Portfolio accounting and balance diagnostics: multiple-testing adjustments

	Real log			Nominal arc		
	(1)	(2)	(3)	(4)	(5)	(6)
Baseline ownership	Nonholder	Holder	H−N	Nonholder	Holder	H−N
Risky-balance term, $\Delta R/\bar{F}$						
Family q, 30 groups	0.0848	0.0902	0.0848	0.2080	0.3813	0.3760
Family q, 58 cells	0.0521	0.0621	0.0521	0.0972	0.3114	0.3114
Combined q, 30 groups	0.1060	0.1127	0.1060	0.2205	0.4330	0.3860
Combined q, 58 cells	0.0651	0.1034	0.0651	0.1537	0.3892	0.3691
Asset term, $-\bar{s}\Delta F/\bar{F}$						
Family q, 30 groups	0.0902	0.1191	0.0935	0.3813	0.3813	0.3760
Family q, 58 cells	0.0554	0.0972	0.0865	0.3114	0.3114	0.3114
Combined q, 30 groups	0.1127	0.1558	0.1298	0.4399	0.4155	0.3915
Combined q, 58 cells	0.0831	0.1537	0.1202	0.3892	0.3892	0.3823
Financial-asset growth, $\Delta F/F_0$						
Family q, 30 groups	0.5041	0.1558	0.5347	0.2249	0.1558	0.2693
Family q, 58 cells	0.5212	0.2150	0.5212	0.4198	0.2150	0.4507
Combined q, 30 groups	0.4668	0.1558	0.5092	0.2205	0.1558	0.2742
Combined q, 58 cells	0.4775	0.2016	0.4964	0.3823	0.1991	0.3892
Risky-balance growth, $\Delta R/F_0$						
Family q, 30 groups	0.1558	0.3860	0.1558	0.3950	0.5960	0.3950
Family q, 58 cells	0.2150	0.4647	0.2150	0.5212	0.6253	0.5212
Combined q, 30 groups	0.1558	0.3860	0.1558	0.4007	0.5960	0.3915
Combined q, 58 cells	0.2016	0.3892	0.2016	0.4829	0.6253	0.4775
Future risky balance, R_1/F_0						
Family q, 30 groups	0.0779	0.5375	0.0779	0.1257	0.1558	0.1257
Family q, 58 cells	0.0627	0.5369	0.0627	0.1584	0.2150	0.1584
Combined q, 30 groups	0.1060	0.5252	0.1060	0.1127	0.1558	0.1127
Combined q, 58 cells	0.0651	0.5245	0.0651	0.1202	0.1537	0.1202

Notes: Benjamini–Hochberg q-values use the complete recorded exploratory families, separately for each clustering convention. The accounting family has 12 tests: two components × two transformations × three status contrasts. The diagnostic family has 18 tests: three balance outcomes × two transformations × three status contrasts. Combined rows pool these 30 tests. The six share-change contrasts are identity replays of existing prospective results, not additional discoveries, and are excluded from these new families. All tests are two-sided CR1, with t_{29} for the primary 30-group inference and t_{57} for the 58-cell sensitivity. Table coefficient stars use unadjusted tests, not these q-values. No first-stage bootstrap is added.

L Additional Survey Definitions

This section retains the detailed survey definitions and broader samples used in the supplementary constructions. The main age-adjusted earnings and prospective portfolio samples are defined in Section 3.

L.1 Public earnings histories

The CPS Annual Social and Economic Supplement supplies annual wage-and-salary income, adjacent-survey person identifiers, industry, education, age, and survey weights. The extract covers interview years 2014–2023. I follow workers aged 25–65 with positive baseline wage earnings to the next ASEC observation, retaining destination earnings of zero. Recorded age must rise by zero, one, or two years. The CPS-to-SIPP crosswalk yields 183,809 linked transitions in 72 candidate industry-by-education-by-age cells. It combines 12 harmonized nonmilitary industries, three education groups, and two age bands, split at age 45. Appendix C provides the mapping, support thresholds, and sample construction.

Retaining destination zeros includes transitions to no measured earnings. Positive-to-positive earnings growth describes workers with measured earnings at both endpoints; it excludes the transition to no measured earnings. Yet a zero can reflect nonemployment, a change in work intensity or arrangement, or the definition of earnings in the survey. It is not automatically a job-loss event. I therefore retain zeros in the baseline and examine their composition explicitly in Section O.

With positive baseline earnings, the bounded arc growth rate accommodates zero destination earnings without an arbitrary offset (Törnqvist, Vartia, and Vartia, 1985):

$$g_{i,t+1} = 2\frac{y_{i,t+1} - y_{i,t}}{y_{i,t+1} + y_{i,t}}. \tag{L.1}$$

It lies in $[-2, 2]$ and maps a transition to zero to -2. Positive-to-positive arc and log growth are reported as separate conditional-earnings objects. Adding a dollar before taking logs would instead impose a unit-dependent distance between zero and small positive earnings.

SIPP provides an independent measurement comparison. The post-redesign survey combines retrospective monthly earnings with annual household balance-sheet information. I use earnings from the first-listed job (job 1, `TJB1_MSUM`), not total earnings or the highest-paying job. Since the 2014 redesign, job slots are ordered chronologically within the reference year, with reporting order breaking ties. The same slot need not identify the same job across waves.[6] I match monthly job-1 earnings to the same month 12 months later, requiring positive origin earnings and allowing a destination zero. The primary reliability comparison retains one fixed transition per person; an overlapping-month construction shows what repeated observations add. Each worker always remains in the same half of a reliability split. Job-or-business class and no-job-spell fields distinguish

[6] See the Census Bureau's *2024 SIPP Users' Guide*, Section 4.2.3, pp. 71–73, and p. 76. A separate main-employer question for retirement coverage in later surveys does not redefine the earnings slots.

arrangements associated with a destination zero. Job-1 industry also determines the SIPP industry assignment. These measures do not summarize every job held by a worker.

Both surveys' distributions use official person weights. Cell support is measured by the Kish effective sample size, $N_{\text{eff}} = (\sum_i w_i)^2 / \sum_i w_i^2$, rather than the number of records (Kish, 1965). Appendix B reports the support rules and alternative estimators. The reliability calculations and the independent CPS resampling used for portfolio inference serve different purposes and are described separately below.

The original pooled CPS assignment retains both 2014 questionnaire subsamples with their supplied weights. Because each subsample independently represents approximately the full population, this convention gives that origin year extra weight; it is not an equal-population-weighted mixture of years. I retain it as the original benchmark and explicitly examine the 2014 survey design in Appendix I.8. The new same-source wage and total-labor-earnings comparison uses only the redesigned 2014 origin subsample, with later years unchanged, following the official guidance for comparison with 2015 onward.

The observable-progression comparison additionally requires equal positive IPUMS validated person identifiers (CPSIDV) at the two earnings endpoints. It rebuilds the matched wage and total-earnings distributions on 88,862 eligible histories, including 83,372 with positive destinations in both measures. This restriction checks longitudinal demographic consistency beyond the original age rule, although some excluded histories may be correctly matched. The original quantile, variance, and local-inference estimates remain original-linkage benchmarks; the validated-link comparison is identified separately in Section N.1 and Appendix I.13.

L.2 Standing portfolios and transitions

SIPP reports the value of directly held stocks and mutual funds and total financial assets. I aggregate available eligible monthly records within each calendar year to form person-year observations. Participation equals one when any eligible monthly stock-and-mutual-fund report is positive. Where financial assets are positive and the numerator is observed, the share is 100 times the stock-and-mutual-fund balance divided by financial assets. The all-worker measure sets the share to zero when this ratio is unavailable. A second share retains only observations with a usable positive-financial-asset ratio, and a third also requires current positive holdings. Keeping these samples separate distinguishes a population portfolio measure from allocation among holders.

These outcomes measure observed stock-and-mutual-fund exposure, not total equity exposure. Mutual funds may hold non-equity securities, SIPP does not observe the allocation inside retirement accounts, and employer-stock compensation cannot be cleanly removed. Changes in a reported balance need not be verified trades. Appendix C.3 documents financial-asset denominators, ratio bounds, eligible-month coverage, and the household-record structure.

The transition sample links consecutive annual observations. It contains 35,112 person-year pairs on 23,170 people: 27,752 begin with zero holdings, and 7,360 begin with positive holdings. The weighted nonholder-to-holder transition rate is 8.79%. Entry is defined using the lagged nonholder

risk set, rather than selecting people who hold stocks at some point in the panel. A transition into positive holdings may nevertheless be a return to holding rather than a first-ever entry.

I also consider sustained entry, which requires positive holdings for a further year, and crossings above nominal balances of $1,000 and $5,000. These distinguish the baseline transition from the appearance of a small or transitory positive balance. Requiring the same assignment cell in both years is a separate migration check, not a condition imposed on the primary sample.

L.3 Separating earnings measurement from portfolio outcomes

The primary design measures cell earnings distributions using CPS origin years through 2017 and destination earnings through 2018. These historical distributions are then assigned to SIPP worker-years in 2019–2024. The resulting level sample has 90,110 observations on 52,115 people in 60 supported cells. Both observations in a portfolio transition must fall in 2019–2024, so transition outcomes occur in 2020–2024. All assigned earnings histories precede both portfolio observations.

This timing eliminates overlap between the earnings-measurement window and the portfolio-outcome window. Using separate surveys also removes mechanical own-observation contamination. Neither feature removes selection into industry, education, or the lagged portfolio state. Historical cell distributions may differ from households' expectations, and annual growth does not identify permanent and transitory risk separately. The broader 2018–2024 portfolio window is retained as a sensitivity analysis, not combined with the strictly later baseline.

M Earnings Measures and Portfolio Estimands

The definitions below accompany the quantile and zero-inclusive comparisons reported in this appendix. Section 3.3 defines the age-adjusted main regressions.

M.1 A comparison at fixed variance

The main comparison allocates a given earnings variance between favorable and adverse outcomes. For earnings growth g with mean μ_c in cell c, define the lower and upper centered second partial moments:

$$V_c^- = \mathbb{E}_c[(g-\mu_c)^2 \mathbf{1}\{g < \mu_c\}], \qquad V_c^+ = \mathbb{E}_c[(g-\mu_c)^2 \mathbf{1}\{g > \mu_c\}]. \tag{M.1}$$

These components sum exactly to total variance, $V_c = V_c^- + V_c^+$. Define upside variance composition as

$$C_c^{\text{up}} = V_c^+ - V_c^-, \qquad V_c^+ = \frac{V_c + C_c^{\text{up}}}{2}, \qquad V_c^- = \frac{V_c - C_c^{\text{up}}}{2}. \tag{M.2}$$

The partial moments use every realization on the corresponding side of the mean. The distinction between favorable and adverse components follows the broader good- and bad-uncertainty literature (Segal, Shaliastovich, and Yaron, 2015); the application here is to individual earnings histories and subsequent household stock market participation.

The entry regression includes (V_c, C_c^{up}) alongside mean growth and its square, the standard deviation, zero incidence, household controls, and fixed effects. Including both V_c and $\sqrt{V_c}$ preserves the original quadratic-in-standard-deviation scale specification. The coefficient on C_c^{up} compares earnings distributions with different allocations of a given variance between favorable and adverse outcomes. Its zero restriction is equivalent to equal *raw-unit* coefficients on V_c^- and V_c^+; an exact change of coordinates leaves fitted values unchanged. It is not a test that variance alone fully describes the distribution, and the separate partial-moment coefficients are not total derivatives when the standard-deviation control also changes.

Upside and downside composition describe the same contrast from opposite directions. If $C_c^{\text{down}} = V_c^- - V_c^+$, then $C_c^{\text{up}} = -C_c^{\text{down}}$. Reversing the orientation reverses the coefficient and interval endpoints but leaves standard errors and two-sided tests unchanged. At fixed total variance, a larger upper component necessarily entails a smaller lower component. This is not the effect of increasing upside risk while holding downside risk fixed, and the two orientations are not independent findings.

I report this comparison on the original 60 cells, on the 59 cells with adequate positive-destination support, and using only positive earnings destinations within those same 59 cells. The last design changes the conditioning event of the earnings distribution, not the SIPP entry sample. Each construction uses its own original cross-cell standard deviation, held fixed in resampling. To describe a comparison within the observed conditional support, I also multiply the coefficient by the weighted standard deviation of composition after projecting out every other regressor in its SIPP specification.

These wage-based estimates are benchmarks for the matched validated-link comparison between wages and total labor earnings. The latter retains 58 common cells and 27,419 SIPP entry transitions, compares zero-inclusive with positive destinations, and reconstructs both raw and observable-progression-adjusted earnings growth using validated longitudinal links. Section N.1 presents that comparison. Its total-earnings magnitudes use a different construction and scale from the headline wage estimate; tests from the original wage or original-linkage designs are not transferred to it.

M.2 Separating the two tails

The quantile comparison describes the two sides without squaring earnings deviations. Let G_c be the distribution of one-year earnings growth in cell c, with quantile $q_{p,c}$. Define the upper and lower tail distances as

$$U_c = q_{0.90,c} - q_{0.50,c}, \qquad L_c = q_{0.50,c} - q_{0.10,c}. \tag{M.3}$$

The upper distance measures the gap between the median and favorable earnings growth; the lower distance measures the gap to adverse growth. They are quantile distances, not probabilities of a gain or loss, and neither is by itself a structural shock.

Their sum, $S_c = L_c + U_c$, is the 90–10 span, a measure of dispersion distinct from variance. Their difference, $D_c = L_c - U_c$, measures the composition of that span toward the downside. Entering

(S_c, D_c) instead of (L_c, U_c) is an exact linear reparameterization. In raw units,

$$b_L L_c + b_U U_c = \frac{b_L + b_U}{2} S_c + \frac{b_L - b_U}{2} D_c.$$

Thus the composition coefficient asks whether shifting distance from the upper to the lower tail is associated with a different portfolio outcome at a given total span. It does not create a second source of identifying variation. Because the empirical regressors are standardized separately, displayed coefficients need not satisfy this identity without first restoring their raw units.

Downside-oriented Kelley shape normalizes the difference by the span:

$$K_c = \frac{L_c - U_c}{L_c + U_c} = \frac{2q_{0.50,c} - q_{0.10,c} - q_{0.90,c}}{q_{0.90,c} - q_{0.10,c}}. \tag{M.4}$$

A positive value denotes a longer downside. Kelley answers a different question from the unscaled composition D_c: it discards the magnitude of the interquantile range. Appendix A defines the alternative Bowley, L-skewness, moment-skewness, and tail-probability measures.

M.3 What it means to hold other risk features fixed

The zero-inclusive distribution combines wage growth with transitions to zero measured earnings. A larger zero mass can move quantiles, the mean, and the variance even if the positive-earnings distribution is unchanged. Once the zero mass reaches ten percent, the lower quantile is at the arc-growth boundary. Lemma 1 in Appendix A.1 states this relationship formally. The resulting tail statistic remains a valid description of the distribution, but it cannot by itself isolate a change in shape holding other distributional features fixed.

I therefore distinguish raw orientation from its incremental component. Across supported cells, the latter is the residual from

$$K_c = a + \gamma p_{0c} + f_2(\mu_c) + h_2(\sigma_c) + \lambda_{e(c)} + \lambda_{a(c)} + u_c, \tag{M.5}$$

where p_{0c} is the probability of zero destination earnings, μ_c and σ_c are mean growth and its standard deviation, and f_2 and h_2 are quadratic functions. Education and age components are included, and the projection is weighted by effective cell support capped at 5,000. Sequential versions remove the zero share and demographic components first, then add mean and dispersion. Each projection is re-estimated within every reliability half and CPS resample.

I control for mean growth because location and Kelley shape can be closely related (Harmenberg, 2021). The residual defines an observable conditional contrast; it does not identify "true skewness" or imply that the removed variation is economically irrelevant. The portfolio application similarly conditions the two tail distances on zero incidence, mean growth, and dispersion. Its question is whether the tails contain additional information within the support of those controls.

M.4 Portfolio comparisons

Write $H_{it} = 1\{A_{it} > 0\}$ for positive reported stock-and-mutual-fund holdings. The entry regressions are estimated among lagged nonholders. The quantile specification is

$$\begin{aligned} H_{it} = \beta_L \widetilde{L}^{\mathrm{CPS}}_{c(i,t-1)} + \beta_U \widetilde{U}^{\mathrm{CPS}}_{c(i,t-1)} + \rho \widehat{p}_{0,c(i,t-1)} + f_2(\widehat{\mu}_{c(i,t-1)}) + h_2(\widehat{\sigma}_{c(i,t-1)}) \\ + X'_{i,t-1}\gamma + \delta_t + \delta_{j(i,t-1)} + \delta_{e(i,t-1)} + \delta_{a(i,t-1)} + \varepsilon_{it}, \qquad H_{i,t-1} = 0. \end{aligned} \tag{M.6}$$

Tildes denote standardization across the supported early-CPS cells, using Kish effective-N weights capped at 1,000. $X_{i,t-1}$ includes a transformed average of monthly job-1 earnings, log total assets, age and its square, sex, and an indicator for any eligible month with nonpositive decoded job-1 earnings. For mean monthly job-1 earnings $\bar{y}^{(1)}$ in unshifted dollars over the retained observations, the income control is $\log(\max\{\bar{y}^{(1)} + 1, 1\})$. Total assets are floored at one dollar before taking logs. The income-control offset is not used in the arc-growth risk measures. The nonpositive-earnings indicator is not a separately observed unemployment spell or a zero of total labor income. Fixed effects cover outcome year and lagged industry, education, and age band. Observations are weighted by the current SIPP person-year weight. Replacing the two tails with span and signed composition leaves fitted values unchanged; a further specification pairs span with incremental Kelley orientation.

I multiply the estimated coefficients in equation (M.6) by 100 to report percentage-point differences in entry per one cross-cell standard deviation of the assigned measure. They are conditional associations, not household responses to exogenous changes in risk. Risk tolerance, lifetime resources, or other characteristics can jointly influence occupation and portfolios (Ranish, 2013; d'Astous and Shore, 2024). Lagging the assignment does not solve this selection problem.

The remaining outcomes distinguish acquisition, retention, and allocation. I estimate exit, $1 - H_{it}$, among $H_{i,t-1} = 1$, and signed participation change, $H_{it} - H_{i,t-1}$, in the full transition sample, controlling lagged participation in the latter. Standing participation and the three share measures use contemporaneous household controls and assignment-cell components with the same pre-period CPS measures. The holder-only share is the relevant standing intensive margin; all-worker and positive-financial-asset shares also incorporate the extensive margin.

The prospective allocation specification instead conditions on *initial* ownership and retains future zero holdings. Let s_{it} denote stocks and mutual funds divided by nonretirement financial assets, and let $H_{i,t-1}$ indicate baseline ownership. I estimate

$$\begin{aligned} s_{it} = (1 - H_{i,t-1})\big(\beta_0 \widetilde{C}^{\mathrm{up}}_{c(i,t-1)} + Z'_{i,t-1}\gamma_0\big) \\ + H_{i,t-1}\big(\beta_1 \widetilde{C}^{\mathrm{up}}_{c(i,t-1)} + Z'_{i,t-1}\gamma_1 + \phi s_{i,t-1}\big) + \varepsilon_{it}. \end{aligned} \tag{M.7}$$

Here Z includes a constant, total variance, the other earnings-distribution controls, household controls, and fixed effects. All coefficients are allowed to differ by initial ownership. The common cross-cell scaling of composition makes $\beta_1 - \beta_0$ a direct test of the portfolio-state difference. Section 4.6

reports the age-adjusted real-log and nominal-arc constructions on the same sample. Baseline holders can subsequently exit; their outcome is not selected on continued participation.

M.5 Inference and scope of the tests

In the original wage and quantile designs, conditional standard errors cluster by SIPP person and CPS assignment cell. The large number of household records does not create a correspondingly large number of independent earnings-risk environments: these designs use at most 60 cells. A null-imposed cell-level wild bootstrap addresses this finite-cell dimension while retaining the two-way covariance structure (Roodman et al., 2019). A separate exercise resamples CPS people and reconstructs the tail measures, adding the resulting coefficient variance to the conditional variance. I call the latter a CPS plug-in sensitivity because it holds SIPP fixed and is not a joint CPS–SIPP bootstrap.

I then allow both samples to vary. In each of 2,000 draws, CPS people are resampled within their original cells, and SIPP observations receive the product of independent mean-one exponential weights for person and assignment cell, multiplied by their survey weights. This crossed reweighting follows Owen and Eckles (2012). Matched CPS-only and SIPP-only draws identify which stage changes the estimates. The sample and coefficient units remain fixed. I report percentile, basic, and original-point normal-reference intervals together to show how inference changes under nonlinear remeasurement. This is a sensitivity to dependence and estimated regressors, not an exact finite-cell test or a full survey-design bootstrap. Section N.6 and Appendix I report the comparison.

The validated-link wage–total comparison uses a separate 1,000-draw reconstruction. CPS multipliers are shared within households; SIPP weights combine sample-unit-lineage and assignment-cell multipliers. Each draw refits the earnings ranks and household-disjoint predictions before rebuilding the distributions and portfolio regressions. Original-point reference tests use t_{57}, alongside percentile and basic intervals and the declared family adjustments. This design carries estimation uncertainty through the observable-progression adjustment and permits household dependence. It is distinct from the 2,000-draw comparisons and does not establish full complex-survey coverage.

In the 58-cell-clustered prospective allocation comparison in equation (M.7), the primary covariance is clustered at the 58 assignment cells with a t_{57} reference, with 30 industry–education groups and a t_{29} reference as a coarser sensitivity. The main tables use the latter group-level convention, as stated in Section 3.3. These tests condition on the estimated CPS moments; they do not include a new first-stage reconstruction. The complete pair of real-log and nominal-arc specifications is reported, with Benjamini–Hochberg adjustments for the two between-state contrasts. The replication output also retains the six-test family covering both state slopes and their difference in each construction.

The four standing outcomes and three transition outcomes are separate displayed testing families. Benjamini–Hochberg q-values adjust within each family and specification (Benjamini and Hochberg, 1995); the two individual tail coefficients also have a separately reported wild-bootstrap adjustment. The three variance-composition comparisons form a separate family, as do the twelve scalar coefficients in the earnings-state exercise. Every table identifies the test used for its stars. These analyses are

exploratory. The displayed adjustments do not account for every specification considered over the project's history.

N Rank-Adjusted and Quantile Portfolio Comparisons

This section reports earnings-progression, own-history, and quantile comparisons. These specifications are distinct from the CSZ-style age-profile adjustment used in Section 4; their coefficient units and inference are stated separately.

N.1 Variance composition and subsequent entry

At a given total earnings variance, distributions tilted toward favorable rather than adverse outcomes are associated with greater subsequent participation among initial nonholders. Table N.1 compares wages with total labor earnings, the sum of wages and net business and farm earnings, on identical CPS histories with validated longitudinal identifiers. The population remains workers with positive origin wages. Both measures have positive origins and nonnegative destinations; a second comparison additionally requires positive destinations in both measures. All four constructions use the same 27,419 subsequent SIPP entry transitions in 58 cells. The total-earnings extension broadens the measured income components without changing the initial worker population or recoding business losses as zero.

At fixed variance, more upside composition necessarily means less downside composition. Reversing the orientation gives equal-magnitude negative coefficients, identical standard errors, and identical two-sided tests; it is not a second empirical confirmation. Nor is this an independent positive upper-tail effect with the lower partial moment held unchanged. Both separate partial-moment coefficients are negative in the original wage benchmark, and the upper one is imprecise (Appendix Table I.1). The difference between the two sides at fixed total variance is the composition contrast.

The assigned distributions contain realized earnings changes, not necessarily unanticipated risk. Favorable changes can reflect progression associated with age and prior earnings, as well as recoveries from earlier losses. I therefore also remove growth predicted by a fixed set of observed origin characteristics before reconstructing the distribution. Prediction uses five household-disjoint CPS folds and does not use SIPP portfolio outcomes. The raw and adjusted comparisons retain validated earnings links, the same 58 cells, identical portfolio observations, and common coefficient units within each pair.

Table N.1 shows that the adjustment weakens the point pattern. The residual-standard-deviation magnitude falls from 0.792 to 0.510 percentage point for zero-inclusive wages and from 0.703 to 0.370 for total earnings. In the positive-destination comparisons it falls from 0.649 to 0.207 and from 0.660 to 0.208 point, respectively. All four adjusted slopes remain positive, but none survives its complete four-test family adjustment: joint-remeasurement q-values range from 0.362 to 0.451.

Thus these estimates do not establish an entry association for earnings composition after observable progression is removed.

Table N.1. Observable earnings progression and stock market participation

	(1)	(2)	(3)	(4)
Earnings	Wages	Total	Wages	Total
Destinations	Zero-inclusive		Both positive	
Panel A. Observable-progression adjusted				
Upside composition, C^{up}	1.793^{*}	1.259	0.380	0.394
	(1.041)	(0.981)	(0.499)	(0.520)
Slope: four-model BH q	0.362	0.409	0.451	0.451
Residual-S.D. magnitude (pp)	0.510	0.370	0.207	0.208
	(0.315)	(0.320)	(0.303)	(0.304)
Magnitude: four-model BH q	0.445	0.498	0.498	0.498
Panel B. Matched raw earnings changes				
Upside composition, C^{up}	3.242^{***}	2.260^{**}	0.974^{**}	1.029^{**}
	(1.067)	(0.858)	(0.403)	(0.412)
Slope: four-model BH q	0.014	0.019	0.019	0.019
Residual-S.D. magnitude (pp)	0.792^{**}	0.703^{**}	0.649^{**}	0.660^{**}
	(0.304)	(0.299)	(0.307)	(0.302)
Magnitude: four-model BH q	0.039	0.039	0.039	0.039
Risk and household controls	Yes	Yes	Yes	Yes
Year and group effects	Yes	Yes	Yes	Yes
Observations	27,419	27,419	27,419	27,419
SIPP people	19,081	19,081	19,081	19,081
SIPP sample-unit lineages	13,849	13,849	13,849	13,849
CPS earnings histories	88,862	88,862	83,372	83,372
Active CPS households	61,480	61,480	58,798	58,798
CPS assignment cells	58	58	58	58
Joint full-pipeline draws	1,000	1,000	1,000	1,000

Notes: Outcome: subsequent entry into reported stock-and-mutual-fund holdings. Equal-positive endpoint CPSIDV restricts this repair to matched 58-cell partial moments; original 60/59-cell estimates and original-link local inference are unchanged. Slopes are percentage points per construction-specific original raw-composition cross-cell S.D. (CPS person-Kish weights capped at 1,000), common across panels. Panel A uses five household-disjoint prediction folds. Magnitudes use each model's conditional composition S.D. Parentheses report 1,000 joint CPS-household and SIPP-lineage-by-cell full-pipeline draws, rebuilding ranks, predictions, moments, regressions, and magnitude S.D.s. Stars use unrounded, unadjusted, approximate two-sided t_{57} tests against zero: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Each raw/adjusted slope/magnitude family has four BH-adjusted tests. Appendix I.13 details controls and weighting; Tables I.10 and I.11 retain conflicting intervals and paired source comparisons. This exploratory sensitivity is not complete survey inference and does not identify unanticipated risk.

Exactly 1,000 matched draws reconstruct the earnings predictions, moments, and portfolio regressions while allowing dependence within CPS households and within SIPP sample-unit lineages and assignment cells. Every adjusted 95% percentile interval includes zero; basic slope intervals exclude it for both zero-inclusive constructions, only barely for total earnings. The raw references likewise have percentile intervals containing zero despite favorable original-point tests and basic intervals. Their draw means fall substantially below the original estimates. Inference for the raw

coefficients therefore depends on the interval method. Appendix Table I.10 reports all three interval constructions and displacement.

The adjusted-minus-raw slope differences are also imprecise, with four-test $q = 0.281$ throughout. Repeating the complete raw–adjusted comparison on the original earnings links gives the same qualitative conclusion; the paired changes due to link validation are imprecise as well (Appendix Table I.11). The smaller adjusted point estimates therefore do not establish that predictable progression explains the original association. Moreover, the adjustment removes only projection on observed covariates. It can absorb measurement-error mean reversion and common realized growth as well as predictable progression; the residuals are not identified earnings shocks. The source of the original association remains unresolved.

Appendix Tables I.1 and I.2 retain the original-linkage wage and matched wage–total benchmarks. Both show similar positive point patterns. In the original wage benchmark, omitting the 2014 ASEC attenuates the associations without reversing their signs. Broader dependence-sensitive inference is inconclusive, and wage–total slope differences are imprecise. These comparisons establish neither equality across earnings definitions nor a stronger total-earnings response. Their separate local and product-reweighting uncertainty calculations are discussed in Section N.6; those tests do not supply inference for the validated estimates.

N.2 Conditioning on workers' own prior earnings

Recent earnings levels and changes could help explain why workers assigned to different historical distributions enter the stock market at different rates. I examine this possibility using two annual totals of each worker's all-job earnings, including signed business income, strictly before the baseline portfolio observation. Requiring complete prior histories with self or proxy interview status leaves 6,639 nonholder transitions in 58 cells, with a destination-weighted entry rate of 7.96%. This is a selected subset of the 27,419-transition reference sample, not a new representative cohort. The comparison without history controls is therefore re-estimated on exactly the same 6,639 observations.

Table N.2 adds both annual earnings levels in inverse-hyperbolic-sine form, a signed arc change, and four indicators for zero and negative annual totals. The dollar unit is fixed at $10,000 in 2022 prices. Every specification retains the original household and earnings-distribution controls, including total variance. The raw and observable-progression-adjusted CPS distributions remain distinct constructions; two personal earnings levels do not estimate a personal risk distribution or elicit beliefs.

Adding the history controls leaves the raw point estimates almost unchanged. For total earnings, the residual-standard-deviation magnitude moves from 0.777 to 0.775 percentage point; with positive CPS destinations in both earnings measures, it moves from 0.729 to 0.727 point. Columns (3)–(4) and (7)–(8) use identical workers within each comparison. These magnitudes use each specification's own residual spread; the table separately reports changes in slopes expressed in common original CPS units. The complete eight-test family of primary matched slope changes has $q = 0.994$. Neither the small point changes nor the imprecise paired differences establish equivalence or rule out sorting.

Table N.2. Earnings composition and entry conditional on prior earnings

	CPS earnings: zero-inclusive				CPS earnings: positive			
	Wages		Total earnings		Wages		Total earnings	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Prior-history controls	No	Yes	No	Yes	No	Yes	No	Yes
Panel A: Raw earnings growth								
Upside composition	4.269**	4.270**	2.530	2.525	1.185	1.184	1.161	1.159
	(2.004)	(1.997)	(1.609)	(1.609)	(0.746)	(0.745)	(0.765)	(0.766)
Composition: BH8 q	0.216	0.217	0.216	0.217	0.216	0.217	0.216	0.217
All 95% CIs positive	–	No	–	No	–	No	–	No
Per own residual SD (pp)	1.024*	1.024*	0.777	0.775	0.772	0.771	0.729	0.727
	(0.544)	(0.542)	(0.537)	(0.536)	(0.537)	(0.535)	(0.530)	(0.530)
Magnitude: BH8 q	0.280	0.280	0.280	0.280	0.280	0.280	0.280	0.280
Matched slope change	–	0.001	–	−0.005	–	−0.001	–	−0.002
		(0.146)		(0.118)		(0.054)		(0.056)
Paired change: BH8 q	–	0.994	–	0.994	–	0.994	–	0.994
Baseline controls and FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Nonholder transitions	6,639; 6,639 people; 5,291 lineages; 58 cells							
Destination years	2020, 2022, 2023, 2024							
Weighted entry rate (%)	7.96							

Outcome: later positive stock-and-mutual-fund holdings among baseline nonholders. All columns use the same self/proxy histories covering 24 prior months; earnings recodes may be imputed. CPS origins require positive wages and totals; negative total destinations are excluded. Positive headers require both CPS destinations positive, without changing recipients. Slopes are percentage points per original raw-source composition SD, common within raw/adjusted pairs and control blocks.

All models retain variance, mean and square, SD, applicable zero incidence, baseline personal controls and fixed effects. Histories add two annual totals strictly before baseline: $\operatorname{asinh}(Y/10{,}000)$ in 2022 dollars, signed arc change and four zero/negative indicators. They are not identified shocks or beliefs. Weights are original destination weights, not history-longitudinal weights.

Parentheses: SDs of 1,000 joint CPS-household/SIPP-lineage-by-cell draws. Stars: unadjusted two-sided original-point t_{57} tests, $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. This approximation does not certify survey coverage. BH8 covers eight raw/adjusted constructions separately by block and target; paired changes have a separate eight-test family. Magnitudes use their own SEs/tests and reflect changing residual spread. Slope changes are augmented minus matched baseline, not mediation shares. "Yes": all three joint point-centered, percentile and basic 95% slope intervals are positive at unrounded bounds, not a new p-value. The positive-reporting screen additionally requires a positive primary augmented point and BH8 $q < 0.05$. Both interval companions retain every method.

The restricted sample does not deliver an inference-stable association beyond recent earnings. All eight history-augmented slopes are positive, but none survives BH8 adjustment; q-values range from 0.217 to 0.870. Every primary 95% percentile interval includes zero. The same-sample baseline coefficients also fail their eight-test family adjustment, so this loss of significance cannot be attributed to adding the controls. The decline-kink and alternative interview/processing samples provide no family-adjusted slope rejection either. Appendix I.16 reports the complete comparisons. The exercise shows that observed recent earnings do not substantially change the estimated point pattern on the matched sample; it does not identify a causal response to risk, establish unanticipated shocks, or recover perceived risk.

Table N.2. Earnings composition and entry conditional on prior earnings (continued)

	CPS earnings: zero-inclusive				CPS earnings: positive			
	Wages		Total earnings		Wages		Total earnings	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Prior-history controls	No	Yes	No	Yes	No	Yes	No	Yes
Panel B: Observable-progression-adjusted growth								
Upside composition	2.880	2.939	1.930	2.044	0.100	0.146	0.359	0.400
	(1.851)	(1.852)	(1.705)	(1.707)	(0.886)	(0.887)	(0.919)	(0.920)
Composition: BH8 q	0.216	0.217	0.350	0.315	0.910	0.870	0.797	0.760
All 95% CIs positive	–	No	–	No	–	No	–	No
Per own residual SD (pp)	0.802	0.817	0.553	0.585	0.054	0.079	0.187	0.208
	(0.548)	(0.547)	(0.540)	(0.539)	(0.534)	(0.533)	(0.530)	(0.530)
Magnitude: BH8 q	0.280	0.280	0.413	0.376	0.920	0.883	0.830	0.796
Matched slope change	–	0.058	–	0.114	–	0.046	–	0.041
		(0.144)		(0.130)		(0.069)		(0.072)
Paired change: BH8 q	–	0.994	–	0.994	–	0.994	–	0.994
Baseline controls and FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Nonholder transitions	6,639; 6,639 people; 5,291 lineages; 58 cells							
Destination years	2020, 2022, 2023, 2024							
Weighted entry rate (%)	7.96							

Outcome: later positive stock-and-mutual-fund holdings among baseline nonholders. All columns use the same self/proxy histories covering 24 prior months; earnings recodes may be imputed. CPS origins require positive wages and totals; negative total destinations are excluded. Positive headers require both CPS destinations positive, without changing recipients. Slopes are percentage points per original raw-source composition SD, common within raw/adjusted pairs and control blocks.

All models retain variance, mean and square, SD, applicable zero incidence, baseline personal controls and fixed effects. Histories add two annual totals strictly before baseline: $\operatorname{asinh}(Y/10{,}000)$ in 2022 dollars, signed arc change and four zero/negative indicators. They are not identified shocks or beliefs. Weights are original destination weights, not history-longitudinal weights.

Parentheses: SDs of 1,000 joint CPS-household/SIPP-lineage-by-cell draws. Stars: unadjusted two-sided original-point t_{57} tests, $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. This approximation does not certify survey coverage. BH8 covers eight raw/adjusted constructions separately by block and target; paired changes have a separate eight-test family. Magnitudes use their own SEs/tests and reflect changing residual spread. Slope changes are augmented minus matched baseline, not mediation shares. "Yes": all three joint point-centered, percentile and basic 95% slope intervals are positive at unrounded bounds, not a new p-value. The positive-reporting screen additionally requires a positive primary augmented point and BH8 $q < 0.05$. Both interval companions retain every method.

N.3 The comparison with conventional skewness

The allocation of variance between its two sides is not the same statistic as a third moment. It nevertheless need not contain separately measurable information beyond conventional skewness. In the original-linkage matched earnings constructions, the conditional correlation between upside composition and standardized skewness ranges from 0.90 to 0.97. Only zero-inclusive total earnings retains positive incremental-composition reliability under both the CPS support metric and the SIPP regression's exact control-profiled metric; the other three constructions fail this outcome-free information requirement.

In that one eligible construction, adding standardized skewness leaves a positive composition coefficient but does not establish an incremental association. Its two-survey p-value is 0.100 and its

two-test q-value is 0.200; the corresponding skewness q-value is 0.472. The pair is jointly informative under the local covariance ($p = 0.038$), but not under the broader dependence sensitivity ($p = 0.145$). A joint rejection tests whether both coefficients vanish, not whether composition adds information beyond skewness. Appendix Table I.12 reports both covariance conventions and the complete two-test family. The evidence distinguishes composition from variance but does not establish an incremental association beyond conventional skewness.

N.4 Lower and upper quantile distances

Table N.3 provides a complementary quantile comparison. In column (1), the lower and upper tail distances enter jointly. A one-standard-deviation wider lower tail is associated with a 2.712-percentage-point lower probability of positive stock-and-mutual-fund holdings one year later. The corresponding upper-tail coefficient is $+1.917$ points. The sample contains 27,752 worker transitions beginning with no reported holdings, with a weighted entry rate of 8.79%. The opposite signs distinguish the favorable and adverse sides of the assigned earnings distribution; the uncertainty is appreciably larger for the upper side.

Column (2) expresses the same comparison in terms of total span and signed composition. The span coefficient is imprecise, while a one-standard-deviation shift toward the downside is associated with 4.456 points less entry. The CPS plug-in p-values are 0.232 for span and 0.014 for composition. The fitted values from columns (1) and (2) agree to 1.1×10^{-8}: the rotation reveals the economic contrast already present in the joint two-tail regression. It does not show that total dispersion is irrelevant, or provide a second independent rejection.

Column (3) pairs total span with incremental Kelley orientation. The orientation coefficient is negative but less precise once uncertainty in the estimated CPS distribution is included (-0.870 point, $p = 0.068$). Normalizing composition by span therefore does not strengthen the result. Figure N.1 shows how adding the CPS estimation component changes the uncertainty around all three representations. In particular, the wider upper-tail and Kelley intervals cross zero.

This contrast is not a recovered variance sign reversal. In the same entry sample, the standardized linear term in cell standard deviation, evaluated at mean standardized dispersion and included with its square, is -0.262 point (SE 1.120; $p = 0.816$). Adding Kelley orientation or the separate tails does not uncover a positive scale coefficient. The empirical content is the difference between the tail associations conditional on dispersion, not an explanation of a positive variance–portfolio-share coefficient.

N.5 Quantile orientation across portfolio margins

Appendix Table C.4 compares the standing outcomes with portfolio transitions using raw and incremental Kelley orientation. Among lagged nonholders, incremental downside orientation is associated with 1.047 points less entry, or 11.9% of the baseline transition rate. Its conditional three-outcome q-value is below 0.001. Raw Kelley has a similar point estimate but less precision: its

Table N.3. Lower and upper earnings tails and stock market participation

	(1) Lower and upper	(2) Span and composition	(3) Span and orientation
Lower distance, $L = q_{0.50} - q_{0.10}$	-2.712^{***}		
	(0.857)		
Upper distance, $U = q_{0.90} - q_{0.50}$	1.917^{*}		
	(1.049)		
Total tail span, $S = L + U$		2.721	-0.755
		(2.279)	(1.254)
Downside composition, $D = L - U$		-4.456^{**}	
		(1.820)	
Incremental Kelley orientation			-0.870^{*}
			(0.477)
Diagnostic component order	L / U	S / D	S / Kelley
Conditional S.E.	0.646 / 0.846	1.850 / 1.530	0.989 / 0.376
CPS coefficient S.D.	0.562 / 0.619	1.331 / 0.986	0.770 / 0.293
Plug-in p	0.002 / 0.068	0.232 / 0.014	0.547 / 0.068
Joint two-component plug-in p	0.002	0.002	0.011
Lagged controls and fixed effects	Yes	Yes	Yes
Observations	27,752	27,752	27,752
People	19,300	19,300	19,300
CPS signal cells	60	60	60

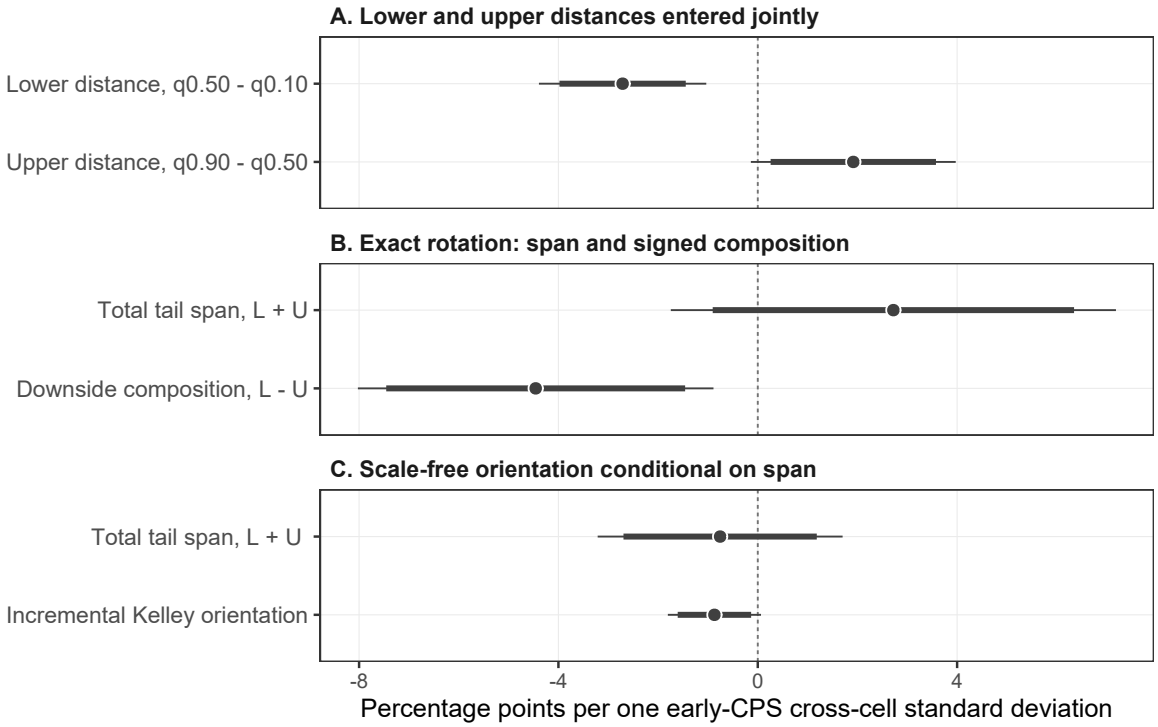
Notes: The dependent variable is a transition from zero to positive reported stock-and-mutual-fund holdings; both annual observations are in 2019–2024. Coefficients are percentage points per one cross-cell standard deviation of the named early-CPS measure. Each column is a separate regression. All columns include lagged zero share, quadratic mean and dispersion, household controls, and year, lagged industry, education, and age-band fixed effects, with current SIPP weights. Parentheses report plug-in standard errors: the conditional variance, clustered by person and lagged CPS cell, plus the coefficient variance from 200 CPS resamples. This is a first-stage sensitivity with SIPP held fixed, not joint inference. Stars use two-sided standard-normal plug-in tests: $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$. Paired diagnostic entries follow the stated component order. In a separate 99,999-draw null-imposed wild-cluster check, the lower and upper distances have two-component BH q-values of 0.019 and 0.087; signed composition has $p = 0.033$. Assignment is endogenous.

unadjusted $p = 0.042$ becomes $q = 0.126$ within the transition family. Separating zero incidence and lower moments changes the precision and interpretation of the comparison rather than revealing a larger raw effect.

The standing-share results for Kelley orientation do not establish an intensive-margin response. Incremental Kelley is negatively associated with the all-worker and positive-financial-asset shares, but both samples include nonholders. The all-worker coefficient is -0.234 point, with conditional $q = 0.039$; after rebuilding the CPS first stage, its p-value rises to 0.109. Among current holders, the coefficient is -0.153 point with SE 0.424 and $q = 0.721$. Exit is also imprecise. Within this quantile construction, the evidence is strongest for acquisition from a zero position. These standing-outcome estimates concern a different earnings statistic and conditioning event from the prospective baseline-status comparison in Table R.1.

Several checks show that the entry association is not confined to a fleeting or very small positive balance. Sustained entry has an incremental-Kelley coefficient of -0.697 point (SE 0.293; six-test

Figure N.1. Tail composition and uncertainty about stock market participation



Notes: Points report the coefficients in columns (1)–(3) of Table N.3, in percentage points per one early-CPS cross-cell standard deviation. Thick intervals are the estimate plus or minus 1.96 conditional two-way-clustered standard errors. Thin intervals use the corresponding CPS plug-in standard error, which adds the coefficient variance across 200 CPS resamples. These are normal-approximation sensitivity intervals, not joint-bootstrap confidence intervals. Components are standardized separately within each panel and enter jointly; the dashed line marks zero.

$q = 0.041$). Crossings above \$1,000 and \$5,000 have coefficients of -1.104 and -0.453 points, with $q < 0.001$ and $q = 0.045$ in the same family. Requiring the same earnings-risk cell in both years yields -0.958 point, and omitting each outcome year in turn leaves five negative estimates between -1.203 and -0.881 points. These tests concern incremental orientation; they are not separate confirmations that both individual tail coefficients are robust.

Household composition also matters because the estimation unit is a worker transition involving household-reported assets. The coefficient is -1.057 point in the stable-household sample and -0.869 point when one eligible worker, ranked by mean lagged job-1 earnings, is retained per household transition. Every leave-one-cell and leave-one-industry coefficient is negative. Appendix Tables C.6, C.9, and C.3 report the full families, alternative covariance estimates, and decision-unit checks. The broader portfolio-level window appears in Appendix Table E.1.

N.6 Uncertainty in the assigned earnings distributions

The following uncertainty calculations concern the original wage and quantile designs; they do not supply inference for the age-adjusted total-earnings or validated-linkage comparisons. Reconstructing the CPS distributions leaves incremental Kelley entry as the strongest plug-in result ($p = 0.0052$;

three-outcome $q = 0.0157$). The lower and upper quantile distances are jointly informative, but the upper distance alone is imprecisely estimated ($p = 0.068$). Thus the evidence for the overall orientation contrast is stronger than that for an independent upper-tail association.

Jointly resampling both surveys exposes sensitivity that the plug-in calculation does not resolve. The remeasured coefficient distributions shift toward zero. Their 95% percentile intervals contain zero, whereas basic intervals exclude it for the lower tail, upper tail, and signed composition. The same disagreement appears for smooth variance components. Neither interval construction is treated as definitive, and a narrower spread after remeasurement is not evidence of greater precision.

For the original wage variance-composition benchmarks, local propagation of moment uncertainty gives family-adjusted $q = 0.015$ and $q = 0.049$ for zero-inclusive and positive-destination earnings. Replacing the SIPP component with the broader crossed-reweighting covariance raises these to 0.066 and 0.105. Appendix I.4 preserves the complete outcome family, remeasurement diagnostics, and covariance comparisons. The conditional associations therefore require a distinction between sampling precision and stability to reconstructing the assigned earnings environment.

N.7 The lower tail and zero earnings

The lower-tail coefficient depends substantially on the support boundary. In 11 of the 60 cells, more than ten percent of CPS destinations have zero wage earnings. The 10th percentile is then -2, and the lower distance equals $2 + q_{0.50,c}$. Across cells, standardized lower distance correlates 0.89 with the zero share and 0.77 with an indicator that the quantile is at the bound. This makes the composition of zero wage earnings central to the disaster interpretation.

Appendix Table C.5 varies the zero-endpoint construction while retaining the baseline portfolio design. Excluding destinations classified as self-employed removes 6.2% of early transitions. Counting a zero only when no weeks were worked removes the 2.3% of transitions that report work without wage earnings. These changes reduce the cross-cell standard deviation of the lower distance from 0.40 to 0.25 and 0.23 arc-growth units. Its portfolio coefficient falls from -2.71 to -0.91 and -0.76 points, respectively, and neither alternative is distinguishable from zero. Restricting the baseline to the 49 cells with an interior 10th percentile yields -1.22 points (SE 1.35).

Signed composition is more stable than either a standalone lower-tail effect or normalized Kelley orientation across these constructions. Its coefficients are -1.73, -1.77, and -4.46 points in the three alternatives, with conditional p-values of 0.048, 0.023, and 0.020. The upper coefficient remains positive but is imprecise after excluding self-employed destinations ($p = 0.13$). Incremental Kelley is imprecise in two of the alternatives ($p = 0.16$ and $p = 0.080$). These comparisons preserve the direction of the signed-composition relation within the baseline assignment, but they do not propagate CPS first-stage uncertainty and do not establish two construction-invariant tail effects.

An additional business-asset comparison does not support an entrepreneurial-substitution interpretation. Among transitions starting with neither business equity nor stock-and-mutual-fund holdings, the lower tail predicts less stock-and-mutual-fund-only entry (-2.99 points; eight-coefficient plug-in $q = 0.0098$). The upper estimate has $q = 0.131$, and neither business-only entry nor the

broader business-entry outcome survives the adjustment. These results also disappear under the self-employment-sensitive tail constructions. The distinction between a CPS self-employment classification and a household's business ownership does not resolve the baseline lower-tail dependence on wage zeros.

A separate earnings-state exercise distinguishes zeros with no reported weeks of work from zeros despite some work. On a common 59-cell sample, allowing the two probabilities to enter separately leaves the unconditional lower–upper pair jointly informative under the CPS plug-in sensitivity (three-model $q = 0.019$). None of the twelve scalar coefficients survives the complete family adjustment. When quantiles are instead estimated only among positive earnings destinations, signed composition is essentially zero, but its incremental reliability is negative. That estimate is uninformative about the absence of an economically relevant positive-earnings contrast. Appendix Table I.5 reports the full family. The exercise rules out neither nonlinear zero-boundary mechanics nor endogenous sorting; it shows that a simple linear distinction between the two zero states does not absorb the baseline joint comparison.

N.8 Sorting, assignment, and predictive scope

Within-person comparisons do not deliver a stable alternative to the cross-cell association. Among 5,789 workers with repeated entry-risk observations, person fixed effects give lower and upper coefficients of -0.58 (SE 1.30) and 4.27 (SE 1.99) under lagged assignment. Assigning the current cell instead gives -0.05 (SE 0.96) and -1.17 (SE 0.75). Changes in assigned tails are themselves imprecise. The timing sensitivity of the upper estimate prevents these results from ruling out time-invariant sorting.

The grouping used to assign risk is consequential as well. Adding occupation to the industry-based grouping reduces the upper estimate to 0.32 and 0.55 point in 69- and 96-cell designs, with incremental reliability of 0.32 and 0.18. Replacing industry with occupation in an 18-cell design reverses it to -1.74 points (SE 0.98). The baseline upper-tail sign is not invariant to the assignment scheme.

A single pre-pandemic comparison is inconclusive. The 6,033 transitions from 2018 to 2019 give lower and upper coefficients of -0.34 and -0.46 point (SE 1.08 and 0.99). This window includes a baseline portfolio observation overlapping the final CPS destination year, and is kept outside the strictly later sample. Its estimates are too imprecise to corroborate or refute the baseline; pooling the windows gives -2.42 and 1.57 points. Split-sample results by liquid assets also do not establish heterogeneous coefficients: formal interaction tests give $p = 0.38$ for the lower tail and $p = 0.89$ for the upper tail. Appendix D reports these comparisons in full.

The decomposition has not demonstrated superior out-of-cell forecasting. Leaving one assignment cell out at a time, the separate tails provide no detectable improvement over total span ($p = 0.571$, $q = 0.695$). Nor do comparisons with entry into bank, bond, retirement, home-equity, and vehicle-equity positions establish stock market specificity. A joint bond contrast has $q = 0.017$ in the all-worker sample but $q = 0.102$ after selecting one worker per household; neither individual tail

contrast survives the full ten-coefficient family. These tests limit forecast and mechanism claims without changing the in-sample conditional comparison.

Within the baseline historical industry-based assignment, composition favoring the upper over the lower side of earnings growth is associated with greater subsequent entry. The quantile and centered-variance representations express this comparison using different features of the distribution. The age-adjusted real-log comparison additionally distinguishes subsequent allocation by initial ownership, with opposite slopes for nonholders and holders. Construction, assignment, and inference determine the scope of these patterns. The next section examines the features of the earnings distributions that make the assigned measures reproducible.

O Reliability of Quantile and Zero-Inclusive Earnings Measures

O.1 Reproducibility and the earnings support

A portfolio association is more informative when different workers generate a similar ranking of earnings environments. I assess this reproducibility by assigning every person globally to one of two halves and estimating the cell statistics separately. Global assignment keeps a worker in the same half after a change in industry or education. If the half-sample correlation is $r_{1/2}$, the full-sample reliability projection is the Spearman–Brown transform (Spearman, 1910; Brown, 1910):

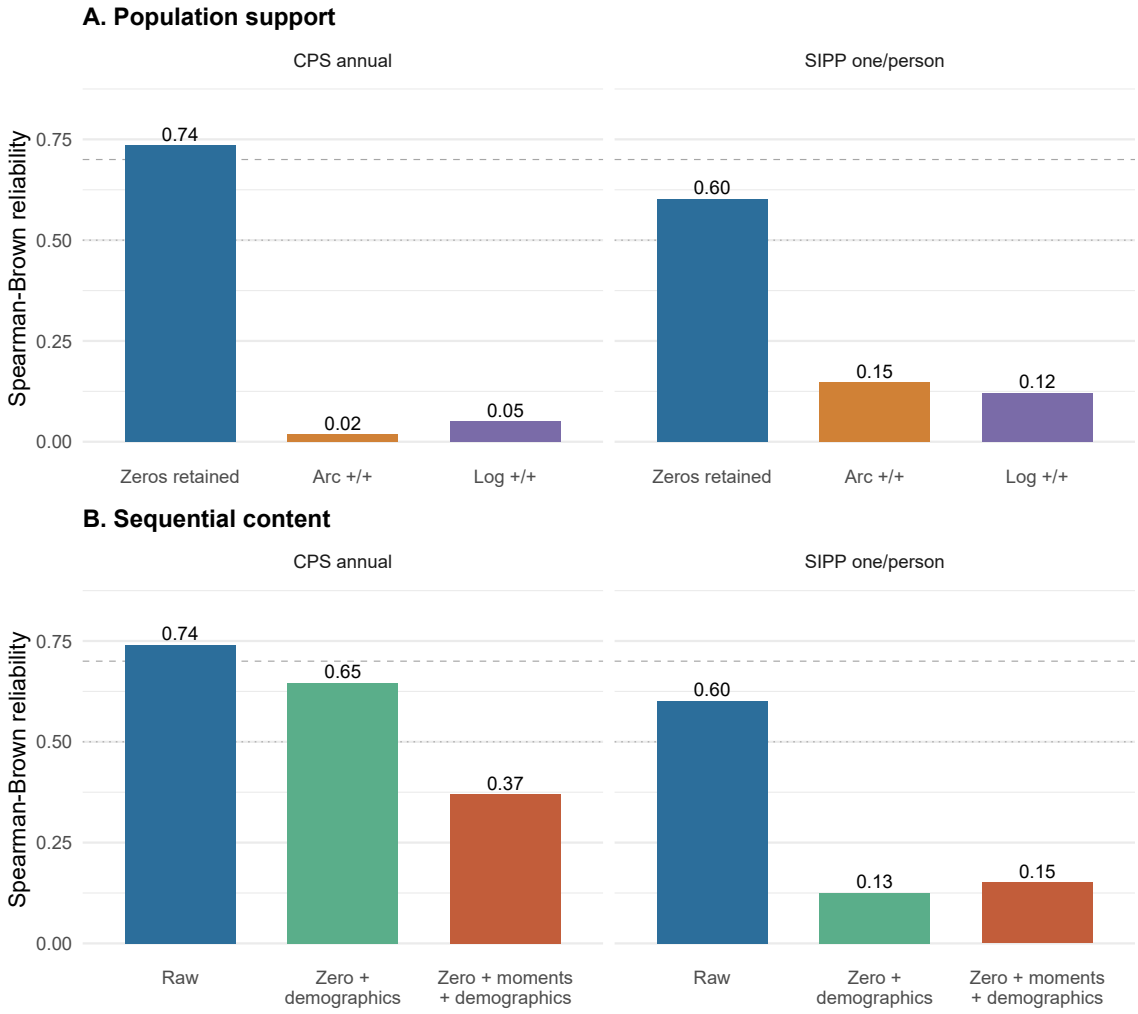
$$\rho_{\mathrm{SB}} = \frac{2r_{1/2}}{1 + r_{1/2}}. \tag{O.1}$$

I apply this transform to the mean correlation across 30 independent-person splits. Rank correlations and early-versus-late correlations provide separate measures of ordering and temporal stability. A high split reliability means that the statistic reproduces a cross-cell ranking; it does not reveal which economic component produces that ranking. Lemma 2 formalizes this distinction.

Retaining destination zeros changes the result sharply. In the companion CPS construction comparison, raw Kelley reliability is 0.018 for positive-to-positive arc growth and 0.735 when destination zeros are retained. The exact crosswalk used for the portfolio application gives 0.740. In the one-transition-per-person SIPP comparison, the corresponding arc-growth reliabilities are 0.147 and 0.602; positive-to-positive log growth has reliability 0.120. Figure O.1 shows that this ordering does not depend on treating overlapping SIPP months as independent evidence: zero-inclusive reliability is 0.622 with repeated months and 0.602 with one transition per person.

Adding zero destinations changes both sample size and the population distribution, placing additional mass at the lower arc-growth boundary. Conditioning it away changes the question to earnings growth among workers with positive measured earnings at both endpoints. Independent worker samples reproduce the zero-inclusive downside ranking within each survey. The economic meaning of that ranking depends on what a zero records.

Figure O.1. Earnings support and the reliability of tail orientation



Notes: Panel A compares raw downside-oriented Kelley reliability when destination zeros are retained with positive-to-positive constructions. Panel B uses the pooled exact CPS crosswalk and matched SIPP. The second stage removes the linear zero share and demographic components; the third also removes quadratic mean and dispersion. Reliability is the Spearman–Brown transform of the mean independent-person half-sample correlation across 30 deterministic splits. Each projection is re-estimated inside each half.

O.2 The economic content of a recorded zero

Table O.1 separates nonwork from work without wage earnings. Among 11,828 CPS annual wage-zero destinations, 64.50% report no work, 35.50% report some work, and 26.02% report work in all 52 weeks. Each component is reproducible across independent worker samples. No-work incidence has reliability 0.840, while worked-without-wage incidence has reliability 0.940, the same as total wage-zero incidence. The raw lower-tail ranking is therefore not simply a ranking of full-year nonemployment.

The SIPP zero refers to a different earnings concept and time interval. Of 25,310 workers in the strict one-transition-per-person anatomy sample, 914 reach zero monthly job-1 earnings. Among

Table O.1. Work arrangements at the zero-earnings endpoint

Endpoint or component	Weighted share (%)	Zero rows	CPS incidence reliability
Panel A. CPS annual wage-and-salary income			
Destination annual wage zero	6.63	11,828	0.940
No work in destination year	64.50		0.840
Worked in destination year	35.50		0.940
Worked all 52 weeks	26.02		0.914
Panel B. SIPP monthly job-1 earnings, one transition per person			
Destination monthly job-1 zero	3.51	914	
Self-employed work arrangement	64.47		
Employer work arrangement	34.62		
Any no-job spell in destination month	2.06		

Overall zero rates condition on positive origin earnings; indented shares condition on a destination zero. Official person weights are used. CPS shares use 183,809 linked annual transitions in all 72 candidate cells; incidence reliabilities use 181,946 transitions in 65 supported cells. Reliability is the Spearman–Brown transform of the mean independent-person half-sample correlation across 30 splits. The strict SIPP anatomy sample selects one eligible 2018–2023 origin transition for each of 25,310 people without conditioning on destination state, giving outcomes in 2019–2024. Arrangement class and no-job indicators refer to the destination month and job 1. Arrangement classes are mutually exclusive, but a no-job spell can coexist with a work arrangement during the month.

these zeros, 64.47% have a self-employment arrangement and 34.62% an employer arrangement in that month. A zero in the first group can represent zero business earnings rather than lost wages. Only 19 zeros, a weighted 2.06%, coincide with any recorded no-job spell. The corresponding weighted share is 0.99% when all eligible overlapping transitions are retained. A work arrangement and a no-job spell can occur within the same month, so the latter is not a mutually exclusive class.

An out-of-time comparison asks whether the CPS ranking also predicts subsequent SIPP job-1 zeros. Across 30 choices of one eligible transition per person, raw Kelley gives a positive coefficient in 29 cases, with a median of 0.297 percentage point. Incremental Kelley gives a positive coefficient in 26 cases, with a median of 0.154 point. No selection passes the four-outcome $q < 0.05$ threshold. Appendix Table B.5 reports the ranges and complete families. The signs are compatible with some shared information about earnings support, but the exercise does not establish a reliable cross-survey prediction.

O.3 Incremental orientation and persistence

Table O.2 separates the stable components of Kelley orientation. In pooled CPS, removing the zero share and demographic components reduces reliability from 0.740 to 0.646; adding quadratic mean and dispersion reduces it to 0.370. The exact early-CPS measure assigned to portfolios has raw reliability of 0.687 and incremental reliability of 0.397. The matched SIPP sequence is 0.602, 0.125, and 0.151. The highly reproducible zero incidence—0.940 in pooled CPS and 0.809 in matched SIPP—is one component of a broader downside environment, rather than an estimate of shape alone.

Time stability is weaker than within-sample reproducibility. The CPS early-versus-late correlation falls from 0.631 for raw Kelley to 0.230 for incremental Kelley; in matched SIPP, it falls from 0.414

Table O.2. Reliability and persistence of earnings-tail orientation

Design	Supported records	Cells	Raw	Zero adjusted	Moment adjusted	Zero share
Panel A. Independent-person reliability						
CPS, pooled	181,946	65	0.740	0.646	0.370	0.940
CPS, early assignment	106,404	60	0.687	0.564	0.397	0.909
SIPP, one per person	50,137	58	0.602	0.125	0.151	0.809
SIPP, repeated months	953,809	60	0.622	0.178	0.070	0.831
Panel B. Early-versus-late Pearson correlation						
CPS, pooled			0.631	0.591	0.230	0.921
SIPP, one per person			0.414	0.153	−0.100	0.691
SIPP, repeated months			0.279	0.066	0.054	0.614

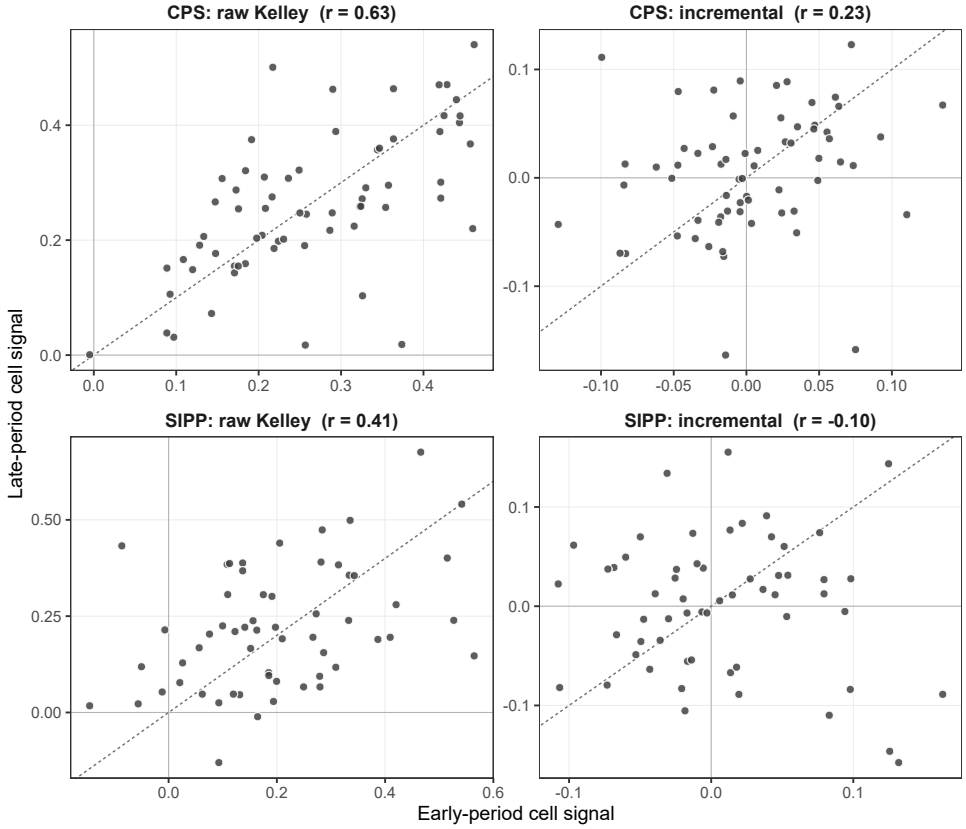
Panel A applies the Spearman–Brown transform to the mean global-person split correlation. Zero-adjusted Kelley removes the linear zero-transition share and education and age components; moment-adjusted Kelley also removes quadratic mean and standard deviation, as in equation (M.5). The zero-share column reports the corresponding statistic for zero incidence itself. Records and cells are post-support counts. Candidate samples contain 183,809 pooled CPS transitions, 108,707 early CPS transitions, 51,152 matched SIPP transitions, and 967,563 overlapping SIPP transitions. Pooled CPS requires Kish support of 200 overall and 100 in each period; the early row requires 200 in the early period. SIPP requires effective-person support of 100 overall and 50 in each period. The one-per-person transition is selected before any destination-positive restriction.

to -0.100. The corresponding rank correlations are 0.634 and 0.326 in CPS and 0.396 and -0.025 in SIPP. Figure O.2 shows the weaker linear persistence of incremental Kelley. Repeated monthly observations do not produce a more stable incremental ranking.

Direct cross-survey transport is weaker still for the incremental component. Over common years, the pooled CPS–SIPP correlations are 0.482 for raw Kelley and 0.053 for incremental Kelley. Alternative splits, one-transition selections, and annual job-1 constructions are reported in Appendix Table B.4. In the broader 2018–2024 portfolio comparison, alternative centered estimators—Bowley, moment skewness, and L-skewness—do not provide a stronger adjusted result; the smallest family q-value is 0.244.

Raw downside orientation contains stable information, much of it shared with zero incidence and lower moments. Incremental orientation is less reliable, less persistent, and less comparable across surveys. The controls may remove economically relevant variation, but the remaining public-data measure represents latent shape imprecisely. Its reliability alone cannot establish a structural interpretation of the portfolio coefficient.

Figure O.2. Persistence of raw and incremental tail orientation



Notes: Each point is one supported cell. The horizontal axis reports the early-period signal and the vertical axis the late-period signal; the dashed line marks equality. Panel titles report Pearson correlations. Raw Kelley uses the zero-inclusive construction. Incremental Kelley is the residual in equation (M.5), re-estimated within each period. CPS uses the exact crosswalk with 65 cells; SIPP uses one transition per person and 58 cells.

P Validated-Link Wage and Total-Earnings Comparisons

This section summarizes the validated-link, zero-inclusive and rank-adjusted comparisons. Their inference conventions are stated separately; the main conclusions using age-adjusted earnings are in Section 7.

At a given total earnings variance, distributions tilted toward favorable rather than adverse outcomes are associated with greater subsequent stock-and-mutual-fund entry in the historical CPS-to-SIPP assignment. In the validated total-earnings comparison, a one-standard-deviation difference in composition remaining after the other regressors corresponds to 0.70 percentage point more entry, or 0.66 when both wage and total earnings destinations remain positive. More upside and less downside describe the same comparison, not independent tail effects.

The positive pattern also extends from wages to a matched measure including net business and farm earnings and persists under explicit treatment of the 2014 survey design. The comparison includes a broader measure of earnings, although the wage–total difference is imprecise and negative total-earnings destinations remain excluded.

In the original-linkage benchmarks, support from local CPS-plus-two-way-SIPP inference is sensitive to broader person-by-cell reweighting. The validated-link reconstruction uses a separate household-and-lineage dependence design. Its raw slopes survive family-adjusted original-point tests, but all 95% percentile intervals include zero, whereas basic intervals exclude it. Wage-zero construction, assignment, and sorting further limit interpretation. The estimates describe conditional differences at a given variance; their uncertainty and observational design preclude using them as identified tail effects or precise life-cycle calibration targets.

In the richer rank-based progression-adjusted entry comparisons, removing observable earnings progression reduces the point magnitudes, and none of the adjusted slopes survives its joint-remeasurement family adjustment. The raw–adjusted differences are imprecise, including when the full comparison is rebuilt using validated longitudinal earnings links. The evidence also fails to establish an incremental composition association beyond conventional skewness. These findings limit the claim to historical earnings variation; they do not identify a portfolio response to unanticipated risk.

Conditioning on workers' own recent annual earnings leaves the raw point estimates almost unchanged on the smaller matched-history sample. Neither the baseline nor the history-controlled coefficients survive their full testing families there, and paired differences remain imprecise. The imprecise coefficient changes cannot establish equality, and observed-history controls cannot eliminate unobserved sorting or identify beliefs.

Conditional-standard-deviation units. For the validated-link comparison, the total-earnings entry magnitude is 0.70 percentage point per conditional standard deviation, or 0.66 with positive earnings destinations. These figures refer to the raw arc-growth construction and its own residual scaling, not the zero-inclusive wage coefficients that lead the main analysis. The accompanying rank-adjusted, linkage, and remeasurement comparisons are retained above and in Appendix I.

Q Portfolio Transitions, Heterogeneity, and Sensitivity

This appendix reports the additional portfolio tests accompanying Table R.11. Except for the wage-only sensitivity comparison, the tests use the age-adjusted total-earnings measures and the original source standard deviations. The real-log and nominal-arc constructions are reported together. The exploratory specifications were fixed after observing the baseline results and before estimating the extensions, using the same public earnings histories.

Q.1 Exit and continuing-position contributions

Let P_{t+1} denote subsequent stock-and-mutual-fund ownership and s_t the initial nonretirement risky share. On the prospective sample, the following identity holds for each observation:

$$s_{t+1} - s_t = \underbrace{-s_t(1 - P_{t+1})}_{\text{exit contribution}} + \underbrace{(s_{t+1} - s_t)P_{t+1}}_{\text{continuing-position contribution}} . \tag{Q.1}$$

For an initial holder, the first term measures the initial share lost at exit; the second is the observed share change if the household continues holding. Both outcomes retain every original observation. Regressing them on the same design, with the same weights, makes their coefficients sum to the share-change coefficient. Cluster-score contributions also add, permitting a direct test of the difference between the two components. A nonholder's exit contribution is identically zero and is not treated as a statistical finding.

Table Q.1 reports the holder components and the direct difference. The real-log share association decomposes into -1.028 percentage points from exits and -0.592 point from continuing positions. Neither component is separately precise, and their difference is imprecise. The decomposition does not establish that exit explains the entire holder-share association. A separate regression among the 5,169 continuing initial holders gives a real-log slope of -0.693 point and an arc slope of $+0.309$ point, both imprecise. This selected-sample comparison retains the initial share but does not correct for selection into continued participation.

The next-year ownership regressions in the main text use the same 31,569 observations and 59-column control design as prospective shares. There are 24,235 initial nonholders and 7,334 initial holders; 2,165 of the latter have no holdings at the next observation. The destination-weighted entry and exit rates on this sample are 9.60% and 30.51%. This entry sample is narrower than the original 27,419-observation entry sample because prospective shares require valid financial-asset denominators. The outcome is ownership in both initial states, so the reported direct contrast is the holder-retention slope minus the nonholder-entry slope. Replacing holder retention by exit reverses its coefficient but leaves its standard error and two-sided test unchanged.

Table Q.1. Exit and continuing-position contributions to portfolio adjustment

	(1) Real log	(2) Nominal arc
A. Initial-holder share adjustment		
Total share change	-1.621^{**}	-0.409
	(0.715)	(0.642)
Exit contribution, $-s_0(1-P_1)$	-1.028	-0.396
	(0.656)	(0.578)
BH q	0.443	0.717
Continuing-position contribution, $(s_1-s_0)P_1$	-0.592	-0.013
	(0.398)	(0.308)
BH q	0.443	0.967
Continuing minus exit contribution	0.436	0.384
	(0.816)	(0.668)
BH q	0.717	0.717
B. Continuing-position comparisons		
Initial nonholders	0.360^{**}	0.264^{**}
	(0.142)	(0.128)
Holder minus nonholder	-0.952^{**}	-0.277
	(0.453)	(0.375)
C. Selected continuing initial holders		
Next-year risky share	-0.693	0.309
	(0.587)	(0.488)
BH q	0.495	0.532
Observations: panels A–B	31,569	31,569
Initial holders / exits	7,334 / 2,165	7,334 / 2,165
Observations: panel C	5,169	5,169

Notes: s_0 and s_1 denote initial and subsequent nonretirement risky shares; P_1 is subsequent participation. Exit and continuing-position components are defined on every original observation and sum exactly to share change. The original state-interacted controls and initial holder share are retained. Panel C conditions on continued ownership and is a selected descriptive comparison. Coefficients are percentage points per fixed source SD of upside composition. Parentheses contain industry–education CR1 standard errors. Stars use unadjusted two-sided $t(G-1)$ tests against zero: $^{*}p<0.10$, $^{**}p<0.05$, $^{***}p<0.01$. Unless indicated otherwise, $G=30$. Inference conditions on the saved CPS moments. BH adjustments cover six initial-holder component/contrast tests in panel A and, separately, two selected-holder tests in panel C. Panel B contains companion comparisons.

Q.2 Work horizon and previous market experience

Table Q.2 examines two implications of the economic interpretation. The first uses within-age-band variation in potential remaining work years. The moderator is $(34.5-\text{age})/10$ for ages 25–44 and $(55-\text{age})/10$ for ages 45–65. Each of the five earnings regressors interacts with this moderator, centered log initial nonretirement financial assets, the centered initial first-listed-job log-income control, and the older-band indicator, separately by initial ownership. Initial financial-asset main effects are added to both blocks; the remaining moderator main effects are already spanned by the original controls. The resulting design has 101 columns and uses the unchanged prospective sample. This comparison controls for observable resources, but age-based work horizon is not an estimate of remaining human capital at a fixed dollar-shock scale.

The holder composition–horizon interactions are imprecise under both transformations. The real-log estimate is 0.034 percentage point per source standard deviation per decade, with a standard error of 1.030; the arc estimate is 0.848 with a standard error of 1.061. The holder-minus-nonholder interaction differences are also imprecise. These tests do not establish the intensive-margin human-capital channel, which is small in the illustration in Section 2.

The second comparison links baseline nonholders to ownership two years before the entry outcome. Missing earlier histories are excluded, not coded as nonownership. The observed-history sample contains 13,141 transitions: 12,031 with earlier nonownership and 1,110 with earlier ownership. I interact the entire original entry design with earlier ownership and report a pooled regression on exactly the same observed histories. Greater real-log upside composition predicts entry even after two observed dates without holdings: the slope is 0.971 percentage point ($p = 0.011$). The directly tested difference from previous holders is imprecise under both transformations, however. The evidence does not establish differential sensitivity by recent market experience or distinguish first-ever entry from reentry over a household's lifetime.

Q.3 Construction, weights, and financial resources

Tables Q.3–Q.5 report the complete sensitivity family. The prospective comparisons replace total earnings by wages on the same CPS histories, replace the nonretirement financial denominator by the broader denominator with its corresponding initial-share control, use equal weights, use initial rather than destination survey weights, and require at least $1,000 of initial nonretirement financial assets. The asset threshold is nominal and set before fitting the extension. Apart from that restriction, each comparison retains the original prospective observations. None adds a future-financial-assets control. The wage-only regressors use their own original source scales, so the total-versus-wage coefficients are standardized construction comparisons rather than equal raw-unit shocks.

The entry comparisons retain all 27,419 original observations and vary only the weights. Initial weights match to every observation. Paired broad-minus-nonretirement contrasts account for the covariance between the two fitted slopes; the estimates are not compared using independent-model standard errors. The source data, full coefficients, model-specific covariance matrices, and cluster influences for every fitted comparison accompany the replication files.

Q.4 Influential groups and portfolio years

Table Q.6 reports the complete deletion exercise for the primary real-log construction. Each regression omits one of the 30 industry–education groups or one of the five portfolio-outcome years. The original entry and prospective-share designs are reestimated with the remaining observations, retaining their control span. Only algebraically redundant nuisance columns are removed when a group or year disappears. The composition and variance terms remain in every regression. Standard errors use the actual number of retained groups and the corresponding t_{G-1} reference distribution.

Table Q.2. Potential work horizon and recent stock market experience

	(1) Real log	(2) Nominal arc
A. Composition by within-band work horizon		
Initial nonholders	-0.126	-0.389^{*}
	(0.184)	(0.203)
Initial holders	0.034	0.848
	(1.030)	(1.061)
Holder-horizon BH q	0.974	0.862
Holder minus nonholder	0.160	1.236
	(1.069)	(1.103)
B. Entry after observed market history		
Previously nonholding	0.971^{**}	0.487
	(0.356)	(0.300)
Previously holding	0.338	5.309^{*}
	(2.895)	(2.871)
Previous holder minus previous nonholder	-0.633	4.821
	(2.934)	(3.023)
History-difference BH q	0.831	0.243
Same-history pooled benchmark	1.018^{**}	0.583^{*}
	(0.461)	(0.322)
Observations: panel A	31,569	31,569
Observed-history observations	13,141	13,141
Previous nonholders / holders	12,031 / 1,110	12,031 / 1,110

Notes: Panel A uses potential work horizon in decades centered at fixed age-band midpoints (34.5 and 55). All five earnings regressors interact with horizon, log initial nonretirement financial assets, the original log-income control and the older-band indicator, separately by initial ownership. Log financial assets also enters each state block; centering uses full-sample survey-weighted means. All companion resource and age-band interactions are retained in the replication files. Panel B joins ownership two years before the entry outcome, retaining observed histories; baseline ownership is zero. The entry design is fully interacted by earlier ownership. Two observed nonholding dates do not establish lifetime nonparticipation. Coefficients are percentage points per fixed source SD of upside composition. Parentheses contain industry–education CR1 standard errors. Stars use unadjusted two-sided $t(G-1)$ tests against zero: $^{*}p<0.10$, $^{**}p<0.05$, $^{***}p<0.01$. Unless indicated otherwise, $G=30$. Inference conditions on the saved CPS moments. Separate BH families contain the two holder-horizon and two direct history-difference tests.

The influence table reports coefficient ranges and sign counts over all deletions, rather than selecting the omissions that give the smallest p-values. These comparisons show how much the associations depend on particular groups and years; they do not provide independent replication or exogenous variation.

Inference and complete families. All new coefficient tests use industry–education CR1 standard errors conditional on the estimated CPS moments. The transition family contains six tests: the two ownership-state slopes and their direct difference under each transformation. The anatomy family contains six tests: two holder components and their direct difference under each transformation. The two selected-continuing-holder comparisons form a separate family. The work-horizon family contains the two holder composition interactions, and the recent-history family contains the two direct earlier-ownership contrasts. The wage-only prospective family contains six ownership-state

Table Q.3. Sensitivity of prospective portfolio allocation

	(1) Initial nonholder	(2) Initial holder	(3) Holder minus nonholder	(4) N
A. Prospective allocation: real log				
Baseline	0.360^{**}	-1.621^{**}	-1.980^{**}	31,569
	(0.142)	(0.715)	(0.737)	
Wage earnings only	0.288^{*}	-1.128^{*}	-1.416^{*}	31,569
	(0.160)	(0.631)	(0.697)	
Wage-family BH q	0.127	0.127	0.127	
Broader financial denominator	0.208^{**}	-1.489^{***}	-1.697^{***}	31,569
	(0.081)	(0.526)	(0.554)	
Equal observation weights	0.427^{**}	-1.333^{*}	-1.760^{**}	31,569
	(0.157)	(0.720)	(0.769)	
Initial survey weights	0.380^{**}	-1.257^{*}	-1.637^{**}	31,569
	(0.154)	(0.705)	(0.730)	
Initial financial assets $\geq \$1,000$	0.299	-1.503^{**}	-1.802^{**}	26,005
	(0.193)	(0.703)	(0.736)	
B. Prospective allocation: nominal arc				
Baseline	0.264^{**}	-0.409	-0.673	31,569
	(0.128)	(0.642)	(0.683)	
Wage earnings only	0.260^{*}	-0.512	-0.772	31,569
	(0.138)	(0.630)	(0.701)	
Wage-family BH q	0.127	0.424	0.335	
Broader financial denominator	0.098	-0.780	-0.877	31,569
	(0.093)	(0.615)	(0.631)	
Equal observation weights	0.329^{**}	-0.958	-1.288^{*}	31,569
	(0.144)	(0.645)	(0.709)	
Initial survey weights	0.204	-0.124	-0.328	31,569
	(0.146)	(0.621)	(0.665)	
Initial financial assets $\geq \$1,000$	0.257	-0.359	-0.616	26,005
	(0.174)	(0.628)	(0.706)	

Notes: The outcome is next-year risky share. Broader-denominator rows include retirement balances in financial wealth and retain the matching initial share. Wage-only rows use wage moments on the same CPS histories and matching fixed wage scales. The asset floor is $1,000 of initial nonretirement financial assets, not future assets. Other controls, assignment and samples remain unchanged except the initial asset floor. Coefficients are percentage points per fixed source SD of upside composition. Parentheses contain industry–education CR1 standard errors. Stars use unadjusted two-sided $t(G-1)$ tests against zero: $^{*}p<0.10$, $^{**}p<0.05$, $^{***}p<0.01$. Unless indicated otherwise, $G=30$. Inference conditions on the saved CPS moments. BH q values cover the complete six-test wage-only prospective family; other rows are sensitivity comparisons.

slopes and differences. Benjamini–Hochberg adjustments are applied within these stated families. Conventional significance stars use the unadjusted two-sided tests; family adjustments are reported separately. All sensitivity and deletion comparisons are reported.

Table Q.4. Survey weighting and subsequent stock market entry

	(1) Real log	(2) Nominal arc
Destination survey weights	0.875^{***}	0.695^{***}
	(0.279)	(0.248)
Equal observation weights	0.898^{***}	0.658^{***}
	(0.265)	(0.232)
Initial survey weights	0.893^{***}	0.608^{**}
	(0.289)	(0.241)
Observations	27,419	27,419
Industry–education clusters	30	30

Notes: The dependent variable is ownership entry on the original initial-nonholder cohort. All rows retain identical observations, source moments, controls and fixed effects; only the observation weights change. Initial weights are joined from the unchanged annual cache without losing observations. Coefficients are percentage points per fixed source SD of upside composition. Parentheses contain industry–education CR1 standard errors. Stars use unadjusted two-sided $t(G-1)$ tests against zero: $^{*}p<0.10$, $^{**}p<0.05$, $^{***}p<0.01$. Unless indicated otherwise, $G=30$. Inference conditions on the saved CPS moments.

Table Q.5. Comparing financial-wealth denominators

	(1) Initial nonholder	(2) Initial holder	(3) Holder minus nonholder
Real log: broader minus nonretirement	-0.152	0.132	0.284
	(0.112)	(0.484)	(0.496)
Nominal arc: broader minus nonretirement	-0.166^{**}	-0.371	-0.205
	(0.071)	(0.368)	(0.362)
Observations	31,569	31,569	31,569

Notes: Each entry subtracts the nonretirement-denominator composition coefficient from its broader-financial-denominator counterpart on identical observations and weights. Broader financial wealth includes retirement balances; each model controls its corresponding initial share. Standard errors use aligned cluster influences, retaining the covariance between the two specifications. Coefficients are percentage points per fixed source SD of upside composition. Parentheses contain industry–education CR1 standard errors. Stars use unadjusted two-sided $t(G-1)$ tests against zero: $^{*}p<0.10$, $^{**}p<0.05$, $^{***}p<0.01$. Unless indicated otherwise, $G=30$. Inference conditions on the saved CPS moments.

Table Q.6. Complete group and year influence diagnostics

	(1) Minimum	(2) Maximum	(3) Positive / negative	(4) Fits	(5) Minimum / maximum N
A. Industry–education-group deletions					
Entry	0.736	1.089	30 / 0	30	23,000 27,347
Nonholder	0.277	0.420	30 / 0	30	25,917 31,496
Holder	-1.959	-1.082	0 / 30	30	25,917 31,496
Difference	-2.347	-1.405	0 / 30	30	25,917 31,496
B. Outcome-year deletions					
Entry	0.788	0.989	5 / 0	5	18,955 23,786
Nonholder	0.298	0.418	5 / 0	5	22,271 27,407
Holder	-2.298	-1.263	0 / 5	5	22,271 27,407
Difference	-2.664	-1.561	0 / 5	5	22,271 27,407

Notes: Ranges summarize every declared deletion for primary age-adjusted total-earnings real-log composition: 30 industry–education-group deletions and five outcome-year deletions per outcome. Entry uses the original entry cohort; the remaining rows use the prospective allocation sample. Ranges are coefficient summaries, not confidence intervals. All remaining observations and the control span are retained; only algebraically redundant nuisance columns may be removed. Group deletions use 29 clusters and $t(28)$; year deletions use 30 clusters and $t(29)$. Coefficients are percentage points per fixed source SD. All 140 individual estimates, standard errors and tests are retained in the replication files. No fit failed; no deletion was selected for favorable results.

R Calendar Dependence, Persistence, and Reporting Quality of Portfolio Transitions

This appendix reports the diagnostics summarized in Section 4.8. They address four questions about the transition evidence in Table R.11: whether the associations depend on particular outcome years, whether they survive richer group-by-year controls, whether they reflect short-lived ownership changes or recent participation behavior, and whether recorded exits are artifacts of survey allocation. The exploratory protocol was fixed after observing the transition results and before estimating these models. All regressions retain the original controls, destination survey weights, fixed source scales, and 30 industry–education CR1 clusters with t_{29} inference conditional on the CPS moments. Ownership coefficients for initial holders measure retention, so exit is their negative; the sequence outcomes in Table R.8 measure exit directly. Stars use the unadjusted coefficient tests, and each declared Benjamini–Hochberg family is stated with its table. The complete designs, covariance matrices, and cluster influences for all 67 models accompany the replication files.

R.1 Allocation and stability tables of the original construction

Tables R.1 and R.2 are the retained next-year share and stability tables of the original construction, with the 58-cell sensitivity rows and the family-adjusted q-values of the screened tests; Tables 5 and 7 in the main text report the same designs with the zero-inclusive wage construction.

R.2 Outcome-period dependence

The portfolio outcomes span 2020–2024, so the associations could reflect pandemic-period conditions. Tables R.3 and R.4 interact each original design with an indicator for the later outcome years, 2023–2024, and report the early and late slopes with their directly tested difference. With real-log composition, prospective entry is 1.185 percentage points in 2020–2022 and 0.683 point in 2023–2024; the difference is -0.502 point ($p = 0.508$). Holder retention is -3.234 and -2.198 points, a difference of 1.036 points ($p = 0.515$). None of the fourteen declared differences is precisely estimated: their family-adjusted q-values are at least 0.789, and the joint tests of equality across the five individual outcome years have p-values that all exceed 0.16. The later slopes are individually less precise, which does not show that the associations are absent after 2022, and the later years are not a randomized control for pandemic conditions.

Table R.5 completes the leave-one-outcome-year comparison for real-log next-year ownership. Across the five omissions, the entry slope ranges from 0.887 to 1.113 points and the retention slope from -3.088 to -2.341 points, each with $p < 0.04$. No single outcome year generates the transition associations.

Table R.1. Earnings-tail composition and subsequent portfolio allocation

Age-adjusted total earnings	(1) Real log	(2) Nominal arc
Outcome: next-year nonretirement risky share		
Upside composition: baseline nonholders	0.360^{**}	0.264^{**}
	(0.142)	(0.128)
58-cell sensitivity p	0.0058	0.0178
Upside composition: baseline holders	-1.621^{**}	-0.409
	(0.715)	(0.642)
58-cell sensitivity p	0.0150	0.4902
Difference: holder minus nonholder	-1.980^{**}	-0.673
	(0.737)	(0.683)
58-cell sensitivity p	0.0044	0.2887
Difference: two-test BH q, 30 groups	0.0237	0.3325
Difference: two-test BH q, 58 cells	0.0089	0.2887
Baseline-nonholder observations	24,235	24,235
Baseline-holder observations	7,334	7,334
Total consecutive-year observations	31,569	31,569
Assignment cells / inference groups	58 / 30	58 / 30
All risk and other controls $\times$ baseline status	Yes	Yes
Baseline share in holder block	Yes	Yes

Notes: Weighted OLS of future stock-and-mutual-fund balances divided by financial assets net of retirement balances. Baselines are 2019–2023; outcomes are 2020–2024. Both dates require positive financial denominators and valid ratios in $[0, 1]$; genuine future zeros, including exits, remain. Coefficients are percentage points per fixed pre-adjustment cross-cell SD of upside composition in total earnings, common across ownership states within each growth definition. Earnings construction and controls follow Sections 3 and 3.3. All coefficients vary by baseline ownership; lagged share additionally enters the holder block. Weights are destination-year person weights. Parentheses give CR1 SEs clustered by 30 industry–education groups. Stars use unadjusted two-sided t_{29} tests: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Sensitivity rows use 58-cell CR1 and t_{57}. BH rows adjust the two growth-specific holder-minus-nonholder tests within each clustering convention. Inference conditions on the estimated CPS moments; no first-stage bootstrap is included.

R.3 Richer fixed effects and prior ownership switching

Table R.6 adds two predeclared sets of fixed effects to the next-year ownership and share regressions, separately by initial ownership. Industry-by-year and education-by-year effects allow industries and education groups to follow separate annual paths, and joint industry–education effects remove persistent differences at the level at which the earnings profiles are estimated. The real-log entry, retention, and holder-share coefficients change by at most 0.06 point under the annual controls. Under the joint effects the entry estimate becomes less precise ($p = 0.0507$), because the fixed exposure then varies only within industry–education groups across age bands and years, whereas the retention coefficient remains -2.713 points ($p = 0.0012$). Full earnings-cell effects would absorb the exposure entirely, and person effects would identify it only from cell movers; neither is estimated. These comparisons do not identify an individual causal response.

Table R.7 uses the 15,535 prospective transitions, 3,718 of them beginning as holders, whose ownership one year before the baseline is observed in the annual history. Missing earlier records are excluded rather than coded as nonholding. Adding an indicator for a switch in ownership between

Table R.2. Stability of the entry and initial-holder associations

	(1) Nonholder entry	(2) Holder retention	(3) Initial-holder share	(4) N
A. Real log composition				
Baseline prospective sample	0.969^{***}	-2.650^{***}	-1.621^{**}	31,569
	(0.302)	(0.952)	(0.715)	
Group-by-year effects	0.988^{***}	-2.707^{**}	-1.599^{**}	31,569
	(0.303)	(1.016)	(0.740)	
Reported components	0.699^{**}	-0.776	-0.788	24,114
	(0.303)	(1.129)	(0.932)	
BH q, 24 screened tests	0.114	0.611	0.607	
Reported and stable households	1.008^{**}	-0.618	-0.593	13,939
	(0.375)	(1.115)	(0.887)	
BH q, 24 screened tests	0.071	0.667	0.611	
B. Nominal arc composition				
Baseline prospective sample	0.770^{***}	-1.848^{**}	-0.409	31,569
	(0.262)	(0.862)	(0.642)	
Group-by-year effects	0.787^{***}	-2.034^{**}	-0.503	31,569
	(0.268)	(0.873)	(0.647)	
Reported components	0.818^{***}	-0.776	0.070	24,114
	(0.212)	(1.074)	(0.837)	
BH q, 24 screened tests	0.005	0.611	0.967	
Reported and stable households	1.194^{***}	-1.266	-0.030	13,939
	(0.278)	(1.098)	(0.725)	
BH q, 24 screened tests	0.004	0.469	0.967	
Industry–education clusters	30	30	30	

Notes: Columns (1)–(2) reestimate the next-year ownership regression of Table R.11 and column (3) the initial-holder block of the next-year share regression of Table R.1; the first row reproduces the published estimates. Row 2 adds industry-by-year and education-by-year effects (group-by-year) by initial ownership on the unchanged sample; joint industry–education effects are in Table R.6. Rows 3–4 keep transitions whose stock and mutual-fund ownership and balance components are reported rather than allocated at both dates, and additionally interviews coded as self-reported with stable household and residence records (Appendix R); initial holders number 7,334, 4,081 and 2,710. Holder retention is next-year ownership among initial holders; exit is its negative with the same standard error. The initial-holder share includes exits. Coefficients are percentage points per fixed source SD of upside composition, with the original controls and destination weights; parentheses are industry–education CR1 standard errors ($G = 30$, t_{29}, conditional on the saved CPS moments). Stars use unadjusted two-sided tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. BH q values adjust the 24 screened tests of Tables R.9–R.10; the fixed-effect row is a sensitivity comparison, not a discovery family.

that earlier year and the baseline, separately by initial ownership, reduces the real-log retention coefficient from -1.830 to -1.016 points: the directly tested change is 0.814 point ($p = 0.0166$), meaning less predicted exit once recent switching is controlled. The entry coefficient moves from 0.989 to 0.905 point, a change of -0.084 point ($p = 0.497$). The initial-holder share coefficient is already weak in this subsample (-0.350 point, $p = 0.742$), so its difference from the full-sample estimate cannot be attributed to the history control. The switch indicator is a proxy for recent participation behavior, not a preference measure.

Table R.3. Early and later outcome years: entry and ownership transitions

	(1) 2020–2022	(2) 2023–2024	(3) Later minus early	(4) Difference BH q	(5) Year-equality p
A. Real log composition					
Original entry cohort	1.027^{**}	0.675	-0.352	0.789	0.613
	(0.438)	(0.418)	(0.637)		
Prospective nonholder entry	1.185^{**}	0.683	-0.502	0.789	0.897
	(0.461)	(0.520)	(0.750)		
Holder retention	-3.234^{**}	-2.198	1.036	0.789	0.862
	(1.229)	(1.308)	(1.572)		
Holder minus nonholder	-4.419^{***}	-2.881^{*}	1.538	0.789	0.806
	(1.334)	(1.427)	(1.587)		
B. Nominal arc composition					
Original entry cohort	0.644	0.757^{**}	0.113	0.841	0.167
	(0.423)	(0.301)	(0.556)		
Prospective nonholder entry	0.678	0.883^{**}	0.205	0.805	0.234
	(0.460)	(0.346)	(0.630)		
Holder retention	-2.429^{*}	-1.577	0.852	0.789	0.266
	(1.196)	(1.270)	(1.700)		
Holder minus nonholder	-3.107^{**}	-2.460^{*}	0.647	0.805	0.718
	(1.412)	(1.350)	(1.844)		

Notes: The original entry cohort has 27,419 transitions; the prospective ownership and share outcomes use the 31,569-transition sample. Each original design is fully interacted with an indicator for 2023–2024 outcomes; columns (1)–(2) are the period slopes, column (3) their directly tested difference and column (4) its q-value in the 14-test family spanning Tables R.3–R.4. Column (5) is the p-value of a joint equality test across the five outcome years ($F_{4,29}$). Later years are not a randomized pandemic control. Holder retention is next-year ownership among initial holders; exit is its negative with the same standard error. The initial-holder share includes exits. Coefficients are percentage points per fixed source SD of upside composition, with the original controls and destination weights; parentheses are industry–education CR1 standard errors ($G = 30$, t_{29}, conditional on the saved CPS moments). Stars use unadjusted two-sided tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

R.4 Persistence of entry and exit

Table R.8 uses the 11,133 prospective transitions with an additional observed annual record, including 2,807 initial holders; 2024 outcomes cannot enter because no further record exists. The outcomes are indicators for complete three-year ownership sequences, so they are joint probabilities conditional on the initial state rather than exit probabilities conditional on entering. Real-log composition predicts temporary entry, $0 \to 1 \to 0$, with a slope of 0.871 point ($p = 0.0037$; twelve-test $q = 0.045$), whereas the persistent-entry slope, $0 \to 1 \to 1$, is 0.091 point ($p = 0.820$). Their directly tested difference is -0.779 point ($p = 0.146$), so the evidence does not establish that the predicted entry is predominantly temporary. Among initial holders, persistent exit, $1 \to 0 \to 0$, has a slope of 2.784 points ($p = 0.087$) and exit followed by reentry 0.710 point ($p = 0.499$); the difference is imprecise ($p = 0.311$). Nominal-arc composition gives temporary and persistent entry slopes of 0.613 and 0.450 point. Annual ownership records cannot establish trading frequency or a preference for gambling, and the selected follow-up sample differs from the full prospective sample.

Table R.4. Early and later outcome years: subsequent risky shares

	(1) 2020–2022	(2) 2023–2024	(3) Later minus early	(4) Difference BH q	(5) Year-equality p
A. Real log composition					
Initial nonholders	0.285	0.452**	0.167	0.789	0.611
	(0.206)	(0.216)	(0.304)		
Initial holders	−2.029***	−0.991	1.038	0.789	0.410
	(0.693)	(1.020)	(0.986)		
Holder minus nonholder	−2.315***	−1.443	0.871	0.789	0.363
	(0.743)	(1.007)	(0.964)		
B. Nominal arc composition					
Initial nonholders	0.141	0.428**	0.288	0.789	0.220
	(0.179)	(0.169)	(0.224)		
Initial holders	−1.139*	0.425	1.564	0.789	0.550
	(0.578)	(1.010)	(0.991)		
Holder minus nonholder	−1.279**	−0.003	1.276	0.789	0.648
	(0.620)	(1.055)	(1.004)		

Notes: The outcome is the next-year nonretirement risky share, including zero holdings. The original entry cohort has 27,419 transitions; the prospective ownership and share outcomes use the 31,569-transition sample. Each original design is fully interacted with an indicator for 2023–2024 outcomes; columns (1)–(2) are the period slopes, column (3) their directly tested difference and column (4) its q-value in the 14-test family spanning Tables R.3–R.4. Column (5) is the p-value of a joint equality test across the five outcome years ($F_{4,29}$). Later years are not a randomized pandemic control. Holder retention is next-year ownership among initial holders; exit is its negative with the same standard error. The initial-holder share includes exits. Coefficients are percentage points per fixed source SD of upside composition, with the original controls and destination weights; parentheses are industry–education CR1 standard errors ($G = 30$, t_{29}, conditional on the saved CPS moments). Stars use unadjusted two-sided tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table R.5. Leave-one-outcome-year ownership regressions

	(1) Nonholder entry	(2) Holder retention	(3) Holder minus nonholder	(4) N
Omitting 2020 outcomes	0.887**	−2.740**	−3.627***	22,271
	(0.350)	(1.126)	(1.207)	
Omitting 2021 outcomes	0.966***	−2.341**	−3.307***	27,407
	(0.272)	(0.969)	(1.099)	
Omitting 2022 outcomes	0.913**	−2.578**	−3.492***	24,679
	(0.355)	(1.003)	(1.171)	
Omitting 2023 outcomes	1.113***	−2.573**	−3.686***	25,602
	(0.344)	(1.172)	(1.252)	
Omitting 2024 outcomes	0.977***	−3.088***	−4.065***	26,317
	(0.343)	(0.990)	(1.097)	

Notes: Each row reestimates the real-log next-year ownership regression of Table R.11 without one outcome year, retaining the control span and 30 clusters; the entry and prospective-share deletion grids are in Table Q.6. These are concentration diagnostics, not independent replications. Holder retention is next-year ownership among initial holders; exit is its negative with the same standard error. The initial-holder share includes exits. Coefficients are percentage points per fixed source SD of upside composition, with the original controls and destination weights; parentheses are industry–education CR1 standard errors ($G = 30$, t_{29}, conditional on the saved CPS moments). Stars use unadjusted two-sided tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table R.6. Richer fixed effects, by initial ownership

	(1) Baseline	(2) Group–year effects	(3) Joint-group effects	(4) Change (2)−(1)	(5) Change (3)−(1)
A. Real log composition					
Nonholder entry	0.969^{***}	0.988^{***}	0.955^{*}	0.019	-0.014
	(0.302)	(0.303)	(0.469)	(0.023)	(0.291)
Holder retention	-2.650^{***}	-2.707^{**}	-2.713^{***}	-0.057	-0.063
	(0.952)	(1.016)	(0.758)	(0.169)	(0.741)
Holder minus	-3.619^{***}	-3.695^{***}	-3.668^{***}	-0.076	-0.049
nonholder ownership	(1.070)	(1.127)	(1.028)	(0.170)	(0.881)
Nonholder share	0.360^{**}	0.364^{**}	0.321	0.004	-0.039
	(0.142)	(0.143)	(0.197)	(0.013)	(0.135)
Initial-holder share	-1.621^{**}	-1.599^{**}	-1.345^{*}	0.022	0.276
	(0.715)	(0.740)	(0.779)	(0.107)	(0.510)
Holder minus	-1.980^{**}	-1.962^{**}	-1.666^{*}	0.018	0.315
nonholder share	(0.737)	(0.769)	(0.834)	(0.106)	(0.521)
B. Nominal arc composition					
Nonholder entry	0.770^{***}	0.787^{***}	0.558	0.018	-0.212
	(0.262)	(0.268)	(0.504)	(0.019)	(0.300)
Holder retention	-1.848^{**}	-2.034^{**}	-3.145^{***}	-0.186	-1.297
	(0.862)	(0.873)	(0.953)	(0.117)	(0.943)
Holder minus	-2.618^{**}	-2.821^{***}	-3.703^{***}	-0.203^{*}	-1.085
nonholder ownership	(0.987)	(1.009)	(1.063)	(0.118)	(0.972)
Nonholder share	0.264^{**}	0.267^{*}	0.099	0.003	-0.165
	(0.128)	(0.131)	(0.258)	(0.013)	(0.160)
Initial-holder share	-0.409	-0.503	-1.097	-0.094	-0.688
	(0.642)	(0.647)	(0.956)	(0.074)	(0.760)
Holder minus	-0.673	-0.769	-1.196	-0.097	-0.524
nonholder share	(0.683)	(0.701)	(1.001)	(0.077)	(0.747)

Notes: Column (1) reproduces the published regressions on 31,569 transitions (7,334 initial holders). Column (2) adds industry-by-year and education-by-year effects (group–year) and column (3) joint industry–education effects (joint group), by initial ownership with the original controls and weights. Columns (4)–(5) directly test the coefficient changes using paired cluster influences. Full earnings-cell effects would absorb the fixed exposure and are not estimated; these are predeclared sensitivity comparisons, not a discovery family. Holder retention is next-year ownership among initial holders; exit is its negative with the same standard error. The initial-holder share includes exits. Coefficients are percentage points per fixed source SD of upside composition, with the original controls and destination weights; parentheses are industry–education CR1 standard errors ($G = 30$, t_{29}, conditional on the saved CPS moments). Stars use unadjusted two-sided tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

R.5 Reporting quality in the public SIPP files

Recorded exits could reflect allocated rather than reported balances. I matched every endpoint of the 31,569 prospective transitions to the six public SIPP releases covering calendar years 2019–2024, 52,652 person-years in all, by panel, person, and reference year. The retained stock and mutual-fund balances reproduce the public amounts exactly at both endpoints, the six component amounts sum to the aggregate, and every link joins adjacent waves of the same panel. The aggregate allocation flag for stock and mutual-fund balances is uninformative, coded 9, for every target observation; the screens therefore use the six component amount flags and the two root ownership flags, whose year-specific

Table R.7. Prior ownership switching on identical observations

	(1) History-support baseline	(2) Adding prior switch	(3) Change (2)−(1)
A. Real log composition			
Nonholder entry	0.989^{*}	0.905^{*}	-0.084
	(0.547)	(0.476)	(0.122)
Holder retention	-1.830^{*}	-1.016	0.814^{**}
	(1.006)	(0.855)	(0.320)
Holder minus nonholder ownership	-2.818^{**}	-1.921^{*}	0.898^{**}
	(1.209)	(0.985)	(0.366)
Nonholder share	0.360	0.320	-0.040
	(0.241)	(0.223)	(0.058)
Initial-holder share	-0.350	0.065	0.416^{**}
	(1.054)	(1.015)	(0.166)
Holder minus nonholder share	-0.711	-0.255	0.456^{**}
	(1.027)	(0.964)	(0.182)
B. Nominal arc composition			
Nonholder entry	0.585^{*}	0.645^{**}	0.060
	(0.339)	(0.281)	(0.084)
Holder retention	-1.197	-0.603	0.593^{**}
	(0.903)	(0.875)	(0.257)
Holder minus nonholder ownership	-1.782^{*}	-1.248	0.533^{*}
	(0.919)	(0.831)	(0.273)
Nonholder share	0.159	0.187	0.028
	(0.192)	(0.174)	(0.041)
Initial-holder share	0.876	1.181	0.304^{**}
	(0.772)	(0.758)	(0.139)
Holder minus nonholder share	0.718	0.993	0.276^{*}
	(0.791)	(0.754)	(0.147)

Notes: Prospective transitions whose ownership one year before the baseline is observed in the annual history (15,535 transitions, 3,718 initial holders); missing earlier records are excluded, not coded as nonholding. Column (2) adds an indicator for an ownership change between that year and the baseline, by initial ownership; column (3) directly tests the change using paired cluster influences. The indicator proxies recent participation behavior, not preferences. Holder retention is next-year ownership among initial holders; exit is its negative with the same standard error. The initial-holder share includes exits. Coefficients are percentage points per fixed source SD of upside composition, with the original controls and destination weights; parentheses are industry–education CR1 standard errors ($G = 30$, t_{29}, conditional on the saved CPS moments). Stars use unadjusted two-sided tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

definitions were taken from the official data dictionaries before any regression was estimated. Root ownership refers to any time during the reference year, whereas balances are year-end, so reported ownership with a reported zero year-end balance is a valid combination.

Of the 2,165 recorded exits, 1,856 involve a reported answer of no stock or mutual-fund ownership during the following year, 253 involve allocated or unknown root ownership at that date, 53 involve verified reported zero components after some ownership during the year, and 3 are otherwise unverified. Weighted exit rates are 46.1% with root-ownership allocation at either endpoint and 28.1% without, and 45.2% with a change in interview type against 27.7% without; allocation of component amounts alone leaves the rate essentially unchanged, 30.7% against 30.4%. Most exits

Table R.8. Persistent and temporary ownership changes in the additional follow-up year

	(1) Persistent change	(2) Temporary change	(3) Persistent minus temporary	(4) One-step change
A. Real log composition				
Nonholder entry sequences	0.091	0.871***	−0.779	0.962**
	(0.396)	(0.276)	(0.522)	(0.441)
BH q, 12 sequence tests	0.874	0.045	0.351	
Holder exit sequences	2.784*	0.710	2.075	3.494*
	(1.574)	(1.037)	(2.013)	(1.747)
BH q, 12 sequence tests	0.262	0.749	0.534	
Exit minus entry sequence	2.693*	−0.161	2.854	2.532
	(1.505)	(1.053)	(1.934)	(1.734)
B. Nominal arc composition				
Nonholder entry sequences	0.450*	0.613**	−0.163	1.063***
	(0.242)	(0.257)	(0.399)	(0.300)
BH q, 12 sequence tests	0.262	0.141	0.848	
Holder exit sequences	0.550	0.812	−0.261	1.362
	(1.447)	(0.720)	(1.633)	(1.600)
BH q, 12 sequence tests	0.848	0.534	0.874	
Exit minus entry sequence	0.100	0.198	−0.098	0.298
	(1.419)	(0.708)	(1.656)	(1.513)

Notes: Prospective transitions with an observed additional annual record (11,133 transitions, 2,807 initial holders); 2024 outcomes cannot enter. Outcomes are indicators for complete ownership sequences from the baseline: persistent entry $0 \to 1 \to 1$ and temporary entry $0 \to 1 \to 0$ for nonholders; persistent exit $1 \to 0 \to 0$ and exit then reentry $1 \to 0 \to 1$ for holders. These are joint probabilities conditional on the initial state, not exit conditional on entering. Column (3) tests the difference directly; column (4) is the one-step change, the sum of columns (1)–(2). Holder rows measure exit sequences, so positive values mean more exit; the last row subtracts the nonholder entry-sequence slope from the holder exit-sequence slope. Coefficients are percentage points per fixed source SD of upside composition, with the original controls and destination weights; parentheses are industry–education CR1 standard errors ($G = 30$, t_{29}, conditional on the saved CPS moments). Stars use unadjusted two-sided tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. The BH family contains the 12 state-specific sequence and difference tests.

have supporting ownership responses, and the descriptive differences do not establish that allocation generates the exit association.

Tables R.9 and R.10 report the two predeclared screens with their available-field baselines. The reported-components screen retains 24,114 transitions, 20,033 initial nonholders and 4,081 initial holders. The stricter screen additionally requires interviews coded as self-reported and stable household and residence records with adjacent waves; the fields needed for that screen are available for 31,088 transitions, and the screen retains 13,939, with 2,710 initial holders. Merely requiring the strict-screen fields does not change the estimates: on the 31,088-transition baseline the real-log coefficients are 0.997, -2.744, and -1.658 points. Real-log entry remains 0.699 and 1.008 points in the two screened samples, and nominal-arc entry 0.818 and 1.194 points, with twenty-four-test q-values of 0.114 and 0.071 for real log and 0.005 and 0.004 for arc. The initial-holder retention and share coefficients are smaller and imprecise in both screened samples, and their confidence intervals include the full-sample magnitudes. The screens change the sample as well as reporting quality, the

changes are descriptive rather than formal cross-sample tests, and the audit does not cover every component of the financial-asset denominator, retirement balances, or respondent continuity.

Table R.9. Reporting-quality screens: next-year ownership

	(1) Full sample	(2) Reported components	(3) Strict-support baseline	(4) Reported and stable
A. Real log composition				
Nonholder entry	0.969***	0.699**	0.997***	1.008**
	(0.302)	(0.303)	(0.304)	(0.375)
BH q, 24 screened tests		0.114		0.071
Holder retention	−2.650***	−0.776	−2.744***	−0.618
	(0.952)	(1.129)	(0.930)	(1.115)
BH q, 24 screened tests		0.611		0.667
Holder minus nonholder	−3.619***	−1.474	−3.741***	−1.627
	(1.070)	(1.235)	(1.043)	(1.250)
BH q, 24 screened tests		0.469		0.444
B. Nominal arc composition				
Nonholder entry	0.770***	0.818***	0.750***	1.194***
	(0.262)	(0.212)	(0.258)	(0.278)
BH q, 24 screened tests		0.005		0.004
Holder retention	−1.848**	−0.776	−1.869**	−1.266
	(0.862)	(1.074)	(0.849)	(1.098)
BH q, 24 screened tests		0.611		0.469
Holder minus nonholder	−2.618**	−1.594	−2.620**	−2.460**
	(0.987)	(1.063)	(0.967)	(1.058)
BH q, 24 screened tests		0.347		0.114
Observations	31,569	24,114	31,088	13,939
Initial holders	7,334	4,081	7,249	2,710

Notes: Column (2) requires reported (or legitimately not-in-universe) stock and mutual-fund ownership and balance components at both dates, using the public SIPP component flags because the aggregate flag is uninformative (coded 9) throughout. Column (4) additionally requires interviews coded as self-reported and stable household and residence records with adjacent waves; column (3) is its available-field baseline, and the column-(2) baseline coincides with column (1). Screens change sample composition as well as reporting quality. Holder retention is next-year ownership among initial holders; exit is its negative with the same standard error. The initial-holder share includes exits. Coefficients are percentage points per fixed source SD of upside composition, with the original controls and destination weights; parentheses are industry–education CR1 standard errors ($G = 30$, t_{29}, conditional on the saved CPS moments). Stars use unadjusted two-sided tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. BH q values adjust the 24 screened tests in columns (2) and (4) of Tables R.9–R.10.

Table R.11 reports the next-year ownership regressions that these diagnostics examine; Section 4.6 discusses them together with the characterization of who exits.

Inference and complete families. Four families receive Benjamini–Hochberg adjustment: the six displayed coefficient tests of Table R.11; the fourteen early-versus-late differences in Tables R.3 and R.4; the twelve state-specific sequence and difference tests in Table R.8; and the twenty-four screened coefficient tests in Tables R.9 and R.10. The fixed-effect, prior-switch, and leave-one-year comparisons are retained in full as sensitivity estimates and are not treated as additional discoveries.

Table R.10. Reporting-quality screens: next-year risky share

	(1) Full sample	(2) Reported components	(3) Strict-support baseline	(4) Reported and stable
A. Real log composition				
Nonholder share	0.360**	0.213	0.366**	0.352**
	(0.142)	(0.141)	(0.144)	(0.161)
BH q, 24 screened tests		0.347		0.119
Initial-holder share	−1.621**	−0.788	−1.658**	−0.593
	(0.715)	(0.932)	(0.704)	(0.887)
BH q, 24 screened tests		0.607		0.611
Holder minus nonholder	−1.980**	−1.001	−2.024***	−0.945
	(0.737)	(0.918)	(0.728)	(0.883)
BH q, 24 screened tests		0.469		0.469
B. Nominal arc composition				
Nonholder share	0.264**	0.237**	0.255*	0.443***
	(0.128)	(0.110)	(0.128)	(0.109)
BH q, 24 screened tests		0.119		0.004
Initial-holder share	−0.409	0.070	−0.411	−0.030
	(0.642)	(0.837)	(0.634)	(0.725)
BH q, 24 screened tests		0.967		0.967
Holder minus nonholder	−0.673	−0.168	−0.666	−0.473
	(0.683)	(0.808)	(0.671)	(0.698)
BH q, 24 screened tests		0.913		0.611
Observations	31,569	24,114	31,088	13,939
Initial holders	7,334	4,081	7,249	2,710

Notes: Column (2) requires reported (or legitimately not-in-universe) stock and mutual-fund ownership and balance components at both dates, using the public SIPP component flags because the aggregate flag is uninformative (coded 9) throughout. Column (4) additionally requires interviews coded as self-reported and stable household and residence records with adjacent waves; column (3) is its available-field baseline, and the column-(2) baseline coincides with column (1). Screens change sample composition as well as reporting quality. Holder retention is next-year ownership among initial holders; exit is its negative with the same standard error. The initial-holder share includes exits. Coefficients are percentage points per fixed source SD of upside composition, with the original controls and destination weights; parentheses are industry–education CR1 standard errors ($G = 30$, t_{29}, conditional on the saved CPS moments). Stars use unadjusted two-sided tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. BH q values adjust the 24 screened tests in columns (2) and (4) of Tables R.9–R.10.

All 67 specifications are reported; implementation and replay records are included in the replication package.

Table R.11. Earnings opportunities and stock market transitions

	(1) Nonholder entry	(2) Holder retention	(3) Holder minus nonholder
Real log composition	0.969^{***}	-2.650^{***}	-3.619^{***}
	(0.302)	(0.952)	(1.070)
BH q	0.010	0.014	0.010
Nominal arc composition	0.770^{***}	-1.848^{**}	-2.618^{**}
	(0.262)	(0.862)	(0.987)
BH q	0.013	0.040	0.015
Initial-state observations	24,235	7,334	31,569
Initial risky share	Zero	Controlled	State-specific
Controls and fixed effects	Yes	Yes	Fully interacted
Industry–education clusters	30	30	30

Notes: The dependent variable is next-year stock/mutual-fund ownership. The two survey-weighted linear probability models use 2019–2023 initial states and 2020–2024 outcomes, on the same denominator-valid sample as the allocation regressions. All earnings moments, household controls and fixed effects differ by initial ownership. Column (3) subtracts the nonholder entry slope from the holder retention slope using their covariance; it is not an exit-minus-entry contrast. Coefficients are percentage points per fixed source SD of upside composition. Parentheses contain industry–education CR1 standard errors. Stars use unadjusted two-sided $t(G-1)$ tests against zero: $^{*}p<0.10$, $^{**}p<0.05$, $^{***}p<0.01$. Unless indicated otherwise, $G=30$. Inference conditions on the saved CPS moments. The six displayed coefficient tests form the reported BH family.

S Asymmetry, Dispersion, and Mechanism Tests

This appendix supports Sections 4.2 and 4.5. It first documents the economic-driver comparisons supporting Section 4.3. The subsequent historical diagnostics re-express the nonlinear regressions in terms of the two partial variances, report the sensitivity of that decomposition to the level at which the dispersion term is evaluated and across the retained diagnostic fits, and report the mechanism tests that were fixed in a written protocol before estimation: the wealth and age heterogeneity of entry, the balance-sheet components among initial holders, total variance on its own, the destinations of incumbents' exits, each transition's realized own earnings change, the characteristics of the households who exit, the magnitude benchmarks for the entry finding, and the reliability of the cell measures together with the zero-inclusive construction. All regressions keep the original controls, destination survey weights, fixed source scales, and 30 industry–education CR1 clusters with t_{29} inference conditional on the CPS moments. Every fit was saved with its design, coefficients, covariance matrix, and cluster influences and independently replayed; the published coefficients were reproduced to 10^{-11} before any derived quantity was computed.

S.1 Economic drivers of earnings composition

What differs across earnings environments. Similar total variance can conceal different distributions of wage outcomes. Table S.1 compares workers aged 25–44 in trade with some college and in transport and utilities with a high-school education. Both cells have total variance close to 0.661. The trade cell has more upper-tail variation, less lower-tail variation, and a smaller frequency of zero wages in the next year: 5.67 rather than 7.75 percent. This pair illustrates the measured variation; it is not a matched comparison of otherwise identical workers.

Transitions out of wage earnings. Across the 58 cells, the zero-wage share is negatively correlated with upside composition (-0.855) and positively correlated with total variance (0.924). A transition from positive to zero wages enters nominal arc growth at -2, contributing to adverse earnings variation. This feature helps account descriptively for why low-composition cells tend to have high variance. It does not make the zero-wage share a sufficient description of the distribution: positive-wage changes and their distances from the cell mean also determine both partial variances. Zero annual wages can reflect nonemployment or a change in income source, rather than an identified dismissal.

Table S.18, Panel B, asks whether this frequency absorbs the participation association. Adding it to the full specification gives a composition coefficient of 2.420 percentage points per source SD (SE 0.911), compared with 1.601 (0.506) in the main regression. The zero-share coefficient is 2.630 points (SE 2.030) per cell SD. Composition therefore predicts starting participation beyond the measured frequency of leaving wage earnings; this comparison does not estimate a causal contribution of job loss.

Table S.1. Earnings distributions with similar total variance

	Trade some college	Transport/utilities high school
Age band	25–44	25–44
Upper partial variance V^+	0.27775	0.25251
Lower partial variance V^-	0.38330	0.40896
Total variance V	0.66105	0.66147
Upside composition C	−0.10555	−0.15646
Zero destination wages (%)	5.67	7.75
Validated earnings histories	2,493	802
Correlations across all 58 cells		
Zero-wage share with composition C	−0.855	
Zero-wage share with variance V	0.924	

Notes: CPS annual wage histories with positive origin wages and nonnegative destination wages; origins 2013–2017 and destinations 2014–2018. Nominal arc growth retains zero destinations, which have growth -2. Survey-weighted industry–education-specific cubic age profiles are removed before centering the moments within the industry–education–age cell. Partial variances and their sum and difference are in squared arc-growth units. The pair is an illustrative comparison selected for similar total variance, not a matched or causal design. The last two rows are unweighted Pearson correlations across all 58 cells; no sampling tests are attached. A zero wage need not mean unemployment or dismissal: a worker can leave wage work for another income source.

Measured cyclical exposure. Table S.21, Panel B, adds two sets of controls. The first is the cell's mean wage growth from 2019 to 2020. The second uses PSID slopes of cell-level earnings moments on national unemployment changes over 1999–2019: mean growth, the lower partial variance, and composition. The composition coefficient is 1.783 points with the 2020 control, 1.612–1.701 with each unemployment exposure separately, and 2.104 with the three exposures jointly. These estimates retain the main sample, controls and inference. They show that the association survives these measures of cyclical exposure; the controls do not exhaust all aggregate-state dependence or identify the source of an individual wage change.

Persistent versus transitory variation. The available earnings histories also permit a diagnostic of persistence, but not an economic decomposition of the two tails. With a permanent component and independent transitory shocks, the covariance of adjacent log-wage changes contains information about the transitory variance. Table S.2 uses this relationship to construct a cell-level proxy from PSID workers with positive wages at three successive biennial waves. The regression adds the proxy and its interactions with composition, variance and the standard deviation to the main participation specification.

The interaction between composition and the transitory-share proxy is 1.238 percentage points (SE 0.440) using nominal log wages and 1.231 (0.437) using real log wages. Excluding PSID histories with an absolute biennial log-wage change above three reduces the interaction to 0.612 point (0.691). This screen removes 443 of 31,665 PSID histories and leaves the SIPP sample unchanged. The sensitivity matters because the proxy depends on variance and autocovariance and can therefore be affected by a small number of extreme wage reports.

Table S.2. Participation and a proxy for transitory earnings risk

PSID proxy construction	(1) Nominal log wages	(2) Real log wages	(3) Nominal wages large-change screen
Composition C (at mean proxy)	1.508^{***}	1.503^{***}	1.353^{**}
	(0.360)	(0.361)	(0.533)
Transitory-share proxy T	0.078	0.068	-0.326
	(0.311)	(0.312)	(0.273)
Composition $\times$ transitory-share proxy	1.238^{***}	1.231^{***}	0.612
	(0.440)	(0.437)	(0.691)
Variance, $\sqrt{V}$, and their interactions with T	Yes	Yes	Yes
Mean growth, baseline controls and FE	Yes	Yes	Yes
Initial-nonholder transitions	27,419	27,419	27,419
Industry–education clusters	30	30	30
PSID three-wave histories	31,665	31,665	31,222
Proxy cells	58	58	58

Notes: Weighted linear probability models of starting stock-or-mutual-fund participation on the common SIPP sample, with the main zero-inclusive CPS wage measures. Coefficients are percentage points per source SD of C or per cross-cell SD of T; the interaction is the change in the composition slope for one SD more T. PSID reference persons are aged 25–64 with positive wages at three successive biennial waves in 1999–2019 (31,665 histories from 8,120 persons before the screen). For adjacent log-wage changes d_1, d_2, the proxy is $T = \max\{0, \min\{1, -2\,\mathrm{Cov}(d_1, d_2)/\,\mathrm{Var}(d)\}\}$, where the variance pools the two changes within each cell. It is standardized across cells using PSID effective-sample-size weights capped at 1,000. Column (3) drops a PSID history if either absolute log change exceeds three, removing 443 histories while leaving the SIPP sample unchanged. The specification includes T and its interactions with composition, variance and the standard deviation. All columns retain mean growth and its square, household controls, fixed effects and outcome-year person weights. Standard errors in parentheses use 30 industry–education CR1 clusters and t_{29}; stars are unadjusted two-sided tests against zero, $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Inference conditions on the estimated CPS and PSID moments. The first two proxy constructions were specified before this test; the large-change screen is an exploratory sensitivity check.

These comparisons establish neither that temporary shocks dominate the composition association nor a fraction attributable to permanent changes. The proxy excludes zero-wage histories and depends on the assumed permanent-plus-transitory process; measurement error and heterogeneous persistence also affect it. Promotions, changes in hours, employment transitions and persistent earnings losses can all alter the observed tails. The empirical evidence describes their combined earnings environment, while the model distinguishes realized resources, expectations and saving within an explicit earnings process.

S.2 Implied marginal associations and their evaluation level

Table S.38 reports local marginal associations implied by the regressions in Tables 2 and 5. The upper- and lower-tail entries use the same estimated coefficients and covariance matrix; they are not separate regressions or independent evidence for the composition finding. Each derivative holds the other partial variance fixed and therefore changes composition and total variance together. They are distinct from the direct linear partial-variance specification in Table 3, which omits the square-root variance term and has constant partial-variance slopes.

Each main regression contains composition $C = V^+ - V^-$, variance $V = V^+ + V^-$, and the standard deviation $\sqrt{V}$, all scaled by fixed cross-cell standard deviations. Because (C, V) is a change of basis of (V^+, V^-), the effect of raising V^+ with V^- fixed is $b_C/s_C + b_V/s_V + b_{\mathrm{SD}}/(2\,\mathrm{SD}_0 s_{\mathrm{SD}})$, and the effect of raising V^- with V^+ fixed replaces b_C by $-b_C$, where b denotes the coefficient on the scaled regressor, s its scale, and SD_0 the level at which the derivative of $\sqrt{V}$ is taken. After scaling both tail derivatives by the source SD of composition, half their difference equals the composition coefficient and does not depend on SD_0; the individual tail effects of Table S.37 do. The mean-level calculations use the survey-weighted mean earnings SD of the state block. For the original positive-support constructions, Table S.3 reports the per-tail effects in cell-SD units and the composition-scaled tail effects at the 25th and 75th percentiles of the assigned SD instead. For original real-log entry, the dispersion channel is indistinguishable from zero at these three levels ($p = 0.75$, 0.40 and 0.49 at the mean, 25th and 75th percentiles). For initial holders under that construction, the variance and SD coefficients are large and offsetting, so the dispersion channel is strong in low-dispersion cells and vanishes in high-dispersion cells; the upper-tail effect remains negative at the reported levels, whereas the lower-tail effect ranges from 0.37 to 2.50 points for retention. The dispersion association with exit depends on the level of dispersion.

Table S.4 evaluates the headline wage equation at the same types of dispersion levels. The upper-tail entry association remains positive and the lower-tail association negative at both quartiles. The dispersion derivative is 0.438 point at the lower quartile ($p = 0.003$), 0.104 at the mean and -0.212 at the upper quartile. Thus the small mean-level derivative does not establish a dispersion null throughout the support. This exploratory evaluation uses the saved coefficients and covariance and was added after inspecting the estimates.

S.3 Stability across the retained diagnostic fits

Tables S.5 and S.6 recompute the tail effects from the saved coefficient vectors and covariance matrices of the fits in Appendix R. The upper-tail entry effect is positive and the lower-tail effect negative in every fit, and the dispersion channel, half the sum of the two, is never distinguishable from zero (standard errors in the replication files); in the real-log reported-components sample the upper tail carries more of the effect, whereas the arc construction is symmetric throughout. The incumbent upper-tail effect is the operative one wherever an incumbent response exists: the full sample, both fixed-effect variants, every outcome-year deletion, and both constructions for retention. In the two reported-information samples and with the prior-switch control, both incumbent tail effects are small and imprecise, with intervals that include the full-sample values.

S.4 Heterogeneity by financial wealth and age

Tables S.7 and S.8 use the retained interaction design of Appendix Q, in which composition, variance, and the standard deviation each interact with within-band work horizon, centered log initial nonretirement financial assets, the centered initial log-income control, and the older-band indicator,

Table S.3. Per-tail effects and sensitivity to the dispersion level

	(1) Upper tail V^+	(2) Lower tail V^-	(3) Upper tail at p25 SD	(4) Upper tail at p75 SD	(5) Lower tail at p25 SD	(6) Lower tail at p75 SD
A. Real log composition (per cell SD of each tail)						
Entry, original cohort	1.026^{**}	-0.914^{**}	1.140^{**}	0.758^{**}	-0.610	-0.993^{***}
	(0.387)	(0.365)	(0.447)	(0.322)	(0.390)	(0.331)
Entry, prospective sample	1.089^{**}	-1.060^{**}	1.233^{**}	0.800^{**}	-0.705	-1.137^{***}
	(0.413)	(0.389)	(0.471)	(0.344)	(0.426)	(0.333)
Holder retention	-4.061^{***}	1.792	-4.933^{***}	-2.801^{**}	0.367	2.498^{**}
	(1.181)	(1.091)	(1.184)	(1.029)	(1.039)	(0.959)
Initial-holder share	-2.243^{**}	1.343	-2.918^{***}	-1.405^{*}	0.323	1.837^{**}
	(0.911)	(0.847)	(0.967)	(0.755)	(0.815)	(0.749)
Nonholder share	0.451^{*}	-0.346^{*}	0.400	0.421^{***}	-0.320	-0.298^{*}
	(0.243)	(0.185)	(0.339)	(0.152)	(0.250)	(0.160)
B. Nominal arc composition (per 0.01 of partial variance)						
Entry, original cohort	0.493^{**}	-0.482^{**}	0.810^{**}	0.598^{**}	-0.579^{**}	-0.792^{**}
	(0.193)	(0.183)	(0.330)	(0.242)	(0.253)	(0.296)
Entry, prospective sample	0.546^{**}	-0.535^{**}	0.904^{**}	0.643^{**}	-0.636^{**}	-0.897^{***}
	(0.203)	(0.196)	(0.332)	(0.261)	(0.279)	(0.302)
Holder retention	-1.792^{***}	0.804	-3.046^{***}	-1.938^{**}	0.651	1.758^{*}
	(0.608)	(0.657)	(0.880)	(0.884)	(0.965)	(0.933)
Initial-holder share	-0.636	-0.062	-1.361^{*}	-0.341	-0.543	0.476
	(0.454)	(0.492)	(0.675)	(0.635)	(0.723)	(0.697)
Nonholder share	0.214^{**}	-0.157	0.300	0.309^{**}	-0.228	-0.219
	(0.100)	(0.095)	(0.185)	(0.117)	(0.146)	(0.151)

Notes: Columns (1)–(2) report the marginal effect of each partial variance holding the other fixed, evaluated at the block's mean SD; panel A scales the tails by their cross-cell SDs under the retained fixed-scale convention (upper 0.0887, lower 0.0908), which is unavailable for the arc construction, so panel B reports effects per 0.01 of partial variance. Columns (3)–(6) evaluate the composition-scaled tail effects of Table S.37 at the 25th and 75th survey-weighted percentiles of the assigned earnings SD instead of its mean; the composition coefficient itself does not depend on this level; columns (3)–(6) are percentage points per fixed source SD of upside composition. Original controls and destination weights; parentheses are industry–education CR1 standard errors ($G = 30$, t_{29}, conditional on the saved CPS moments). Stars use unadjusted two-sided tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

separately by initial ownership. The share rows are the retained fits; the ownership rows refit the same 101-column design with next-year ownership, as declared in the protocol addendum. Figure S.1 plots the ownership rows for initial nonholders. The entry coefficient declines across the wealth quartiles in both constructions, from 1.97 to 1.27 points with real log growth and from 0.68 to 0.18 points with arc growth; the wealth interactions themselves are imprecise ($p = 0.178$ and $p = 0.291$). The nonholder-share point estimates decline similarly, and the real-log wealth interaction is precise ($p = 0.031$). The point-estimate gradient is consistent with a participation threshold, although the wealth interactions are imprecise.

The age gradients have opposite signs. Real log growth gives an entry coefficient of 1.49 points at ages 25–44 and 0.67 at ages 45–65; the older-minus-younger difference is -0.82 point ($p = 0.227$). Nominal arc growth gives 0.33 and 1.51 points, respectively, with a difference of 1.18 points ($p = 0.036$). All four entry coefficients are positive; it is their ordering across ages that changes. These

Table S.4. Wage-tail entry associations at alternative dispersion levels

	(1) 25th percentile of earnings SD	(2) Weighted mean earnings SD	(3) 75th percentile of earnings SD
Earnings SD	0.704	0.764	0.830
Upper partial variance V^+	2.040^{***}	1.705^{***}	1.389^{**}
	(0.594)	(0.561)	(0.538)
Lower partial variance V^-	-1.163^{**}	-1.498^{***}	-1.814^{***}
	(0.443)	(0.474)	(0.510)
Dispersion component	0.438^{***}	0.104	-0.212
	(0.135)	(0.116)	(0.136)
Nonholder transitions	27,419	27,419	27,419
Industry–education clusters	30	30	30

Notes: Algebraic evaluations of the same retained zero-inclusive wage entry equation; no regression is refitted. Each tail is increased with the other fixed, in units of one fixed source SD of composition. The dispersion component is half the sum of the two tail derivatives. Because the equation includes both variance and its square root, that derivative depends on the evaluation level. Levels are the outcome-weighted mean and empirical quartiles of the assigned earnings SD among the same initial nonholders. The composition coefficient, half the upper-minus-lower difference, remains 1.601 points at every level. This sensitivity was added after inspection of the retained coefficients: it is an ex-post diagnostic, not a prespecified testing family or a set of independent findings. Parentheses use the saved 30-group CR1 covariance, conditional on the CPS moments and evaluation levels. Stars use unadjusted two-sided t_{29} tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

conditional interaction estimates are not the separate age-band regressions of Section 4.7, and neither uses the zero-inclusive wage construction. On positive earnings endpoints, nominal arc growth equals twice the hyperbolic tangent of half nominal log growth, compressing extreme changes; real log growth additionally removes inflation. The resulting centered partial moments need not rank cells identically. This explains why the constructions can differ, but does not identify the source of the observed age reversal. The figure does not establish a common age gradient. For initial holders under real log growth, the retention point estimate is more negative in the younger band (-5.37 against -1.66 points), although the age difference is imprecise. The retention point estimates are nearly constant across wealth, while the holder-share point estimates become more negative; both wealth interactions are imprecise.

S.5 Balance-sheet components

Table S.9 decomposes the accounting components of Appendix K into tail effects. Among initial holders with real log growth, a fatter upper tail lowers the risky-balance contribution by 4.57 points ($p = 0.011$) and a fatter lower tail raises it by an imprecise 2.79 points ($p = 0.136$); the denominator contribution offsets part of each. The incumbent response therefore runs through risky balances, with the same tail asymmetry as the share change. The arc components are imprecise.

Table S.5. Tail effects across the retained diagnostic fits: real log composition

	(1) Entry upper tail	(2) Entry lower tail	(3) Retention upper tail	(4) Retention lower tail	(5) Share upper tail	(6) Share lower tail
Baseline	0.993^{**}	-0.944^{**}	-3.704^{***}	1.596	-2.046^{**}	1.196
	(0.377)	(0.346)	(1.077)	(0.972)	(0.831)	(0.755)
Industry-year and education-year effects	1.014^{**}	-0.962^{***}	-3.857^{***}	1.557	-2.052^{**}	1.145
	(0.386)	(0.339)	(1.170)	(0.992)	(0.871)	(0.749)
Joint industry–education effects	0.939	-0.971^{**}	-3.544^{***}	1.882^{**}	-1.707^{*}	0.982
	(0.569)	(0.465)	(0.993)	(0.698)	(0.869)	(0.842)
Reported components (24,114)	0.924^{**}	-0.474	-0.629	0.922	-0.635	0.941
	(0.437)	(0.327)	(1.312)	(1.192)	(1.137)	(0.890)
Reported and stable households (13,939)	1.275^{**}	-0.741^{**}	-0.694	0.543	-0.444	0.743
	(0.498)	(0.362)	(1.364)	(1.147)	(1.036)	(0.906)
Observed prior year (15,535)	1.163	-0.814	-2.504^{*}	1.155	-0.528	0.173
	(0.760)	(0.517)	(1.280)	(0.983)	(1.259)	(0.975)
Observed prior year, prior-switch control	1.051	-0.759	-1.407	0.624	0.033	-0.098
	(0.682)	(0.453)	(1.090)	(0.817)	(1.208)	(0.938)
Omitting 2020 outcomes	0.941^{**}	-0.833^{*}	-3.575^{**}	1.905^{*}	—	—
	(0.440)	(0.414)	(1.401)	(1.046)		
Omitting 2021 outcomes	0.988^{***}	-0.944^{***}	-3.592^{***}	1.090	—	—
	(0.334)	(0.296)	(1.111)	(0.983)		
Omitting 2022 outcomes	0.897^{*}	-0.929^{**}	-3.158^{**}	1.999^{*}	—	—
	(0.460)	(0.447)	(1.202)	(1.084)		
Omitting 2023 outcomes	1.154^{**}	-1.073^{***}	-3.808^{***}	1.337	—	—
	(0.465)	(0.343)	(1.329)	(1.129)		
Omitting 2024 outcomes	1.002^{***}	-0.952^{**}	-4.452^{***}	1.724^{*}	—	—
	(0.361)	(0.397)	(1.141)	(1.002)		

Notes: Tail effects recomputed from the saved coefficient vectors and CR1 covariance matrices of the retained diagnostic fits of Appendix R; the baseline row reproduces Table S.37. Fixed-effect variants and year deletions keep all 31,569 transitions; the reporting screens and the observed-prior-year sample are the subsamples defined there. Year deletions were run for real-log ownership only. The upper-tail effect is the marginal effect of the upper partial variance V^{+} with V^{-} fixed, scaled to one source SD of composition; the lower-tail effect is the marginal effect of V^{-} with V^{+} fixed, scaled the same way. Half their difference is the composition coefficient and half their sum is the dispersion channel, the effect of total variance at fixed composition, evaluated at the state block's survey-weighted mean earnings SD. Coefficients are percentage points per fixed source SD of upside composition, with the original controls and destination weights; parentheses are industry–education CR1 standard errors ($G = 30$, t_{29}, conditional on the saved CPS moments). Stars use unadjusted two-sided tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

S.6 Total variance on its own

Table S.10 refits the original positive-support designs without composition, and without composition and the standard deviation. For original real-log entry, one cross-cell SD of variance changes entry by 0.12, -0.19, and -0.23 points in the three versions, none distinguishable from zero. For holder retention under that construction, the full model's negative dispersion effect (-2.09 points, $p = 0.010$) comes from the offsetting variance and SD terms; without composition it is -1.24 points ($p = 0.115$), and a single linear variance term gives -0.07 points ($p = 0.945$). Original real-log composition is

Table S.6. Tail effects across the retained diagnostic fits: nominal arc composition

	(1) Entry upper tail	(2) Entry lower tail	(3) Retention upper tail	(4) Retention lower tail	(5) Share upper tail	(6) Share lower tail
Baseline	0.777^{**}	-0.762^{**}	-2.552^{***}	1.145	-0.906	-0.088
	(0.288)	(0.279)	(0.865)	(0.935)	(0.646)	(0.701)
Industry-year and education-year effects	0.791^{**}	-0.783^{***}	-2.762^{***}	1.305	-1.000	0.006
	(0.295)	(0.281)	(0.904)	(0.920)	(0.666)	(0.686)
Joint industry–education effects	0.528	-0.588	-3.277^{***}	3.013^{**}	-1.380^{*}	0.814
	(0.486)	(0.549)	(0.811)	(1.184)	(0.793)	(1.187)
Reported components (24,114)	0.846^{***}	-0.790^{***}	-1.017	0.534	-0.197	-0.336
	(0.269)	(0.217)	(1.154)	(1.107)	(0.889)	(0.867)
Reported and stable households (13,939)	1.249^{***}	-1.139^{***}	-1.284	1.248	-0.186	-0.126
	(0.330)	(0.270)	(1.145)	(1.175)	(0.706)	(0.838)
Observed prior year (15,535)	0.696	-0.473	-1.712^{*}	0.681	0.486	-1.267
	(0.440)	(0.345)	(0.860)	(1.087)	(0.762)	(0.834)
Observed prior year, prior-switch control	0.714^{*}	-0.575^{*}	-0.920	0.287	0.893	-1.469^{*}
	(0.375)	(0.285)	(0.795)	(1.052)	(0.747)	(0.817)

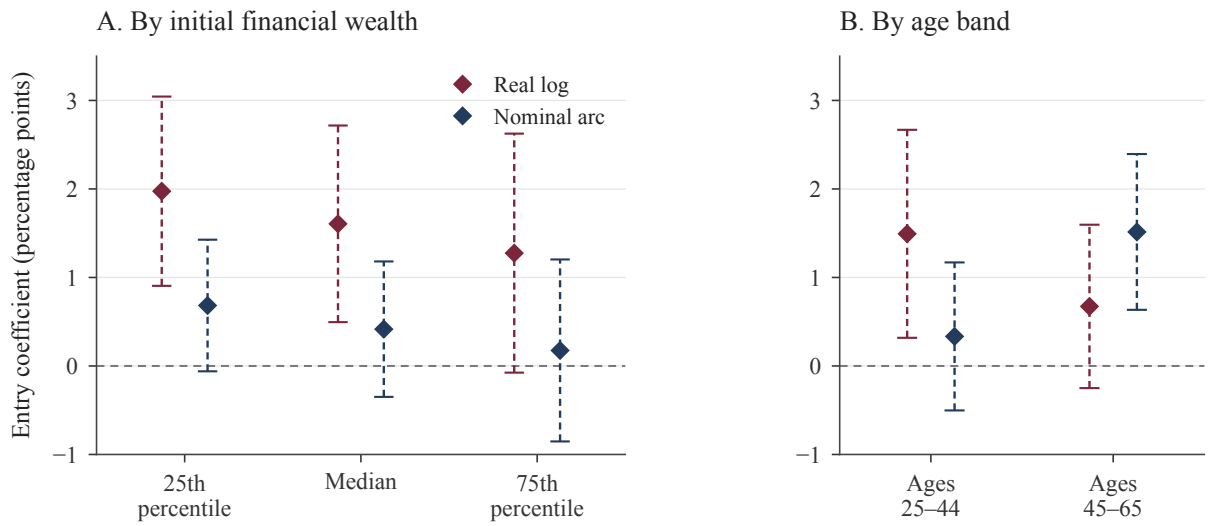
Notes: Tail effects recomputed from the saved coefficient vectors and CR1 covariance matrices of the retained diagnostic fits of Appendix R; the baseline row reproduces Table S.37. Fixed-effect variants and year deletions keep all 31,569 transitions; the reporting screens and the observed-prior-year sample are the subsamples defined there. The upper-tail effect is the marginal effect of the upper partial variance V^{+} with V^{-} fixed, scaled to one source SD of composition; the lower-tail effect is the marginal effect of V^{-} with V^{+} fixed, scaled the same way. Half their difference is the composition coefficient and half their sum is the dispersion channel, the effect of total variance at fixed composition, evaluated at the state block's survey-weighted mean earnings SD. Coefficients are percentage points per fixed source SD of upside composition, with the original controls and destination weights; parentheses are industry–education CR1 standard errors ($G = 30$, t_{29}, conditional on the saved CPS moments). Stars use unadjusted two-sided tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

nearly uncorrelated with variance across cells (-0.07). The headline wage construction differs: its correlation is -0.81, and Table S.4 reports its local dispersion derivatives.

S.7 Exit destinations and business entry

Table S.11 uses the December business values and tenure fields of the six public SIPP files, matched to both endpoints of every prospective transition. Of the 2,165 exits by initial holders, 50 coincide with business entry, 73 with a home purchase and no business entry, and 2,042 with neither. The exits associated with upside composition are exits to neither destination: the upper-tail effect for that category is 4.11 points ($p = 0.001$; twelve-test $q = 0.008$) with real log growth and 2.78 points ($p = 0.005$) with arc growth, whereas the business and housing destinations have small or negative upper-tail effects. The base rates of the two private destinations are low, 0.7% and 1.0% of initial holders, so those tests have little power, but the sign pattern does not support a reading in which incumbents liquidate public equity to fund private upside. Among initial nonholders, stock entry without business entry carries the entire entry response (0.86 points for the upper tail and -0.92

Figure S.1. Entry response to earnings asymmetry by financial wealth and age



Notes: Composition coefficients for next-year stock-and-mutual-fund entry among the 24,235 initial nonholders, in percentage points per fixed pre-adjustment source standard deviation, from the interaction design in which composition, variance, and the standard deviation each interact with within-band work horizon, centered log initial nonretirement financial assets, the centered initial log-income control, and the older-band indicator, separately by initial ownership. Panel A evaluates the coefficient at the survey-weighted quartiles of log financial assets among initial nonholders, with the other moderators at their centering points; Panel B evaluates it for the two age bands. Burgundy diamonds use real log growth and navy diamonds nominal arc growth, both original constructions. Bars are 95% CR1/t_{29} intervals using 30 industry–education clusters, conditional on estimated CPS moments. Tables S.7 and S.8 report the estimates and interactions.

for the lower tail), and business entry itself responds weakly and only through the upper tail (0.33 points, $p = 0.057$). Stock entry and business entry are parallel rather than competing responses.

S.8 Realized own earnings

Table S.12 asks whether the associations reflect anticipation of an earnings environment or a reaction to the household's own realized earnings, following the protocol addendum for this test. Each transition's own annual earnings are the twelve-month sums of monthly earnings in the public SIPP files, and the realized change between the baseline and outcome years is the log change for the real-log construction and the arc change for the arc construction, winsorized at the 1st and 99th survey-weighted percentiles and standardized within the initial-ownership block; it is observed for 23,441 nonholder and 7,131 holder transitions under the log definition and for 23,992 and 7,264 under the arc definition. Estimating the prospective specifications on these rows gives the same qualitative pattern. Adding the realized change as a control moves no tail effect by more than 0.04 point: the entry effects are 0.964 and -0.898 points with real log growth and 0.772 and -0.759 points with arc growth, and the upper-tail retention effects are -3.637 and -2.529 points. The realized change itself is imprecisely associated with entry (0.276 and 0.308 point per standard deviation) and positively related to retention (1.107 and 1.039 points, $p = 0.071$ in both constructions). Interacting the realized change with composition, variance, and the standard deviation gives composition

Table S.7. Composition effects by financial wealth

	(1) 25th percentile	(2) Median	(3) 75th percentile	(4) Interaction per log unit
A. Real log composition				
Next-year entry (initial nonholders)	1.974^{***}	1.606^{***}	1.275^{*}	-0.243
	(0.523)	(0.543)	(0.660)	(0.176)
Holder retention	-5.359^{**}	-5.348^{**}	-5.336^{**}	0.010
	(2.181)	(2.094)	(2.063)	(0.292)
Nonholder next-year share	0.780^{***}	0.493	0.235	-0.190^{**}
	(0.271)	(0.296)	(0.357)	(0.083)
Initial-holder next-year share	-3.270^{**}	-3.522^{**}	-3.767^{**}	-0.212
	(1.189)	(1.327)	(1.513)	(0.263)
B. Nominal arc composition				
Next-year entry (initial nonholders)	0.683^{*}	0.416	0.175	-0.177
	(0.364)	(0.374)	(0.503)	(0.164)
Holder retention	-1.078	-0.666	-0.268	0.345
	(1.106)	(1.169)	(1.418)	(0.432)
Nonholder next-year share	0.361^{**}	0.137	-0.065	-0.148^{*}
	(0.155)	(0.188)	(0.266)	(0.080)
Initial-holder next-year share	0.470	0.447	0.424	-0.019
	(0.634)	(0.805)	(1.044)	(0.272)

Notes: Composition coefficients from the retained work-horizon design (Appendix Q), in which every earnings regressor interacts with horizon, centered log initial nonretirement financial assets, the centered initial log-income control and the older-band indicator, separately by initial ownership. Ownership rows refit that design with next-year ownership; share rows are the retained share fits. Columns (1)–(3) evaluate the composition coefficient at the survey-weighted quartiles of centered log financial assets within the state block, with the other moderators at their centering points (younger band, mean income, band-midpoint horizon); column (4) is the interaction coefficient per log unit of assets. Coefficients are percentage points per fixed source SD of upside composition, with the original controls and destination weights; parentheses are industry–education CR1 standard errors ($G = 30$, t_{29}, conditional on the saved CPS moments). Stars use unadjusted two-sided tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

interactions of 0.112 point ($p = 0.538$) and 0.380 point ($p = 0.070$) for entry, -0.337 and 0.340 point for retention, and -0.856 point ($p = 0.042$) and -0.060 point for the holder share; none survives the six-test family adjustment (smallest $q = 0.211$). The measured realized earnings change does not explain the entry association. These controls do not establish anticipation: realized earnings can themselves be affected by the earnings environment, and the specifications do not identify a direct effect or a bound. The interaction evidence is insufficient to locate the holder response among households with realized gains.

S.9 Who exits

Tables S.13–S.17 characterize the exits associated with upside composition, following the protocol written for this question. Table S.13 compares initial holders who exit with those who stay, and initial nonholders who enter with those who do not. Exiters hold small positions (median stock and mutual-fund balance $12,000 against $25,000; median risky share 0.375 against 0.456), have

Table S.8. Composition effects by age band

	(1) Ages 25–44	(2) Ages 45–65	(3) Older minus younger
A. Real log composition			
Next-year entry (initial nonholders)	1.493^{**}	0.673	-0.820
	(0.575)	(0.451)	(0.664)
Holder retention	-5.369^{**}	-1.656	3.713
	(2.288)	(1.696)	(2.359)
Nonholder next-year share	0.405	0.191	-0.214
	(0.314)	(0.171)	(0.323)
Initial-holder next-year share	-3.064^{**}	-0.260	2.804^{*}
	(1.129)	(1.076)	(1.477)
B. Nominal arc composition			
Next-year entry (initial nonholders)	0.334	1.514^{***}	1.181^{**}
	(0.409)	(0.430)	(0.537)
Holder retention	-1.412	-2.955	-1.543
	(1.225)	(1.844)	(2.133)
Nonholder next-year share	0.068	0.526^{***}	0.458^{**}
	(0.211)	(0.167)	(0.202)
Initial-holder next-year share	0.489	-0.997	-1.486
	(0.597)	(1.261)	(1.380)

Notes: Same designs as Table S.7. Columns (1)–(2) evaluate the composition coefficient for the two age bands with wealth, income and horizon at their centering points; column (3) is the older-band interaction. Coefficients are percentage points per fixed source SD of upside composition, with the original controls and destination weights; parentheses are industry–education CR1 standard errors ($G = 30$, t_{29}, conditional on the saved CPS moments). Stars use unadjusted two-sided tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

lower income and financial assets, and draw their nonretirement financial assets down to 40% of the baseline level at the median; business entry and home purchase do not differ. In the subsample with an observed earlier year, 49.6% of exiters had entered in the previous year against 22.8% of stayers, and in the subsample with an observed later year 30.9% of exiters re-enter and 48.6% of entrants exit again. Table S.14 augments the retained prospective designs on the prior-history subsample with a block-specific indicator for a different ownership status one year before the baseline and its interactions with the earnings moments. Among initial holders, a fatter upper tail lowers retention by 4.91 points among recent entrants ($p = 0.012$) and moves it by 0.48 point in the opposite direction among established holders ($p = 0.73$); the difference is -5.39 points ($p = 0.028$; four-test $q = 0.110$). For the risky share (Table S.15), established holders' upper-tail effect is 1.05 points ($p = 0.47$) with real log growth and 1.84 points ($p = 0.064$) with arc growth, against -1.89 and -0.98 points for recent entrants. Table S.16 does not detect a difference by initial risky share: the upper-tail retention effect is -3.81 points for holders at or above the median risky share and -3.85 points below it (difference $p = 0.98$). Table S.17 decomposes the retained persistence fits into tail effects: with real log growth, a fatter upper tail raises entry followed by exit by 0.96 point ($p = 0.005$) and persistent entry by 0.17 point ($p = 0.71$); among holders, persistent exit responds to the lower tail (-3.41 points, $p = 0.018$) and temporary exit to the dispersion channel (1.34 points, $p = 0.002$). The

Table S.9. Tail effects on the balance-sheet components among initial holders

	(1) Composition	(2) Upper tail V^+	(3) Lower tail V^-	(4) Dispersion channel
A. Real log composition				
Share change	-1.621^{**}	-2.046^{**}	1.196	-0.425
	(0.715)	(0.831)	(0.755)	(0.344)
Risky-balance contribution	-3.679^{**}	-4.569^{**}	2.789	-0.890
	(1.595)	(1.690)	(1.818)	(0.732)
Denominator contribution	2.058^{*}	2.523^{**}	-1.594	0.465
	(1.050)	(1.131)	(1.173)	(0.474)
B. Nominal arc composition				
Share change	-0.409	-0.906	-0.088	-0.497^{**}
	(0.642)	(0.646)	(0.701)	(0.208)
Risky-balance contribution	-1.244	-2.279	0.208	-1.036^{**}
	(1.339)	(1.359)	(1.483)	(0.479)
Denominator contribution	0.835	1.373	-0.296	0.539^{*}
	(0.847)	(0.878)	(0.927)	(0.313)

Notes: The retained code104 accounting designs (Appendix K) replayed with the retained routine and decomposed into tail effects; the two components sum exactly to the share change. Share change is in share percentage points; contributions are per 100 of the midpoint financial-asset denominator. The upper-tail effect is the marginal effect of the upper partial variance V^+ with V^- fixed, scaled to one source SD of composition; the lower-tail effect is the marginal effect of V^- with V^+ fixed, scaled the same way. Half their difference is the composition coefficient and half their sum is the dispersion channel, the effect of total variance at fixed composition, evaluated at the state block's survey-weighted mean earnings SD. Coefficients are percentage points per fixed source SD of upside composition, with the original controls and destination weights; parentheses are industry–education CR1 standard errors ($G = 30$, t_{29}, conditional on the saved CPS moments). Stars use unadjusted two-sided tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

arc construction gives the same signs with less precision. The original construction links upper-tail risk more strongly to exits among recent entrants. Longer-standing holders show no statistically detectable upper-tail retention response, and the real-log share interval permits a modest decline. The temporary-entry point estimate exceeds the persistent-entry estimate, but their composition contrast is imprecise; the wage recency comparison is also imprecise (Section 4.6).

S.10 Tests engaging the closest precedents

Tables S.18 and S.19 report the tests written before estimation in the design note of 2026-09-17 to engage the closest precedents: the earnings moments entered separately and jointly, as Catherine, Sodini, and Zhang (2024) enter their co-moments; the cell share of histories ending at zero wages, related to but distinct from the dismissal risk studied by Ahn, Jo, and Lam (2026); the entry response by education; the exit hazard by tenure and the one-year exits inside the reporting screens, the facts that Galaasen and Raja (2024) document in Norwegian registers; and the recent-entry exit design excluding the down-market year 2022. The last comparison tests dependence on that market decline, without ruling out household-specific investment experiences.

Table S.10. Total variance on its own: nested restrictions of the published designs

	(1) Full model	(2) Without composition	(3) Linear variance only
A. Real log composition			
Entry, original cohort	0.121	−0.190	−0.234
	(0.381)	(0.389)	(0.308)
Entry, prospective sample	0.049	−0.299	−0.350
	(0.395)	(0.402)	(0.298)
Holder retention	−2.089**	−1.243	−0.068
	(0.758)	(0.765)	(0.979)
Nonholder share	0.103	−0.026	0.067
	(0.265)	(0.264)	(0.142)
Initial-holder share	−0.842	−0.325	0.574
	(0.681)	(0.656)	(0.585)
B. Nominal arc composition			
Entry, original cohort	0.035	0.016	−0.081
	(0.436)	(0.441)	(0.390)
Entry, prospective sample	0.033	0.010	−0.143
	(0.463)	(0.459)	(0.396)
Holder retention	−3.036**	−3.187***	−1.376
	(1.134)	(1.040)	(1.410)
Nonholder share	0.175	0.167	0.200
	(0.229)	(0.237)	(0.161)
Initial-holder share	−2.145**	−2.179**	−0.417
	(0.896)	(0.849)	(0.977)

Notes: Each cell is the marginal effect of one cross-cell SD of total earnings variance, evaluated at the block's mean SD, in percentage points of the outcome. Column (1) is the published design with composition, variance and SD; column (2) drops composition; column (3) drops composition and SD, leaving a single linear variance term. Same rows, weights, controls and clusters throughout. Original controls and destination weights; parentheses are industry–education CR1 standard errors ($G = 30$, t_{29}, conditional on the saved CPS moments). Stars use unadjusted two-sided tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

S.11 Inference, skewness, cyclicality, sorting, and the ownership benchmark

Tables S.20–S.23 report the tests written before estimation in the design note of 2026-09-17 for the remaining referee concerns: wild-cluster bootstrap inference at 30 groups; composition against the third standardized moment of the same residuals, with the split-half reliability of both; the entry design with cell-level cyclical-exposure controls, the Covid shock of 2020 and the PSID exposures of 1999–2019; the 1996 PSID risk-tolerance gambles as a direct test of sorting; and the SCF 2019 benchmark of the SIPP ownership measure.

Table S.22 completes the comparison with conventional skewness while retaining the headline dispersion controls. It reports the existing composition-only, skewness-only and joint shape specifications for all four constructions on the common entry sample. The joint wage specification is imprecise because the measures are closely related conditional on dispersion. The partial-variance representation supplies an additive and relatively reproducible description of earnings asymmetry; these comparisons do not identify a separate shape dimension beyond skewness.

Table S.11. Exit destinations and entry alternatives

		Real log composition			Nominal arc composition		
	(1)	(2)	(3) Upper	(4) Lower	(5)	(6) Upper	(7) Lower
	Events	Comp.	tail	tail	Comp.	tail	tail
A. Initial holders: next-year exit by destination							
Exit, all	2,165	2.650^{***}	3.704^{***}	-1.596	1.848^{**}	2.552^{***}	-1.145
		(0.952)	(1.077)	(0.972)	(0.862)	(0.865)	(0.935)
Exit with business entry	50	-0.350	-0.258	0.442^{**}	-0.145	-0.064	0.226
		(0.218)	(0.242)	(0.210)	(0.212)	(0.199)	(0.232)
Exit with home purchase	73	0.028	-0.143	-0.199^{*}	-0.131	-0.164	0.099
		(0.125)	(0.171)	(0.110)	(0.117)	(0.117)	(0.131)
Exit to neither	2,042	2.972^{***}	4.105^{***}	-1.838^{*}	2.124^{**}	2.779^{***}	-1.469
		(0.992)	(1.077)	(1.044)	(0.947)	(0.904)	(1.056)
B. Initial nonholders: next-year entry and business entry							
Stock entry, all	2,450	0.969^{***}	0.993^{**}	-0.944^{**}	0.770^{***}	0.777^{**}	-0.762^{**}
		(0.302)	(0.377)	(0.346)	(0.262)	(0.288)	(0.279)
Business entry (any)	470	0.184	0.329^{*}	-0.039	0.142	0.177	-0.107
		(0.116)	(0.166)	(0.109)	(0.106)	(0.121)	(0.103)
Stock entry with business entry	79	0.079	0.133	-0.024	0.017	0.039	0.005
		(0.054)	(0.093)	(0.038)	(0.045)	(0.062)	(0.031)
Stock entry without business entry	2,371	0.890^{***}	0.860^{**}	-0.920^{**}	0.753^{***}	0.739^{***}	-0.767^{***}
		(0.311)	(0.370)	(0.347)	(0.250)	(0.264)	(0.273)

Notes: December business values (TVAL_BUS) and tenure (ETENURE) from the public SIPP files, joined to both endpoints of the 31,569 prospective transitions; every transition has both endpoints observed. Business entry is a positive business value at the outcome year-end after none at baseline; home purchase is owner-occupancy at the outcome year-end after renting or rent-free occupancy at baseline. Panel A splits initial holders' exits into three mutually exclusive destinations (business entry takes precedence over home purchase) whose coefficients sum to the total; panel B splits initial nonholders' stock entry by concurrent business entry. Holder rows measure exit, so positive values mean more exit. Comp. is the composition coefficient. The upper-tail effect is the marginal effect of the upper partial variance V^{+} with V^{-} fixed, scaled to one source SD of composition; the lower-tail effect is the marginal effect of V^{-} with V^{+} fixed, scaled the same way. Half their difference is the composition coefficient and half their sum is the dispersion channel, the effect of total variance at fixed composition, evaluated at the state block's survey-weighted mean earnings SD. Coefficients are percentage points per fixed source SD of upside composition, with the original controls and destination weights; parentheses are industry–education CR1 standard errors ($G = 30$, t_{29}, conditional on the saved CPS moments). Stars use unadjusted two-sided tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Two twelve-test BH families (tail effects by destination or type, both constructions) are reported in the replication files.

S.12 Two further sources, the 2014-panel hold-out, and equity compensation

Table S.24 reports the SCF 2019 levels test and the ownership levels in the SIPP 2014 panel (reference years 2013–2016); Table S.25 reports that panel's pre-registered one-year entry hold-out and the churn of its ownership reports against the 2019–2024 panels, as pre-specified in the design note of 2026-09-17; Table S.26 reports the entry design after removing the workers most likely to hold equity compensation.

Table S.12. Realized own earnings: tail effects with and without the realized change

	No control		Realized change as a control			Interacted
	(1) Upper tail	(2) Lower tail	(3) Upper tail	(4) Lower tail	(5) Realized change	(6) Comp. × change
A. Real log composition; realized log change in own annual earnings						
Entry (initial nonholders)	0.962^{**}	-0.898^{**}	0.964^{**}	-0.898^{**}	0.276	0.112
	(0.370)	(0.341)	(0.369)	(0.340)	(0.251)	(0.180)
BH q			0.028	0.028		0.828
Holder retention	-3.646^{***}	1.478	-3.637^{***}	1.440	1.107^{*}	-0.337
	(1.125)	(0.989)	(1.136)	(0.990)	(0.590)	(0.700)
BH q			0.028	0.209		0.828
Initial-holder share	-2.019^{**}	1.119	-2.020^{**}	1.123	-0.138	-0.856^{**}
	(0.886)	(0.761)	(0.884)	(0.759)	(0.410)	(0.402)
BH q			0.051	0.209		0.211
B. Nominal arc composition; realized arc change in own annual earnings						
Entry (initial nonholders)	0.772^{***}	-0.761^{***}	0.772^{***}	-0.759^{***}	0.308	0.380^{*}
	(0.276)	(0.270)	(0.277)	(0.270)	(0.261)	(0.202)
BH q			0.028	0.028		0.211
Holder retention	-2.533^{***}	1.128	-2.529^{***}	1.101	1.039^{*}	0.340
	(0.846)	(0.946)	(0.853)	(0.948)	(0.554)	(0.846)
BH q			0.028	0.278		0.828
Initial-holder share	-0.840	-0.169	-0.840	-0.167	-0.075	-0.060
	(0.631)	(0.697)	(0.630)	(0.694)	(0.395)	(0.432)
BH q			0.231	0.811		0.890
Rows with realized change	23,441 / 7,131		23,992 / 7,264			

Notes: Protocol C3. Own annual earnings are twelve-month sums of monthly earnings (TPEARN) in the public SIPP files; the realized change between the baseline and outcome years is the log change (real-log construction) or the arc change (arc construction), winsorized at the survey-weighted 1st and 99th percentiles and standardized within the initial-ownership block. Columns (1)–(2) refit the retained prospective design on the rows with the realized change observed; columns (3)–(5) add the block-specific realized change as a control; column (6) interacts it with composition, variance and the standard deviation and reports the composition interaction per SD of the realized change. The bottom row gives nonholder / holder rows with the realized change observed (real log; arc). Tail effects are defined as in Table S.37. Coefficients are percentage points per fixed source SD of upside composition, with the original controls and destination weights; parentheses are industry–education CR1 standard errors ($G = 30$, t_{29}, conditional on the saved CPS moments). Stars use unadjusted two-sided tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. BH families: the twelve tail tests of columns (3)–(4) for entry, retention and holder share; the six composition interactions of column (6).

S.13 Magnitude benchmarks

Table S.27 states the entry finding relative to base rates and to the other determinants of entry, using the same estimated specifications. The entry rates of initial nonholders around the wealth points of Figure S.1 are 6.4, 7.3, and 11.9 percent, so the composition effects of 1.97, 1.61, and 1.27 points raise entry by 31, 22, and 11 percent. Across the 58 cells, the 10th and 90th percentiles of upside composition among initial nonholders are -1.18 and 1.11 source standard deviations; with the Table S.36 coefficient the move raises the predicted entry rate by 2.0 points, 23 percent of the 8.82 percent base, whereas the same move in total variance, using the linear variance-only coefficient of

Table S.13. Who exits and who enters: baseline characteristics

	(1) Holders who exit	(2) Holders who stay	(3) Nonholders who enter	(4) Nonholders who stay out
Transitions	2,165	5,169	2,450	21,785
Share of the initial-ownership block (weighted)	0.305	0.695	0.096	0.904
Stock and mutual-fund balance, median ($)	12,000	25,000	0	0
Nonretirement financial assets, median ($)	48,000	79,796	14,000	4,525
Risky share, median	0.375	0.456	—	—
Log income, mean	8.30	8.59	8.23	7.91
Age, mean	43.8	44.6	43.2	43.6
Unemployment spell at baseline, share	0.068	0.067	0.071	0.076
Realized log change in own earnings, median	0.036	0.048	0.051	0.042
Business entry, share	0.023	0.025	0.030	0.018
Home purchase, share	0.033	0.022	0.027	0.026
Next-year financial assets / baseline, median	0.39	1.13	2.88	1.10
Ownership status differed one year before baseline, share	0.496	0.228	0.324	0.065
Ownership status reverts one year after outcome, share	0.309	0.780	0.486	0.937

Notes: Initial holders (7,334) and initial nonholders (24,235) of the prospective sample, split by the next-year transition; destination-weighted medians, means or shares of baseline characteristics. Balances and assets are nominal dollars; the risky share is the stock and mutual-fund balance over nonretirement financial assets. The realized change is the log change in own annual earnings between the baseline and outcome years (Appendix S, protocol C3). The last two rows use the subsamples with an observed year before the baseline (Appendix R; 1,020 exiters and 2,698 stayers; 1,117 entrants and 10,700 non-entrants) and with an observed year after the outcome (779 exiters, 2,028 stayers, 846 entrants, 7,480 non-entrants): for holders who exit, the share who had entered in the previous year and the share who re-enter the following year; for nonholders who enter, the share who had exited in the previous year and the share who exit again the following year.

Table S.10, changes it by -0.4 point. In the prospective ownership regression, one standard deviation of log financial assets among initial nonholders (2.18) raises entry by 3.25 points and one standard deviation of log income (1.85) by 1.04 points, against 0.97 point for composition. In the persistence fits, the temporary-entry coefficient is 0.91 of the any-entry coefficient for composition and 0.85 for the upper tail; the persistent-minus-temporary difference itself is imprecise ($p = 0.146$).

S.14 Measurement reliability and the zero-inclusive construction

Tables S.28 and S.30 support Section 4.10, using the common construction described below. The cell moments of every construction are recomputed from the retained 88,862 validated CPS histories with the bridge conventions of Appendix J: group-specific weighted cubic age profile, centered partial variances, 58 cells; the positive-support moments reproduce the retained moment file to 10^{-15}. The zero-inclusive construction uses nominal arc growth of total earnings or wages with a zero destination contributing -2; the pooled construction adds histories for interview years 2019–2022 linked from the same IPUMS extract with the retained rules (person link to the next March, ages 25–65 at

Table S.14. Incumbent retention effects by recent entry

	(1) Composition	(2) Upper tail V^+	(3) Lower tail V^-	(4) Dispersion channel	(5) Rows
A. Real log composition					
Established holders (owners one year before)	0.723	0.480	−0.965	−0.243	2,602
	(1.119)	(1.404)	(1.083)	(0.565)	
Recent entrants (nonowners one year before)	−4.662***	−4.912**	4.413**	−0.249	1,116
	(1.665)	(1.844)	(1.828)	(0.773)	
Recent minus established	−5.385**	−5.392**	5.378**	−0.007	
	(2.125)	(2.324)	(2.316)	(0.931)	
BH q, 4 tests		0.110			
B. Nominal arc composition					
Established holders (owners one year before)	0.008	−0.165	−0.182	−0.173	2,602
	(1.078)	(1.141)	(1.133)	(0.361)	
Recent entrants (nonowners one year before)	−1.916	−2.547	1.285	−0.631	1,116
	(1.787)	(1.640)	(2.000)	(0.389)	
Recent minus established	−1.924	−2.382	1.467	−0.458	
	(2.213)	(2.277)	(2.220)	(0.400)	
BH q, 4 tests		0.434			

Notes: Prior-history subsample of Appendix R (15,535 transitions with an observed year before the baseline; 3,718 initial holders). The retained prospective design is augmented with a block-specific indicator for a different ownership status one year before the baseline (for holders, entry since the previous year) and its interactions with composition, variance and the standard deviation; the nonholder block carries the symmetric indicator for recent exit. Rows evaluate the holder-block tail effects for established holders, for recent entrants, and their difference. The outcome is next-year ownership (holder retention). Coefficients are percentage points per fixed source SD of upside composition, with the original controls and destination weights; parentheses are industry–education CR1 standard errors ($G = 30$, t_{29}, conditional on the saved CPS moments). Stars use unadjusted two-sided tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Tail effects are defined as in Table S.37. The BH family contains the four holder-block upper-tail differences (recent entry and position size, both constructions; Table S.16).

baseline, positive baseline wage, nonnegative next wage, age change of zero to two years, same sex, race and household), which reproduce the retained histories exactly; 72,231 of the 74,491 added histories fall in the 58 cells. Reliability is the correlation across the 58 cells of a moment computed on two random halves of households, converted by the Spearman–Brown formula and averaged over 30 splits; temporal stability is the correlation of the moment across the early and late halves of the origin years. The remeasurement resamples CPS households with replacement 499 times, recomputes the moments with the age profile refitted, standardizes them with the fixed cross-cell scales, and refits the original-cohort entry design; the SIPP sample is held fixed, so the intervals reflect CPS sampling error only, whereas the joint remeasurement of Section 4.9 also resamples SIPP. The estimates in Table S.29 come from the retained designs with the five earnings regressors replaced; the replication files contain all fitted specifications. With the pre-determined zero-inclusive total-earnings construction, a linear variance term on its own is -0.27 point ($p = 0.64$) for entry, and the six composition tests

Table S.15. Incumbent risky-share effects by recent entry

	(1) Composition	(2) Upper tail V^+	(3) Lower tail V^-	(4) Dispersion channel	(5) Rows
A. Real log composition					
Established holders (owners one year before)	1.039	1.051	−1.028	0.012	2,602
	(1.240)	(1.431)	(1.224)	(0.486)	
Recent entrants (nonowners one year before)	−2.053	−1.888	2.218*	0.165	1,116
	(1.342)	(1.696)	(1.211)	(0.609)	
Recent minus established	−3.092*	−2.939	3.245*	0.153	
	(1.696)	(2.018)	(1.725)	(0.804)	
B. Nominal arc composition					
Established holders (owners one year before)	2.156**	1.836*	−2.475**	−0.319	2,602
	(0.942)	(0.955)	(1.005)	(0.273)	
Recent entrants (nonowners one year before)	−0.740	−0.981	0.498	−0.241	1,116
	(1.532)	(1.485)	(1.627)	(0.280)	
Recent minus established	−2.895	−2.817	2.973	0.078	
	(1.900)	(1.898)	(1.992)	(0.417)	

Notes: Prior-history subsample of Appendix R (15,535 transitions with an observed year before the baseline; 3,718 initial holders). The retained prospective design is augmented with a block-specific indicator for a different ownership status one year before the baseline (for holders, entry since the previous year) and its interactions with composition, variance and the standard deviation; the nonholder block carries the symmetric indicator for recent exit. Rows evaluate the holder-block tail effects for established holders, for recent entrants, and their difference. The outcome is the next-year risky share. Coefficients are percentage points per fixed source SD of upside composition, with the original controls and destination weights; parentheses are industry–education CR1 standard errors ($G = 30$, t_{29}, conditional on the saved CPS moments). Stars use unadjusted two-sided tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Tail effects are defined as in Table S.37. Exploratory addendum to protocol E1; no family is declared.

of the total-earnings and pooled-wage constructions, the initially specified family, excluding the subsequently added pre-portfolio wage construction, form one Benjamini–Hochberg family (entry $q = 0.066$, retention $q = 0.066$, holder share $q = 0.36$ for total earnings); the nine-test family below covers all three zero-inclusive constructions. Three subsequent exercises examine measurement uncertainty and extend the testing family.

First, the remeasurement of the wage construction (Table S.30, 499 draws): 99.4 percent of draws are positive, with a descriptive 2.5th–97.5th percentile range of 0.13 to 1.51 points. These ranges are not confidence intervals for the population coefficient. The table's "main tables" heading denotes the positive-support real-log specification.

Second, the SIMEX exercise in Table S.31: for noise multiples ζ of 0.5, 1, 1.5 and 2, 200 household cluster-bootstrap draws each perturb the observed moments by $\sqrt{\zeta}$ times the draw's deviation and the entry design is refitted; extrapolating the draw means to $\zeta = -1$ by a quadratic in ζ gives 2.82 points for wages, 1.80 for total earnings and 1.64 for pooled wages, and a linear extrapolation gives 1.96, 1.27 and 1.63, against point estimates of 1.60, 1.04 and 1.23. The draw means fall with ζ as

Table S.16. Incumbent tail effects by position size

	(1) Composition	(2) Upper tail V^+	(3) Lower tail V^-	(4) Dispersion channel	(5) Rows
A. Real log composition					
Holder retention: Risky share at or above the holder median	-2.515^{**}	-3.805^{***}	1.225	-1.290^{*}	3,770
	(1.092)	(1.256)	(1.364)	(0.727)	
Holder retention: Risky share below the holder median	-2.907^{**}	-3.854^{**}	1.960	-0.947	3,564
	(1.314)	(1.547)	(1.301)	(0.563)	
Holder retention: Below minus above	-0.392	-0.048	0.735	0.343	
	(1.435)	(1.750)	(1.768)	(1.017)	
BH q, 4 tests		0.978			
B. Nominal arc composition					
Holder retention: Risky share at or above the holder median	-1.031	-1.795	0.267	-0.764^{**}	3,770
	(1.218)	(1.236)	(1.299)	(0.352)	
Holder retention: Risky share below the holder median	-2.823^{**}	-3.500^{***}	2.145^{*}	-0.677^{**}	3,564
	(1.131)	(1.207)	(1.119)	(0.275)	
Holder retention: Below minus above	-1.792	-1.705	1.879	0.087	
	(1.573)	(1.705)	(1.506)	(0.337)	
BH q, 4 tests		0.434			

Notes: Full prospective sample (7,334 initial holders). The retained prospective ownership design is augmented with a block-specific indicator for a baseline risky share below the destination-weighted median among initial holders and its interactions with composition, variance and the standard deviation; the nonholder block carries the symmetric indicator for nonretirement financial assets below the nonholder median. Coefficients are percentage points per fixed source SD of upside composition, with the original controls and destination weights; parentheses are industry–education CR1 standard errors ($G = 30$, t_{29}, conditional on the saved CPS moments). Stars use unadjusted two-sided tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Tail effects are defined as in Table S.37. The BH family is shared with Table S.14.

the classical attenuation model predicts, with an implied conditional signal share of about 0.12 for wages and 0.21 for total earnings; under that model the rational extrapolant is several times larger than the quadratic one, so the extrapolation to zero noise is not identified from these draws and no standard error is attached. SIMEX is therefore reported as evidence of attenuation, not as a corrected estimate.

Third, a supplementary nine-test Benjamini–Hochberg family over the composition tests of the three zero-inclusive constructions gives $q = 0.033$ for wage entry, $q = 0.068$ for wage retention and $q = 0.59$ for the wage holder share. The protocol anticipated that the pooled construction, the more reliable of the two wage constructions, would give the tighter remeasurement interval; it is the pre-determined constructions whose intervals exclude zero, whereas the pooled construction's includes it (-0.06 to 1.90 points, 96.4 percent of draws positive). Collinearity helps explain this difference: composition and variance correlate across cells at -0.86 in the pooled construction (-0.81 for wages and -0.67 for total earnings over 2013–2017) because both are driven by the share

Table S.17. Temporary and persistent transitions: tail decomposition of the persistence fits

	(1) Composition	(2) Upper tail V^+	(3) Lower tail V^-	(4) Dispersion channel
A. Real log composition				
Entry followed by exit (temporary entry)	0.871^{***}	0.964^{***}	-0.778^{**}	0.093
	(0.276)	(0.315)	(0.338)	(0.175)
Entry that persists	0.091	0.172	-0.011	0.080
	(0.396)	(0.461)	(0.457)	(0.230)
Any entry	0.962^{**}	1.135^{**}	-0.789	0.173
	(0.441)	(0.441)	(0.528)	(0.206)
Exit followed by re-entry (temporary exit)	0.710	2.046	0.627	1.336^{***}
	(1.037)	(1.294)	(0.897)	(0.404)
Exit that persists	2.784^{*}	2.161	-3.408^{**}	-0.623
	(1.574)	(2.087)	(1.358)	(0.790)
Any exit	3.494^{*}	4.207^{*}	-2.781^{*}	0.713
	(1.747)	(2.199)	(1.463)	(0.661)
B. Nominal arc composition				
Entry followed by exit (temporary entry)	0.613^{**}	0.645^{**}	-0.581^{**}	0.032
	(0.257)	(0.263)	(0.281)	(0.091)
Entry that persists	0.450^{*}	0.544^{*}	-0.356	0.094
	(0.242)	(0.296)	(0.254)	(0.132)
Any entry	1.063^{***}	1.189^{***}	-0.937^{**}	0.126
	(0.300)	(0.326)	(0.345)	(0.151)
Exit followed by re-entry (temporary exit)	0.812	1.518^{*}	-0.105	0.707^{***}
	(0.720)	(0.836)	(0.675)	(0.240)
Exit that persists	0.550	0.596	-0.504	0.046
	(1.447)	(1.648)	(1.470)	(0.586)
Any exit	1.362	2.114	-0.609	0.753
	(1.600)	(1.736)	(1.622)	(0.511)

Notes: Retained persistence fits of Appendix R (11,133 transitions with an observed year after the outcome; 8,326 initial nonholders and 2,807 initial holders), decomposed into tail effects from their saved coefficient vectors and CR1 covariance matrices, evaluated at each block's survey-weighted mean earnings SD; no model is refitted. Temporary and persistent changes sum to the one-step change by construction. Coefficients are percentage points per fixed source SD of upside composition, with the original controls and destination weights; parentheses are industry–education CR1 standard errors ($G = 30$, t_{29}, conditional on the saved CPS moments). Stars use unadjusted two-sided tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Tail effects are defined as in Table S.37. The retained twelve-test persistence family of Appendix R applies; no new family is declared.

of histories that end in zero earnings, so the composition coefficient at fixed variance has a large conditional standard error.

Inference and complete families. The twelve tail-effect tests in Table S.37 form one Benjamini–Hochberg family. The destination tables use two further twelve-test families, the realized-earnings diagnostic uses a twelve-test family for the controlled tail effects and a six-test family for the composition interactions, the exit characterization uses a four-test family for the holder-block upper-tail differences by recent entry and position size, and the zero-inclusive module uses a six-test family for its composition coefficients and a supplementary nine-test family covering all three zero-inclusive

Table S.18. Size versus shape, dismissal risk, and education: entry design

	(1) Zero-inclusive wages	(2) Zero-inclusive total earnings	(3) Original real log	(4) Original nominal arc
A. Moments entered separately and jointly (percentage points of entry per source SD)				
Variance alone (no composition, no SD column)	−0.457	−0.269	−0.234	−0.081
	(0.584)	(0.571)	(0.308)	(0.390)
Composition alone (no variance, no SD column)	1.091**	0.843*	0.769**	0.650***
	(0.532)	(0.434)	(0.300)	(0.233)
Composition, full design	1.601***	1.042**	0.875***	0.695***
	(0.506)	(0.405)	(0.279)	(0.248)
Size effect at fixed composition (dispersion channel), full design	0.104	0.060	0.061	0.008
	(0.116)	(0.150)	(0.192)	(0.101)
B. The cell share of histories ending at zero wages (zero-inclusive constructions)				
Zero share alone (per cell SD of the share)	−1.163	−0.505		
	(0.963)	(0.760)		
Composition, full design plus zero share	2.420**	1.851***		
	(0.911)	(0.593)		
Zero share, full design plus zero share	2.630	1.828**		
	(2.030)	(0.767)		
Cross-cell correlation of zero share with composition	−0.85	−0.85		
Cross-cell correlation of zero share with variance	0.92	0.84		
C. Entry response by education (composition per source SD; zero-inclusive wages and original real log)				
Workers without a college degree	1.978***		1.093***	
	(0.477)		(0.246)	
College graduates	0.128		−0.244	
	(1.602)		(0.745)	
Difference, college minus non-college	−1.850		−1.337*	
	(1.795)		(0.738)	
Entry rate (%), non-college / college	5.4 / 13.9		5.4 / 13.9	
Transitions, non-college / college	16,503 / 10,916		16,503 / 10,916	

Notes: Weighted linear probability models of next-year stock-or-mutual-fund entry among the 27,419 baseline nonholders of Table 2, with its controls, fixed effects and weights; standard errors clustered by the 30 industry–education groups in parentheses, $t(29)$. Panel A, as Catherine, Sodini, and Zhang (2024) do for their co-moments: variance alone drops composition and the SD column; composition alone drops variance and the SD column; the size effect is half the sum of the two tail effects (Section S.2). Panel B: cell share of validated histories whose destination wage is zero, standardized across the 58 cells with Kish-capped weights, alone (with mean growth and its square, controls and fixed effects) and with the full design. Panel C: risk columns interacted with a college-degree indicator; the difference row is the interaction contrast. Design frozen in lift/precedent_engagement_design_2026-09-17.md (code195). *, **, ***: $p < 0.10, 0.05, 0.01$.

Table S.19. Exit by tenure, reporting screens, and the down-market year

	Exit rate (%)	Transitions	Persons
A. Exit hazard by tenure among baseline holders (outcome-year weights)			
Entered the year before (nonholder one year earlier)	46.6	1,116	
Held for two years	33.0	170	170
Held for three years or more	15.9	567	567
Held one year earlier, earlier status unobserved	22.1	1,215	1,215
All baseline holders	30.5	7,334	5,299
B. One-year exits inside the reporting screens of Section 4.8			
Full prospective sample: entrants, out again the next year	46.7	805	805
Full prospective sample: established holders who exit	20.8	2,602	2,025
Reported ownership and balance components: entrants, out again the next year	43.1	400	400
Reported ownership and balance components: established holders who exit	20.1	1,633	1,290
Self-reported interviews, stable records: entrants, out again the next year	41.4	214	214
Self-reported interviews, stable records: established holders who exit	17.4	1,145	913

	Recent entrants	Established holders	Difference
C. Upper-tail exit response of initial holders (percentage points of retention per composition SD)			
Original real log, all outcome years	-4.912^{**}	0.480	-5.392^{**}
	(1.844)	(1.404)	(2.324)
Original real log, excluding 2022 outcomes	-6.683^{***}	1.810	-8.493^{***}
	(2.130)	(1.837)	(2.765)
Zero-inclusive wages, all outcome years	-0.779	-0.944	0.165
	(3.958)	(2.648)	(5.347)
Zero-inclusive wages, excluding 2022 outcomes	2.694	-0.466	3.160
	(3.952)	(2.870)	(6.018)

Notes: Panel A: ownership histories reconstructed from the prospective person-year pairs (no conflicting records); exit is not holding stocks or mutual funds in the outcome year, outcome-year weights; the first row uses the prior-history definition of Section 4.6; holders whose status one year earlier is unobserved (4,572 rows) are included only in the last row. Panel B: one-year exits inside the saved samples of the two reporting screens of Table 7; entrants are baseline nonholders who hold in the outcome year and are observed one year later in the same screened sample. Panel C: the recent-entry interaction design of Table S.14 (15,535 transitions; 12,730 excluding 2022, the only outcome year with a negative calendar-year U.S. stock market return; the nominal-arc rows are in the replication files). Standard errors clustered by the 30 industry–education groups in parentheses; $t(29)$. Code195. *, **, ***: $p < 0.10, 0.05, 0.01$.

Table S.20. Wild-cluster bootstrap inference for the headline coefficients

	Estimate (SE)	Analytic p	WCR p	WCR p, Webb	WCU p	95% interval
A. Entry design (percentage points per composition SD; original cohort, 27,419 transitions)						
Zero-inclusive wages: composition	1.601 (0.506)	0.004	0.039	0.036	0.012	[0.39, 2.81]
Zero-inclusive wages: upper tail	1.705 (0.561)	0.005	–	–	0.009	[0.45, 2.95]
Zero-inclusive wages: lower tail	−1.498 (0.474)	0.004	–	–	0.016	[−2.70, −0.29]
Zero-inclusive total earnings: composition	1.042 (0.405)	0.015	0.080	0.078	0.026	[0.12, 1.97]
Zero-inclusive total earnings: upper tail	1.102 (0.461)	0.023	–	–	0.033	[0.07, 2.13]
Zero-inclusive total earnings: lower tail	−0.982 (0.401)	0.020	–	–	0.051	[−1.97, 0.00]
Original real log: composition	0.875 (0.279)	0.004	0.022	0.015	0.009	[0.25, 1.50]
Original real log: upper tail	0.936 (0.353)	0.013	–	–	0.020	[0.18, 1.69]
Original real log: lower tail	−0.814 (0.325)	0.018	–	–	0.047	[−1.62, −0.01]
Original nominal arc: composition	0.695 (0.248)	0.009	0.019	0.017	0.021	[0.12, 1.27]
Original nominal arc: upper tail	0.703 (0.274)	0.016	–	–	0.027	[0.08, 1.32]
Original nominal arc: lower tail	−0.686 (0.260)	0.013	–	–	0.032	[−1.31, −0.07]
B. Prospective ownership design (next-year ownership; composition by initial state)						
Zero-inclusive wages, nonholders (entry)	1.646 (0.599)	0.010	0.052	0.045	0.022	[0.21, 3.08]
Zero-inclusive wages, holders (retention)	−3.720 (1.708)	0.038	0.118	0.110	0.085	[−8.17, 0.73]
Original real log, nonholders (entry)	0.969 (0.302)	0.003	0.021	0.013	0.008	[0.26, 1.68]
Original real log, holders (retention)	−2.650 (0.952)	0.009	0.020	0.019	0.015	[−4.76, −0.54]

Notes: Cluster wild bootstrap over the 30 industry–education groups, 1,999 draws. WCR imposes the null by refitting without the composition column and resampling the restricted residuals with Rademacher (or Webb six-point) cluster weights; WCU resamples the unrestricted residuals and centres the bootstrap t at the estimate; the interval inverts the WCU distribution. Restricted bootstraps are defined for the composition coefficient only. Analytic p: CR1 $t(29)$. Dispersion-channel rows (bootstrap p 0.42–0.94) are in the replication files. Panel B: the prospective design of Table 5, next-year ownership. Code198.

Table S.21. Composition, skewness, and cyclical exposure: entry design

	(1) Zero-inclusive wages	(2) Zero-inclusive total earnings	(3) Original real log	(4) Original nominal arc
A. Composition and the third standardized moment of the same residuals (pp of entry per SD)				
Skewness alone (no composition, variance or SD column)	0.151	0.374	0.294	0.746***
	(0.330)	(0.343)	(0.433)	(0.247)
Composition, with skewness	1.320**	1.099	2.008***	−1.238
	(0.619)	(0.777)	(0.474)	(0.843)
Skewness, with composition	−0.267	−0.245	−1.371**	1.856**
	(0.354)	(0.542)	(0.556)	(0.832)
Cross-cell correlation of skewness with composition	0.05	0.62	0.90	0.95
Partial correlation given variance and SD	0.91	0.94	0.90	0.95
Split-half reliability, skewness / composition	0.39 / 0.73	0.39 / 0.62	0.29 / 0.28	0.21 / 0.24
B. Composition with cyclical-exposure controls (pp per source SD; columns (1) and (3))				
None (Table 2)	1.601***		0.875***	
	(0.506)		(0.279)	
Covid cell shock: 2020 mean wage growth	1.783***		0.905***	
	(0.528)		(0.277)	
PSID exposure of mean growth to the unemployment change	1.612***		0.989***	
	(0.479)		(0.290)	
PSID exposure of the lower partial variance to the unemployment change	1.701***		0.877***	
	(0.533)		(0.275)	
PSID exposure of composition to the unemployment change (cyclical skewness analogue)	1.662***		0.941***	
	(0.487)		(0.292)	
The three unemployment exposures jointly	2.104***		1.152***	
	(0.452)		(0.268)	

Notes: Panel A: cell skewness is the weighted third standardized moment of the age-adjusted residuals that define the partial variances, standardized across the 58 cells; the head-to-head rows omit the variance and SD columns because, conditional on them, skewness is nearly a linear function of composition (partial correlation row). Reliability: Spearman–Brown from the 30 household-half splits of Appendix S. Panel B: the Covid shock is the cell's mean zero-inclusive arc wage growth from 2019 to 2020 in the IPUMS links (the version differenced from the cell's raw 2013–2017 mean gives 1.97 for wages and is in the replication files); PSID exposures are cell-by-cell slopes of the biennial cell moments (1999–2019, heads 25–64) on the two-year change in the national unemployment rate; all standardized across cells and added to the retained entry design. Further specifications (zero-share shock, market-return exposures) are in the replication files. Standard errors clustered by the 30 industry–education groups in parentheses; $t(29)$. Codes 199 and 200. *, **, ***: $p < 0.10, 0.05, 0.01$.

Table S.22. Composition and conventional skewness with common dispersion controls

	(1) Zero-inclusive wages	(2) Zero-inclusive total earnings	(3) Original real log	(4) Original nominal arc
Composition without skewness	1.601^{***}	1.042^{**}	0.875^{***}	0.695^{***}
	(0.506)	(0.405)	(0.279)	(0.248)
Skewness without composition	0.912^{**}	0.710^{*}	0.390	0.784^{***}
	(0.414)	(0.362)	(0.487)	(0.262)
Composition with skewness	2.585	1.364	2.050^{***}	-1.166
	(1.670)	(1.385)	(0.500)	(0.862)
Skewness with composition	-0.726	-0.260	-1.452^{**}	1.826^{**}
	(1.195)	(1.103)	(0.644)	(0.851)
Variance and SD controls	Yes	Yes	Yes	Yes
Other baseline controls and FE	Yes	Yes	Yes	Yes
Nonholder transitions	27,419	27,419	27,419	27,419
Industry–education clusters	30	30	30	30

Notes: Existing weighted entry specifications on the common sample of Table 2, with outcome-year SIPP person weights, mean growth and its square, baseline household controls and fixed effects retained throughout. Each row pair reports its stated shape regressor; the last two pairs come from the same joint model. Composition coefficients are percentage points per fixed source SD of composition; skewness coefficients use their own fixed cross-cell SD. All rows retain both variance and SD. These specifications therefore differ from the comparison without dispersion controls in Table S.21, Panel A. The complete four-construction comparison is retained. Parentheses give CR1 standard errors clustered by 30 industry–education groups, conditional on the estimated CPS moments. Stars use unadjusted two-sided t_{29} tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. The table presents saved models, with no new estimation or test selection.

constructions; the persistence decomposition and the reliability and remeasurement exercises declare no new family. The level, stability, nested-restriction, and heterogeneity comparisons are reported in full. The replication files contain the pre-estimation design note and protocol, fitted designs, and sources for the reported estimates.

S.15 External validation in the PSID: details

Tables S.32 and S.33 report the parts of the PSID exercise of Section 4.7 that are summarized rather than tabulated in the main text: the agreement of the cell composition across the two surveys in both measurement windows, the pooled and recent-window two-year transitions for all ages and the within-household and within-sibling ownership estimates (Table S.32); holder retention by age band in both surveys and the recent-entrant split of the PSID holder block (Table S.33); Table S.35 preserves the complete grid of entry and ownership estimates by construction, age band and window in both surveys. The design note, the panel construction and every fit are in the replication files.

Table S.34 reports the implied upper- and lower-tail associations, standard errors, and tests for the survey comparisons. Both age bands and all three constructions are reported under their existing survey-specific specifications.

Table S.23. Risk tolerance, sorting, and the ownership benchmark

	(1)	(2)	(3)	(4)
	Tolerance score (1–4)	Most tolerant two categories		
A. 1996 PSID risk tolerance on the assigned cell's wage moments (3,296 heads; cell clusters)				
Composition (per source SD)	0.053	2.433		
	(0.066)	(2.832)		
Variance (per source SD)	0.160***	6.542***		
	(0.055)	(2.233)		
	Without categories	With categories	Most tolerant category	
B. PSID designs of Section 4.7 with tolerance categories (19,399 transitions; wages)				
Ownership level	0.604	0.563	4.932***	
	(1.159)	(1.217)	(1.697)	
Two-year entry (nonholders)	−1.073	−1.081	0.939	
	(0.740)	(0.745)	(1.021)	
Two-year retention (holders)	0.667	0.746	5.663***	
	(1.718)	(1.749)	(1.659)	
	SCF: stocks or stock funds	SCF: any equity incl. retirement accounts	SIPP: stocks or mutual funds	
C. Ownership rates, aged 25–64 (percent; SCF 2019 households versus SIPP baseline person-years)				
All	18.9	55.4	23.4	
Aged 25–44	17.3	54.6	22.7	
Aged 45–64	20.3	56.1	24.3	
College degree	33.8	75.7	33.7	
No college degree	10.1	43.4	12.7	

Notes: Panel A: the 1996 PSID hypothetical income-gamble sequence (Barsky, Juster, Kimball and Shapiro categories, least to most tolerant) for heads later observed in the 1999–2019 panel, regressed on the zero-inclusive wage composition and variance of their first 1999+ cell with age, age squared, sex and education controls; the score column reports score points per source SD, the indicator column percentage points. Panel B: the code181 designs with three tolerance-category indicators added; the third column is the most-tolerant indicator. Panel C: SCF 2019 summary extract, five implicates averaged, households with a head aged 25–64; SIPP rows are the prospective analysis sample of workers with assigned cells, baseline ownership of stocks or mutual funds, baseline weights. The public SIPP records retirement-account balances but not their asset composition, so the SCF's equity-inclusive rate has no SIPP counterpart. Codes 201 and 203. *, **, ***: $p < 0.10, 0.05, 0.01$.

S.16 Main tables of the original construction

Tables S.36 and S.37 are the retained entry and tail-effect tables of the original construction, real log and nominal arc growth on positive endpoints, on which Sections 4.5, 4.9 and 5 and the appendix diagnostics are built; Table 2 in the main text reports the zero-inclusive wage entry design, and Table S.38 in this appendix reports its implied marginal associations together with those of the prospective portfolio regressions. Benjamini–Hochberg q-values for the declared families appear in the table rows here.

Table S.24. Ownership levels in two further sources: SCF 2019 and the SIPP 2014 panel

	(1)	(2)	(3)	(4)
	Wages: composition	Wages: variance	Original: composition	Original: variance
A. SCF 2019, working heads aged 25–64 (pp per source SD; Rubin SE over five implicates)				
Directly held stocks or stock funds, with industry effects	3.09^{*}	8.93	0.37	-6.28
	(1.80)	(14.13)	(0.96)	(8.61)
Directly held stocks or stock funds, without industry effects	2.60	-19.31^{**}	-1.24	2.72
	(1.70)	(9.26)	(0.84)	(10.27)
Any equity, including retirement accounts, with industry effects	5.04^{**}	-37.94^{**}	1.36	-3.39
	(2.40)	(16.76)	(1.48)	(11.18)
Any equity, including retirement accounts, without industry effects	4.19^{*}	-29.78^{*}	-0.25	25.92^{***}
	(2.16)	(14.85)	(1.60)	(9.22)
Ownership rate (%); households; coarse cells	20.2 / 60.9	3,373	37	
	Aged 25–44	Aged 45–65	All	
B. SIPP 2014 panel, ownership of stocks or mutual funds, December 2013–2016 (composition per source SD)				
Zero-inclusive wages	1.80^{***}	1.02	-0.42	
	(0.49)	(1.25)	(0.58)	
Zero-inclusive total earnings	1.59^{***}	-1.63	-0.23	
	(0.33)	(1.10)	(0.49)	
Original real log	1.43^{**}	-0.12	-0.03	
	(0.62)	(0.55)	(0.62)	
Original nominal arc	0.97^{**}	0.37	0.03	
	(0.38)	(0.42)	(0.47)	
Ownership rate (%); person-years	11.1; 37,264	18.8; 36,903	14.8; 74,167	

Notes: Panel A: SCF 2019 summary extract and public file; coarse cells are the head's seven-way industry code (the summary extract's industry field has two working bins) by education by age band, 37 cells with CPS moments recomputed at that level; weighted linear probability models with age, age squared, sex, education, log income, signed log net worth and family-structure controls; standard errors clustered by coarse cell and combined across the five implicates; professional and business services are dropped because they straddle two SCF categories (mapping in the receipt). Panel B: the 2014 SIPP panel (reference years 2013–2016) built from the public monthly files with the project's cell definitions, controls (log income, log nonretirement financial assets, age, age squared, sex, unemployment months, education, industry, age-band and year effects), December person weights and CR1 by the 30 industry–education groups. Codes 209 and 210. *, **, ***: $p < 0.10, 0.05, 0.01$.

Table S.25. The 2014-panel entry hold-out and the quality of its ownership reports

	Aged 25–44	Aged 45–65	All	Retained 2019–2023, 25–44
A. SIPP 2014 panel, one-year entry 2013–2016 (pre-registered hold-out; composition per source SD)				
Zero-inclusive wages	−0.50	0.52	−0.08	2.65***
	(0.57)	(0.92)	(0.39)	(0.53)
Zero-inclusive total earnings	0.06	0.13	0.25	2.95***
	(0.51)	(0.88)	(0.34)	(0.35)
Original real log	0.47	−0.23	0.18	2.33***
	(0.46)	(0.41)	(0.32)	(0.48)
Original nominal arc	0.57*	0.54	0.35	0.93***
	(0.30)	(0.39)	(0.28)	(0.31)
Entry rate (%); transitions	5.5; 18,710	8.1; 18,048	6.7; 36,758	
Hold-out minus retained, wages 25–44	−3.14 (SE 0.78)			
PSID two-year entry, 25–44, 2013–2017 baselines, by construction in row order	2.90***	2.29***	1.50***	0.88***
	(0.79)	(0.61)	(0.49)	(0.22)
	2014 panel	2019–2024		
B. Ownership-report churn, adults aged 25–65 (percent)				
Ownership rate	13.0–14.3	23.4 (analysis sample)		
Holders who exit within a year	39.3	30.5		
Entrants out again the next year	54.1	46.7		
Holders of three years or more who exit	19.5	15.9		

Notes: Panel A: the 2014 SIPP panel entry design of Table S.24 (same cells, controls, weights and inference), pre-registered with the wage construction and ages 25–44 as the primary test; column (4) repeats the retained 2019–2023 design on the same band; the PSID row gives the two-year entry responses of Table S.33 for 2013–2017 baselines. Panel B compares the 2014 panel's ownership reports (December records, adults aged 25–65) with the 2019–2024 panels of Table S.19; the 2019–2024 ownership rate is that of the analysis sample. Codes 206 and 210. *, **, ***: $p < 0.10, 0.05, 0.01$.

Table S.26. Entry response after removing workers likely to hold equity compensation

	(1) Wages: composition	(2) Wages: upper / lower	(3) Original: composition	(4) Original: upper / lower
Full entry sample	1.60^{***}	1.70^{***} / -1.50^{***}	0.88^{***}	0.94^{**} / -0.81^{**}
	(0.51)	(0.56) / (0.47)	(0.28)	(0.35) / (0.33)
transitions; groups; entry rate (%)	27,419; 30; 8.8			
Dropping finance, information, professional services	0.90^{*}	1.21^{*} / -0.60	0.66^{*}	0.77^{*} / -0.55
	(0.50)	(0.63) / (0.44)	(0.33)	(0.39) / (0.36)
transitions; groups; entry rate (%)	21,393; 21; 7.7			
Dropping the top income quartile	1.13^{*}	1.02 / -1.24^{**}	0.84^{***}	0.65^{*} / -1.02^{***}
	(0.62)	(0.66) / (0.60)	(0.26)	(0.33) / (0.29)
transitions; groups; entry rate (%)	20,660; 30; 6.8			
Dropping college graduates in those industries	1.68^{***}	2.09^{***} / -1.28^{***}	0.82^{**}	1.02^{***} / -0.62^{*}
	(0.49)	(0.57) / (0.46)	(0.31)	(0.33) / (0.36)
transitions; groups; entry rate (%)	24,307; 27; 7.6			
Dropping the top financial-wealth quartile	1.10^{**}	0.89 / -1.30^{***}	0.77^{***}	0.52 / -1.02^{***}
	(0.46)	(0.54) / (0.39)	(0.22)	(0.38) / (0.22)
transitions; groups; entry rate (%)	20,125; 30; 6.4			
Dropping those industries and the top income quartile	-0.35	-0.54 / 0.16	0.42	0.05 / -0.80^{***}
	(0.81)	(0.88) / (0.77)	(0.25)	(0.36) / (0.24)
transitions; groups; entry rate (%)	16,653; 21; 6.2			

Notes: Replays of the retained entry design of Table 2 on subsamples. The equity-compensation industries are the project's information, finance–insurance–real estate and professional–business services groups (22.6 percent of weight, 9 of the 30 inference groups); quartiles are weighted quartiles of the baseline income and financial-asset controls. Tail effects per composition SD as in Table S.38. Standard errors clustered by the remaining industry–education groups in parentheses. Code207. *, **, ***: $p < 0.10, 0.05, 0.01$.

Table S.27. Magnitude benchmarks for the entry finding

	(1) Entry rate (%)	(2) Composition effect (pp)	(3) Relative effect (%)
A. Entry response relative to the local entry rate, by initial financial wealth (real log)			
25th percentile of financial assets (20th–30th percentile band)	6.4	1.97	31
Median (45th–55th percentile band)	7.3	1.61	22
75th percentile (70th–80th percentile band)	11.9	1.27	11
B. Cross-cell spread among initial nonholders (10th to 90th percentile of the assigned moment)			
	Spread (source SD)	Entry difference (pp)	Relative (%)
Upside composition (Table S.36 coefficient)	2.29	2.00	23
Total variance (linear variance-only coefficient, Table S.10)	1.87	-0.44	-5
C. Entry gradients in the prospective ownership regression (percentage points per standard deviation)			
	SD among nonholders	Effect (pp per SD)	Composition / gradient
Log nonretirement financial assets	2.18	3.25	0.30
Log income	1.85	1.04	0.93
Upside composition (fixed source SD)	1.00	0.97	1.00
D. Temporary share of the induced entry (persistence fits)			
		Temporary / any entry	
Composition coefficient		0.91	
Upper-tail effect		0.85	

Notes: No model is refitted. Panel A: destination-weighted next-year entry rates among initial nonholders in the percentile bands of log nonretirement financial assets around the evaluation points of Figure S.1, the retained composition effects at those points (Table S.7, ownership rows), and their ratio. Panel B: destination-weighted 10th and 90th percentiles of the assigned composition and variance z-scores across initial nonholders, the implied difference in predicted entry using the stated coefficient, and its ratio to the 8.82 percent entry rate of the original cohort. Panel C: nonholder-block coefficients of the retained prospective ownership regression (Table R.11) times the destination-weighted standard deviation of each regressor among initial nonholders. Panel D: the temporary-entry coefficient over the any-entry coefficient in the persistence fits (Table S.17); the persistent-minus-temporary difference test is imprecise ($p = 0.146$).

Table S.28. Split-half reliability and temporal stability of the cell measures

	(1) Histories	(2) Half-sample r	(3) Spearman–Brown	(4) Rank r	(5) Early vs. late r
Positive support, real log (main tables): composition	83,372	0.177	0.283	0.238	0.178
Positive support, real log (main tables): variance	83,372	0.509	0.670	0.527	0.456
Positive support, arc: composition	83,372	0.144	0.238	0.191	0.021
Positive support, arc: variance	83,372	0.637	0.777	0.663	0.666
Positive support, arc, wages: composition	83,372	0.132	0.215	0.167	-0.028
Positive support, arc, wages: variance	83,372	0.665	0.798	0.691	0.670
Zero-inclusive, arc, wages, 2013–2017: composition	88,862	0.579	0.731	0.583	0.523
Zero-inclusive, arc, wages, 2013–2017: variance	88,862	0.843	0.914	0.841	0.729
Zero-inclusive, arc, 2013–2017: composition	88,862	0.459	0.624	0.429	0.322
Zero-inclusive, arc, 2013–2017: variance	88,862	0.762	0.864	0.758	0.663
Zero-inclusive, arc, wages, 2013–2021: composition	161,093	0.747	0.854	0.752	0.845
Zero-inclusive, arc, wages, 2013–2021: variance	161,093	0.916	0.956	0.906	0.914

Notes: Each construction's cell moments are computed with the retained bridge conventions (group-specific cubic age profile removed; centered partial variances; 58 industry–education–age cells). Half-sample correlation: the correlation across cells between the moment computed on two random halves of households, averaged over 30 splits; each split's correlation is converted by the Spearman–Brown formula to the reliability of the full-sample moment and the results are averaged; rank correlation is the Spearman analogue. Early versus late: correlation between the moment computed on the first and second halves of the origin years (2013–2015 versus 2016–2017; 2013–2017 versus 2018–2021 for the pooled construction). Positive-support constructions use the 83,372 histories with positive earnings at both endpoints; zero-inclusive constructions add the 5,490 validated histories with zero destination wages (3,498 with zero total earnings) and, for the pooled construction, 72,231 histories linked for interview years 2019–2022.

Table S.29. Entry, retention, and share with reliable measures of earnings asymmetry

	(1) Positive support real log (main tables)	(2) Zero-inclusive wages 2013–2017	(3) Zero-inclusive total earnings 2013–2017	(4) Zero-inclusive wages 2013–2021
Split-half reliability of composition	0.28	0.73	0.62	0.85
A. Entry, original cohort (27,419 initial nonholders)				
Composition	0.875^{***}	1.601^{***}	1.042^{**}	1.233
	(0.279)	(0.506)	(0.405)	(0.770)
Upper tail V^+	0.936^{**}	1.705^{***}	1.102^{**}	1.311
	(0.353)	(0.561)	(0.461)	(0.865)
Lower tail V^-	-0.814^{**}	-1.498^{***}	-0.982^{**}	-1.156
	(0.325)	(0.474)	(0.401)	(0.691)
Dispersion channel	0.061	0.104	0.060	0.078
	(0.192)	(0.116)	(0.150)	(0.143)
B. Holder retention, prospective sample (7,334 initial holders)				
Composition	-2.650^{***}	-3.720^{**}	-2.522^{**}	-4.939^{**}
	(0.952)	(1.708)	(1.126)	(2.182)
Upper tail V^+	-3.704^{***}	-4.982^{**}	-3.750^{**}	-6.590^{**}
	(1.077)	(2.136)	(1.470)	(2.537)
Lower tail V^-	1.596	2.457^{*}	1.293	3.288^{*}
	(0.972)	(1.427)	(0.983)	(1.916)
Dispersion channel	-1.054^{**}	-1.262^{*}	-1.229^{**}	-1.651^{***}
	(0.383)	(0.619)	(0.543)	(0.543)
C. Initial-holder next-year share				
Composition	-1.621^{**}	-0.514	-0.602	-2.115^{*}
	(0.715)	(0.941)	(0.650)	(1.175)
Upper tail V^+	-2.046^{**}	-1.090	-1.243	-2.732^{*}
	(0.831)	(1.153)	(0.899)	(1.477)
Lower tail V^-	1.196	-0.062	-0.040	1.497
	(0.755)	(0.839)	(0.544)	(0.936)
Dispersion channel	-0.425	-0.576	-0.641^{*}	-0.617
	(0.344)	(0.364)	(0.360)	(0.386)

Notes: Column (1) reproduces the retained designs with the main tables' measure (Table S.36 column (1), Tables R.11 and R.1). Columns (2)–(4) replace the five earnings regressors by the same moments computed from nominal arc growth that keeps transitions to zero earnings: wages and total earnings on the retained 2013–2017 histories, which predate every portfolio observation, and wages on 2013–2021 histories, which overlap the portfolio years. Row 1: Spearman–Brown split-half reliability of the composition across the 58 cells (Table S.28). Same samples, weights, controls and clusters as the retained designs; every fit is independently replayed. Percentage points per fixed source SD of composition; CR1 standard errors clustered by 30 industry–education groups in parentheses (t_{29}, conditional on the CPS moments); $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Tail effects are marginal effects of one partial variance with the other fixed, scaled to one source SD of composition, and the dispersion channel is half their sum, as in Table S.37; q-values in Appendix S.

Table S.30. Remeasurement of the entry coefficient: household cluster bootstrap of the CPS histories

	(1) Point estimate	(2) Draw mean	(3) Draw SD	(4) 2.5th percentile	(5) 97.5th percentile	(6) Share of draws > 0
Positive support, real log (main tables)						
Composition	0.875	0.443	0.225	0.030	0.855	0.982
Upper tail V^+	0.936	0.430	0.283	-0.075	1.014	0.934
Lower tail V^-	-0.814	-0.455	0.248	-0.930	0.048	0.042
Dispersion channel	0.061	-0.013	0.142	-0.309	0.267	0.453
Zero-inclusive, arc, wages, 2013–2017						
Composition	1.601	0.846	0.348	0.125	1.511	0.994
Upper tail V^+	1.705	0.914	0.388	0.133	1.676	0.990
Lower tail V^-	-1.498	-0.777	0.326	-1.394	-0.146	0.008
Dispersion channel	0.104	0.068	0.086	-0.090	0.254	0.790
Zero-inclusive, arc, 2013–2017						
Composition	1.042	0.595	0.279	0.031	1.105	0.976
Upper tail V^+	1.102	0.648	0.305	0.039	1.212	0.976
Lower tail V^-	-0.982	-0.541	0.273	-1.054	0.002	0.028
Dispersion channel	0.060	0.054	0.077	-0.099	0.202	0.754
Zero-inclusive, arc, wages, 2013–2021						
Composition	1.233	0.889	0.485	-0.057	1.899	0.964
Upper tail V^+	1.311	0.893	0.545	-0.169	2.004	0.946
Lower tail V^-	-1.156	-0.886	0.436	-1.731	-0.019	0.022
Dispersion channel	0.078	0.004	0.093	-0.169	0.198	0.505

Notes: 499 draws per construction. In each draw the CPS households of the construction's histories are resampled with replacement, the cell moments are recomputed (age profile refitted) and standardized with the fixed cross-cell scales, and the original-cohort entry design (Table S.36) is refitted; the SIPP side is held fixed. Percentage points per fixed source SD of composition. The point estimate for the positive-support real log construction is the retained Table S.36 coefficient reproduced by the same code; the retained joint remeasurement of Section 4.9 gives the same direction for that construction. Draw means lie below the point estimates because each draw adds a second layer of sampling noise to the regressors; the percentile columns are descriptive ranges of the draws, not confidence intervals. Table S.31 traces this attenuation.

Table S.31. SIMEX attenuation curves of the entry coefficients under added CPS sampling noise

	(1) Point ($\zeta = 0$)	(2) $\zeta = 0.5$	(3) $\zeta = 1$	(4) $\zeta = 1.5$	(5) $\zeta = 2$	(6) SIMEX quadratic	(7) SIMEX linear
Positive support, real log (main tables)							
Composition	0.875	0.605	0.437	0.336	0.275	1.579	1.093
Upper tail V^+	0.936	0.608	0.426	0.306	0.246	1.775	1.178
Lower tail V^-	-0.814	-0.601	-0.448	-0.366	-0.305	-1.383	-1.009
Dispersion channel	0.061	0.004	-0.011	-0.030	-0.029	0.196	0.085
Zero-inclusive, arc, wages, 2013–2017							
Composition	1.601	1.115	0.850	0.687	0.580	2.817	1.955
Upper tail V^+	1.705	1.194	0.912	0.749	0.627	2.974	2.078
Lower tail V^-	-1.498	-1.037	-0.788	-0.625	-0.534	-2.660	-1.832
Dispersion channel	0.104	0.079	0.062	0.062	0.046	0.157	0.123
Zero-inclusive, arc, 2013–2017							
Composition	1.042	0.750	0.574	0.477	0.410	1.795	1.266
Upper tail V^+	1.102	0.802	0.623	0.526	0.451	1.865	1.333
Lower tail V^-	-0.982	-0.697	-0.525	-0.427	-0.369	-1.726	-1.199
Dispersion channel	0.060	0.052	0.049	0.049	0.041	0.069	0.067
Zero-inclusive, arc, wages, 2013–2021							
Composition	1.233	1.178	0.919	0.654	0.607	1.639	1.629
Upper tail V^+	1.311	1.204	0.929	0.649	0.597	1.836	1.731
Lower tail V^-	-1.156	-1.151	-0.910	-0.658	-0.617	-1.442	-1.526
Dispersion channel	0.078	0.027	0.009	-0.005	-0.010	0.197	0.102

Notes: Simulation–extrapolation (Cook and Stefanski, 1994) with bootstrap pseudo-errors. For each noise multiple ζ, 200 household cluster-bootstrap draws of the CPS histories give perturbed cell moments equal to the observed moments plus $\sqrt{\zeta}$ times the draw's deviation for composition, variance and mean growth (SD is the square root of the perturbed variance); the moments are standardized with the fixed cross-cell scales and the original-cohort entry design (Table S.36) is refitted; columns (2)–(5) report the mean over draws. Columns (6)–(7) extrapolate the means, with the point estimate at $\zeta = 0$, to $\zeta = -1$ (no sampling noise) by a quadratic and by a linear fit in ζ; the extrapolated values depend on the extrapolant and carry no standard error, so they document attenuation rather than provide corrected estimates. Percentage points per fixed source SD of composition.

Table S.32. PSID external validation: measure, windows, retention and identification

	(1) Original construction positive-support real log	(2) Zero-inclusive wages 2013–2017	(3) Zero-inclusive total earnings 2013–2017
A. Cell composition: PSID versus CPS (58 cells)			
Correlation, PSID 1999–2019	-0.19 ($p = 0.143$)	0.33 ($p = 0.011$)	0.27 ($p = 0.040$)
Correlation, PSID 2013–2019	-0.32 ($p = 0.016$)	0.33 ($p = 0.012$)	0.20 ($p = 0.124$)
Split-half reliability, PSID / CPS	0.36 / 0.28	0.68 / 0.73	0.77 / 0.62
Corrected for both reliabilities	-0.62	0.47	0.39
PSID 1999–2007 versus 2009–2019 cells	0.14	0.31	0.43
Variance: correlation with the CPS cells	0.47	0.69	0.70
B. PSID two-year transitions, all ages, CPS moments assigned (pp per source SD)			
Entry, baselines 1999–2017	0.082	-0.183	-0.494
	(0.312)	(0.386)	(0.357)
Retention, baselines 1999–2017	0.240	-1.447	-0.779
	(0.786)	(1.581)	(1.197)
Entry, baselines 2013–2017	-0.225	-0.189	-0.733
	(0.383)	(0.559)	(0.491)
Retention, baselines 2013–2017	1.725	-3.719	-1.999
	(1.944)	(5.051)	(3.997)
C. PSID baseline ownership, all ages (pp per source SD)			
Cross-section with controls	0.423	1.356^{*}	0.826
	(0.524)	(0.663)	(0.506)
Within household (person fixed effects)	0.033	0.981^{*}	0.714^{*}
	(0.235)	(0.497)	(0.359)
Within sibling group (sibling fixed effects)	0.440	-0.573	0.306
	(0.387)	(0.937)	(0.613)

Notes: Complements Tables 6 and S.33. Panel A: the paper's moments computed on PSID reference persons' biennial labor-income growth (total labor income for columns (1) and (3), wages for column (2); group-specific cubic age adjustment; 50,013 growth pairs) against the CPS moments of the manuscript. Panel B: the fully interacted prospective design fit to 45,840 two-year transitions (38,162 nonholder and 7,678 holder baselines, 2,405 entries, 2,670 exits) and to the 2013–2017 baselines alone (14,756 transitions). Panel C: baseline ownership with household controls, wave and industry effects, with person fixed effects, or with fixed effects for the 1,563 groups of reference persons sharing a mother or father identifier (18,910 rows). Percentage points per fixed source SD of composition; CR1 standard errors clustered by 30 industry–education groups in parentheses (t_{29}, conditional on the CPS moments); $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Design frozen before estimation; the window and age splits were specified after the pooled result and are exploratory.

Table S.33. PSID external validation: holder retention by age band and by recency of entry

	(1) Original construction positive-support real log	(2) Zero-inclusive wages 2013–2017	(3) Zero-inclusive total earnings 2013–2017
A. Holder retention by age band (pp per source SD)			
SIPP, ages 25–44 (annual)	-5.460^{**}	-6.880^{**}	-7.274^{**}
	(1.978)	(3.275)	(3.358)
PSID, ages 25–44 (2009–2017 baselines)	9.349^{**}	11.222^{***}	7.856^{**}
	(4.060)	(3.878)	(3.126)
SIPP, ages 45–65 (annual)	-2.741	-9.818^{*}	-9.384^{***}
	(2.166)	(5.208)	(2.429)
PSID, ages 45–65 (2009–2017 baselines)	-3.428^{**}	-17.976^{***}	-7.804^{*}
	(1.664)	(3.838)	(3.970)
B. PSID upper-tail retention effect by recency of entry, baselines 1999–2017			
Established holders (owners at the previous wave)	1.451	-1.265	-1.425
	(0.984)	(2.450)	(1.297)
Recent entrants (nonowners at the previous wave)	1.956	-1.850	-0.303
	(1.874)	(3.834)	(2.706)

Notes: Complements Tables 6 and S.32. Panel A: the fully interacted prospective design within age bands, in the PSID (two-year transitions, 2009–2017 baselines, CPS moments assigned) and in the retained SIPP design of Table R.11 (annual transitions). Panel B: PSID holder block interacted with an indicator for entry since the previous wave, upper-tail effect evaluated for each group; the SIPP counterparts are 0.48 among established holders and -4.91 among recent entrants with the original real-log construction (Table S.14). Percentage points per fixed source SD of composition; parentheses give CR1 standard errors clustered by industry–education group. Panel A contains 29 groups and uses t_{28} tests; Panel B contains 30 groups and uses t_{29} tests. Stars use unadjusted two-sided tests conditional on the CPS moments: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Exploratory; the age split was specified after the pooled result.

Table S.34. Upper- and lower-tail entry associations across surveys

	(1) Zero-inclusive wages	(2) Zero-inclusive total earnings	(3) Original real log
Ages 25–44			
SIPP: upper tail	3.186^{***}	3.342^{***}	3.578^{***}
	(0.681)	(0.452)	(0.510)
SIPP: lower tail	-2.110^{***}	-2.550^{***}	-1.083^{*}
	(0.424)	(0.297)	(0.633)
PSID: upper tail	4.585^{***}	2.810^{***}	0.808
	(0.992)	(0.605)	(0.617)
PSID: lower tail	-2.543^{***}	-1.458^{***}	-1.807^{***}
	(0.511)	(0.382)	(0.574)
Ages 45–65			
SIPP: upper tail	2.340^{**}	3.397^{***}	-0.042
	(0.889)	(1.037)	(0.771)
SIPP: lower tail	-2.013^{***}	-2.529^{***}	-0.889^{**}
	(0.704)	(0.746)	(0.431)
PSID: upper tail	0.968	0.285	-0.477
	(1.412)	(1.306)	(0.958)
PSID: lower tail	0.084	0.334	-0.405
	(1.243)	(0.895)	(0.474)
SIPP observations, ages 25–44	13,639	13,639	13,639
SIPP observations, ages 45–65	13,780	13,780	13,780
PSID observations, ages 25–44	14,306	14,306	14,306
PSID observations, ages 45–65	10,109	10,109	10,109
Industry–education clusters	29	29	29

Notes: Saved tail effects from the same age-band entry equations as Table 6. SIPP transitions are annual, with baselines 2019–2023; PSID transitions are biennial, with baselines 2009–2017. Outcomes, weighting and controls retain each survey's original specification. Each partial variance is increased with the other fixed, scaled to that construction's fixed source SD of composition and evaluated at the equation's weighted mean earnings SD. All three constructions and both age bands are shown; the original full comparison remains in Table S.35. Parentheses give CR1 standard errors clustered by the 29 industry–education groups. Stars use saved unadjusted two-sided t_{28} tests, conditional on CPS moments: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. These exploratory age and window comparisons followed the pooled PSID results; unequal significance does not establish an age or survey difference.

Table S.35. Entry and ownership by age band, construction and survey

	(1) Zero-inclusive wages	(2) Zero-inclusive total earnings	(3) Original real log	(4) Original nominal arc
A. SIPP entry, Table 2 design refit within age bands (annual transitions, 2019–2024)				
All ages 25–65	1.601^{***}	1.042^{**}	0.875^{***}	0.695^{***}
	(0.506)	(0.405)	(0.279)	(0.248)
Ages 25–44	2.648^{***}	2.946^{***}	2.331^{***}	0.927^{***}
	(0.526)	(0.350)	(0.483)	(0.311)
Ages 45–65	2.177^{***}	2.963^{***}	0.424	1.380^{***}
	(0.772)	(0.880)	(0.530)	(0.338)
B. PSID two-year entry, ages 25–44, by baseline window (CPS moments assigned)				
Baselines 1999–2007	1.154	−0.195	−0.107	−0.334
	(1.767)	(1.255)	(0.819)	(0.391)
Baselines 2009–2017	3.564^{***}	2.134^{***}	1.308^{***}	0.631^{**}
	(0.728)	(0.448)	(0.377)	(0.287)
Baselines 2013–2017	2.903^{***}	2.290^{***}	1.497^{***}	0.880^{***}
	(0.793)	(0.606)	(0.488)	(0.220)
C. PSID two-year entry, ages 45–65, by baseline window				
Baselines 2009–2017	0.442	−0.024	−0.036	0.626
	(1.302)	(1.077)	(0.634)	(0.598)
Baselines 2013–2017	−0.985	-1.758^{*}	−0.835	0.059
	(1.185)	(0.989)	(0.501)	(0.470)
D. Baseline stock market ownership by age band (levels, pp per source SD)				
SIPP, ages 25–44	3.109^{*}	2.791^{**}	1.803	1.815^{***}
	(1.710)	(1.024)	(1.110)	(0.618)
SIPP, ages 45–65	3.083	−1.388	0.112	1.808^{*}
	(2.425)	(2.258)	(1.293)	(0.922)
PSID, ages 25–44	6.317^{***}	4.893^{***}	2.026^{***}	1.993^{***}
	(0.809)	(0.615)	(0.515)	(0.434)
PSID, ages 45–65	1.801	−0.047	−0.238	1.570^{**}
	(2.163)	(1.624)	(0.829)	(0.721)

Notes: Complete construction-by-age-band grid behind Table 6 and Section 4.5, preserved for reference. Panel A refits the retained SIPP entry design of Table 2 within age bands with each construction's fixed source scales. Panels B and C fit the fully interacted prospective design to PSID two-year transitions in imputed direct stock ownership with the CPS moments assigned, by baseline window and age band; the windows and bands were specified after the pooled result and are exploratory. Panel D regresses baseline ownership on the assigned moments with household controls, wave and industry effects; SIPP ownership includes stocks and mutual funds, while PSID ownership excludes IRAs. Percentage points per fixed source SD of composition; parentheses give CR1 standard errors clustered by industry–education group. Age-band estimates contain 29 groups and use t_{28} tests; the all-ages SIPP row contains 30 groups and uses t_{29} tests. Stars use unadjusted two-sided tests conditional on the CPS moments: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table S.36. Age-adjusted earnings composition and subsequent stock market participation

Earnings Growth	(1) Total Real log	(2) Wages Real log	(3) Total Nominal arc	(4) Wages Nominal arc
Upside composition	0.875^{***}	0.782^{**}	0.695^{***}	0.740^{***}
	(0.279)	(0.307)	(0.248)	(0.263)
Conditional-SD magnitude (pp)	0.573	0.524	0.534	0.582
Group-specific cubic age adjustment	Yes	Yes	Yes	Yes
Risk and baseline controls	Yes	Yes	Yes	Yes
Industry, education, age-band, year FE	Yes	Yes	Yes	Yes
Nonholder transitions	27,419	27,419	27,419	27,419
Observed entry rate (%)	8.82	8.82	8.82	8.82
Assignment cells / inference groups	58 / 30	58 / 30	58 / 30	58 / 30

Notes: Weighted linear probability models of next-year stock-or-mutual-fund entry among baseline nonholders, using outcome-year SIPP person weights. Baselines are 2019–2023; outcomes are 2020–2024. The CPS source comprises 83,372 validated histories, ages 25–65, with positive wage and total-earnings endpoints; origins are 2013–2017 and destinations end in 2018. Total earnings add wages, net business earnings and net farm earnings. Survey-weighted group-specific origin-age cubics are removed before constructing centered moments in 58 industry–education–age cells. Composition is upper minus lower partial variance. Coefficients are percentage points per fixed pre-adjustment source SD for each construction; the conditional-SD row instead uses residual composition after controls. Variance, SD, mean growth and its square are controlled; Section 3.3 specifies household controls and FE. Stars use unadjusted two-sided zero-coefficient tests with CR1 standard errors clustered by 30 industry–education groups and t_{29}: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Inference conditions on CPS moments. Table J.1 reports 58-cell CR1/t_{57} sensitivity, every raw/adjusted model, and joint CPS–SIPP remeasurement.

Table S.37. Which tail moves participation: effects of the upper and lower partial variances

	(1) Composition	(2) Upper tail V^+	(3) Lower tail V^-	(4) Dispersion channel
A. Real log composition				
Entry, original cohort	0.875^{***}	0.936^{**}	-0.814^{**}	0.061
	(0.279)	(0.353)	(0.325)	(0.192)
BH q, 12 tail tests		0.034	0.034	
Entry, prospective sample	0.969^{***}	0.993^{**}	-0.944^{**}	0.025
	(0.302)	(0.377)	(0.346)	(0.199)
Holder retention	-2.650^{***}	-3.704^{***}	1.596	-1.054^{**}
	(0.952)	(1.077)	(0.972)	(0.383)
BH q, 12 tail tests		0.021	0.165	
Initial-holder share	-1.621^{**}	-2.046^{**}	1.196	-0.425
	(0.715)	(0.831)	(0.755)	(0.344)
BH q, 12 tail tests		0.034	0.165	
B. Nominal arc composition				
Entry, original cohort	0.695^{***}	0.703^{**}	-0.686^{**}	0.008
	(0.248)	(0.274)	(0.260)	(0.101)
BH q, 12 tail tests		0.034	0.034	
Entry, prospective sample	0.770^{***}	0.777^{**}	-0.762^{**}	0.008
	(0.262)	(0.288)	(0.279)	(0.107)
Holder retention	-1.848^{**}	-2.552^{***}	1.145	-0.704^{**}
	(0.862)	(0.865)	(0.935)	(0.263)
BH q, 12 tail tests		0.034	0.252	
Initial-holder share	-0.409	-0.906	-0.088	-0.497^{**}
	(0.642)	(0.646)	(0.701)	(0.208)
BH q, 12 tail tests		0.206	0.901	
Observations	27,419 / 31,569			
Industry–education clusters	30			

Notes: Each row re-expresses a published regression: entry on the original cohort (Table S.36), next-year ownership and next-year share on the prospective sample (Tables R.11 and R.1). Because the regressions contain both composition $C = V^+ - V^-$ and variance $V = V^+ + V^-$, the effect of each partial variance holding the other fixed is a linear combination of the saved coefficients and covariance. The upper-tail effect is the marginal effect of the upper partial variance V^+ with V^- fixed, scaled to one source SD of composition; the lower-tail effect is the marginal effect of V^- with V^+ fixed, scaled the same way. Half their difference is the composition coefficient and half their sum is the dispersion channel, the effect of total variance at fixed composition, evaluated at the state block's survey-weighted mean earnings SD. Holder retention is next-year ownership among initial holders; exit is its negative. Coefficients are percentage points per fixed source SD of upside composition, with the original controls and destination weights; parentheses are industry–education CR1 standard errors ($G = 30$, t_{29}, conditional on the saved CPS moments). Stars use unadjusted two-sided tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. The BH family contains the twelve tail tests for original entry, holder retention and holder share under both constructions.

Table S.38. Implied marginal associations of upper and lower partial variance

	(1) Composition	(2) Upper tail V^+	(3) Lower tail V^-	(4) Local variance association
A. Zero-inclusive wage composition (headline construction)				
Entry, original cohort	1.601^{***}	1.705^{***}	-1.498^{***}	0.104
	(0.506)	(0.561)	(0.474)	(0.116)
Entry, prospective sample	1.646^{**}	1.733^{**}	-1.558^{***}	0.088
	(0.599)	(0.656)	(0.562)	(0.121)
Holder retention	-3.720^{**}	-4.982^{**}	2.457^{*}	-1.262^{*}
	(1.708)	(2.136)	(1.427)	(0.619)
Initial-holder share	-0.514	-1.090	-0.062	-0.576
	(0.941)	(1.153)	(0.839)	(0.364)
B. Original real log composition				
Entry, original cohort	0.875^{***}	0.936^{**}	-0.814^{**}	0.061
	(0.279)	(0.353)	(0.325)	(0.192)
Entry, prospective sample	0.969^{***}	0.993^{**}	-0.944^{**}	0.025
	(0.302)	(0.377)	(0.346)	(0.199)
Holder retention	-2.650^{***}	-3.704^{***}	1.596	-1.054^{**}
	(0.952)	(1.077)	(0.972)	(0.383)
Initial-holder share	-1.621^{**}	-2.046^{**}	1.196	-0.425
	(0.715)	(0.831)	(0.755)	(0.344)
Observations	27,419 / 31,569			
Industry–education clusters	30			

Notes: Columns (2)–(4) are local marginal associations, not separately estimated coefficients. Each row re-expresses one regression: entry on the original cohort (Table 2), and next-year ownership and next-year share on the prospective sample (Tables 5 and R.11). Because the regressions contain both composition $C = V^+ - V^-$ and variance $V = V^+ + V^-$, the effect of each partial variance holding the other fixed is a linear combination of the saved coefficients and covariance. The upper-tail effect is the marginal effect of the upper partial variance V^+ with V^- fixed, scaled to one source SD of composition; the lower-tail effect is the marginal effect of V^- with V^+ fixed, scaled the same way. Half their difference is the composition coefficient and half their sum is the dispersion channel, the effect of total variance at fixed composition, evaluated at the state block's survey-weighted mean earnings SD. Holder retention is next-year ownership among initial holders; exit is its negative. Panel A uses the zero-inclusive wage construction, Panel B the original real-log construction, each with its own fixed source scales, the original controls and destination weights; parentheses are industry–education CR1 standard errors ($G = 30$, t_{29}, conditional on the CPS moments). Stars use unadjusted two-sided tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$; the declared test families and their Benjamini–Hochberg q-values are in Appendix S. Unrestricted wild-cluster bootstrap p-values for the wage entry tail effects are 0.009 (upper), 0.016 (lower) and 0.417 (dispersion channel); for holder retention (composition) the restricted bootstrap p is 0.118 with wages and 0.020 with the original construction (Appendix Table S.20).

T Realized Earnings and Cross-Panel Comparisons

T.1 Entry conditional on realized annual earnings

Table T.1 examines whether controlling for observed earnings changes the entry association for the headline wage construction. It uses the same 23,261 nonholder transitions as the joint-outcome decomposition below, preserving the earnings moments, source scales, weights, controls and fixed effects. Column (1) reproduces the matched-sample baseline. For real annual personal earnings E_t in 2024 dollars, define $a_t = \operatorname{asinh}(E_t/10{,}000)$ and $g = a_{t+1} - a_t$. Column (2) adds a_t and g; column (3) also adds $\max(g, 0)$, allowing the gain and loss slopes to differ. The fixed transformation retains the 113 negative and 189 zero destination-earnings observations. No subsequent financial-asset control enters.

Table T.1. Stock market participation conditional on realized earnings

	(1) Baseline	(2) Annual earnings	(3) Gain/loss slopes
Upside composition	1.518**	1.574**	1.569**
	(0.587)	(0.586)	(0.576)
Change from column (1)	—	0.056*	0.051
		(0.032)	(0.043)
Baseline controls and effects	Yes	Yes	Yes
Annual baseline earnings	No	Yes	Yes
Realized earnings change	No	Yes	Yes
Separate gain/loss slopes	No	No	Yes
Outcome mean (percent)	9.68	9.68	9.68
Observations	23,261	23,261	23,261
Industry–education clusters	30	30	30

Notes: Weighted linear probability models of subsequent stock-or-mutual-fund entry among the same baseline nonholders. Coefficients are percentage points per fixed source standard deviation of zero-inclusive wage composition. All columns retain variance, its square root, mean growth and its square, the headline household controls and fixed effects, and outcome-year person weights. Annual personal labor earnings are twelve-month sums of TPEARN, deflated to 2024 dollars. Baseline annual earnings must be positive; zero and negative destination earnings remain. Let $a_t = \operatorname{asinh}(E_t/10{,}000)$ and $g = a_{t+1} - a_t$. Column (2) adds a_t and g; column (3) also adds $\max(g, 0)$. The coefficient changes retain cross-model covariance from paired cluster influences. Column (3) minus column (2) is -0.005 point (SE 0.026, $p = 0.838$). CR1 standard errors clustered by industry–education are in parentheses. Stars use unadjusted two-sided t_{29} tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. The two added composition tests and three direct-change tests have separate BH adjustments in the replication output. This post hoc diagnostic conditions on realized earnings; it does not identify mediation or a direct causal effect.

The composition coefficient is 1.518 points in the baseline, 1.574 with the annual earnings controls, and 1.569 with separate gain and loss slopes. The respective changes from baseline are 0.056 point (SE 0.032, $p = 0.089$) and 0.051 point (SE 0.043, $p = 0.244$). Their standard errors use paired industry–education cluster influences, preserving covariance between specifications. The third-minus-second difference is -0.005 point (SE 0.026, $p = 0.838$). These changes concern the entire added earnings block, including baseline annual earnings, rather than growth alone.

This two-specification diagnostic was fixed in writing before estimating it, after the earlier joint-outcome results had been examined. The two added composition tests form one Benjamini–Hochberg family (both $q = 0.012$); the three coefficient-change tests form a separate family. All three specifications and their differences are reported. The entry association remains after controlling for observed annual earnings. Because conditioning on realized earnings can remove a channel or induce selection, the comparison leaves resource mechanisms, unobserved sorting, and perceived risk unresolved.

T.2 Entry with substantial earnings gains

Table T.2 keeps the headline zero-inclusive wage moments, their fixed CPS scales, and the main entry regression's controls and industry–education clustering. The entry sample is intersected with the prospective allocation sample and with complete annual personal earnings at both dates. Of the 27,419 headline transitions, 23,701 belong to both portfolio samples; 23,261 also have twelve monthly earnings records at each date and positive baseline earnings. The entry coefficient is 1.656 points in the portfolio-sample intersection and 1.518 points in the final matched sample, compared with 1.601 points in the headline sample.

Table T.2. Stock market participation and realized earnings gains

	(1) Total entry	(2) Entry with gain	(3) Other entry	(4) (2)−(3)
Upside composition	1.518**	0.546***	0.972*	-0.427
	(0.587)	(0.176)	(0.543)	(0.555)
Outcome mean (percent)	9.68	2.68	7.00	-4.32
Observations	23,261	23,261	23,261	23,261
Industry–education clusters	30	30	30	30
Baseline controls and effects	Yes	Yes	Yes	Yes

Notes: Joint outcomes among initial nonholders with twelve observed earnings months at both dates and positive baseline annual earnings. A substantial gain is at least 25 percent growth in real annual personal labor earnings (TPEARN), deflated by annual CPI-U. Columns (2) and (3) retain all nonholders, not selected gain groups; their outcomes and coefficients sum to column (1). Column (4) uses their full covariance. Coefficients are percentage points per fixed source SD of zero-inclusive wage composition, conditional on variance, SD, mean growth and its square and the headline entry controls and effects. Outcome-year person weights. CR1 standard errors clustered by industry–education appear in parentheses; stars use unadjusted two-sided t_{29} tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. The headline sample has 27,419 transitions (coefficient 1.601); the prospective nonholder intersection has 23,701 (coefficient 1.656). This matched-sample joint-outcome decomposition is not causal mediation and does not order earnings and purchases within the year. The complete three-test mechanism family is retained in the replication files.

Annual earnings sum the SIPP's monthly personal labor earnings, not just earnings from the first-listed job. Each calendar-year total is deflated by the annual CPI-U. A substantial gain is real earnings growth of at least 25 percent. Entry with such a gain and other entry are mutually exclusive joint outcomes defined for every baseline nonholder. Nonentrants have zero for both outcomes; the regressions do not condition on having realized a gain. There are 2,365 entry events, of which 633 coincide with a substantial gain and 1,732 do not. The two component coefficients add exactly to

total entry; the reported difference uses their joint covariance. No subsequent financial-asset control enters this exercise.

The three-test family consists of gain-entry, other-entry, and their difference. Their conventional p-values are 0.004, 0.084, and 0.448; Benjamini–Hochberg q-values are 0.013, 0.126, and 0.448. This exploratory decomposition was specified before estimation. The gain-entry component is positive, but its difference from other entry is imprecise and it does not identify a causal earnings channel.

The gain-entry coefficient is 0.546 point (SE 0.176), against a 2.68 percent frequency of that event. It accounts for 36 percent of the matched entry coefficient; other entry contributes 0.972 point (SE 0.543). Neither outcome holds resources fixed: the second includes smaller earnings gains as well as unchanged or declining earnings. Moreover, the identity $\Pr(E \cap G \mid C, X) = \Pr(E \mid C, X) \Pr(G \mid E, C, X)$ shows why a positive joint association need not mean that composition changes the earnings gains of entrants. Here E denotes entry, G a substantial gain, C composition, and X the controls. This decomposition neither establishes a distinct risk channel nor shows that realized wealth explains away composition. Annual earnings and portfolio transitions cover the same interval, so it also does not establish their within-year ordering.

T.3 Earnings before and after first observed entry

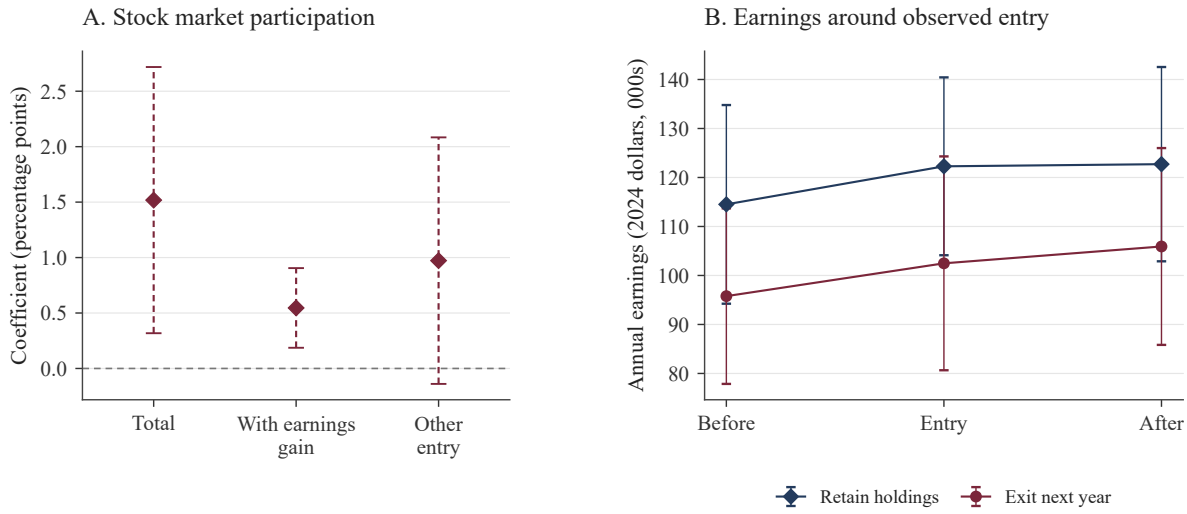
The earnings profiles select each person's first observed entry in the matched decomposition sample before imposing follow-up availability. There are 2,333 such entrants, of whom 828 have subsequent ownership observed and 802 also have a complete earnings profile. Retainers hold stocks or mutual funds at the next annual observation; exiters do not. These profiles therefore describe a selected balanced cohort, not every entrant in the main sample. They use the entry-year person weight, the same people at all three dates, and real earnings expressed in 2024 dollars.

Table T.3. Earnings changes around stock market participation

	(1) Retain	(2) Exit	(3) Exit minus retain
Entry year: $E_t - E_{t-1}$	7.768*	6.697	-1.071
	(4.314)	(5.456)	(8.004)
Following year: $E_{t+1} - E_t$	0.438	3.440	3.001
	(3.790)	(4.404)	(5.202)
Entrants in comparison	429	373	802
Industry–education clusters	30	30	30

Notes: Weighted mean changes in annual personal labor earnings (TPEARN), in thousands of 2024 dollars, deflated by annual CPI-U. Date t is the first observed transition from no stock or mutual-fund holdings to positive holdings; columns (1) and (2) separate entrants who retain holdings at $t + 1$ from those who exit. Column (3) is the direct difference, with the covariance between group means retained. All means and contrasts use the same two-group weighted regression on 802 entrants with complete earnings and ownership histories at $t - 1$, t, and $t + 1$; the counts show the entrants contributing to each comparison. The first observed entry is selected before screening for follow-up availability. Entry years are 2020–2023 and entry ages 25–64. Entry-year person weights are used at every date. These selected-entrant profiles describe earnings changes and are not causal estimates or a mediation analysis. Parentheses contain CR1 standard errors clustered by baseline industry–education. Stars use unadjusted two-sided t_{29} tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Both direct change contrasts and their multiplicity-adjusted results are retained in the replication files.

Figure T.1. Stock market participation and realized earnings



Notes: Panel A decomposes entry into entry with at least 25 percent growth in real annual personal labor earnings and all other entry. Both joint outcomes retain all 23,261 matched initial nonholder transitions; their coefficients sum to the total-entry coefficient. Estimates are percentage points per fixed CPS standard deviation of zero-inclusive wage composition, with the controls and inference in Table T.2. Panel B plots weighted mean annual personal labor earnings (TPEARN), in thousands of 2024 dollars, around the first observed entry. The 802 entrants with complete three-date histories comprise 429 who retain holdings in the following year and 373 who exit. Entry years are 2020–2023 and entry ages 25–64. Entry-year person weights are used at every date. Bars are pointwise 95% CR1/t_{29} confidence intervals with baseline industry–education clustering. Table T.3 reports the changes and direct group contrasts. The earnings profiles are descriptive, not causal responses or mediation estimates.

Both groups experience mean earnings gains at entry. In the following year, earnings rise by $438 among retainers and $3,440 among exiters. Neither change nor their difference is precisely estimated. The observed means show no earnings reversal among exiters. The profiles describe levels rather than innovations, and neither within-year ordering nor the earnings paths of entrants lost to follow-up are observed.

T.4 Common definitions across the SIPP panels

The common-definition comparisons in Table T.4 use the same wage moments, based on earnings origins in 2013–2017 and destinations through 2018, and ages 25–44 in both panels. These source years precede the later panel's portfolios but overlap the earlier panel's transitions. The comparisons preserve the earlier panel's original holdout result and examine four definitions applied symmetrically: annual records with gross wealth, annual records with a liquid-asset control, ownership answers or positive balances, and December worker snapshots. In each panel, the regression includes the same earnings moments, log income and assets, unemployment proxy, demographic controls and separate industry, education and year effects.

Table T.4. Earnings composition and entry under common SIPP definitions

	(1) Annual	(2) Alternative assets	(3) Ownership answers	(4) December
2014 panel	-0.153	-0.249	-0.225	0.296
	(0.603)	(0.595)	(0.592)	(0.612)
Observations	17,810	17,810	17,775	13,220
Industry–education clusters	29	29	29	29
Later panel	2.732***	2.739***	2.285***	2.681***
	(0.539)	(0.553)	(0.528)	(0.696)
Observations	13,584	13,584	13,491	10,210
Industry–education clusters	29	29	29	29
Later minus earlier	2.885***	2.988***	2.510***	2.384***
	(0.568)	(0.585)	(0.511)	(0.710)
Common controls and effects	Yes	Yes	Yes	Yes

Notes: Coefficients are percentage points per fixed CPS standard deviation of zero-inclusive wage composition, conditional on variance, its square root, and mean earnings growth and its square. Baseline nonholders aged 25–44 are followed for one year. Baseline years are 2013–2015 in the 2014 panel and 2019–2023 in the later panel. Each column applies identical definitions to both panels. Controls are baseline income, assets, age and age squared, sex, the nonpositive job-1-earnings indicator, and industry, education and outcome-year effects. Column (1) uses annual worker records, positive stock/fund balances to define ownership, and gross assets as the asset control. Columns (2)–(4) each change one element of column (1): column (2) uses $\log(1 + \max(A, 0))$, where A is the annual mean of `TVAL_BANK` + `TVAL_STMF` + `TVAL_ESAV`; column (3) also treats affirmative stock/fund ownership answers as ownership; column (4) uses December records. Annual definitions aggregate eligible worker-month records and use outcome-year mean person weights; column (4) uses December weights. Assignment cells at both dates must belong to the common set of 58 retained cells. Parentheses contain CR1 standard errors clustered by industry–education. Difference standard errors use aligned group influence vectors to retain cross-panel covariance. All reported tests use 29 clusters and two-sided t_{28} inference, conditional on estimated CPS moments. Stars use unadjusted p-values: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. These symmetric comparisons are an exploratory follow-up to the original 2014-panel holdout test.

The common annual construction aggregates eligible monthly worker records in the same way in both panels. Ownership is a positive stock-and-mutual-fund balance in at least one eligible month, with an observed nonnegative balance required. Income and gross assets use eligible-month means; survey weights use the next year's eligible-month mean. Industry and education are modal categories, breaking ties with the earliest record. Both endpoints must match the fixed CPS cells. The unemployment proxy is any nonpositive job-1 earnings observation, with missing earnings not coded as zero. The alternative definitions change only the stated components, using the same rule in both panels.

The primary common-definition samples contain 17,810 earlier-panel transitions and 13,584 later-panel transitions, with entry rates of 5.56 and 9.12 percent. The later-minus-earlier estimates use paired industry–education cluster influences, retaining covariance from the groups shared by both panels, with t_{28} inference. All four period contrasts remain significant. The twelve-test composition family comprises the eight panel coefficients and four period differences; its largest adjusted q-value among the differences is below 0.01. These comparisons rule out the tested coding differences as a sufficient explanation of the entry discrepancy. They do not identify a calendar effect or make the populations and reporting processes identical.

T.5 Comparing populations with similar starting resources

The common definitions leave open whether the entry coefficients differ because the panels contain workers with different starting resources. I reweight the two primary common-definition samples toward an overlapping population, using baseline information alone. The samples, earnings moments, outcome controls and source units remain unchanged. Any change in the coefficient gap reflects the jointly balanced baseline characteristics, not the contribution of financial resources alone. This comparison was specified after observing the period discrepancy, before estimating the new weights or entry regressions.

Following Li, Morgan, and Zaslavsky (2018), I estimate a weighted logit for membership in the later panel, normalizing each period's survey weights to sum to one-half. Let $e(Z)$ be this probability given baseline characteristics Z. Earlier-panel records receive the multiplier $e(Z)$ and later-panel records receive $1 - e(Z)$, each applied to the original survey weight. The resulting population emphasizes workers whose observed characteristics are represented in both periods; it is not the full pooled population. No observations are trimmed.

The logit includes industry–education group indicators, age and its square, sex, the unemployment proxy, and three resource measures: monthly earnings from the first-listed job (job 1), gross assets, and liquid assets. Resources are expressed in 2024 dollars and transformed as $\operatorname{asinh}(E/1{,}000)$ for monthly earnings and $\operatorname{asinh}(A/10{,}000)$ for each asset measure. The specification includes their squares, pairwise products and zero-value indicators, together with interactions of composition with each transformed resource and with age. Monthly earnings undo the one-dollar addition used in the original income control before deflation. The weighted-logit score equations equate the two periods' weighted means of the included functions; they do not equate the entire joint distributions. Neither future earnings nor future assets enter the weights.

The weighting retains all 29 groups and closely balances all 49 included functions. Kish effective sample sizes fall from 12,885 to 12,043 in the earlier panel and from 9,554 to 9,136 in the later panel, retaining 93.5 and 95.6 percent, respectively. The maximum distance between the two weighted liquid-asset distribution functions falls from 0.166 to 0.032; the largest remaining distance across age and the three resources is 0.035. Thus the comparison does not depend on a small subset of the original samples.

The reweighted coefficients are -0.028 point in the earlier panel and 2.595 points in the later panel. Their 2.623-point difference is only 0.262 point smaller than the original 2.885-point gap. The change is imprecisely estimated (SE 0.294, $p = 0.381$): balancing the observed baseline characteristics does not account for the period discrepancy. This comparison does not distinguish remaining reporting differences from unobserved population differences or changes in the relationship between earnings environments and entry.

Inference for this comparison uses a paired delete-one-group jackknife over the 29 shared industry–education groups (MacKinnon, Nielsen, and Webb, 2023). Each deletion removes the same group from both periods, renormalizes the survey weights, refits the logit and overlap weights, and reestimates the original and reweighted entry equations. The joint covariance of the four coefficients therefore

Table T.5. Balancing starting characteristics across the SIPP panels

	(1) Original weights	(2) Overlap weights	(3) (2)−(1)
Earlier panel	-0.153	-0.028	0.125
	(1.765)	(1.934)	(0.246)
Later panel	2.732*	2.595*	-0.137
	(1.470)	(1.494)	(0.201)
Later minus earlier	2.885**	2.623*	-0.262
	(1.293)	(1.431)	(0.294)
Earlier observations	17,810	17,810	—
Later observations	13,584	13,584	—
Shared industry–education groups	29	29	29
Earlier Kish effective observations	12,885	12,043	—
Later Kish effective observations	9,554	9,136	—

Notes: Coefficients report the percentage-point association between a one-standard-deviation increase in zero-inclusive wage composition and subsequent entry; the CPS scale is held fixed. Both cohorts use the common annual definitions of Table T.4, initial nonholders aged 25–44, and unchanged risk moments, nominal household controls and fixed effects. Earlier baselines are 2013–2015; later baselines are 2019–2023. Column (2) multiplies the original survey weights by $e(Z)$ for earlier observations and $1 - e(Z)$ for later observations, where $e(Z)$ is a pooled period logit with equal initial mass for each period. Its 49 baseline resource, demographic and group functions are specified in Appendix T.5. No observation is trimmed and no later resource or entry outcome enters this weight model. The largest weighted resource/age CDF distance after weighting is 0.035. The target is the equal-period-mixture overlap population, not the complete pooled population. Every displayed standard error uses the same full-procedure paired cluster jackknife: each of the 29 groups is deleted from both cohorts, and the period logit, weights and four regressions are refit. Covariance is $(28/29)$ times the centered deletion cross-products; full-sample point estimates are retained. Stars use unadjusted two-sided t_{28} tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. The primary quantity is the bottom-right gap change. BH q-values for the earlier overlap slope, later overlap slope, overlap gap and gap change are 0.988, 0.187, 0.187, 0.508, respectively. The historical CR1 standard errors for the original earlier, later and gap estimates are 0.603, 0.539 and 0.568, respectively; they are not used for this table's stars. The original failed holdout is unchanged. This post hoc comparison does not identify a causal resource effect.

retains dependence across periods and specifications, including that from estimated weights. All new tests use t_{28} critical values. The original point estimates reproduce Table T.4; their standard errors here use the same jackknife as the reweighted estimates, rather than the earlier table's CR1 estimator. The principal comparison tests the change in the later-minus-earlier coefficient gap.

T.6 A quantitative benchmark with endogenous saving

The analytical example holds saving fixed to isolate the participation threshold. This benchmark allows saving and the stock share to be chosen jointly. Households live from ages 25 to 84, work through age 64, and retire with deterministic income of 0.7 times mean working income. They have CRRA utility with coefficient 5, discount factor 0.96, zero initial financial wealth, and no borrowing or short sales. The safe gross return is 1.02; the stock return has mean 1.06 and standard deviation 0.18, independently of income, represented by five-point normal quadrature. The recurring participation cost is 0.002 times mean working income. All remaining wealth is consumed at the final age. These

are illustrative parameters fixed before computing portfolio outcomes, not estimates fitted to the SIPP.

The income experiment uses the shape of age-adjusted CPS dollar wage changes from the same 88,862 pre-portfolio histories. Both endpoints are deflated to 2018 dollars before computing changes and removing group-specific cubic age profiles. I discretize the pooled residual distribution with 31 weighted quantile means, standardize it, and symmetrize its support. Minimum-relative-entropy reweighting produces two reflected probability laws with mean zero and variance one, whose composition-to-variance ratios are -0.158678 and $+0.158678$. Their difference is one cross-cell standard deviation of the dollar-change composition ratio. The laws also have identical even moments. This comparison is not measured in the empirical regression's arc-composition units.

Working income is $y = 1 + 0.15z$, where z follows the standardized law. The 15 percent income-level standard deviation is an illustrative scale motivated by the transitory-risk magnitude in Gomes and Michaelides (2005), not an estimated conversion of dollar-change variance or an equivalence to their log-income process. The resulting support is 0.466–1.534 times mean working income. Quantile discretization retains 52.5 percent of the original residual variance before standardization; the experiment preserves the moments of the resulting model laws, not the full empirical distribution or literal zero-income states. The baseline income process is independent over time. Two fixed comparisons introduce persistence through a 0.5 probability of retaining the current income state, or remove the participation cost.

Table T.6 reports the complete comparison. In the refined baseline, upside orientation lowers average saving at ages 25–44 from 0.918 to 0.871 times annual mean income, a 5.14 percent decline, and raises the lifetime consumption equivalent by 0.414 percent. Participation is nearly universal under both laws, however, and entry among the small remaining nonholder population falls rather than rises. The persistence and zero-cost comparisons change the entry sign. None is a quantitative explanation of the empirical 1.6-point coefficient: the model does not reproduce limited participation, and its composition units differ from the regression's. Endogenous precautionary saving can therefore alter the entry response obtained with fixed resources.

The primary cash, saving and stock-share grids contain 801, 801 and 31 points; the fixed refinement uses 1,601, 1,601 and 61. Refinement changes the young participation contrast by 0.095 percentage point and the entry contrast by 0.423 point, with maximum relative lifetime-value change of 0.0382 percent. Budget feasibility, probability conservation, terminal consumption and forward–backward value consistency hold. Both persistence cases exceed the predeclared upper-saving-bound mass tolerance by small amounts, indicated in the table; they are retained with that numerical qualification. Consumption equivalents scale every date-and-state consumption allocation under the downside law to match utility under the comparison law. They are model welfare comparisons, not estimated treatment effects or compensated wealth transfers.

Table T.6. An illustrative fixed-moment composition counterfactual

Calibration	Income law	Ownership (%)	Entry (%)	Cond. share (%)	Saving (income units)	V_{25}	CE (%)
Primary, coarse	Downside	98.73	93.56	100.00	0.917	-6.8869	0.000
	Symmetric	98.65	92.01	100.00	0.896	-6.8322	0.199
	Upside	98.72	91.05	100.00	0.870	-6.7740	0.414
Primary, refined	Downside	98.68	93.30	100.00	0.918	-6.8842	0.000
	Upside	98.58	90.37	100.00	0.871	-6.7715	0.414
Persistence $\rho = 0.5$	Downside†	98.70	50.26	99.99	1.175	-7.4999	0.000
	Upside†	98.51	53.56	99.98	1.068	-7.2089	0.994
No participation cost	Downside	99.19	95.98	100.00	0.919	-6.8330	0.000
	Upside	99.37	97.60	100.00	0.873	-6.7201	0.417

Notes: The model has endogenous consumption and saving, CRRA coefficient 5, discount factor 0.96, and a recurring participation cost of 0.002 annual-income units. Upside and downside income laws have identical means, variances, and even centered moments; their composition-to-variance ratios differ by one cross-cell standard deviation of age-adjusted CPS dollar wage changes. This is not the paper's arc-composition standard deviation. The symmetric law uses the same empirical support. All laws have mean working income one and an externally set income-level standard deviation of 0.15. This scale is illustrative, not an estimated conversion of wage-change variance. Ownership, entry and conditional share summarize ages 25–44; initial age-25 entry from imposed zero holdings is excluded. Saving is post-consumption financial saving, before returns, divided by mean annual working income. V_{25} is expected lifetime utility before the initial income draw. CE is the constant-consumption increase that equates downside utility to the reported law's utility, within each calibration and grid. The model is not fitted to portfolio moments and nearly saturates participation. † The persistence cases exceed the frozen upper-saving-bound mass tolerance of 10^{-8} (actual maxima 2.43×10^{-8} and 3.24×10^{-8}); they are retained with this numerical flag. No significance tests are applied.

U Earnings Opportunities and Sustained Market Access

U.1 Joint entry and duration outcomes

The duration extension follows initial nonholders across three or four annual ownership observations. Write $D_0 = 0$ for baseline nonownership and D_1 for ownership at the potential entry date. With one follow-up, entry is $E = D_1$, sustained entry is $S = D_1 D_2$, and interrupted entry is $T = E - S$. With two follow-ups, $S = D_1 D_2 D_3$ requires ownership at both later dates; an interruption followed by reentry remains in T. Nonentrants have zero for both joint outcomes. Thus $S + T = E$ observation by observation. Separate regressions use exactly the same rows, weights and design matrix, so their coefficients and cluster influences obey the same identity.

The one-follow-up wage sample contains 9,236 initial-nonholder transitions and 873 entries; 452 have ownership at the next annual date. The two-follow-up sample contains 3,254 transitions and 337 entries; 128 retain ownership at both dates. These are different complete-follow-up populations, not alternative horizons observed for every baseline nonholder. Missing observations are not exits. Worker-sample eligibility, as well as survey response, determines whether a later ownership observation enters the annual history. Table U.8 shows that eligible follow-up is positively related to upside composition.

The estimated joint probabilities also permit a direct comparison of conditional survival. Let $p_E = x'\widehat{\beta}_E$ and $p_S = x'\widehat{\beta}_S$ at a stated covariate profile x, with direction vector v. The local derivative of fitted conditional survival is

$$g(v, x) = \frac{v'\widehat{\beta}_S}{p_E} - \frac{p_S \, v'\widehat{\beta}_E}{p_E^2}.$$

Its uncertainty uses the full joint cluster covariance of both fitted models, including uncertainty in p_E and p_S. This is a local derivative of the ratio of fitted joint probabilities, not the ratio of the two coefficients or an average individual derivative. No fitted probability is clipped. The reported mean and asset profiles have admissible probabilities; individual linear predictions need not lie in the unit interval.

The extension was specified as an exploratory family before its new estimates were examined. Tables U.1–U.5 report all nine designs and all four economic directions. The two-follow-up specification has a positive sustained-entry association. The difference from interrupted entry and the direct survival gradient are too imprecise to establish shorter participation spells. Entry estimates also weaken under the reporting-component screen, and follow-up eligibility is selected.

U.2 Earnings paths and pre-entry financial resources

The seven resource specifications use the same 802 first observed entrants with complete annual earnings before entry, at entry and at the next observation. All include pre-entry annual earnings and nonretirement financial assets. The asset variable is measured before entry; no future financial assets

are conditioned on. Tables U.9–U.13 retain all 37 reported slopes and contrasts, the complete nine-test focal family, paired changes in the composition coefficient and the raw earnings-path comparison.

A substantial gain raises real annual personal earnings by at least 25 percent at entry. A sustained gain keeps the subsequent income level at least 25 percent above the original baseline; it does not require another 25-percent growth realization. Of 238 substantial gains, 162 are sustained and 76 are not. The latter category includes 51 full reversals to or below baseline. The adjusted retention difference between sustained and nonsustained gains is -0.006 percentage point (SE 4.326), while the composition–asset interaction is 0.511 point (SE 4.377). These imprecise estimates leave open whether earnings reversals or low pre-entry financial assets explain participation duration, including economically meaningful resource effects.

U.3 What tail composition does not determine about duration

The threshold example distinguishes an earnings opportunity from the persistence of the resources it generates. Holding the stationary income distribution fixed, let a renewal process retain the current state with probability ρ and redraw it with probability $1-\rho$. If only the upper state permits marginal households to participate and its stationary probability is q, entry from the lower state is $(1-\rho)q$, whereas continuation from the upper state is $\rho+(1-\rho)q$. More persistence changes duration without changing the stationary tail composition. Table U.16 reports the complete four-case example. It is a feasibility argument, not an estimate of earnings persistence or a universal right-tail comparative static.

The endogenous-saving benchmark in Appendix T reinforces this distinction. Table U.14 follows entrants under all nine previously solved policy cases. In the refined primary case, one-year survival is 97.478 percent under the upside law and 96.079 percent under the downside law. Evaluating the same policies at earnings-only cash reverses the ordering to 88.581 and 92.842 percent. Carried wealth can sustain participation after an entry-year opportunity; the earnings-only calculation is a policy evaluation, not a newly optimized economy. Even that calculation predicts high participation, so saving alone does not explain the benchmark's distance from the empirical entry margin.

Table U.15 reports the corresponding earnings and financial resources. Modeled exiters experience large earnings declines yet retain assets worth many annual participation fees. Their exits reflect the consumption–portfolio tradeoff, not literal fee insolvency. The benchmark still predicts nearly universal participation and a negative upside entry contrast. Sustained access depends on resource persistence and carried wealth as well as tail orientation.

Table U.1. Entry and duration: 1 of 5

	(1) Entry	(2) Sustained entry	(3) Interrupted entry	(4) Sustained − interrupted	(5) Survival gradient
Wage construction, one follow-up ($N = 9,236$)					
Upside composition	2.086***	0.978*	1.108**	-0.131	-0.959
	(0.697)	(0.501)	(0.512)	(0.735)	(4.036)
Upper partial variance	2.517***	1.037*	1.480**	-0.443	-2.699
	(0.808)	(0.530)	(0.613)	(0.812)	(4.453)
Lower partial variance	-1.655**	-0.919*	-0.737	-0.182	-0.781
	(0.648)	(0.511)	(0.459)	(0.723)	(3.974)
Dispersion component	0.431*	0.059	0.372**	-0.313	-1.740
	(0.224)	(0.141)	(0.175)	(0.226)	(1.231)
Total-earnings construction, one follow-up ($N = 9,236$)					
Upside composition	1.294**	0.451	0.842*	-0.391	-2.268
	(0.580)	(0.461)	(0.421)	(0.665)	(3.623)
Upper partial variance	1.632**	0.480	1.152**	-0.671	-3.824
	(0.598)	(0.414)	(0.483)	(0.672)	(3.671)
Lower partial variance	-0.956	-0.423	-0.533	0.110	0.713
	(0.644)	(0.544)	(0.409)	(0.715)	(3.884)
Dispersion component	0.338	0.029	0.309*	-0.280	-1.555
	(0.223)	(0.147)	(0.152)	(0.199)	(1.075)

Notes: All quantities are percentage points on their fixed CPS scales. Columns (1)–(4) report joint-outcome slopes; column (5) reports the local conditional-survival derivative at the weighted mean design. Composition holds variance fixed. Each partial-variance direction raises that tail with the other fixed and therefore changes total variance; both are scaled to the composition standard deviation. Dispersion is half their sum. All controls, weights and samples are identical across joint outcomes within a panel. The reported-components screen requires reported portfolio components at all three dates, within the reporting-flag support. The 2022 exclusion removes first follow-ups in 2022; it is not a market-state interaction. The asset specifications add the baseline nonretirement-financial-asset measure, with all five risk-moment interactions in the final panel. Standard errors in parentheses use CR1 clustering by 30 industry–education groups, conditional on the assigned CPS moments. Stars use unadjusted two-sided t_{29} tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. The contrasts and gradients use full joint covariance. The nine mean-profile composition survival tests and the high-minus-low asset-profile test form a ten-test BH family; all have $q = 0.959$.

Table U.2. Entry and duration: 2 of 5

	(1) Entry	(2) Sustained entry	(3) Interrupted entry	(4) Sustained − interrupted	(5) Survival gradient
Positive-support real-log construction ($N = 9,236$)					
Upside composition	0.905*	0.147	0.758***	-0.611	-3.410
	(0.472)	(0.341)	(0.256)	(0.377)	(2.039)
Upper partial variance	1.221**	0.285	0.936***	-0.652	-3.666
	(0.467)	(0.389)	(0.284)	(0.495)	(2.691)
Lower partial variance	-0.590	-0.009	-0.580*	0.571	3.154
	(0.565)	(0.390)	(0.325)	(0.442)	(2.379)
Dispersion component	0.316	0.138	0.178	-0.040	-0.256
	(0.215)	(0.187)	(0.166)	(0.281)	(1.514)
Wage construction, two follow-ups ($N = 3,254$)					
Upside composition	3.315**	1.894**	1.421	0.473	7.296
	(1.430)	(0.786)	(1.053)	(1.187)	(5.674)
Upper partial variance	3.953**	1.983**	1.970*	0.014	5.990
	(1.592)	(0.927)	(1.138)	(1.333)	(6.550)
Lower partial variance	-2.676*	-1.805**	-0.872	-0.933	-8.601
	(1.328)	(0.703)	(1.011)	(1.127)	(5.266)
Dispersion component	0.638*	0.089	0.549**	-0.460	-1.305
	(0.325)	(0.243)	(0.226)	(0.338)	(1.769)

Notes: All quantities are percentage points on their fixed CPS scales. Columns (1)–(4) report joint-outcome slopes; column (5) reports the local conditional-survival derivative at the weighted mean design. Composition holds variance fixed. Each partial-variance direction raises that tail with the other fixed and therefore changes total variance; both are scaled to the composition standard deviation. Dispersion is half their sum. All controls, weights and samples are identical across joint outcomes within a panel. The reported-components screen requires reported portfolio components at all three dates, within the reporting-flag support. The 2022 exclusion removes first follow-ups in 2022; it is not a market-state interaction. The asset specifications add the baseline nonretirement-financial-asset measure, with all five risk-moment interactions in the final panel. Standard errors in parentheses use CR1 clustering by 30 industry–education groups, conditional on the assigned CPS moments. Stars use unadjusted two-sided t_{29} tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. The contrasts and gradients use full joint covariance. The nine mean-profile composition survival tests and the high-minus-low asset-profile test form a ten-test BH family; all have $q = 0.959$.

Table U.3. Entry and duration: 3 of 5

	(1) Entry	(2) Sustained entry	(3) Interrupted entry	(4) Sustained − interrupted	(5) Survival gradient
Wage construction, reporting-flag support ($N = 7,799$)					
Upside composition	2.002***	1.005*	0.998*	0.007	-0.589
	(0.654)	(0.563)	(0.512)	(0.854)	(4.273)
Upper partial variance	2.351***	1.093*	1.258**	-0.165	-1.543
	(0.759)	(0.610)	(0.598)	(0.940)	(4.697)
Lower partial variance	-1.654**	-0.916	-0.738	-0.178	-0.366
	(0.618)	(0.568)	(0.469)	(0.837)	(4.182)
Dispersion component	0.349	0.089	0.260	-0.172	-0.955
	(0.225)	(0.175)	(0.162)	(0.251)	(1.230)
Wage construction, reported components ($N = 6,242$)					
Upside composition	0.921	0.359	0.562	-0.203	-2.578
	(0.665)	(0.673)	(0.556)	(1.041)	(8.234)
Upper partial variance	0.993	0.220	0.773	-0.553	-5.433
	(0.758)	(0.772)	(0.614)	(1.171)	(9.196)
Lower partial variance	-0.849	-0.498	-0.351	-0.147	-0.276
	(0.643)	(0.622)	(0.532)	(0.962)	(7.653)
Dispersion component	0.072	-0.139	0.211	-0.350	-2.854
	(0.226)	(0.194)	(0.142)	(0.254)	(1.942)

Notes: All quantities are percentage points on their fixed CPS scales. Columns (1)–(4) report joint-outcome slopes; column (5) reports the local conditional-survival derivative at the weighted mean design. Composition holds variance fixed. Each partial-variance direction raises that tail with the other fixed and therefore changes total variance; both are scaled to the composition standard deviation. Dispersion is half their sum. All controls, weights and samples are identical across joint outcomes within a panel. The reported-components screen requires reported portfolio components at all three dates, within the reporting-flag support. The 2022 exclusion removes first follow-ups in 2022; it is not a market-state interaction. The asset specifications add the baseline nonretirement-financial-asset measure, with all five risk-moment interactions in the final panel. Standard errors in parentheses use CR1 clustering by 30 industry–education groups, conditional on the assigned CPS moments. Stars use unadjusted two-sided t_{29} tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. The contrasts and gradients use full joint covariance. The nine mean-profile composition survival tests and the high-minus-low asset-profile test form a ten-test BH family; all have $q = 0.959$.

Table U.4. Entry and duration: 4 of 5

	(1) Entry	(2) Sustained entry	(3) Interrupted entry	(4) Sustained − interrupted	(5) Survival gradient
Wage construction, excluding 2022 follow-up ($N = 6,868$)					
Upside composition	1.632**	1.228*	0.404	0.824	4.404
	(0.676)	(0.628)	(0.453)	(0.860)	(4.921)
Upper partial variance	2.022**	1.391**	0.631	0.760	3.996
	(0.745)	(0.647)	(0.551)	(0.943)	(5.363)
Lower partial variance	-1.242*	-1.065	-0.177	-0.888	-4.813
	(0.668)	(0.661)	(0.401)	(0.866)	(4.952)
Dispersion component	0.390*	0.163	0.227	-0.064	-0.409
	(0.209)	(0.185)	(0.165)	(0.282)	(1.559)
Wage construction, baseline-asset control ($N = 8,145$)					
Upside composition	1.969***	0.890*	1.079*	-0.188	-1.235
	(0.695)	(0.516)	(0.592)	(0.867)	(4.427)
Upper partial variance	2.322***	0.964*	1.358*	-0.394	-2.315
	(0.796)	(0.560)	(0.670)	(0.944)	(4.801)
Lower partial variance	-1.616**	-0.817	-0.799	-0.017	0.154
	(0.648)	(0.526)	(0.552)	(0.862)	(4.398)
Dispersion component	0.353	0.074	0.279*	-0.206	-1.081
	(0.210)	(0.169)	(0.161)	(0.255)	(1.266)

Notes: All quantities are percentage points on their fixed CPS scales. Columns (1)–(4) report joint-outcome slopes; column (5) reports the local conditional-survival derivative at the weighted mean design. Composition holds variance fixed. Each partial-variance direction raises that tail with the other fixed and therefore changes total variance; both are scaled to the composition standard deviation. Dispersion is half their sum. All controls, weights and samples are identical across joint outcomes within a panel. The reported-components screen requires reported portfolio components at all three dates, within the reporting-flag support. The 2022 exclusion removes first follow-ups in 2022; it is not a market-state interaction. The asset specifications add the baseline nonretirement-financial-asset measure, with all five risk-moment interactions in the final panel. Standard errors in parentheses use CR1 clustering by 30 industry–education groups, conditional on the assigned CPS moments. Stars use unadjusted two-sided t_{29} tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. The contrasts and gradients use full joint covariance. The nine mean-profile composition survival tests and the high-minus-low asset-profile test form a ten-test BH family; all have $q = 0.959$.

Table U.5. Entry and duration: 5 of 5

	(1) Entry	(2) Sustained entry	(3) Interrupted entry	(4) Sustained − interrupted	(5) Survival gradient
Wage construction, baseline-asset interactions ($N = 8,145$)					
Upside composition	2.095***	1.017*	1.078*	-0.061	-0.618
	(0.666)	(0.521)	(0.580)	(0.877)	(4.491)
Upper partial variance	2.430***	1.103*	1.327*	-0.224	-1.481
	(0.784)	(0.581)	(0.654)	(0.957)	(4.875)
Lower partial variance	-1.760***	-0.931*	-0.829	-0.102	-0.246
	(0.604)	(0.512)	(0.543)	(0.866)	(4.431)
Dispersion component	0.335	0.086	0.249	-0.163	-0.864
	(0.213)	(0.169)	(0.159)	(0.250)	(1.239)

Notes: All quantities are percentage points on their fixed CPS scales. Columns (1)–(4) report joint-outcome slopes; column (5) reports the local conditional-survival derivative at the weighted mean design. Composition holds variance fixed. Each partial-variance direction raises that tail with the other fixed and therefore changes total variance; both are scaled to the composition standard deviation. Dispersion is half their sum. All controls, weights and samples are identical across joint outcomes within a panel. The reported-components screen requires reported portfolio components at all three dates, within the reporting-flag support. The 2022 exclusion removes first follow-ups in 2022; it is not a market-state interaction. The asset specifications add the baseline nonretirement-financial-asset measure, with all five risk-moment interactions in the final panel. Standard errors in parentheses use CR1 clustering by 30 industry–education groups, conditional on the assigned CPS moments. Stars use unadjusted two-sided t_{29} tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. The contrasts and gradients use full joint covariance. The nine mean-profile composition survival tests and the high-minus-low asset-profile test form a ten-test BH family; all have $q = 0.959$.

Table U.6. Baseline financial assets and participation duration

	(1) Entry	(2) Sustained entry	(3) Interrupted entry	(4) Sustained − interrupted	(5) Survival gradient
A. Lower asset quartile ($1,233)					
Upside composition	1.766*	0.834	0.932	-0.098	-0.767
	(1.006)	(0.712)	(0.748)	(1.059)	(6.643)
Upper partial variance	2.134*	0.993	1.141	-0.148	-1.114
	(1.230)	(0.860)	(0.888)	(1.243)	(7.784)
Lower partial variance	-1.398	-0.675	-0.723	0.047	0.420
	(0.833)	(0.617)	(0.643)	(0.946)	(5.930)
Dispersion component	0.368	0.159	0.209	-0.050	-0.347
	(0.303)	(0.230)	(0.204)	(0.312)	(1.937)
B. Upper asset quartile ($21,973)					
Upside composition	2.297***	1.130*	1.168*	-0.038	-0.467
	(0.713)	(0.556)	(0.653)	(0.981)	(4.591)
Upper partial variance	2.612***	1.171*	1.441*	-0.270	-1.569
	(0.813)	(0.613)	(0.720)	(1.062)	(4.929)
Lower partial variance	-1.983***	-1.088*	-0.895	-0.193	-0.634
	(0.661)	(0.545)	(0.619)	(0.961)	(4.515)
Dispersion component	0.314	0.041	0.273*	-0.232	-1.102
	(0.200)	(0.165)	(0.154)	(0.249)	(1.126)

Notes: The interacted wage specification uses 8,145 initial-nonholder transitions. Pre-entry nonretirement financial assets are in 2024 dollars, transformed as $\operatorname{asinh}(A/10{,}000)$ and standardized on all 23,701 matched baseline observations before follow-up restrictions. These assets include deposits and other nonretirement assets, not only immediately liquid cash. Each profile fixes assets at the indicated baseline quartile, holds other controls at common weighted means, and rebuilds all five moment interactions. Partial-tail directions, local gradients and units are as in Tables U.1–U.5. Standard errors in parentheses use CR1 clustering by 30 industry–education groups, conditional on the assigned CPS moments. Stars use unadjusted two-sided t_{29} tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table U.7. Differences in conditional survival across asset profiles

	(1) Upper minus lower quartile	(2) p-value
Upside composition	0.300	0.959
	(5.783)	
Upper partial variance	-0.455	0.949
	(7.081)	
Lower partial variance	-1.055	0.819
	(4.567)	
Dispersion component	-0.755	0.603
	(1.434)	

Notes: Paired differences between the local survival gradients at the upper and lower baseline-asset quartiles in Table U.6. These are differences in gradients, not differences in survival levels. Standard errors in parentheses use CR1 clustering by 30 industry–education groups, conditional on the assigned CPS moments. Stars use unadjusted two-sided t_{29} tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. The composition contrast belongs to the ten-test survival family ($q = 0.959$); the other directions are supplementary.

Table U.8. Full-sample entry and follow-up availability

	(1) Entry	(2) Eligible follow-up
Upside composition	1.601***	1.820**
	(0.506)	(0.780)
Observations	27,419	22,868

Notes: Column (1) repeats the headline entry specification. Column (2) uses initial-nonholder transitions with entry years through 2023 and predicts whether next-year ownership is observed in the annual worker sample. Nonobservation includes both survey loss and loss of worker-sample eligibility. The positive association indicates that the complete-follow-up population is selected; missing ownership is never treated as exit. Units are percentage points per one-standard-deviation increase in wage composition. Standard errors in parentheses use CR1 clustering by 30 industry–education groups, conditional on the assigned CPS moments. Stars use unadjusted two-sided t_{29} tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table U.9. Resources and retention among first observed entrants

	(1) Baseline resources	(2) Asset interactions	(3) Earnings changes	(4) Earnings-gain paths
Upside composition	-4.961	-4.714	-4.711	-5.126
	(5.304)	(5.729)	(5.215)	(5.167)
Upper partial variance	-6.439	-5.953	-6.145	-6.672
	(5.682)	(6.405)	(5.534)	(5.475)
Lower partial variance	3.483	3.474	3.277	3.580
	(5.122)	(5.285)	(5.086)	(5.046)
Pre-entry financial assets	-1.118	—	—	—
	(1.804)	—	—	—
Composition × assets	—	0.511	—	—
	—	(4.377)	—	—
Entry-year earnings change	—	—	1.817	—
	—	—	(4.279)	—
Follow-up earnings change	—	—	0.608	—
	—	—	(2.615)	—
Sustained earnings gain	—	—	—	-2.352
	—	—	—	(4.504)
Nonsustained earnings gain	—	—	—	-2.346
	—	—	—	(5.034)
Sustained minus nonsustained gain	—	—	—	-0.006
	—	—	—	(4.326)
Pre-entry asset control	Yes	Yes	Yes	Yes
Observations	802	802	802	802

Notes: Outcome: next-year ownership among 802 first observed entrants with complete three-date earnings. All columns add pre-entry annual earnings and financial assets to equation (9); future assets are excluded. Column (2) includes all five risk–asset interactions. Its risk slopes are evaluated at standardized assets zero: the transformed-asset mean of the 23,701 baseline scaling observations, not the entrant mean. Composition and assets use the fixed standard-deviation scales of Table U.6; partial-tail directions hold the other tail fixed and use the CPS composition SD, not their own SD. Column (3) reports slopes per unit change in $\operatorname{asinh}(E/10{,}000)$ for annual earnings E in 2024 dollars. A substantial gain raises entry earnings at least 25 percent above baseline; sustained gains maintain that level at follow-up. Column (4) compares these gain paths with no substantial gain. Asset-row dashes denote undisplayed slopes, not omitted controls. Standard errors in parentheses use CR1 clustering by 30 industry–education groups, conditional on the assigned CPS moments. Stars use unadjusted two-sided t_{29} tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. The selected-entrant comparisons do not identify causal mediation.

Table U.10. Retention gradients at different pre-entry asset levels

	(1) Lower quartile	(2) Upper quartile	(3) Upper − lower
Upside composition	-5.116	-4.449	0.666
	(7.558)	(5.453)	(5.707)
Upper partial variance	-6.071	-5.875	0.196
	(8.923)	(5.819)	(6.780)
Lower partial variance	4.160	3.023	-1.137
	(6.393)	(5.296)	(4.665)

Notes: Directional slopes from the asset-interacted retention regression in Table U.9, column (2). The lower and upper asset quartiles are $1,233 and $21,973 in 2024 dollars, defined before selecting entrants. The final column uses paired covariance from the same fitted model. All columns use the same 802 entrants. Standard errors in parentheses use CR1 clustering by 30 industry–education groups, conditional on the assigned CPS moments. Stars use unadjusted two-sided t_{29} tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table U.11. Earnings-gain outcomes among first observed entrants

	(1) Substantial entry-year gain	(2) Sustained gain	(3) Nonsustained gain
Upside composition	-7.007*	-7.715	0.708
	(3.463)	(4.920)	(2.932)
Upper partial variance	-9.879**	-9.562	-0.316
	(4.098)	(5.662)	(3.205)
Lower partial variance	4.135	5.867	-1.732
	(2.993)	(4.289)	(2.706)
Observations	802	802	802
Outcome mean (percent)	30.56	21.26	9.30
Composition: sustained − nonsustained		-8.423	
		(7.323)	

Notes: Each column is an indicator outcome among the same 802 first observed entrants. Substantial gains and their persistence are defined as in Table U.9. Columns (2) and (3) add exactly to column (1); the final comparison uses their full joint covariance. These are conditional-on-entry earnings outcomes, not the initial-nonholder joint entry/gain outcomes in Table T.2. Coefficients are percentage points on the fixed CPS scales. Standard errors in parentheses use CR1 clustering by 30 industry–education groups, conditional on the assigned CPS moments. Stars use unadjusted two-sided t_{29} tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table U.12. Joint reporting of the resource-mechanism tests

	(1) Estimate	(2) p-value	(3) BH q-value
Retention: composition	-4.961	0.357	0.999
	(5.304)		
Retention: baseline assets	-1.118	0.540	0.999
	(1.804)		
Retention: composition × assets	0.511	0.908	0.999
	(4.377)		
Retention: entry earnings change	1.817	0.674	0.999
	(4.279)		
Retention: follow-up earnings change	0.608	0.818	0.999
	(2.615)		
Retention: sustained minus nonsustained gain	-0.006	0.999	0.999
	(4.326)		
Substantial gain: composition	-7.007*	0.052	0.471
	(3.463)		
Sustained gain: composition	-7.715	0.128	0.575
	(4.920)		
Nonsustained gain: composition	0.708	0.811	0.999
	(2.932)		
Adding earnings changes: composition change	0.250	0.656	—
	(0.556)		
Adding gain paths: composition change	-0.165	0.613	—
	(0.323)		

Notes: The first nine rows are the complete declared focal family from Tables U.9 and U.11. BH values adjust this nine-test family only. Stars continue to use unadjusted marginal tests. The last two rows subtract the baseline composition slope from the earnings-change and gain-path specifications using paired cluster influences; they are supplementary comparisons, not members of the nine-test family. All models use the same 802 entrants. Standard errors in parentheses use CR1 clustering by 30 industry–education groups, conditional on the assigned CPS moments. Stars use unadjusted two-sided t_{29} tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table U.13. Earnings paths and subsequent ownership

	(1) Entrants	(2) Retention (percent)	(3) Baseline earnings	(4) Entry earnings	(5) Follow-up earnings
Nonsustained substantial gain	76	50.42	114,160	217,446	100,125
Sustained substantial gain	162	46.97	56,901	117,010	143,702
No substantial entry gain	564	55.15	119,508	97,736	107,943

Notes: Counts are unweighted; retention and earnings means use the entry-year person weight and the same 802 entrants at all three dates. Earnings are annual personal earnings in 2024 dollars. Of the 76 entrants with nonsustained substantial gains, 51 return to or below baseline earnings. Failure to sustain a 25-percent elevation is not necessarily a full reversal. The adjusted retention comparison is reported in Table U.9; the raw means here have no coefficient stars.

Table U.14. Participation durability with endogenous saving

	(1) One-year survival	(2) Two-year survival	(3) Income-only holding	(4) Participation
A. Entry ages 26–44				
No recurring cost, downside	96.128	93.543	95.980	99.191
No recurring cost, upside	97.701	95.703	97.601	99.372
Persistent income, downside	98.652	98.087	93.466	98.703
Persistent income, upside	98.901	98.407	87.665	98.509
Primary, downside	96.080	92.725	93.213	98.728
Primary, symmetric	96.838	94.037	90.906	98.649
Primary, upside	97.548	95.422	89.781	98.719
Primary (refined), downside	96.079	92.696	92.842	98.684
Primary (refined), upside	97.478	95.350	88.581	98.581
B. Entry ages 45–64				
No recurring cost, downside	96.299	94.818	95.981	99.990
No recurring cost, upside	97.732	96.219	97.601	99.993
Persistent income, downside	99.132	98.544	95.073	99.891
Persistent income, upside	99.312	98.982	94.108	99.921
Primary, downside	96.129	93.582	93.691	99.979
Primary, symmetric	96.902	94.595	93.260	99.982
Primary, upside	97.677	95.821	91.651	99.985
Primary (refined), downside	96.127	93.528	93.727	99.979
Primary (refined), upside	97.681	95.814	91.469	99.985

Notes: Deterministic percentages from the nine previously solved life-cycle policy cases. Entrants are followed for two full years; the artificial age-25 initial condition is excluded. Two-year survival requires uninterrupted participation. The income-only column evaluates the same next-year policy at cash equal to earnings, holding age and income state fixed; it is not a reoptimized no-saving economy or welfare counterfactual. Participation averages ages 25–44 in Panel A and 45–64 in Panel B; the survival and earnings-only columns follow entry cohorts aged 26–44 and 45–64, respectively. Participation is not conditional on recent entry. Both persistent-income cases retain their small upper-saving-grid boundary flags. No policy was resolved or calibrated to these duration outcomes, and no statistical stars apply. The earlier benchmark retains nearly universal participation and a negative upside-minus-downside entry contrast.

Table U.15. Carried wealth and earnings among modeled entrants

	(1) Entry income	(2) Entry saving	(3) Survivor wealth	(4) Exiter wealth	(5) Exiter income change
A. Entry ages 26–44					
No recurring cost, downside	1.0223	0.1533	0.1630	0.1488	-0.5400
No recurring cost, upside	1.0131	0.1355	0.1441	0.1237	-0.5236
Persistent income, downside	1.0144	0.1707	0.1816	0.1331	-0.4920
Persistent income, upside	1.0043	0.1489	0.1584	0.1135	-0.4459
Primary, downside	1.0272	0.1636	0.1739	0.1616	-0.5462
Primary, symmetric	1.0267	0.1612	0.1715	0.1532	-0.5343
Primary, upside	1.0251	0.1520	0.1618	0.1350	-0.5063
Primary (refined), downside	1.0276	0.1650	0.1754	0.1630	-0.5466
Primary (refined), upside	1.0261	0.1555	0.1655	0.1369	-0.4962
B. Entry ages 45–64					
No recurring cost, downside	1.0223	0.1734	0.1847	0.1628	-0.5350
No recurring cost, upside	1.0131	0.1570	0.1670	0.1428	-0.5215
Persistent income, downside	1.0234	0.2092	0.2224	0.1476	-0.5035
Persistent income, upside	1.0173	0.1942	0.2064	0.1366	-0.4890
Primary, downside	1.0258	0.1878	0.1996	0.1841	-0.5438
Primary, symmetric	1.0234	0.1814	0.1928	0.1736	-0.5367
Primary, upside	1.0227	0.1758	0.1869	0.1622	-0.5253
Primary (refined), downside	1.0258	0.1879	0.1997	0.1843	-0.5439
Primary (refined), upside	1.0228	0.1758	0.1870	0.1624	-0.5261

Notes: All values are in units of mean annual working income. Carried wealth is the next-year financial wealth of entrants who retain ownership or exit at the first follow-up. Exiter income change is next-year minus entry income. The recurring cost is 0.002 except in the no-cost cases. The refined upside exiter carries wealth of 0.1369, about 68.5 annual fees; modeled exit is not literal inability to pay the fee. Cases and qualifications are as in Table U.14. These are deterministic accounting quantities, not estimated effects.

Table U.16. Tail opportunities and persistence in the threshold example

	(1) Entry (percent)	(2) One-year survival	(3) Two-year survival	(4) Expected spell	(5) Aggregate retention
$q = 0.5,\ \rho = 0.0$	0.000	—	—	—	100.000
$q = 0.5,\ \rho = 0.5$	0.000	—	—	—	100.000
$q = 0.2,\ \rho = 0.0$	20.000	20.000	4.000	1.250	86.667
$q = 0.2,\ \rho = 0.5$	10.000	60.000	36.000	2.500	93.333

Notes: The zero-mean innovation has variance one and takes values $H = \sqrt{(1-q)/q}$ and $L = -\sqrt{q/(1-q)}$, with stationary upper-state probability q. The resource gap is 1.5, so only the upside law ($q = 0.2$) can cross it. Renewal persistence ρ retains the current state with probability ρ and otherwise redraws from the stationary law. When only the upper state crosses the threshold, as under $q = 0.2$ here, marginal-household entry is $(1-\rho)q$ and one-year survival is $\rho + (1-\rho)q$. Two-year survival is the square of the latter, and expected spell length is its inverse exit probability. Survival columns are percentages; expected spells count participation decisions. Aggregate retention gives equal population mass to marginal and always-participating established households. Under the symmetric law marginal households never enter, so conditional survival and spell length are undefined, not zero. Changing persistence leaves the stationary mean, variance and composition unchanged.

V Retirement Contributions and Expenditure

V.1 Data, timing, and specification

The PSID extension links the public family files for 2005–2019 to the existing reference-person panel. Family quantities are merged by wave and family identifier, and adjacent observations are matched to the same reference person. Baseline years are 2005–2017 and destination years are 2007–2019. The spending sample contains 33,353 transitions for 9,183 people: 28,569 transitions begin without non-IRA stockholdings and 4,784 begin with holdings. The contribution and stock-flow samples contain 33,056 and 32,379 transitions, respectively. Destination ownership is not an inclusion condition. All samples retain both age bands and all supported industry–education cells.

The six outcomes are common broad cash expenditure, selected nondurable and service spending, furnishings and household equipment, food, net non-IRA stock acquisition, and any private IRA/annuity contribution. The broad expenditure sum includes food, housing, transportation, education, childcare, health, clothing, trips and recreation. Housing already includes repairs and furnishings; health includes insurance. Computing is excluded in every wave. This is a cash-outlay aggregate: mortgage and vehicle financing payments are not consumption-service flows.

The selected nondurable/service basket includes food, utilities, telephone and internet, auto insurance, vehicle repairs, gasoline, parking, public and other transportation, education, childcare, health, clothing and trips. It excludes shelter payments, home insurance, property taxes, imputed rent, home repairs, furnishings, vehicle financing and leases, mixed recreation, and additional vehicle expenses. It is not an exhaustive nondurable-consumption measure. Furnishings and equipment provide a narrower durable-expenditure proxy, not a durable stock or service flow. Food is out-of-pocket spending and does not add food stamps. The expenditure baskets overlap and are not additive components of the broad sum.

The official spending components are annualized but combine usual current spending and previous-calendar-year recall periods. Monetary values are expressed in 2019 dollars using the annual CPI-U. Furnishings and income use the previous-calendar-year deflator; the mixed spending aggregates and reported-period stock flows use the interview-year deflator. Genuine negative imputations and zeros are retained. The primary transforms are $\operatorname{asinh}(x/10{,}000)$ for broad expenditure, selected nondurables/services and food, and $\operatorname{asinh}(x/1{,}000)$ for furnishings and net stock acquisition. The transforms are fixed across waves and ownership groups; their coefficients are not exact percentage changes.

Net stock acquisition subtracts reported sales from purchases outside IRAs, respecting the survey's pre-2013 branching and later combined transaction gate. A valid no-transaction response establishes zero; an unknown required amount makes the flow missing. The private IRA/annuity contribution outcome is the survey's yes/no contribution gate. It does not measure the amount contributed, total saving, or the account's equity allocation, and it excludes some employer-plan saving. The flow questions start in January of the destination year minus two and end at the destination interview.

Their windows therefore overlap the beginning of the baseline interview year; they are not wholly subsequent or exactly two-year annualized flows.

Every specification allows all slopes and fixed effects to differ by initial non-IRA ownership. Baseline controls are the transformed earlier outcome, age and age squared, sex, family money income, liquid financial wealth, homeownership, family size, education and the age-band indicator. The specifications include industry and destination-wave fixed effects and the five earnings-moment controls used elsewhere in the paper. Baseline family weights are used throughout; future financial assets are not conditioned on. The spending samples require destination family income even in the primary models so that the additional-income comparison is paired. Standard errors use CR1 clustering by the 30 industry–education groups and t_{29} inference, conditional on the saved CPS moments.

The primary exposure is the zero-inclusive wage construction; the original positive-support real-log construction supplies the comparison on identical outcome samples. The CPS source period overlaps or postdates some PSID observations. The estimates therefore describe exploratory external associations; the timing precludes forward validation, and the assigned histories measure neither beliefs nor exogenous earnings-risk shocks.

V.2 Complete outcome family

The bounded family was recorded before estimation: six wage models, six original-construction comparisons, four wage spending models with realized destination income, and four wage dollar-level spending models. The last four retain the exact primary design, including the transformed earlier outcome. Tables V.1–V.12 report all 320 state-specific slopes, direct ownership-state contrasts and baseline-weight-share averages. Each partial-variance calculation raises one tail with the other fixed, scaled to the fixed composition standard deviation; the standard-deviation term is evaluated at each ownership state's weighted mean. Composition is half the upper-minus-lower difference and dispersion half their sum.

The contribution result extends the stock market participation evidence to retirement-saving participation. Among initial stockholders, the wage composition coefficient is 6.356 percentage points (SE 2.191), against a contribution rate of 37.65 percent. The direct holder–nonholder difference is 6.141 points (SE 2.325). The upper and lower partial-variance associations are 6.198 and -6.515 points, respectively. The original-construction holder composition estimate is -0.596 point (SE 1.455); it does not reproduce the contribution finding. Realized-income conditioning was declared for spending, not for contributions, so the contribution estimates do not identify a realized-earnings-independent saving mechanism.

The primary broad-spending estimates are small and imprecise. In dollars, the baseline-weight-share average is \$452 per year, with a 95-percent interval from $-\$610$ to \$1,513, against mean expenditure of \$56,966. The selected nondurable/service and food averages are \$196 and \$39. Broad expenditure is imprecisely related to composition despite the positive contribution-participation

association. The estimates leave expenditure responses and cross-outcome differences unresolved and do not identify precautionary saving.

Furnishings show a positive transformed association among initial holders: 0.114 asinh unit (SE 0.041), with a holder–nonholder contrast of 0.107 (SE 0.046). Conditioning on realized family income leaves the holder coefficient at 0.107 (SE 0.039). The dollar-level holder estimate is $188 (SE $266), and the original-construction composition estimate is imprecise. This outcome covers furnishings and equipment, a narrow component of durable expenditure. Net non-IRA stock acquisition is also imprecisely related to composition. The evidence therefore does not establish a transfer from taxable equities to retirement accounts.

V.3 Paired comparisons and multiplicity

Table V.13 reports all 16 changes in the composition slope after conditioning on realized family income. The paired standard errors use the covariance of the two fitted models, not independent-model approximations. These conditional comparisons do not identify mediation. Table V.14 reports the declared Benjamini–Hochberg adjustments for six primary average-composition tests and, separately, twelve primary ownership-state composition tests. Conventional table stars continue to use the unadjusted coefficient tests. The complete samples, model matrices, covariance estimates, cluster influences and source-variable maps are retained in the replication package.

Table V.1. Wage composition and household expenditure: initial ownership groups

	(1) Broad expenditure	(2) Selected nondurables and services	(3) Furnishings and equipment	(4) Food
Outcome units	Asinh units	Asinh units	Asinh units	Asinh units
A. Initial nonholders				
Upside composition	-0.001	0.001	0.007	0.002
	(0.008)	(0.008)	(0.013)	(0.005)
Upper partial variance	-0.003	0.001	0.000	0.004
	(0.009)	(0.008)	(0.015)	(0.006)
Lower partial variance	-0.001	-0.001	-0.013	-0.001
	(0.008)	(0.008)	(0.012)	(0.005)
Dispersion component	-0.002	0.000	-0.007*	0.001
	(0.002)	(0.001)	(0.004)	(0.002)
B. Initial holders				
Upside composition	0.014	0.012	0.114***	0.009
	(0.016)	(0.020)	(0.041)	(0.017)
Upper partial variance	0.019	0.014	0.135***	0.010
	(0.017)	(0.021)	(0.043)	(0.018)
Lower partial variance	-0.010	-0.009	-0.093**	-0.007
	(0.015)	(0.018)	(0.039)	(0.015)
Dispersion component	0.005	0.003	0.021***	0.001
	(0.003)	(0.004)	(0.006)	(0.003)
Nonholder observations	28,569	28,569	28,569	28,569
Holder observations	4,784	4,784	4,784	4,784
Total observations	33,353	33,353	33,353	33,353
Distinct reference persons	9,183	9,183	9,183	9,183
Industry–education clusters	30	30	30	30

Notes: Zero-inclusive wage construction. The PSID sample, baseline controls, ownership interactions and family weights are as in Table 8. Composition holds total variance fixed. Each partial-variance direction raises that tail with the other fixed, scaled to the fixed composition standard deviation; the standard-deviation term is evaluated at the state-specific weighted mean. Dispersion is half the sum of the two tail directions. Differences and baseline-weight-share averages use the full joint covariance. Monetary outcomes use $\operatorname{asinh}(\text{2019 dollars}/s)$: $s = 1{,}000$ for furnishings and net stock acquisition, and $s = 10{,}000$ for other expenditure. These coefficients are not exact percentage changes. Standard errors in parentheses use CR1 clustering by 30 industry–education groups, conditional on the saved CPS moments. Stars use unadjusted two-sided t_{29} tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table V.2. Wage composition and household expenditure: ownership-state comparisons

	(1) Broad expenditure	(2) Selected nondurables and services	(3) Furnishings and equipment	(4) Food
Outcome units	Asinh units	Asinh units	Asinh units	Asinh units
A. Holder minus nonholder				
Upside composition	0.015	0.011	0.107**	0.006
	(0.018)	(0.024)	(0.046)	(0.019)
Upper partial variance	0.022	0.014	0.136**	0.007
	(0.020)	(0.026)	(0.050)	(0.020)
Lower partial variance	-0.008	-0.008	-0.079*	-0.006
	(0.017)	(0.023)	(0.043)	(0.018)
Dispersion component	0.007*	0.003	0.028***	0.000
	(0.004)	(0.004)	(0.007)	(0.003)
B. Baseline-weight-share average				
Upside composition	0.002	0.003	0.027**	0.004
	(0.007)	(0.006)	(0.012)	(0.004)
Upper partial variance	0.001	0.003	0.025*	0.005
	(0.008)	(0.006)	(0.013)	(0.006)
Lower partial variance	-0.003	-0.002	-0.028**	-0.002
	(0.007)	(0.005)	(0.011)	(0.004)
Dispersion component	-0.001	0.001	-0.002	0.001
	(0.001)	(0.001)	(0.003)	(0.002)
Nonholder observations	28,569	28,569	28,569	28,569
Holder observations	4,784	4,784	4,784	4,784
Total observations	33,353	33,353	33,353	33,353
Distinct reference persons	9,183	9,183	9,183	9,183
Industry–education clusters	30	30	30	30

Notes: Zero-inclusive wage construction. The PSID sample, baseline controls, ownership interactions and family weights are as in Table 8. Composition holds total variance fixed. Each partial-variance direction raises that tail with the other fixed, scaled to the fixed composition standard deviation; the standard-deviation term is evaluated at the state-specific weighted mean. Dispersion is half the sum of the two tail directions. Differences and baseline-weight-share averages use the full joint covariance. Monetary outcomes use $\operatorname{asinh}(\text{2019 dollars}/s)$: $s = 1{,}000$ for furnishings and net stock acquisition, and $s = 10{,}000$ for other expenditure. These coefficients are not exact percentage changes. Standard errors in parentheses use CR1 clustering by 30 industry–education groups, conditional on the saved CPS moments. Stars use unadjusted two-sided t_{29} tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table V.3. Wage composition and saving-related flows: initial ownership groups

	(1) Net non-IRA stock acquisition	(2) IRA/annuity contribution
Outcome units	Asinh units	Percentage points
A. Initial nonholders		
Upside composition	-0.005	0.215
	(0.009)	(0.331)
Upper partial variance	-0.011	0.364
	(0.012)	(0.471)
Lower partial variance	-0.001	-0.066
	(0.008)	(0.259)
Dispersion component	-0.006	0.149
	(0.004)	(0.187)
B. Initial holders		
Upside composition	-0.069	6.356***
	(0.085)	(2.191)
Upper partial variance	-0.084	6.198**
	(0.093)	(2.352)
Lower partial variance	0.055	-6.515***
	(0.078)	(2.105)
Dispersion component	-0.014	-0.158
	(0.013)	(0.424)
Nonholder observations	28,217	28,337
Holder observations	4,162	4,719
Total observations	32,379	33,056
Distinct reference persons	9,081	9,148
Industry–education clusters	30	30

Notes: Zero-inclusive wage construction. The PSID sample, baseline controls, ownership interactions and family weights are as in Table 8. Composition holds total variance fixed. Each partial-variance direction raises that tail with the other fixed, scaled to the fixed composition standard deviation; the standard-deviation term is evaluated at the state-specific weighted mean. Dispersion is half the sum of the two tail directions. Differences and baseline-weight-share averages use the full joint covariance. Monetary outcomes use $\operatorname{asinh}(\text{2019 dollars}/s)$: $s = 1{,}000$ for furnishings and net stock acquisition, and $s = 10{,}000$ for other expenditure. These coefficients are not exact percentage changes. Stock acquisition is purchases minus sales outside IRAs; IRA/annuity participation records any contribution, not equity exposure or total saving. Both flow windows partly overlap the baseline interview year and are not annualized. Standard errors in parentheses use CR1 clustering by 30 industry–education groups, conditional on the saved CPS moments. Stars use unadjusted two-sided t_{29} tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table V.4. Wage composition and saving-related flows: ownership-state comparisons

	(1) Net non-IRA stock acquisition	(2) IRA/annuity contribution
Outcome units	Asinh units	Percentage points
A. Holder minus nonholder		
Upside composition	-0.064	6.141**
	(0.090)	(2.325)
Upper partial variance	-0.073	5.834**
	(0.100)	(2.568)
Lower partial variance	0.055	-6.449***
	(0.083)	(2.196)
Dispersion component	-0.009	-0.307
	(0.015)	(0.550)
B. Baseline-weight-share average		
Upside composition	-0.016	1.365***
	(0.012)	(0.407)
Upper partial variance	-0.023	1.456***
	(0.014)	(0.462)
Lower partial variance	0.009	-1.274***
	(0.011)	(0.387)
Dispersion component	-0.007**	0.091
	(0.003)	(0.127)
Nonholder observations	28,217	28,337
Holder observations	4,162	4,719
Total observations	32,379	33,056
Distinct reference persons	9,081	9,148
Industry–education clusters	30	30

Notes: Zero-inclusive wage construction. The PSID sample, baseline controls, ownership interactions and family weights are as in Table 8. Composition holds total variance fixed. Each partial-variance direction raises that tail with the other fixed, scaled to the fixed composition standard deviation; the standard-deviation term is evaluated at the state-specific weighted mean. Dispersion is half the sum of the two tail directions. Differences and baseline-weight-share averages use the full joint covariance. Monetary outcomes use $\operatorname{asinh}(\text{2019 dollars}/s)$: $s = 1{,}000$ for furnishings and net stock acquisition, and $s = 10{,}000$ for other expenditure. These coefficients are not exact percentage changes. Stock acquisition is purchases minus sales outside IRAs; IRA/annuity participation records any contribution, not equity exposure or total saving. Both flow windows partly overlap the baseline interview year and are not annualized. Standard errors in parentheses use CR1 clustering by 30 industry–education groups, conditional on the saved CPS moments. Stars use unadjusted two-sided t_{29} tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table V.5. Original construction: household expenditure: initial ownership groups

	(1) Broad expenditure	(2) Selected nondurables and services	(3) Furnishings and equipment	(4) Food
Outcome units	Asinh units	Asinh units	Asinh units	Asinh units
A. Initial nonholders				
Upside composition	-0.006	-0.004	-0.006	-0.004
	(0.004)	(0.003)	(0.007)	(0.003)
Upper partial variance	-0.007	-0.003	-0.006	-0.002
	(0.005)	(0.005)	(0.010)	(0.004)
Lower partial variance	0.004	0.006**	0.007	0.007**
	(0.003)	(0.003)	(0.007)	(0.003)
Dispersion component	-0.002	0.001	0.001	0.002
	(0.002)	(0.002)	(0.005)	(0.002)
B. Initial holders				
Upside composition	0.010	0.007	0.027	0.003
	(0.009)	(0.010)	(0.017)	(0.007)
Upper partial variance	0.012	0.010	0.046**	0.005
	(0.010)	(0.012)	(0.021)	(0.008)
Lower partial variance	-0.008	-0.004	-0.008	-0.001
	(0.009)	(0.009)	(0.018)	(0.007)
Dispersion component	0.002	0.003	0.019*	0.002
	(0.003)	(0.004)	(0.009)	(0.002)
Nonholder observations	28,569	28,569	28,569	28,569
Holder observations	4,784	4,784	4,784	4,784
Total observations	33,353	33,353	33,353	33,353
Distinct reference persons	9,183	9,183	9,183	9,183
Industry–education clusters	30	30	30	30

Notes: Original positive-support real-log construction. The PSID sample, baseline controls, ownership interactions and family weights are as in Table 8. Composition holds total variance fixed. Each partial-variance direction raises that tail with the other fixed, scaled to the fixed composition standard deviation; the standard-deviation term is evaluated at the state-specific weighted mean. Dispersion is half the sum of the two tail directions. Differences and baseline-weight-share averages use the full joint covariance. Monetary outcomes use $\operatorname{asinh}(\text{2019 dollars}/s)$: $s = 1{,}000$ for furnishings and net stock acquisition, and $s = 10{,}000$ for other expenditure. These coefficients are not exact percentage changes. Standard errors in parentheses use CR1 clustering by 30 industry–education groups, conditional on the saved CPS moments. Stars use unadjusted two-sided t_{29} tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table V.6. Original construction: household expenditure: ownership-state comparisons

	(1) Broad expenditure	(2) Selected nondurables and services	(3) Furnishings and equipment	(4) Food
Outcome units	Asinh units	Asinh units	Asinh units	Asinh units
A. Holder minus nonholder				
Upside composition	0.016	0.012	0.033	0.007
	(0.010)	(0.011)	(0.020)	(0.009)
Upper partial variance	0.020	0.013	0.052*	0.007
	(0.012)	(0.014)	(0.026)	(0.009)
Lower partial variance	-0.012	-0.010	-0.015	-0.008
	(0.010)	(0.010)	(0.020)	(0.009)
Dispersion component	0.004	0.002	0.018	-0.001
	(0.004)	(0.004)	(0.011)	(0.003)
B. Baseline-weight-share average				
Upside composition	-0.003	-0.002	0.000	-0.003
	(0.003)	(0.003)	(0.005)	(0.002)
Upper partial variance	-0.004	-0.001	0.004	-0.001
	(0.004)	(0.004)	(0.008)	(0.004)
Lower partial variance	0.002	0.004*	0.004	0.005**
	(0.003)	(0.002)	(0.006)	(0.002)
Dispersion component	-0.001	0.002	0.004	0.002
	(0.002)	(0.002)	(0.004)	(0.002)
Nonholder observations	28,569	28,569	28,569	28,569
Holder observations	4,784	4,784	4,784	4,784
Total observations	33,353	33,353	33,353	33,353
Distinct reference persons	9,183	9,183	9,183	9,183
Industry–education clusters	30	30	30	30

Notes: Original positive-support real-log construction. The PSID sample, baseline controls, ownership interactions and family weights are as in Table 8. Composition holds total variance fixed. Each partial-variance direction raises that tail with the other fixed, scaled to the fixed composition standard deviation; the standard-deviation term is evaluated at the state-specific weighted mean. Dispersion is half the sum of the two tail directions. Differences and baseline-weight-share averages use the full joint covariance. Monetary outcomes use $\operatorname{asinh}(\text{2019 dollars}/s)$: $s = 1{,}000$ for furnishings and net stock acquisition, and $s = 10{,}000$ for other expenditure. These coefficients are not exact percentage changes. Standard errors in parentheses use CR1 clustering by 30 industry–education groups, conditional on the saved CPS moments. Stars use unadjusted two-sided t_{29} tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table V.7. Original construction: saving-related flows: initial ownership groups

	(1) Net non-IRA stock acquisition	(2) IRA/annuity contribution
Outcome units	Asinh units	Percentage points
A. Initial nonholders		
Upside composition	0.008	0.032
	(0.007)	(0.299)
Upper partial variance	0.005	0.147
	(0.010)	(0.433)
Lower partial variance	-0.010	0.083
	(0.006)	(0.303)
Dispersion component	-0.002	0.115
	(0.005)	(0.225)
B. Initial holders		
Upside composition	0.001	-0.596
	(0.038)	(1.455)
Upper partial variance	-0.013	-0.696
	(0.041)	(1.858)
Lower partial variance	-0.015	0.496
	(0.039)	(1.270)
Dispersion component	-0.014	-0.100
	(0.011)	(0.646)
Nonholder observations	28,217	28,337
Holder observations	4,162	4,719
Total observations	32,379	33,056
Distinct reference persons	9,081	9,148
Industry–education clusters	30	30

Notes: Original positive-support real-log construction. The PSID sample, baseline controls, ownership interactions and family weights are as in Table 8. Composition holds total variance fixed. Each partial-variance direction raises that tail with the other fixed, scaled to the fixed composition standard deviation; the standard-deviation term is evaluated at the state-specific weighted mean. Dispersion is half the sum of the two tail directions. Differences and baseline-weight-share averages use the full joint covariance. Monetary outcomes use $\operatorname{asinh}(\text{2019 dollars}/s)$: $s = 1{,}000$ for furnishings and net stock acquisition, and $s = 10{,}000$ for other expenditure. These coefficients are not exact percentage changes. Stock acquisition is purchases minus sales outside IRAs; IRA/annuity participation records any contribution, not equity exposure or total saving. Both flow windows partly overlap the baseline interview year and are not annualized. Standard errors in parentheses use CR1 clustering by 30 industry–education groups, conditional on the saved CPS moments. Stars use unadjusted two-sided t_{29} tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table V.8. Original construction: saving-related flows: ownership-state comparisons

	(1) Net non-IRA stock acquisition	(2) IRA/annuity contribution
Outcome units	Asinh units	Percentage points
A. Holder minus nonholder		
Upside composition	-0.006	-0.628
	(0.040)	(1.648)
Upper partial variance	-0.018	-0.843
	(0.044)	(2.137)
Lower partial variance	-0.005	0.413
	(0.038)	(1.429)
Dispersion component	-0.012	-0.215
	(0.012)	(0.768)
B. Baseline-weight-share average		
Upside composition	0.007	-0.086
	(0.008)	(0.236)
Upper partial variance	0.002	-0.011
	(0.009)	(0.322)
Lower partial variance	-0.011	0.160
	(0.009)	(0.257)
Dispersion component	-0.004	0.075
	(0.005)	(0.171)
Nonholder observations	28,217	28,337
Holder observations	4,162	4,719
Total observations	32,379	33,056
Distinct reference persons	9,081	9,148
Industry–education clusters	30	30

Notes: Original positive-support real-log construction. The PSID sample, baseline controls, ownership interactions and family weights are as in Table 8. Composition holds total variance fixed. Each partial-variance direction raises that tail with the other fixed, scaled to the fixed composition standard deviation; the standard-deviation term is evaluated at the state-specific weighted mean. Dispersion is half the sum of the two tail directions. Differences and baseline-weight-share averages use the full joint covariance. Monetary outcomes use $\operatorname{asinh}(\text{2019 dollars}/s)$: $s = 1{,}000$ for furnishings and net stock acquisition, and $s = 10{,}000$ for other expenditure. These coefficients are not exact percentage changes. Stock acquisition is purchases minus sales outside IRAs; IRA/annuity participation records any contribution, not equity exposure or total saving. Both flow windows partly overlap the baseline interview year and are not annualized. Standard errors in parentheses use CR1 clustering by 30 industry–education groups, conditional on the saved CPS moments. Stars use unadjusted two-sided t_{29} tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table V.9. Household expenditure conditional on realized income: initial ownership groups

	(1) Broad expenditure	(2) Selected nondurables and services	(3) Furnishings and equipment	(4) Food
Outcome units	Asinh units	Asinh units	Asinh units	Asinh units
A. Initial nonholders				
Upside composition	-0.002	0.000	0.006	0.002
	(0.007)	(0.006)	(0.013)	(0.005)
Upper partial variance	-0.001	0.003	0.002	0.005
	(0.007)	(0.006)	(0.015)	(0.006)
Lower partial variance	0.004	0.004	-0.010	0.001
	(0.006)	(0.006)	(0.012)	(0.004)
Dispersion component	0.002	0.004**	-0.004	0.003
	(0.001)	(0.001)	(0.003)	(0.002)
B. Initial holders				
Upside composition	0.009	0.008	0.107**	0.006
	(0.016)	(0.019)	(0.039)	(0.016)
Upper partial variance	0.015	0.011	0.129***	0.007
	(0.018)	(0.021)	(0.042)	(0.017)
Lower partial variance	-0.004	-0.005	-0.086**	-0.004
	(0.014)	(0.018)	(0.038)	(0.015)
Dispersion component	0.005	0.003	0.022***	0.002
	(0.003)	(0.004)	(0.005)	(0.003)
Nonholder observations	28,569	28,569	28,569	28,569
Holder observations	4,784	4,784	4,784	4,784
Total observations	33,353	33,353	33,353	33,353
Distinct reference persons	9,183	9,183	9,183	9,183
Industry–education clusters	30	30	30	30

Notes: Zero-inclusive wage construction. The PSID sample, baseline controls, ownership interactions and family weights are as in Table 8. Composition holds total variance fixed. Each partial-variance direction raises that tail with the other fixed, scaled to the fixed composition standard deviation; the standard-deviation term is evaluated at the state-specific weighted mean. Dispersion is half the sum of the two tail directions. Differences and baseline-weight-share averages use the full joint covariance. Monetary outcomes use $\operatorname{asinh}(\text{2019 dollars}/s)$: $s = 1{,}000$ for furnishings and net stock acquisition, and $s = 10{,}000$ for other expenditure. These coefficients are not exact percentage changes. These specifications additionally condition on realized destination family income, using the same observations as the primary models; they are not causal mediation estimates. Standard errors in parentheses use CR1 clustering by 30 industry–education groups, conditional on the saved CPS moments. Stars use unadjusted two-sided t_{29} tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table V.10. Household expenditure conditional on realized income: ownership-state comparisons

	(1) Broad expenditure	(2) Selected nondurables and services	(3) Furnishings and equipment	(4) Food
Outcome units	Asinh units	Asinh units	Asinh units	Asinh units
A. Holder minus nonholder				
Upside composition	0.012	0.008	0.101**	0.004
	(0.017)	(0.022)	(0.043)	(0.017)
Upper partial variance	0.016	0.008	0.127**	0.002
	(0.020)	(0.024)	(0.046)	(0.019)
Lower partial variance	-0.008	-0.009	-0.076*	-0.005
	(0.016)	(0.021)	(0.041)	(0.016)
Dispersion component	0.004	0.000	0.026***	-0.001
	(0.004)	(0.004)	(0.006)	(0.003)
B. Baseline-weight-share average				
Upside composition	0.000	0.001	0.025**	0.003
	(0.006)	(0.005)	(0.012)	(0.004)
Upper partial variance	0.002	0.005	0.026*	0.005
	(0.006)	(0.006)	(0.013)	(0.005)
Lower partial variance	0.003	0.002	-0.024**	0.000
	(0.006)	(0.004)	(0.011)	(0.004)
Dispersion component	0.002*	0.004***	0.001	0.003
	(0.001)	(0.001)	(0.003)	(0.002)
Nonholder observations	28,569	28,569	28,569	28,569
Holder observations	4,784	4,784	4,784	4,784
Total observations	33,353	33,353	33,353	33,353
Distinct reference persons	9,183	9,183	9,183	9,183
Industry–education clusters	30	30	30	30

Notes: Zero-inclusive wage construction. The PSID sample, baseline controls, ownership interactions and family weights are as in Table 8. Composition holds total variance fixed. Each partial-variance direction raises that tail with the other fixed, scaled to the fixed composition standard deviation; the standard-deviation term is evaluated at the state-specific weighted mean. Dispersion is half the sum of the two tail directions. Differences and baseline-weight-share averages use the full joint covariance. Monetary outcomes use $\operatorname{asinh}(\text{2019 dollars}/s)$: $s = 1{,}000$ for furnishings and net stock acquisition, and $s = 10{,}000$ for other expenditure. These coefficients are not exact percentage changes. These specifications additionally condition on realized destination family income, using the same observations as the primary models; they are not causal mediation estimates. Standard errors in parentheses use CR1 clustering by 30 industry–education groups, conditional on the saved CPS moments. Stars use unadjusted two-sided t_{29} tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table V.11. Household expenditure in dollar levels: initial ownership groups

	(1) Broad expenditure	(2) Selected nondurables and services	(3) Furnishings and equipment	(4) Food
Outcome units	2019 dollars	2019 dollars	2019 dollars	2019 dollars
A. Initial nonholders				
Upside composition	504	124	3	53
	(449)	(273)	(38)	(91)
Upper partial variance	654	271	-21	80
	(490)	(302)	(45)	(107)
Lower partial variance	-354	22	-27	-27
	(421)	(250)	(34)	(87)
Dispersion component	150*	147***	-24*	27
	(83)	(52)	(12)	(34)
B. Initial holders				
Upside composition	225	508	188	-23
	(1,729)	(937)	(266)	(303)
Upper partial variance	1,132	612	378	-29
	(1,975)	(1,129)	(274)	(341)
Lower partial variance	682	-404	3	18
	(1,555)	(852)	(265)	(275)
Dispersion component	907**	104	190***	-5
	(412)	(350)	(45)	(65)
Nonholder observations	28,569	28,569	28,569	28,569
Holder observations	4,784	4,784	4,784	4,784
Total observations	33,353	33,353	33,353	33,353
Distinct reference persons	9,183	9,183	9,183	9,183
Industry–education clusters	30	30	30	30

Notes: Zero-inclusive wage construction. The PSID sample, baseline controls, ownership interactions and family weights are as in Table 8. Composition holds total variance fixed. Each partial-variance direction raises that tail with the other fixed, scaled to the fixed composition standard deviation; the standard-deviation term is evaluated at the state-specific weighted mean. Dispersion is half the sum of the two tail directions. Differences and baseline-weight-share averages use the full joint covariance. The outcome is in annual 2019 dollars; the earlier outcome control retains its primary asinh transform. These four declared scale checks use the same rows and design as the primary spending regressions. Standard errors in parentheses use CR1 clustering by 30 industry–education groups, conditional on the saved CPS moments. Stars use unadjusted two-sided t_{29} tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table V.12. Household expenditure in dollar levels: ownership-state comparisons

	(1) Broad expenditure	(2) Selected nondurables and services	(3) Furnishings and equipment	(4) Food
Outcome units	2019 dollars	2019 dollars	2019 dollars	2019 dollars
A. Holder minus nonholder				
Upside composition	-279	384	185	-77
	(1,730)	(950)	(275)	(335)
Upper partial variance	478	341	399	-108
	(1,961)	(1,082)	(283)	(365)
Lower partial variance	1,037	-426	30	45
	(1,573)	(916)	(275)	(314)
Dispersion component	757*	-42	214***	-32
	(409)	(320)	(48)	(61)
B. Baseline-weight-share average				
Upside composition	452	196	38	39
	(519)	(296)	(54)	(83)
Upper partial variance	744	335	54	59
	(585)	(368)	(60)	(104)
Lower partial variance	-159	-58	-21	-18
	(470)	(244)	(51)	(73)
Dispersion component	292**	139	16	21
	(110)	(98)	(12)	(35)
Nonholder observations	28,569	28,569	28,569	28,569
Holder observations	4,784	4,784	4,784	4,784
Total observations	33,353	33,353	33,353	33,353
Distinct reference persons	9,183	9,183	9,183	9,183
Industry–education clusters	30	30	30	30

Notes: Zero-inclusive wage construction. The PSID sample, baseline controls, ownership interactions and family weights are as in Table 8. Composition holds total variance fixed. Each partial-variance direction raises that tail with the other fixed, scaled to the fixed composition standard deviation; the standard-deviation term is evaluated at the state-specific weighted mean. Dispersion is half the sum of the two tail directions. Differences and baseline-weight-share averages use the full joint covariance. The outcome is in annual 2019 dollars; the earlier outcome control retains its primary asinh transform. These four declared scale checks use the same rows and design as the primary spending regressions. Standard errors in parentheses use CR1 clustering by 30 industry–education groups, conditional on the saved CPS moments. Stars use unadjusted two-sided t_{29} tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table V.13. Changes after conditioning on realized family income

	(1) Broad expenditure	(2) Selected nondurables and services	(3) Furnishings and equipment	(4) Food
Initial nonholders	-0.001	-0.001	-0.001	-0.001
	(0.003)	(0.003)	(0.003)	(0.002)
Initial holders	-0.005	-0.004	-0.007	-0.003
	(0.003)	(0.003)	(0.004)	(0.002)
Holder minus nonholder	-0.003	-0.003	-0.006	-0.003
	(0.004)	(0.004)	(0.005)	(0.003)
Baseline-weight-share average	-0.002	-0.001	-0.002	-0.001
	(0.003)	(0.002)	(0.002)	(0.001)
Observations	33,353	33,353	33,353	33,353
Industry–education clusters	30	30	30	30

Notes: Paired changes in the wage-composition coefficient: the model with destination family income minus the primary model. Each comparison uses identical observations, baseline controls, weights and exposures. Uncertainty uses the difference between the two models' cluster influences, retaining their covariance. Units are asinh units with the fixed scales in Table V.1. The average uses baseline ownership weight shares. Standard errors in parentheses use CR1 clustering by 30 industry–education groups, conditional on the saved CPS moments. Stars use unadjusted two-sided t_{29} tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table V.14. Composition associations: marginal and family-adjusted inference

	(1) Average p	(2) Average q	(3) Nonholder p	(4) Nonholder q	(5) Holder p	(6) Holder q
Broad expenditure	0.812	0.812	0.896	0.937	0.374	0.774
Selected nondurables and services	0.640	0.768	0.937	0.937	0.553	0.774
Furnishings and equipment	0.031	0.092	0.627	0.774	0.010	0.058
Food	0.405	0.607	0.645	0.774	0.596	0.774
Net non-IRA stock acquisition	0.195	0.390	0.579	0.774	0.422	0.774
IRA/annuity contribution	0.002	0.013	0.521	0.774	0.007	0.058

Notes: The six primary wage-composition average tests form one Benjamini–Hochberg family; the twelve primary ownership-state composition tests form a separate family. The nonholder and holder q columns belong to that same twelve-test family, not separate six-test families. Marginal p values use two-sided CR1/t_{29} inference. The coefficients and their conventional, unadjusted stars appear in Tables V.1 and V.3; this table does not replace their declared tests. Direct holder–nonholder contrasts were not members of these two families.

W Career Earnings Calibration

This appendix describes the earnings process used in the life-cycle model, its fit to CPS and PSID moments, and the identification of cross-cell earnings laws. Financial estimation holds these laws fixed.

W.1 The career earnings process

For a worker of education type h, equation (15) combines the education age profile, group wage level, group growth adjustment and persistent and temporary earnings components. The persistent component evolves as

$$\eta_t = \rho\,\eta_{t-1} + \zeta_t + J^+\,I_t^+ - J_b^-\,I_t^- - m_b, \qquad \zeta_t \sim N(0, s_0^2),$$

where I_t^+ and I_t^- indicate mutually exclusive promotions and falls. In ten-year age band b, their probabilities are $p_b^+ \exp(k^+\eta_{t-1}/\sigma_\eta)$ and $p_b^- \exp(-k^-\eta_{t-1}/\sigma_\eta)$, each capped at 0.45; the remaining probability is an ordinary year with neither event. The grid scale σ_η approximates the stationary standard deviation. The fall size takes a separate value after 55. The compensating drift m_b is set by bisection so that each band's discretized chain has stationary mean zero. This normalization does not set every state's conditional innovation mean to zero. Each mixture component is discretized on seventeen states with Gauss–Hermite nodes and mean-preserving linear binning, which transmits the small Gaussian component and the compensating drift exactly up to edge clamping.

Temporary earnings are a two-component normal mixture discretized on ten nodes. They affect current wages without becoming the persistent state. Wage income is lost with hazard $\delta_b \exp(-\phi\eta/\bar{\eta})$, capped at 0.6, where $\bar{\eta}$ is the top of the grid, and regained with an age-specific hazard capped at 0.95. Persistent earnings states continue to evolve outside wage work. The survey observes $\log W^{\text{obs}} = \log W + u$, where u is an independent, symmetric two-component reporting-error mixture. Households act on true resources; u enters measurement and the researcher's initial-state filter, not the budget.

The joint fit to one-year CPS and multi-horizon PSID changes disciplines the split of measured non-persistent variation between real transitory shocks and reporting error. It leaves 11% (non-college) and 13% (college) of measured non-persistent one-year variance as real, with PSID reporting-error variance assumed to be one quarter of its CPS value. Twenty-eight parameters per education type are fitted by least squares to one-year CPS arc wage changes pooled by ten-year age band (zero shares and upper and lower partial variances, zero shares by origin-wage tercile, the quantiles of positive changes and the variance of log wage levels) and to the PSID moments of prime-age heads with origin wages above \$5,000: the variance and ratio V^+/V^- of residualized two-, four- and six-year log real wage changes by origin band, and shares of destinations with no wage and with a wage of \$1–5,000. These are earnings-calibration targets; no coefficient relating earnings risk to portfolios is used. Tables W.1 and W.2 report the parameters and the PSID fit.

For calibration, simulated log wage changes are centered within origin-age bands, whereas empirical PSID moments remove wave means and an education-specific age cubic from log wage levels. A fixed-parameter evaluation applying the age cubic to simulated levels uses wages above \$5,000 and PSID age weights. The largest changes are 0.00949 for the partial-variance ratio (0.12 empirical standard error) and 0.00161 for the variance (0.08 empirical standard error). Tables W.1 and W.2 report the calibration and its fitting moments.

Table W.1. The career earnings process: fitted parameters

	Non-college	College
Persistence of the career component, ρ	0.946	0.944
Standard deviation of the Gaussian innovation, s_0	0.024	0.037
Promotion size, J^+ (log points)	0.328	0.522
Fall size, J^- (log points)	0.742	0.419
Fall size at ages 55–64, J^-_{55} (log points)	1.150	1.346
Promotion hazard, ages 25–34	0.117	0.082
Promotion hazard, ages 35–44	0.039	0.019
Promotion hazard, ages 45–54	0.013	0.009
Promotion hazard, ages 55–64	0.046	0.021
Fall hazard, ages 25–34	0.036	0.010
Fall hazard, ages 35–44	0.032	0.029
Fall hazard, ages 45–54	0.023	0.030
Fall hazard, ages 55–64	0.035	0.043
State dependence of the promotion hazard, k^+	0.248	0.140
State dependence of the fall hazard, k^-	0.148	0.931
Re-entry hazard shift, r	1.926	1.996
Fit loss (half the sum of squared scaled residuals)	102.1	142.4

Notes: Parameters of the career process of Appendix W.1, fitted by education type jointly with the transitory, exit, re-entry and reporting-error parameters to the CPS one-year moments through the paper's pipeline and to the PSID moments of Table W.2; jump sizes in log points, hazards annual. Fitted parameters, not regression estimates.

Promotions of 0.33 (non-college) and 0.52 (college) log points arrive at 12% and 8% a year at ages 25–34 and at 1–5% a year later, and their hazards rise with the current state (for non-college workers, up to 2.4 times at the top of the grid). Ordinary falls of 0.74 and 0.42 log points arrive at 1–4% a year at the center of the grid and more often at low states (for college workers, 2.5 times as often one standard deviation below the center); late-career falls of 1.15 and 1.35 log points arrive at 3.5–4.3% a year. For college workers the model reproduces the direction of the decline of V^+/V^- with age and most of its size, overshooting it at two years (1.09 to 0.66 against 1.22 to 0.84) and reaching two-thirds of it at six (1.07 to 0.58 against 1.16 to 0.43); for non-college workers it reproduces less than half of the six-year decline (0.15 against 0.33). It misses the young college right skew and the deepest late-career left tails (0.73 and 0.58 against 0.54 and 0.43 at six years).

W.2 Cell laws

Each industry–education–age cell c modifies its education-type base law through four parameters $(a_c^{\text{exit}}, b_c, s_c, a_c)$. The first multiplies the wage-work departure hazard by $e^{a_c^{\text{exit}}}$ before the 0.6 cap. The

Table W.2. PSID moments of h-year changes in log real wages by origin age band: model and data

	25–34		35–44		45–54		55–58	
	Model	PSID	Model	PSID	Model	PSID	Model	PSID
Non-college								
2-year V^+/V^-	0.90	0.94	0.88	0.82	0.90	0.91	0.77	0.74
2-year variance	0.22	0.24	0.21	0.20	0.20	0.20	0.26	0.19
4-year V^+/V^-	0.88	0.91	0.87	0.84	0.84	0.85	0.74	0.65
4-year variance	0.29	0.29	0.26	0.26	0.25	0.26	0.33	0.30
6-year V^+/V^-	0.88	0.87	0.87	0.81	0.82	0.80	0.73	0.54
6-year variance	0.35	0.33	0.31	0.27	0.31	0.30	0.40	0.44
College								
2-year V^+/V^-	1.09	1.22	1.01	0.93	0.95	0.80	0.66	0.84
2-year variance	0.21	0.22	0.19	0.16	0.19	0.18	0.31	0.26
4-year V^+/V^-	1.09	1.17	0.99	0.89	0.84	0.75	0.59	0.47
4-year variance	0.27	0.28	0.23	0.22	0.26	0.25	0.46	0.46
6-year V^+/V^-	1.07	1.16	0.98	0.96	0.73	0.67	0.58	0.43
6-year variance	0.33	0.31	0.27	0.26	0.36	0.35	0.58	0.59

Notes: The ratio of upper to lower partial variance, V^+/V^-, and the variance of residualized h-year changes in log real wages of PSID heads aged 25–58 at the origin with wages above \$5,000 at both dates (1999–2019, wave means and an age cubic removed within type), by origin band, against the model's moments measured with PSID reporting error. PSID bootstrap standard errors by person are 0.03–0.10 for the ratios and 0.01–0.07 for the variances. Fitted moments, not regression estimates.

second changes deterministic growth: with younger and older coefficients b_{g0} and b_{g1},

$$B_g(t) = \begin{cases} b_{g0}(t-35), & t < 45, \\ 10b_{g0} + b_{g1}(t-45), & t \geq 45. \end{cases}$$

The function is continuous at 45. The scale parameter s_c multiplies temporary log-earnings nodes and reporting-error variation by e^{s_c}. The fourth parameter a_c tilts both temporary-shock probabilities and persistent jump probabilities in the baseline.

Temporary-shock tilt. Let e_j and p_j^0 be the ten base temporary-shock nodes and probabilities. Write $\mu_0 = \sum_j p_j^0 e_j$, $\sigma_0^2 = \sum_j p_j^0 (e_j - \mu_0)^2$ and $x_j = \mathrm{clip}((e_j - \mu_0)/\sigma_0, -4, 4)$. The cell probabilities and rescaled nodes are

$$\begin{aligned} p_j^c &= \frac{p_j^0 \exp\{a_c \operatorname{sign}(x_j) x_j^2\}}{\sum_k p_k^0 \exp\{a_c \operatorname{sign}(x_k) x_k^2\}}, \\ \epsilon_j^c &= e^{s_c} e_j - \log\left(\sum_k p_k^c \exp(e^{s_c} e_k)\right) + \log\left(\sum_k p_k^0 \exp(e_k)\right). \end{aligned} \tag{W.1}$$

Positive a_c raises the relative probability of upper nodes. The location correction preserves the expected temporary wage multiplier, $\sum_j p_j^c e^{\epsilon_j^c} = \sum_j p_j^0 e^{e_j}$. It does not preserve temporary-shock

variance. It also does not hold expected career earnings fixed when growth, employment risk and persistent dynamics change.

Persistent-jump tilt. For each ten-year band b within cell age band c, define

$$\widetilde{p}^{+}_{cb} = \chi_{cb} p^{+}_{b} e^{a_c}, \qquad \widetilde{p}^{-}_{cb} = \chi_{cb} p^{-}_{b} e^{-a_c},$$
$$\chi_{cb} = \frac{p^{+}_{b}(J^{+})^2 + p^{-}_{b}(J^{-}_{b})^2}{p^{+}_{b} e^{a_c}(J^{+})^2 + p^{-}_{b} e^{-a_c}(J^{-}_{b})^2}. \tag{W.2}$$

The normalization preserves the raw jump second moment before caps and state dependence. Each $\widetilde{p}$ is capped at 0.45, multiplied by the corresponding state factor $\exp(k^{+}\eta/\sigma_\eta)$ or $\exp(-k^{-}\eta/\sigma_\eta)$, and capped again at 0.45. The stationary-mean drift is then recomputed for the discretized chain. These steps do not preserve the variance of the complete innovation or the stationary earnings distribution. The baseline applies the same a_c to both components; the persistent-only and temporary-only allocations are alternative assumptions about where cell asymmetry resides.

Joint calibration and identification. All four cell adjustments are fitted jointly to the measured zero-wage share, mean growth and upper and lower partial variances. Targets are shrunk toward their education-type and broad-age pool, with moment-specific empirical-Bayes weights that fall as estimated sampling variance rises. For mean growth, the target adds the cell's shrunk deviation from the empirical pool to the model pool's mean; the other three targets use the shrunk moments directly. The model passes its measured wage histories, including reporting error, through the paper's age adjustment and partial-variance calculation. Hence the fit disciplines the same earnings objects used in the empirical analysis while leaving portfolio-risk coefficients untargeted.

Figure W.1 plots model moments against the CPS: the correlations are 0.99 for the two partial variances and 0.98 for composition and the zero share. Mean growth (not plotted) is matched less closely (correlation 0.92) because the cell growth shift is bounded at two percentage points a year, a bound reached in 26 of the 58 cells. Exit, growth and scale carry much of the cross-cell variation in composition (correlations with cell composition -0.8, -0.45 and -0.4). The one-year moments do not precisely identify a_c: independently drawn, identically distributed temporary log shocks have a symmetric difference even when their levels are skewed, and tilting rare persistent jumps makes little difference over one year. Under the baseline allocation the fitted tilts are small (median absolute value 0.09 across the 58 cells, correlation 0.13 with cell composition); when the tilt is allocated entirely to persistent jump hazards, 43 of the 58 cells sit at their bounds. This persistent allocation also changes the stationary variance of the persistent component by factors between 0.4 and 3.7 across cells and age bands (median 1.04); the corresponding factor is at most 1.18 under the baseline allocation. Matching the cell moments therefore does not identify a unique persistent-versus-temporary source of their asymmetry, and the cell-versus-pooled portfolio comparison changes the full earnings environment rather than a_c alone.

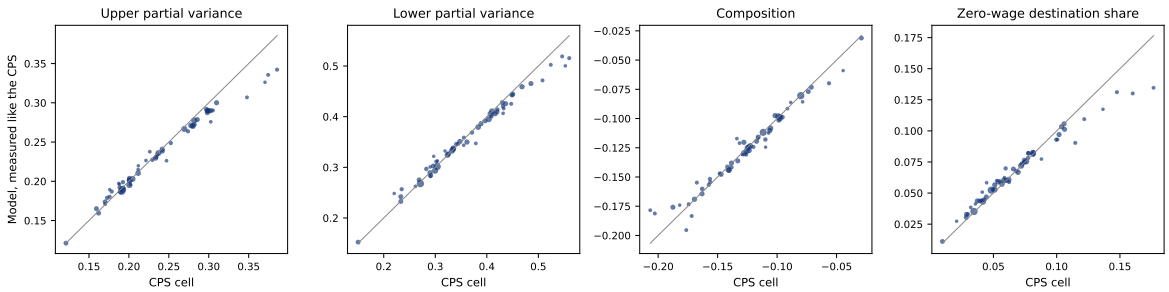


Figure W.1. Cell moments in the model and in the CPS

Each point is one of the 58 industry–education–age cells, with size increasing in the number of CPS records. The model moment is measured on the simulated one-year law with the paper's pipeline (Section 3); the CPS moment is the unshrunk cell value. Because the laws are fitted to empirical-Bayes-shrunk values, small cells lie off the diagonal, on the side of their education pool. Upper and lower partial variance, composition, and the share of destinations with no wage.

W.3 Persistence across earnings cells

The PSID shows whether asymmetry differences across cells persist over longer horizons. Table W.3 assigns PSID heads to their cell at the origin, forms observation-weighted terciles of the paper's cell composition measure, and reports the top-minus-bottom gap in partial-variance moments of two-, four- and six-year changes about each cell's own mean change, and in the share without a wage, with bootstrap standard errors by person. The gap in the paper's zero-inclusive composition measure is about 80% as large at six years as at two and five standard errors from zero, and it is carried by the 2.3-percentage-point gap in the probability of having no wage, which is flat in the horizon; among workers with a wage at both dates the gap is small and its horizon profile cannot be determined. The three allocations of the tilt produce the same continuing-wage gaps in $V^+ - V^-$ to within 0.003, so the allocation is not identified by these moments either; all three carry the nonemployment gap, at about three times the PSID's size because the cell exit shifts match the CPS one-year cell zero shares, whose cross-cell dispersion the PSID does not reproduce.

The continuing-wage comparisons use the calibration moments defined above. Applying the age-cubic adjustment to simulated levels changes the top-minus-bottom composition gap by up to 0.00163; the spread across the three allocations remains below 0.003. These moments leave the persistent allocation of cell asymmetry weakly identified.

W.4 Persistence of wage-work interruptions

In the PSID, interruptions of wage work often last several years, especially for non-college heads. Of prime-age heads with a wage above $5,000 who have no wage two years later, 43% (non-college) and 57% (college) have such a wage again four years after the origin and 46% and 61% six years after it. Those who do earn 0.15 and 0.22 log points less than the continuously employed, measured from the origin (0.08 and 0.05 at six years). In the calibrated law, a lost wage is usually a one-year interruption with no lasting loss: the fitted re-entry hazard is at its cap of 0.95 a year before 45 (before 55 for college workers) and 0.3–0.7 later. The resource schedule also partly insures a wage

Table W.3. Is cross-cell composition persistent? Top minus bottom tercile of cells in the PSID and in the model

	2 years	4 years	6 years
Panel A. Paper's zero-inclusive composition (arc change, zeros at -2)			
PSID	0.061	0.048	0.048
	(0.008)	(0.009)	(0.009)
Model, asymmetry split across both components	0.106	0.109	0.105
Model, transitory component only	0.107	0.109	0.105
Model, persistent jump hazards only	0.111	0.116	0.112
Panel B. Share without a wage at the destination (pp)			
PSID	-2.2	-2.4	-2.4
	(0.3)	(0.4)	(0.4)
Model, asymmetry split across both components	-6.2	-7.4	-8.3
Model, transitory component only	-6.2	-7.5	-8.3
Model, persistent jump hazards only	-6.2	-7.5	-8.4
Panel C. Continuing-wage $V^+ - V^-$ (wages above \$5,000 at both dates)			
PSID	0.014	0.003	0.018
	(0.007)	(0.009)	(0.011)
Model, asymmetry split across both components	0.008	0.012	0.014
Model, transitory component only	0.008	0.013	0.014
Model, persistent jump hazards only	0.009	0.015	0.015
Panel D. Continuing-wage V^+/V^-			
PSID	0.11	0.00	0.08
	(0.06)	(0.06)	(0.07)
Model, asymmetry split across both components	0.06	0.06	0.06
Model, transitory component only	0.05	0.07	0.06
Model, persistent jump hazards only	0.07	0.09	0.07

Notes: Cells are ranked by the paper's composition measure and PSID heads (1999–2019, origin age 25–58, origin wage above \$5,000) are assigned to their cell at the origin year. Entries are differences between the top and bottom observation-weighted terciles of cells, with bootstrap standard errors by person in parentheses: partial-variance moments of h-year changes about each cell's own mean change (Panels A, C and D) and the share without a wage (Panel B). Panel A uses the paper's zero-inclusive arc change with zero destination wages placed at -2; Panel B is the share of origin workers with no wage h years later, in percentage points; Panels C and D use heads with wages above \$5,000 at both dates. Model rows apply the same construction to simulated panels under the three allocations of the cell asymmetry tilt; they differ from each other by less than one PSID standard error in every panel (at most 0.007 in Panel A, 0.1 percentage point in Panel B, 0.003 in Panel C and 0.02 in Panel D), so the allocation is not identified by these moments. The model's zero-share gap is larger than the PSID's because the cell exit-hazard shifts match the CPS one-year cell zero shares. Descriptive moments, not regression estimates.

interruption: the transfer floor and the possibility of other own earnings leave \$3,000 plus 62% of the persistent wage in expectation, before taxes. The model therefore understates the persistence of observed wage-work interruptions.

X Conditional Life-Cycle Model: Environment and Methods

This appendix describes the model environment, observed-state initialization, entry opportunities, participation costs and conditional predictions used in Section 6. Appendix W documents the career

earnings calibration. Appendix Y reports financial estimation, fit and comparisons outside the fitting criterion. Appendix Z considers permanent participation types.

X.1 Observed states and sample

The model studies financial transitions from workers' observed portfolios. The decision unit is the individual, and the initial distribution of wealth and ownership is an input. The prospective SIPP frame contains 31,569 worker-year pairs. Restricting initial ages to 25–63 leaves 30,890 pairs; requiring complete positive annual employer wages at both dates leaves 27,530; and applying the reported risky-component screen at both dates leaves 21,188. The estimation sample contains 17,606 initial nonholders and 3,582 holders, representing 14,497 people and 11,341 household lineages. Both endpoints require positive gross nonretirement financial assets.

Ownership means a positive stock-and-mutual-fund balance. The equity share is that balance divided by gross nonretirement financial assets, so exits have zero shares. The asset measure excludes retirement accounts and does not subtract liabilities. The reporting screen checks risky amounts and ownership routing but does not verify every safe-asset component of the denominator. Annual employer wages sum monthly earnings across disclosed employee-job lines, requiring twelve distinct months and complete job-count coverage. Suppressed active jobs make wages incomplete. Total labor earnings, including business profits and losses, remain a separate variable. Reference-year CPI converts each endpoint to 2018 dollars.

At the initial year-end, worker i has observed age, industry, education, assets A_{it}, share α_{it}, ownership D_{it} and annual wage w_{it}^{obs}. The latent persistent earnings state η_{it} has posterior

$$\pi_i(e) \propto \Pr_g(\eta_t = e \mid \text{wage work at age } t) f_g(\log w_{it}^{\text{obs}} \mid \eta_t = e, t), \tag{X.1}$$

where the likelihood integrates real transitory earnings and the reporting-error mixture. Wealth and ownership enter the financial state and budget; they do not enter this earnings filter. Reporting-error variance is calibrated to one-quarter of its CPS value, following the annual-survey convention in the PSID comparison. This is a SIPP measurement assumption. Preference, cost and opportunity parameters are common across workers. Workers observe their persistent earnings and work states when choosing; the posterior describes the researcher's uncertainty, not household learning.

X.2 Timing, budgets and recursive preferences

At each subsequent date, the inherited portfolio earns its return and the worker receives disposable resources. A worker who did not hold stocks in the previous period then receives an investment opportunity with probability q; a previous holder has access with probability one. An open opportunity allows the worker to observe the recurring-cost offer and choose whether to hold. A closed opportunity restricts the current portfolio to safe assets, while leaving consumption and saving choices available. Let m be cash before the cost, d prior ownership and $K_t = \kappa_R \xi_t$ the potential recurring cost. The

budget and next cash position are

$$
\begin{aligned}
c_t + A_t + D_t K_t &= M_t, \\
M_{t+1} &= A_t\{R_{f,t+1} + \alpha_t(R^s_{t+1} - R_{f,t+1})\} + Y_{t+1}, \qquad \text{(X.2)} \\
c_t > 0, \quad A_t \geq 0, \quad D_t \in \{0,1\}, &\qquad 0 \leq \alpha_t \leq D_t, \qquad \alpha_t A_t \geq \varepsilon D_t.
\end{aligned}
$$

The minimum funded equity position is $\varepsilon = \$1$. It prevents an unfunded account from preserving access. This constraint applies to model choices; the empirical sample has no corresponding balance cutoff. Model ownership and shares are jointly determined, without independent reporting flips.

Write z for the persistent earnings and wage-work state, $\rho = 1 - 1/\psi$ and $\omega = 1 - \gamma$. Let $V_{jt}(z, x)$ be optimized value conditional on current holding choice j and cash x after any participation payment. The positive, consumption-equivalent recursion is

$$
\begin{aligned}
V_{jt}(z,x) &= \max_{(A,\alpha)\in\mathcal{A}_j(x)} \left[(x-A)^\rho + \beta s_t \left\{ \mathbb{E}\big[\bar{V}_{t+1}(z', j, AR(\alpha) + Y')^\omega \mid z\big] \right\}^{\rho/\omega} \right]^{1/\rho}, \\
\bar{V}_t(z,d,m) &= \Big\{ [1-q(d)] V_{0t}(z,m)^\omega \qquad \text{(X.3)} \\
&\qquad + q(d)\mathbb{E}_\xi \Big[\max\big\{ V_{0t}(z,m), V_{1t}(z, m - \kappa_R \xi) \big\}^\omega \Big] \Big\}^{1/\omega}.
\end{aligned}
$$

Here $q(d) = q + (1-q)d$, and $\mathcal{A}_j(x)$ imposes the budget and funded-position restrictions, and an infeasible participating branch is excluded. The first expectation integrates next earnings and portfolio returns; $\bar{V}$ already incorporates the next opportunity and cost draws. The second equation integrates open and closed opportunities in the same risk certainty equivalent. The opportunity is observed before the worker chooses whether to participate; it is not an adjustment to entry applied after solving an unrestricted problem. An exit exposes the worker to the opportunity restriction in the following year, so q also affects the continuation value of current holders. The primitive opportunity and cost draws are independent across dates and of earnings and stock returns. Survival s_t multiplies discounted continuation. Working life ends at 65. At age 84, $\bar{V}_{84}(z,d,m) = m$: terminal resources are consumed, with no portfolio choice or bequest motive. The recursion omits the usual flow multiplier $1-\beta$. Multiplying all values, including the terminal value, by $(1-\beta)^{1/\rho}$ restores that normalization with the same β; cross-model comparisons also depend on the terminal and resource conventions.

X.3 Earnings and resource calibration

The earnings block has 17 persistent states, 30 industry–education career groups and 58 supported industry–education–age cells. The empirical age bands are 25–44 and 45–65; model workers switch adjustments at 45 and retire at 65. Each career group has a four-parameter adjustment in each band: a departure hazard, a growth shift, a transitory scale adjustment and an asymmetry tilt. These adjustments modify one of two base processes, for college and noncollege workers. High-school and some-college groups share the noncollege base but retain distinct cell adjustments. Of the 60

group–age combinations implied by the 30 career groups, two lack a supported cell estimate; each uses its other age band's adjustment. Thus 58 counts observed earnings environments, 30 counts career groups, and 17 counts numerical persistent states. They are distinct objects. The empirical cell definition and support requirements are described in Section 3; subdividing cells does not refine the numerical solution and would require estimating earnings dynamics with thinner data.

For each career group, the law supplies persistent-state transition probabilities, temporary earnings amounts and probabilities, and departure and re-entry probabilities for wage work. Equation (15) maps employed states to gross wages; the resource conventions below convert those wages, or the income received outside wage work, into disposable resources Y'. Together these objects determine the joint distribution of Y' and next earnings state z' in the first expectation of equation (X.3). Solving the recursion gives consumption, saving and participation rules for that law. Drawing outcomes from the same law then generates cash and state transitions at which the rules are evaluated. Reporting error affects the initial-state likelihood in equation (X.1) and the measurement of earnings moments, but is absent from the resource budget.

The CPS and PSID earnings moments discipline these laws before financial estimation. The four cell adjustments jointly target shrunk zero-wage shares, mean growth and upper and lower partial variances. They need not correspond one for one to those moments. Appendix W.2 gives the tilt formulas, normalization conventions and the weak identification of the persistent-versus-temporary allocation. Earnings parameters remain fixed in the financial fit; portfolio-risk regression coefficients enter neither stage's objective.

For the pooled counterfactual, all eight entries of each career group's two-age-band adjustment matrix are set to zero. Education-specific base processes remain non-Gaussian; the base age profiles $f_h(t)$ and group wage levels ℓ_g remain in place, while the cell growth adjustment $B_g(t)$ is removed. Observed initial wages, wealth, ownership and cell-filtered persistent-state posteriors are identical to the cell calculation. Saving and portfolio policies are solved again under pooled future laws using the same preferences, participation costs and opportunity probability. The empirical projection uses the same original CPS composition and variance measures in both cases. Consequently, pooled predictions may retain an association with those measures through initial states and retained wage levels. Pooling is neither setting every worker's earnings equal nor refitting the model to the financial targets. Section 6 reports the resulting fixed-parameter comparison.

Patience β and risk aversion are estimated; $\psi = 0.5$ remains fixed. The decision problem uses stationary return expectations: the safe gross real return is $R_f = 1.02$, and ordinary stock returns are lognormal with mean 1.06 and standard deviation 0.18, mixed with a one-percent probability of gross return 0.515. The mixture mean is 1.05455. In this decision law, stock returns are independent of earnings and cost shocks. Survival equals one before retirement and $1-\min\{0.00005\exp(0.085t), 0.5\}$ thereafter, where t denotes age.

For observed financial transitions, I apply destination-calendar-year market and safe-return proxies to inherited portfolios and evaluate the life-cycle policies at the resulting cash. The dated safe return is $R_{f,t+1}$ in equation (X.2); future return expectations remain the stationary decision law.

Monthly broad-market and Treasury-bill return proxies from the French data library are compounded within each calendar year and converted to real returns with the same annual CPI endpoint ratio used for SIPP balances.[7] These common proxies do not recover each individual's bond and mutual-fund exposure. Calendar returns and initial portfolios are inputs to the comparison, not predicted model outcomes. Table Y.5 reports the return factors and retirement-resource calibration.

Working resources are $(1 - 0.30 - q_{et})w$, where w is own gross wages and q_{et} is an age–education employee-contribution rate. PSID reference persons aged 25–64 with positive own wages supply required and voluntary employee contributions, including employee payments to defined-benefit plans. Within four age bands and two education groups, the rate is the ratio of weighted own-contribution totals to own-wage totals on the same complete-case sample, using normalized within-wave weights for 2013, 2015, 2017 and 2019. Known nonparticipation contributes zero; unresolved responses remain missing. Rates are interpolated between age-band midpoints and held constant beyond the end midpoints, ranging from 2.01 to 5.44%. Dollar and payment-frequency responses are annualized; percentage responses use own annual wages. These annualized current-job contributions may differ from payments made during a particular calendar year. Applying reference-person rates and a common tax wedge to all workers also requires approximation. Employer payments are not deducted. Without wage work, pre-tax resources contain $3,000 plus 30 percent of the persistent wage and, with probability 0.35, additional own labor income of 90 percent of that wage.

Retirement resources use own Social Security and non-VA pension income of PSID reference persons, by education and income-year age. Nonrecipients and workers remain in the averages, with zero benefits retained. Weighted means within ages 65–74 and 75–84 give each of the four survey waves equal total weight; the income reference years are 2012, 2014, 2016 and 2018. The common 30-percent tax wedge converts gross benefits to disposable resources. Spouse benefits, IRA withdrawals, annuity income and other retirement receipts do not enter this schedule. The latter three are retained as separate coverage diagnostics. The reference-person population need not represent every adult. Contribution deductions and retirement benefits are calibrated separately; the model does not track an individual retirement account or its depletion. Numerical budgets use thousands of 2018 dollars; the observed wage supplied to the earnings filter is in dollars.

X.4 Entry opportunities and recurring costs

Let $d = D_{t-1}$ and let $\xi_t = \exp(\sigma_\kappa \nu_t - \sigma_\kappa^2/2)$, where ν_t is an iid standard normal draw observed when participation is available. Potential cost is

$$K_t = \kappa_R \xi_t, \qquad \Pr(O_t = 1 \mid D_{t-1} = d) = q(d). \tag{X.4}$$

The worker pays this cost only when choosing a funded stock position. Its offered mean is κ_R and its variance is $\kappa_R^2[\exp(\sigma_\kappa^2) - 1]$. There is no additional cash fee after nonholding. The opportunity

[7] Kenneth R. French Data Library, https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/Data_Library/f-f_factors.html.

restriction does not consume resources: a closed opportunity changes the feasible portfolio set. It is a parsimonious way to allow intermittent access or consideration, rather than an estimate of attention, trust or time spent acquiring financial information. No household has a permanent access type in this specification.

To describe selection, suppress age and earnings-state arguments. Let $V_0(m)$ be the optimized value of nonparticipation at cash m, and let $V_1(x)$ be the optimized value of funded participation at cash x after paying its cost. Define

$$x^*(m) = \inf\{x \in \mathcal{X}_1 : V_1(x) \geq V_0(m)\}, \qquad B(m) = m - x^*(m), \tag{X.5}$$

where $\mathcal{X}_1$ is the feasible cash set for funded participation and an empty set gives $x^* = +\infty$. Monotonicity of the participating value makes $B(m)$ the largest cost consistent with choosing stocks. Equality at the threshold has probability zero under a continuous cost draw.

Write $F_\kappa(u) = \Pr(\xi_t \leq u)$ and $H_\kappa(u) = \mathbb{E}[\xi_t \mathbf{1}\{\xi_t \leq u\}]$. For $u > 0$,

$$F_\kappa(u) = \Phi\left(\frac{\log u + \sigma_\kappa^2/2}{\sigma_\kappa}\right), \qquad H_\kappa(u) = \Phi\left(\frac{\log u - \sigma_\kappa^2/2}{\sigma_\kappa}\right), \tag{X.6}$$

and both functions equal zero for $u \leq 0$. With $u = B(m)/\kappa_R$, the participation probability conditional on cash and the other state variables is $q(d)F_\kappa(u)$. Expected payment, including zero for nonparticipants, is

$$\mathbb{E}[K_t D_t \mid m, d] = q(d)\kappa_R H_\kappa(u). \tag{X.7}$$

The formulas condition on the suppressed earnings state. The reported selected-payment summaries use simulated paths with next-year wage work and positive financial assets. Within each origin, the sum of payments on those paths and the count of participating paths are each divided by the number of eligible paths. The survey-weighted payment sum is then divided by the survey-weighted participation sum, separately for initial nonholders and holders. Thus selection into participation determines the weights in the mean payment; the calculation does not average origin-specific entrant means with unchanged survey weights. If an origin has no eligible simulated path, the full-origin selected mean is undefined, and the supported-origin mean is reported explicitly. Workers participate when the draw is low enough, so the mean paid cost differs from the mean potential cost. These are resource-equivalent access costs, not observed brokerage fees.

X.5 Conditional financial moments

The participation and allocation targets are next-year entry rates in three initial nonholder wealth terciles, retention rates in two initial-holder share bins, and the mean next-year equity share of all initial holders, including exits at zero. Weighted empirical quantiles determine the cutoffs, with ties assigned to the lower bin. Model predictions use those fixed empirical memberships. Aggregate entry and exit are weighted averages of the bin rates and are reported as diagnostics rather than

additional targets. Mean one-year changes in real gross nonretirement financial wealth for initial nonholders and holders use the same financial-transition sample. Their magnitudes enter financial estimation, and their positive signs are constrained. Together these define eight financial targets. Wealth changes use thousands of 2018 dollars, and rates and shares use proportions. Open and closed opportunity branches are integrated into the outcome and eligibility masses before forming each origin's conditional prediction. Conditioning branches separately and then weighting their conditional means would change the moment when eligibility differs across branches.

The empirical origins already satisfy destination eligibility. Predictions retain their weights without selecting the origin distribution a second time. Let $\mathcal{E}_{i1}$ denote positive next-year wage work and financial assets, and let $\mathcal{I}_i$ contain the observed origin and its earnings posterior. The calendar-return input is fixed in each expectation and eligibility probability. For outcome h_j and empirical bin $\mathcal{B}_j$, the model moment is

$$m_j(\theta) = \frac{\sum_{i \in \mathcal{B}_j} w_i \dfrac{\mathbb{E}_\theta[h_j \mathbf{1}\{\mathcal{E}_{i1}\} \mid \mathcal{I}_i]}{\Pr_\theta(\mathcal{E}_{i1} \mid \mathcal{I}_i)}}{\sum_{i \in \mathcal{B}_j} w_i}. \tag{X.8}$$

Here w_i is the destination sampling weight. Zero model eligibility makes the conditional prediction undefined; such origins are not deleted. For wealth growth, $h_j = A_{i,t+1} - A_{it}$. A holder moment conditions on initial holding, not on continued ownership at the destination.

Uncertainty in the empirical moments is measured with 299 common exponential household-lineage multiplier draws, recomputing the quantile cutoffs and moments while retaining the survey weights. This is lineage resampling conditional on those weights. Appendix Y states the entry, holder and wealth covariance blocks, their relative priority, the positive-growth constraints, parameter bounds and bounded search. Earnings-risk regression coefficients do not enter fitting or numerical acceptance.

X.6 Numerical solution and model comparisons

Backward induction uses 93, 141 and 201 cash nodes for the coarse, fine and accuracy calculations. Each grid includes the funded-position boundary and extends to $100 million. Ordinary return quadrature uses three, five and seven nodes, each supplemented by the crash state. Portfolio choice searches a coarse share grid, the feasible endpoints and the best interval with scalar refinement. Saving respects the funded constraint inside the value calculation. Conditional cost integration uses five Gauss–Legendre nodes; participation probabilities use the analytic lognormal CDF. Return quadrature integrates future portfolio returns in the decision problem. First-year financial moments instead enumerate the finite earnings law and integrate participation costs at each origin's observed calendar return. The fitting criterion therefore does not depend on a simulation seed.

The selected financial parameters must be a feasible, converged terminal point. Fixed-parameter calculations at 141 and 201 cash nodes check both wealth signs and the stability of all eight fitted financial moments. Appendix Y gives the component-specific tolerances. The numerical checks verify

budgets, funded ownership, consistent shares, positive probabilities of eligibility and coverage by the cash grid.

The cells-versus-pooled comparison uses the same observed origins, cell-filtered earnings posteriors, measured retirement schedule and calendar return inputs. The pooled laws remove the cell adjustments while retaining education-specific wage profiles and group wage-level shifts; each law has its own optimal policies. The common regression design includes the five CPS risk regressors, monthly-income and gross-wealth controls, age and age squared, sex, unemployment, and year, industry, education and age-band effects. Empirical inference clusters by industry–education group, conditional on the saved CPS moments. Projecting both predictions on this design measures the contribution of the cell adjustments to each regression coefficient. Those adjustments change mean resources and risks together, so the comparison does not hold mean resources fixed. Let p_i^{cell} and p_i^{pool} be the conditional entry probabilities under the two earnings laws, X the common full-rank design and W its weight matrix. The selector e_C picks the standardized composition column. The net projection reported in Table 13 is

$$\Delta_C = 100\, e_C'(X'WX)^{-1}X'W\big(p^{\text{cell}} - p^{\text{pool}}\big). \tag{X.9}$$

The composition, variance, mean-growth and volatility columns use fixed pre-adjustment CPS-cell means and standard deviations; the fifth risk regressor is squared standardized mean growth. Thus the composition coefficient is already scaled to one source standard deviation, without rescaling to the SIPP sample. Write U and L for upper and lower partial variances, $C = U - L$, $V = U + L$ and $S = \sqrt{V}$. Let b_C, b_V, b_S denote the coefficients on the standardized columns, and s_C, s_V, s_S their fixed source standard deviations. At the common sample-weighted mean volatility $\bar{S}$, the four rows of Table 13 are

$$\begin{aligned} \mathcal{D}(\bar{S}) &= \frac{s_C}{s_V} b_V + \frac{s_C}{2\bar{S}s_S} b_S, \\ T_C &= 100 b_C, \qquad T_D = 100\mathcal{D}(\bar{S}), \\ T_U &= 100[b_C + \mathcal{D}(\bar{S})], \qquad T_L = 100[-b_C + \mathcal{D}(\bar{S})]. \end{aligned} \tag{X.10}$$

Holding mean growth and the remaining controls fixed, T_U and T_L are local derivatives with respect to one partial variance while the other stays fixed, evaluated at $\bar{S}$ and multiplied by $100 s_C$. Their half-difference is the composition contrast T_C; their half-sum is the dispersion contrast T_D, the scaled local variance derivative at fixed composition. Dispersion includes both the variance and square-root-variance terms. These derivatives are evaluated at mean volatility, rather than averaged across observations. The identities also hold for the cell-minus-pooled column by linearity. The model comparison changes all cell earnings adjustments together. These regression contrasts therefore do not isolate changes to individual tails or describe exact finite changes in partial variances.

Model projections carry no coefficient significance stars. A separate 201-node, seven-return-node calculation checks each of four net projections against the fine-grid value, using an absolute tolerance of 0.05 percentage points. The return-node counts refer to the stationary decision problem; both resolutions apply the same dated returns to observed transitions.

Untargeted comparisons examine age profiles and ownership at successive annual observations. Multi-year comparisons use the same complete-case cohorts and condition on positive wages and assets at every required date. They measure recorded annual ownership, not continuous market access. Each observed follow-up applies its destination-year return proxies. A separate windfall calculation adds $150,000 to the next cash position of observed age-44 nonholders and follows paired portfolios through ages 45–47 without resetting assets. This hypothetical experiment uses the stationary return process because no future calendar realization is specified. It conditions on future wage work but does not select on post-windfall assets. The Swedish lottery estimate supplies an external economic benchmark, with a different population and estimand.

X.7 Earnings laws and portfolio decisions

The cell-law comparison combines two economic responses. Different earnings paths change the resources available to finance participation, and households choose different saving and portfolio rules when they face those paths. To separate these responses, I cross the cell and pooled earnings laws with the optimal policies under each law, holding financial parameters fixed. All four calculations start from the same observed portfolios and cell-filtered distributions over initial earnings states. They share the measured retirement schedule and dated return proxies, as well as identical state and cash grids. A policy specifies the value functions and saving and share rules; participation is evaluated at realized cash under the common opportunity and cost distributions, integrating both sources of uncertainty.

Let $p_i^{\ell q}$ be the conditional entry probability when earnings follow law $\ell \in \{C, P\}$ and choices use policy $q \in \{C, P\}$, where C denotes cells and P denotes pooled. Each calculation conditions within origin i on future wage work and positive assets under that calculation. The symmetric decomposition is

$$\begin{aligned} \phi_i^Y &= \tfrac{1}{2}\big[(p_i^{CC} - p_i^{PC}) + (p_i^{CP} - p_i^{PP})\big], \\ \phi_i^q &= \tfrac{1}{2}\big[(p_i^{CC} - p_i^{CP}) + (p_i^{PC} - p_i^{PP})\big], \\ \phi_i^Y + \phi_i^q &= p_i^{CC} - p_i^{PP}. \end{aligned} \tag{X.11}$$

This averages the two possible orders of changing the law and the policy. Applying the common weighted projection to each component preserves the identity for composition and every tail contrast. The earnings-law component includes state transitions and changes in conditional support as well as realized resources. The policy component changes saving and portfolio choices together. Because several features change within each component, neither identifies a perception effect, a mean-preserving risk rotation, or causal mediation. The four corner projections and both signed contributions appear in Table 14.

X.8 Entry followed by sustained ownership

I also estimate earnings-risk gradients for entry followed by ownership at subsequent annual observations. The primary cohort contains 5,459 initial nonholders observed at two future dates; the

secondary cohort contains 1,801 observed at three. Initial ages are 25–62 and 25–61, respectively, so all follow-up dates precede the model's retirement age. Every required date meets the same wage-work, positive-assets and reported-risky-component screens. In the model, future eligibility requires wage work and positive assets at each date; funded stock positions determine ownership without an additional reporting event. Let $\mathcal{E}_i^h = \bigcap_{s=1}^h \mathcal{E}_{is}$ denote this future eligibility. For $h \in \{2, 3\}$, the model predicts

$$p_i^{\ell,h} = \frac{\mathbb{E}_\ell\left[\left(\prod_{s=1}^h D_{i,t+s}\right)\mathbf{1}\{\mathcal{E}_i^h\} \mid \mathcal{I}_i\right]}{\Pr_\ell(\mathcal{E}_i^h \mid \mathcal{I}_i)}. \tag{X.12}$$

Cell and pooled economies evolve saving and ownership without resetting the individual's portfolio. They use the same dated return sequence and cell-filtered initial earnings posterior. The data indicator and each model probability are projected on identical regressors and weights within the relevant cohort. The CPS source scaling is fixed; local tail derivatives use that cohort's weighted mean volatility. These smaller cohorts have their own empirical coefficients, rather than inheriting the coefficient from a larger sample. Empirical inference uses CR1 standard errors and the t_{29} reference distribution for 30 industry–education clusters. Model projections carry no sampling stars or confidence intervals.

Two fixed path seeds assess simulation variability, beginning with 512 paths per origin. Origins with no eligible path are retried with 2,048 and then 8,192 paths. If support remains unresolved, the origin remains in the weighted regression sample, with its conditional probability bounded by $[0, 1]$. The resulting bounds propagate through the linear projection rather than treating that origin as a nonparticipant or deleting it. Both seeds and any unresolved bounds are reported. No sustained outcome or coefficient enters financial estimation. Holdings at annual observations do not establish continuous ownership between interviews.

Y Opportunity Access: Estimation and Additional Results

This appendix gives the financial estimation, participation diagnostics and additional experiments for the opportunity model in Section 6. The earnings and resource calibration is common to the opportunity model and the permanent-type alternative in Appendix Z. Observed initial assets and ownership are inputs to both specifications.

Y.1 Financial targets and estimation

Let $r_E(\theta)$ contain the three entry-rate residuals by initial nonholder wealth tercile, $r_H(\theta)$ the two retention-rate residuals and next-year share residual for initial holders, and $r_W(\theta)$ the two wealth-change residuals by initial ownership. The next-share moment includes exits at zero. Financial wealth is gross of liabilities and excludes retirement accounts; its change includes returns on inherited assets. Empirical group memberships and survey weights remain fixed during estimation.

The five estimated parameters are $\theta = (\gamma, \beta, \kappa_R, \sigma_\kappa, q)$. Entry across wealth groups helps discipline the frequency of opportunities and the cost of investing relative to available resources. Retention helps

separate the recurring fee from the additional restriction after nonholding. Shares and accumulation discipline preferences jointly with access. These relationships are not one-to-one identifications of parameters by moments. All parameters can affect several outcomes through the dynamic problem.

With $\widehat{\Sigma}_j$ denoting the marginal covariance block, the criterion is

$$\begin{aligned} Q(\theta) = {} & r_E(\theta)'[\widehat{\Sigma}_E + s_E^2 I_3]^{-1} r_E(\theta) \\ & + 0.1\, r_H(\theta)'[\widehat{\Sigma}_H + s_E^2 I_3]^{-1} r_H(\theta) \\ & + r_W(\theta)'[\widehat{\Sigma}_W + s_W^2 I_2]^{-1} r_W(\theta), \qquad s_E = 0.0025, \quad s_W = 2. \end{aligned} \tag{Y.1}$$

Rates and shares enter as proportions and wealth changes as thousands of 2018 dollars. The covariance blocks use 299 common household-lineage multiplier draws. Within-block covariance is retained and cross-block covariance omitted. The holder block receives one-tenth weight; the wealth regularizer permits broad dollar discrepancies while penalizing their size. The weights reflect economic priorities chosen during model development. The criterion is neither efficient SMM nor a specification-test statistic.

For $d \in \{0, 1\}$, mean financial-wealth growth $g_d(\theta)$ uses the within-origin conditioning in equation (X.8). Optimization imposes $g_d(\theta) \geq 0.5$, a \$500 positive buffer for both initial ownership groups. Final acceptance instead requires positive growth at both numerical resolutions, with wealth differences below \$250. The buffer facilitates the search; it is not a target of exactly \$500 or a restriction imposed on every subgroup.

Table 11 reports the estimates. Parameter bounds are $\gamma \in [2, 20]$, $\beta \in [0.60, 0.99]$, $\kappa_R \in [\$1, \$1{,}000]$, $\sigma_\kappa \in [0.1, 2.5]$ and $q \in [0.01, 1]$. The recurring-cost bound constrains the offered mean, rather than each lognormal offer or each selected payment. It is an exploratory plausibility restriction, not an externally identified fee cap. The elasticity of intertemporal substitution remains $\psi = 0.5$.

Table Y.1 records the bounded searches. Two 93-node starts precede a 141-node refinement. After the initial refinement reached its evaluation budget, one continuation with unchanged targets, bounds and numerical stopping rules was allowed before examining the earnings-risk or dynamic comparisons. This continuation converged after 61 objective calls. The selected estimate is its feasible terminal point, not an unconverged best visited point. A separate 201-node evaluation checks accuracy; global optimality is not claimed. The financial parameters remain fixed in the subsequent comparisons, apart from explicitly labeled preference sensitivities. No earnings-risk, age, duration, windfall or account-size outcome enters their estimation.

Y.2 Fit, accumulation and earnings-law accounting

Table 12 reports all eight financial targets and Table 13 applies the empirical design to the same 17,606 initial nonholders. Cell and pooled calculations share financial parameters, observed origins, cell-filtered initial earnings posteriors, retirement resources and dated returns. The pooled calculation is an accounting comparison, rather than a separately fitted model. Its holder wealth growth is

Table Y.1. Opportunity access: bounded financial search

Search	Calls	Q	Stopping status
Final continuation	61	9.770	Converged
Coarse start 1	49	41.411	Converged
Coarse start 2	52	14.109	Converged
Initial fine search	100	10.507	Budget limit

Notes: Coarse searches use 93 cash nodes and three ordinary-return nodes; fine searches use 141 and five. A finite continuation was declared after the first fine searches exhausted their budgets and before untargeted diagnostics. Only natively converged, feasible terminals passing the 201/seven-node financial screens were selectable. The criterion is $Q_E + 0.1Q_H + Q_W$. All outcomes, including budget stops, are retained; global optimality is not claimed.

approximately $-\$483$, so positive growth in the fitted cell-law economy does not extend to every counterfactual.

The model predicts positive annual financial-wealth growth for both initial ownership groups. It nevertheless overpredicts nonholder accumulation and underpredicts holder accumulation. The nonholder discrepancy is especially large in the middle and upper wage groups: predicted growth is about $8,700 and $17,500, compared with roughly $200 and $9,200 in the data. Holder subgroups can have negative growth even though the overall holder mean is positive. Financial-wealth changes include portfolio returns; holder saving net of returns remains negative. These distinctions prevent an aggregate direction restriction from being interpreted as a fit to every saving margin.

The cell-law model produces positive composition and upper-tail entry gradients and a negative lower-tail gradient. The pooled upper-tail projection is negative. Their difference has the empirical opposing-tail pattern but remains smaller than the matched empirical gradients. Composition and dispersion are the half-difference and half-sum of the tail derivatives in equation (X.10); the four rows are not four independent validation moments.

Table 14 attributes about 0.27 percentage point of composition to earnings laws and -0.03 to decision rules. Policies dampen the upper-tail contribution and reinforce the negative lower-tail contribution. Each corner conditions on its own future wage work and positive assets. The accounting includes changes in transitions and conditional support, rather than identifying pure beliefs, a mean-preserving tail rotation or causal mediation.

Y.3 Participation payments and ownership histories

The offered recurring cost averages about $524. Actual participants select offers they are willing and able to pay. Among supported entrant origins, the mean selected payment is about $235; among retained holders it is about $359. These are means of model resource costs, not observed brokerage fees or medians of offered costs. Closing an opportunity charges no fee.

The simulated first-year payment and account-size comparisons have eligible paths for 17,605 of the 17,606 initial nonholders, covering 99.994% of survey weight. Full-origin entrant means are therefore undefined in that finite simulation. Table Y.2 identifies the supported-origin means, and the account-size comparison retains the missing origin's possible contribution. This small support

Table Y.2. Opportunity access: offered and paid participation costs

Cost concept	2018 dollars
Potential recurring mean	524
Entrants: selected recurring payment	235
Retained holders: selected recurring payment	359

Notes: The recurring cost is consumed only upon funded stockholding. The entrant payment mean covers 17,605 supported origins, or 99.994% of origin survey weight; one unresolved origin prevents a full-sample selected mean. Retained-holder payments have complete support. Selected payments are simulation summaries, whereas the potential mean is a parameter. The opportunity restriction itself consumes no money.

gap does not explain the substantial underproduction of small entrant positions: approximately one quarter of observed entrants hold $1,000 or less, whereas the model's upper bound is about one tenth of one percent. Positions close to the model's $1 funding minimum are negligible. Preserving access through token holdings is not driving aggregate retention in these calculations, but the model does not match the observed distribution of small accounts.

Previous ownership provides a separate comparison. Among the 408 initial nonholders with an affirmative, quality-screened holding observation in the preceding two years, observed re-entry is about 27%, compared with about 10% in the opportunity model. The opportunity restriction treats all current nonholders alike at the same state, regardless of more distant ownership history. Among the other 17,198 nonholders, predicted and observed entry are approximately 6.3% and 5.7%. This second group includes missing or allocated histories and is not a sample of proven lifetime nonparticipants.

Y.4 Sustained ownership and windfalls

Table Y.3. Opportunity access: outcomes outside the financial objective

Outcome	Data	Model
Entry and holding at 2 annual follow-ups (%)	3.48	4.19–4.22
Entry and holding at 3 annual follow-ups (%)	2.27	3.32–3.34
Re-entry: reported prior holders (%)	27.16	9.83
Entry: no reported prior holding (%)	5.73	6.27
Windfall ownership response, year 0 (pp)	–	7.74
Windfall ownership response, year 1 (pp)	–	12.79
Windfall ownership response, year 2 (pp)	–	15.25
Entrants with a position at most $1,000 (%)	24.22	0.007–0.101

Notes: Sustained outcomes use the common 5,459- and 1,801-origin cohorts and show the range across two simulation seeds. Annual observations do not establish continuous ownership. Reported-history groups contain 408 prior holders and 17,198 other nonholders; the latter are not classified as lifetime nonparticipants. The hypothetical $150,000 windfall starts from 423 observed age-44 nonholders, arrives at age 45, and uses stationary stochastic returns; it differs from the Swedish lottery estimand. The small-position interval bounds one unresolved origin and is not a sampling interval. No row selects parameters.

Sustained ownership is evaluated on the common cohorts of 5,459 and 1,801 initial nonholders described in Appendix X. The model overpredicts ownership at all three successive follow-up dates:

about 3.3% compared with 2.3% in the data. The corresponding empirical composition gradients are imprecise. Both path seeds remain part of the comparison; differences between them measure simulation variability rather than sampling uncertainty. Recorded annual holdings do not establish continuous market participation between observations.

The $150,000 windfall experiment follows 423 observed age-44 nonholders through ages 45–47. It raises ownership by about 7.7 percentage points immediately and 15.2 after two years. Initial nonholders who fail to receive an opportunity immediately can enter later while retaining some of the additional resources. The paired experiment uses stationary stochastic returns and conditions on future wage work without selecting on post-windfall assets. The comparison with Swedish lottery evidence concerns economic scale, not an identical population, conditioning rule or estimand.

Y.5 Decision rules and the life-cycle experiments

The fixed-state policy comparisons integrate current access and cost uncertainty at common cash, age and earnings states. They do not apply the future-work and positive-asset conditioning used in financial estimation. After nonholding, the holding probability cannot exceed the opportunity probability at the same state. A prior holder has access for certain but still chooses whether a funded risky position justifies its recurring cost. The policy atlas covers ages 30, 45 and 60, three persistent earnings states, both ownership histories and cash between $5,000 and $300,000 in two industry–education groups. Both numerical grids and all states are retained in the accompanying results.

The life-cycle projections start from 452 observed age-25 origins in 29 earnings cells. The first new decision occurs at age 26. Paths condition on survival, with no further selection on work or positive assets. Initial wealth, ownership, shares and filtered earnings states are inputs. Observed age profiles summarize different cross-sections and are descriptive context, not matched-cohort validation targets. The model accumulates wealth during working life and draws it down strongly after retirement. It has no bequest motive and consumes all remaining financial resources at age 84.

The sensitivity exercises use the two largest age-25 cells by initial survey weight: college workers in education and health, and in professional and business services. They answer conditional economic questions about those cells. They are not population policy effects. The complete family contains 39 scenarios per specification, with the following comparisons.

Access and recurring costs. Opening access after every nonholding year, removing the recurring fee, and removing both restrictions distinguish their roles. Additional exercises halve or double the recurring mean and make its offer deterministic. Saving and portfolio rules are reoptimized in each case. At age 45, ownership in the two-cell cohort rises from about 42% to 66% with open access, to 98% without the fee and to nearly 100% when both restrictions are removed. Wealth changes much less. Removing the fee alone does not guarantee an immediate opportunity; over repeated years, inexpensive holding allows access to accumulate and be preserved.

Earnings risk, insurance and retirement resources. The family pools earnings laws, compresses or removes employed transitory variation, replaces its log distribution with Gaussian support, reduces nonwork resources, freezes persistent earnings transitions, and reduces retirement income. Transitory compression preserves conditional mean dollar earnings. Gaussianization concerns employed transitory innovations, not the entire earnings process. Insurance replaces one quarter, one half or all of working income with the initial cohort's group-by-age mean. It is balanced at that group and age, not at every current persistent state; no general equilibrium financing is modeled. Full insurance sharply reduces precautionary accumulation and produces zero ownership at age 45 in these two cells. Reducing earnings risk therefore need not raise participation when saving adjusts. Retirement-income cuts leave payroll contributions unchanged and are not a complete pension reform.

Returns, preferences and portfolio restrictions. Belief-only changes alter expected stock returns while leaving actual returns unchanged. Separate experiments change both expected and actual premia or scale return dispersion around its mixture mean. Risk aversion, patience and intertemporal substitution are varied one at a time. Lower risk aversion can increase the conditional risky share while reducing ownership through lower accumulation: with $\gamma = 4$, age-45 ownership is about 23% and the conditional share is almost one. Fixed 50% shares, age-dependent shares and no equity provide portfolio-rule comparisons; consumption, saving and feasible participation are still optimized. These experiments distinguish ownership from conditional allocation rather than assuming risk preferences are irrelevant to entry.

Resource shifts and initial wealth. The family includes a $500 transfer, temporary changes equal to 1% of current income, permanent 1% changes in future employed wage resources, and zero initial financial wealth. Permanent wage shifts leave nonwork and retirement resources unchanged. The zero-wealth exercise changes the financial starting point rather than estimating an initial wealth distribution. Paired paths distinguish ownership, first entry and financial accumulation. Reported entry flows use the initial cohort denominator when near-universal ownership makes the remaining nonholder risk set negligible.

Y.6 Responses to realized earnings

A realized earnings gain changes current resources and the assets carried forward. This exercise illustrates the response to a current resource change, separately from the comparison of future earnings laws in the main text. Figure Y.1 gives an unexpected earnings gain or loss equal to 25% of a fixed current reference gross wage immediately before the age-45 decision. The current cash change incorporates the existing tax and employee-contribution wedge; every future earnings law is unchanged. All three specified initial cash positions are shown. Favorable realizations increase participation and adverse realizations reduce it. The effect can last after the transitory earnings realization has disappeared because the household carries different savings and participation histories forward.

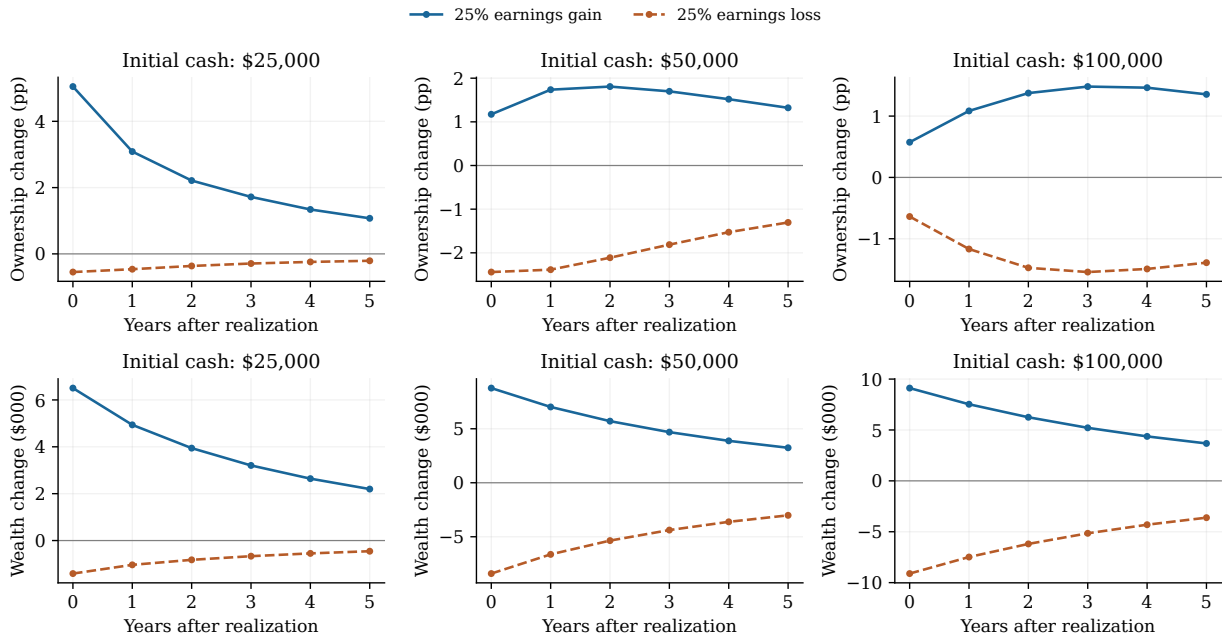


Figure Y.1. Participation after a transitory earnings gain or loss

Notes: Controlled model paths for an initially nonholding age-45 college graduate in education and health. Each panel compares a gain or loss equal to 25% of the fixed current reference gross wage with the same-state baseline, with unchanged future laws. Initial cash before the imposed gain or loss is $25,000, $50,000 or $100,000. Ownership differences are percentage points; wealth differences are thousands of real 2018 dollars. Paths use stationary returns and follow the fixed initial cohort without conditioning on subsequent employment or positive assets. Curves use the final 301-node calculation and paired simulation draws; they are not estimated empirical event studies. Figure Y.2 reports the separate distribution-news exercise; the numerical-accuracy discussion below retains the finer-grid sensitivity.

At $50,000 of initial cash, the favorable realization raises ownership by about 1.2 percentage points immediately and 1.8 points two years later. Resources arrive before the household has necessarily received an opportunity to invest, so some entry occurs later. This is an earnings-resource experiment, not a shock to the probability of future upside outcomes. Mean-preserving tail-news experiments answer that separate question: their responses are small and state dependent, and a favorable composition change does not raise entry at every cash position. The model supports a resource-to-entry mechanism, not a universal positive response to upside news. The full news results appear in Figure Y.2.

Y.7 Distribution news at unchanged current resources

The realized-income experiments in Figure Y.1 change current resources. A separate exercise announces a change in next year's employed-wage distribution at unchanged current cash. Upper-tail and lower-tail spreads, and two opposite composition rotations, preserve expected gross and net wages within each persistent destination. Work probabilities, persistent transitions and subsequent resource laws remain unchanged. The threshold is expected employed next wages. These dollar partial variances differ from the empirical zero-inclusive arc-growth partial variances.

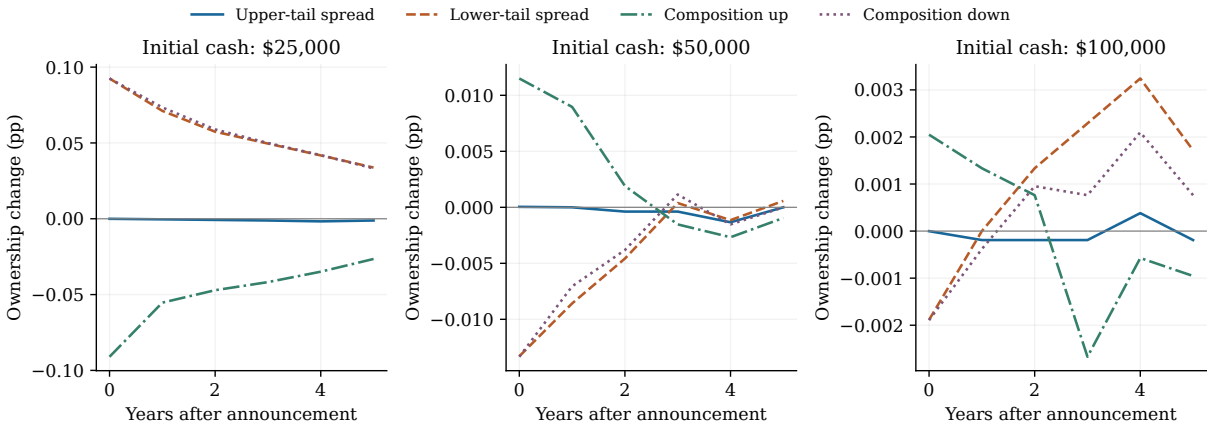


Figure Y.2. Distribution news at unchanged current cash

Notes: Opportunity model, age 45, initially employed college workers in education and health, persistent state 8. The four changes affect the transitory employed-wage distribution at age 46 only and preserve each persistent destination's expected gross and net wage. Initial cash is $25,000, $50,000 or $100,000. Upper and lower spreads each change the corresponding dollar partial variance by the same amount; composition rotations keep total variance fixed. These are conditional model responses, not empirical estimates. All directions and cash states are retained.

The announcement response depends on current cash. More upside composition slightly raises entry at $50,000 and $100,000 but reduces it at $25,000. At low cash, a lower-tail spread increases saving and can move a household toward participation. The isolated upper-tail spread has almost no effect at the implemented dose. Thus the model does not imply that every mean-preserving increase in upside composition raises entry at fixed resources. The full cell-law comparison and realized-income experiments answer broader and different questions.

The common partial-variance change is about 0.8% of baseline employed-wage variance. The realized earnings shocks are substantially larger resource changes. Their larger participation response does not establish that anticipation is unimportant per unit of risk. Policy-only and resource-only paths, with their interaction, remain in the accompanying results. At the announcement date, the resource-only response is zero by construction; later responses include realized earnings as well as changed choices.

Y.8 Controlled composition and the accumulation mechanism

The experiments in Section 6.7 change transitory employed earnings for one year (age 26), five years (ages 26–30), or every remaining working year (ages 26–64). Retirement resources are unchanged from age 65. The annual perturbation at each shared age and state is identical across durations. For each earnings group, age and persistent destination state, let μ be the baseline mean dollar wage and let U and L be its upper and lower partial variances around μ. The two laws satisfy

$$(U^s, L^s) = (U + sd, L - sd), \qquad s \in \{-1, +1\}, \quad d \geq 0. \tag{Y.2}$$

Each tail keeps its probability mass and conditional mean. Within-tail deviations are expanded or contracted around that tail's conditional mean, with a square-root scale chosen to deliver equation (Y.2). The transfer is 10% of the smaller adjustable within-tail variance, reduced by the same amount in both directions when positivity or preservation of tail membership requires it. Unavailable rotations leave the state unchanged. Symmetry therefore concerns dollar partial-variance changes, not equal and opposite movements of every wage outcome. The wage-to-disposable-income map is unchanged. Fixed destination means and variances, together with unchanged persistent and employment transitions, preserve every current state's conditional disposable-income mean and variance and its expected discounted disposable resources. Arc-growth moments and risk-adjusted human-capital values are not constrained to remain fixed.

The cohort consists of 452 observed age-25 origins in 29 earnings groups, including 395 initial nonholders and 57 holders. Initial financial assets, ownership, risky shares, earnings-state posteriors and survey weights are common across scenarios. Paths condition on survival without selecting on future employment or positive financial assets. The first new choice occurs at age 26, after earnings arrive. For the one-year change, no future earnings change remains at any reported choice, so all matched-state rules equal baseline. For the five-year change, rules coincide from the age-30 choice after the last affected income. The persistent experiment changes earnings through age 64; policies anticipate this entire schedule even though the plots display only ages 26–35. Simulated paths can continue to differ after the laws coincide because they inherit different assets and ownership histories.

For each direction, the full response changes both earnings and the optimized decision rule. A law-only comparison feeds changed earnings into the baseline rule, which continues to adjust consumption and investment to realized resources. A rule-only comparison applies the changed rule to baseline earnings, allowing the resulting financial states to evolve. Both use the same initial conditions, earnings draws, returns, opportunities and cost offers. The two separate baseline-reference effects need not add to the full response because their interaction can be nonzero. Averaging the two change orders allocates that interaction symmetrically. These crossed paths are accounting comparisons, not separately optimized economies under a common law.

The budget identity separates accumulation from participation payments. Let G_t be investment gains on inherited assets and K_t the paid recurring fee. With identical initial financial wealth, the difference between any two paths satisfies

$$\Delta A_h = \sum_{t=1}^{h} \big(\Delta Y_t + \Delta G_t - \Delta c_t - \Delta K_t\big). \tag{Y.3}$$

Figure Y.5 shows that higher consumption accounts for almost all of the first-year upside wealth decline: wealth falls by \$40.86, consumption rises by \$40.81, and paid fees rise by \$0.03. Initial investment gains are identical. Later wealth differences also reflect returns on the assets carried forward. The opposite rotation produces lower consumption and higher wealth. This accounting isolates expenditure and return contributions, but does not assign a unique preference motive to each consumption choice.

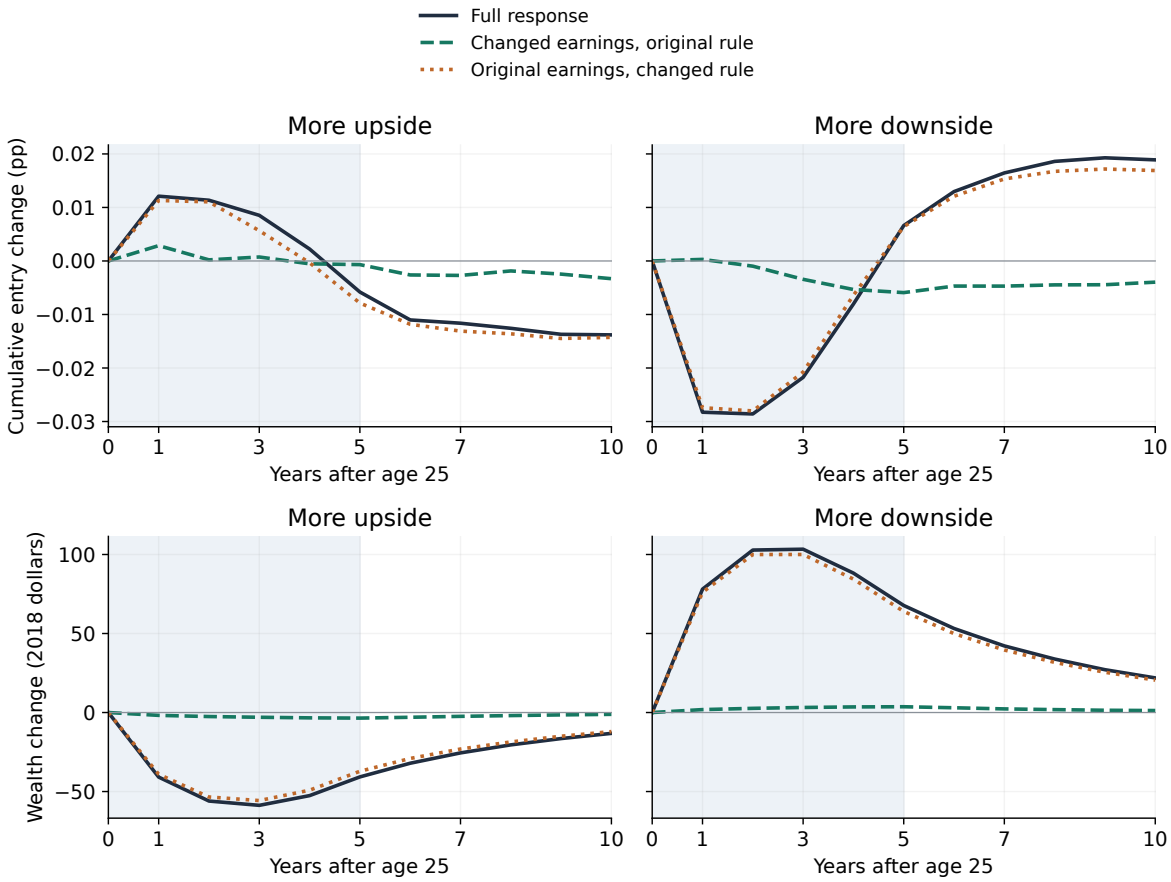


Figure Y.3. Earnings composition, decision rules and subsequent entry

Notes: Controlled composition changes for 395 initial nonholders, drawn from 452 observed age-25 origins in 29 earnings groups. Each direction transfers the same amount between upper and lower dollar partial variances within each destination state, preserving conditional expected disposable earnings and total dollar variance. Earnings changes last through age 30; paths continue to age 35. Shading marks the five affected income years. Paths condition on survival without future employment or positive-asset selection. Cumulative first entry is the fraction of the initial nonholders who have ever entered, regardless of subsequent exit. The full response changes both the earnings law and the optimized rule. Broken lines separately change earnings with the original rule or change the rule with original earnings; their interaction is not included in either separate contrast. Financial wealth differences are real 2018 dollars and entry differences are percentage points. Curves use the 201-node calculation and common simulation draws.

The matched-state comparison in Figure Y.4 holds cash, earnings and work state, inherited ownership, the opportunity and the cost offer fixed along baseline paths. Its choices do not feed back into later probe states. The no-stock alternative removes current participation and its fee while preserving future investment opportunities. Its saving response supports lower buffer demand under the upside law, without separating pure prudence from every future portfolio-option motive. No separate path changes saving alone while freezing all participation and allocation decisions.

For the five-year illustration, both numerical grids use four paired batches of 1,024 paths per origin. The 201-node calculation refines the 141-node solution, and paired batch differences measure simulation uncertainty rather than empirical or parameter uncertainty. The first-year law-only entry effect is very small and changes sign across grids. The larger matched-state saving response and the

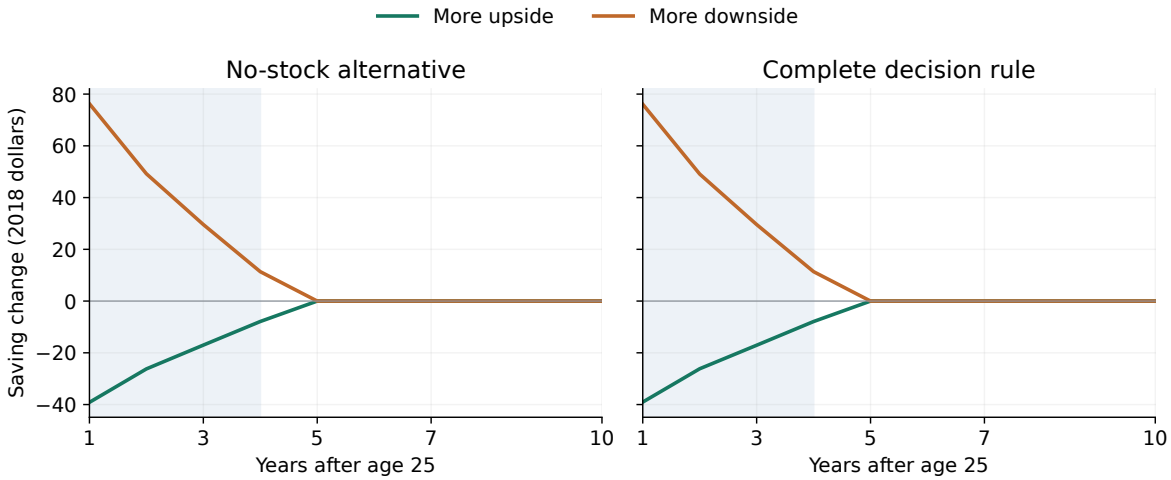


Figure Y.4. Saving responses at identical current resources

Notes: Both earnings-law rules are evaluated at the same cash, earnings and employment state, inherited ownership, opportunity and cost offer along baseline paths for the 395 initial nonholders. Probe choices do not alter subsequent states. The no-stock alternative permits only safe saving today and retains future investment opportunities. The complete rule also chooses participation and the risky share. Differences in end-of-period assets equal differences in saving because inherited assets and current returns are identical. Units are real 2018 dollars. Future laws, and hence rules at identical states, coincide from age 30 onward; the plotted differences are then zero. Curves use the 201-node calculation.

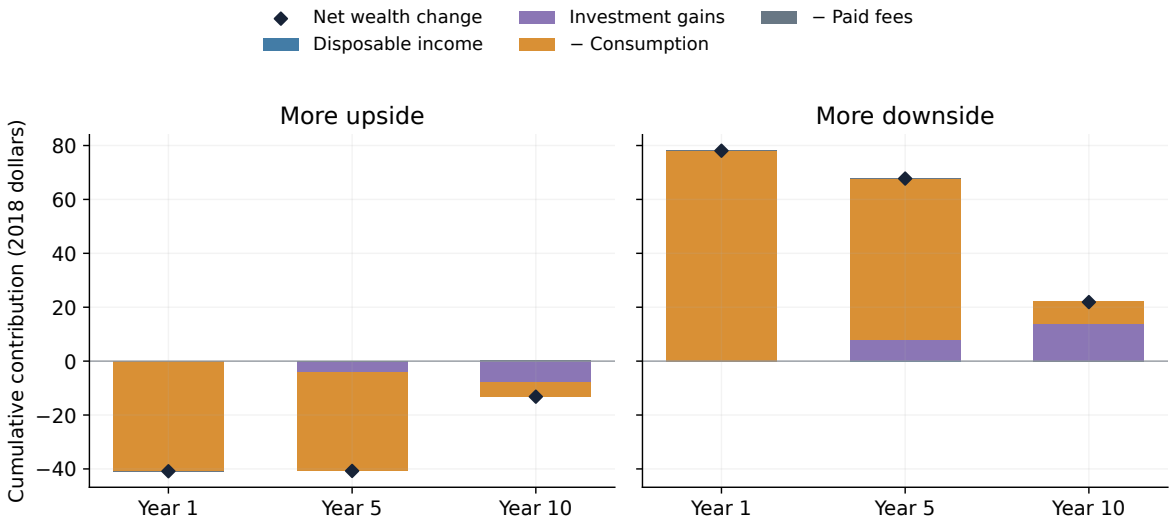


Figure Y.5. Consumption and fees in the wealth response to composition

Notes: Cumulative budget contributions to the five-year full wealth response for the 395 initial nonholders in Figure Y.3, in real 2018 dollars. Positive terms increase wealth and negative terms reduce it. Investment gains are returns net of inherited principal, and paid fees enter only when the household holds a funded stock position. The four signed terms sum to the net wealth response, marked by a diamond. Conditional expected disposable earnings are fixed analytically; small simulated income differences reflect finite simulation draws. Both rotations, all displayed horizons and the 201-node calculation are shown.

later full entry decline retain their signs. After ten years, the upside full cumulative-entry response is -0.0138 percentage point; keeping the original rule gives -0.0033 point with a paired simulation standard error of 0.0018 point. These small quantities illustrate the mechanism and are not scaled to explain the empirical composition coefficient.

Duration, entry flows and common-state choices. Figure Y.6 reports annual first-entry mass: the fraction of the initial nonholder cohort entering for the first time that year. It uses the original cohort denominator, rather than the shrinking set that has never entered. It therefore differs from an annual hazard. Under persistent upside, the positive first-entry differences in years one and two total 0.0195 percentage point; the negative differences in years three through ten total -0.0237 point. Their sum is the near-zero year-ten cumulative response. Earlier entrants remain counted after a later exit. Mean wealth, by contrast, includes entrants and those who never enter and can fall without changing either household's entry indicator.



Figure Y.6. Duration and the timing of first entry

Notes: Annual first-entry mass differences for the 395 initial nonholders in Figure 5. The denominator remains the original cohort, conditional on survival; this is not entry conditional on still never having entered. Years are elapsed years after the observed age-25 origin. Both mirrored directions and all three durations are shown. The one-year entry signs are grid-sensitive.

Persistent upside lowers saving at identical current resources throughout the displayed decade. Its matched-state holding probability among baseline current nonholders nevertheless rises by 0.0151 percentage point in year one and 0.0166 point in year ten. The corresponding five-year response is 0.0113 point in year one and zero from year five (Figure Y.8). This demonstrates opposing saving and entry-choice margins, but does not quantify their separate contributions to cumulative first entry. The rule-only path below is different again: changed choices alter later wealth and ownership, so the path can reduce entry even while the isolated same-state choice response is positive.

Resources, decision rules and their interaction. Figures Y.9 and Y.10 apply the same four-corner accounting to each duration and direction. Let $F(L, P)$ denote an outcome generated by earnings law L and rule P, with B denoting baseline and s the perturbed law or its optimized rule. The law-only

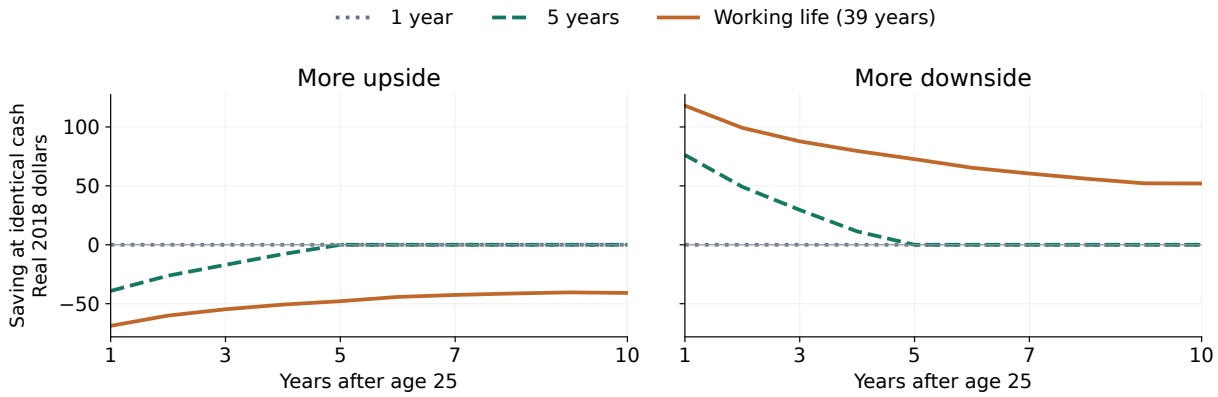


Figure Y.7. Duration and desired saving at the same current resources

Notes: Changed rules evaluated on the same baseline cash, age, earnings and work states for the initial nonholder cohort. The plotted no-stock alternative has no current stock position or participation payment, while retaining future investment opportunities. Probe choices do not change later probe states. One-year rules equal baseline at every reported choice; five-year rules coincide from year five, after the last affected income; persistent rules anticipate changed working-life earnings through age 64. Amounts are real 2018 dollars.



Figure Y.8. Duration and the willingness to enter at the same current resources

Notes: Difference in the current holding probability under the changed rule, evaluated on baseline current-nonholder states within the fixed cohort of 395 initial nonholders. Cash, age, earnings and work state, prior holding, the opportunity and fee draw are common. Earlier entrants who subsequently exited can enter this risk set; this is not a never-entrant first-entry hazard. Probe choices do not feed into future states. Both earnings directions and all three durations use the finer grid.

response is $F(s, B) - F(B, B)$ and the rule-only response is $F(B, s) - F(B, B)$. Their interaction is the full difference $F(s, s) - F(B, B)$ minus these two responses. The rule-only term includes the financial states produced by changed saving and portfolio choices; it does not hold cash fixed over time.

The common-draw resource paths coincide exactly at shared dates: retaining the baseline rule produces identical first-year outcomes for all durations, and identical outcomes through year five for the five-year and persistent regimes. Differences between their early full responses therefore arise from rules anticipating different future earnings schedules and the financial states those rules generate. Persistent upside's year-ten wealth gap is almost entirely a rule-path response, while its

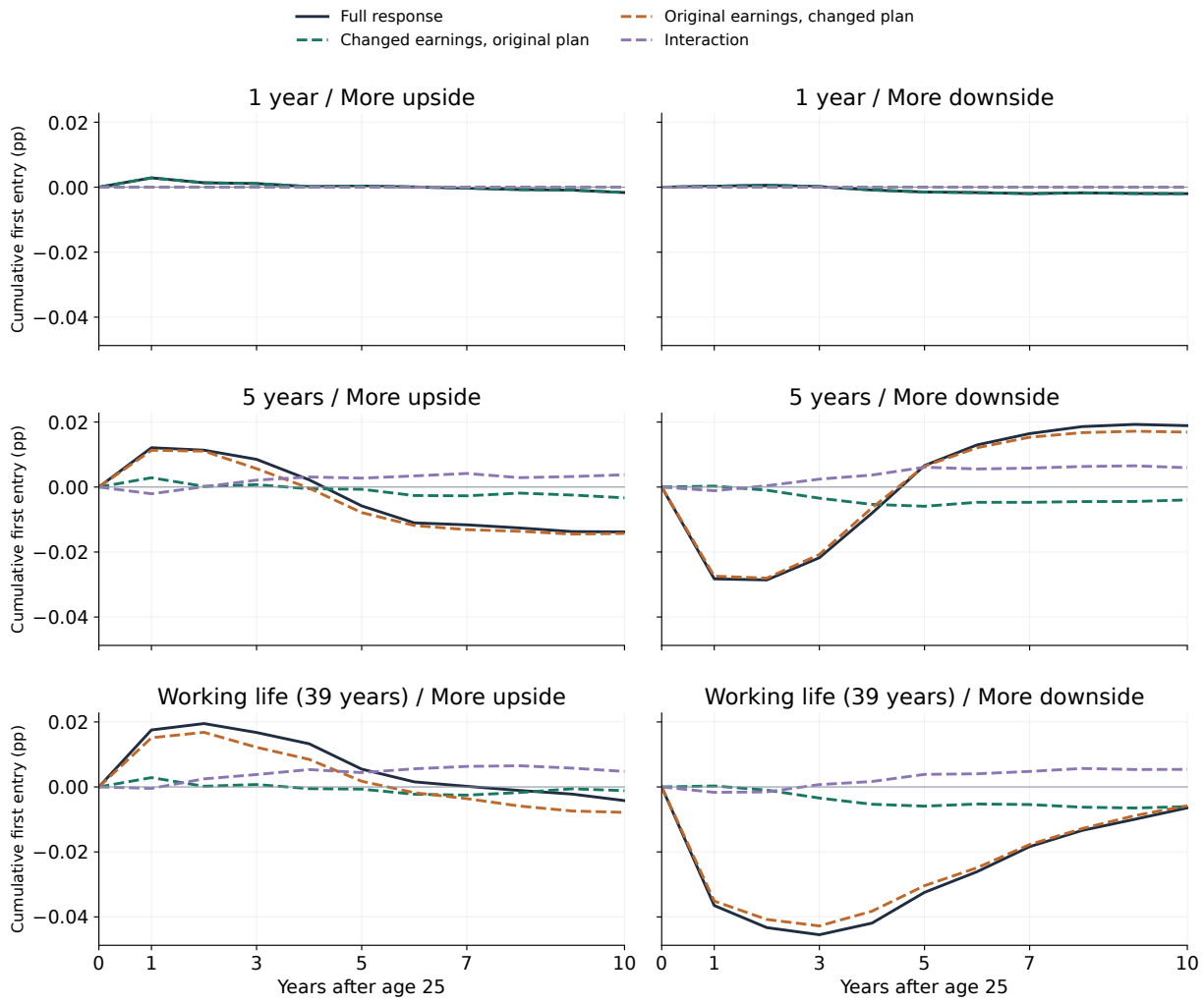


Figure Y.9. Duration and the resource–rule accounting for cumulative entry

Notes: Cumulative first-entry differences in percentage points for the 395 initial nonholders. Rows distinguish one year, five years and the remaining working life; columns distinguish more upside and more downside. Full response equals resources-only plus rule-only plus their interaction. Resources-only feeds changed earnings into the original rule; rule-only applies the changed rule to baseline earnings and lets financial states evolve. These paths are not separately optimized economies under a common law. The one-year rule-only response and interaction are exactly zero.

cumulative-entry gap is near zero. These are different outcomes and need not have proportional responses.

For persistent upside at year ten, equation (Y.3) allocates the $-\$191.95$ wealth gap to $-\$162.51$ from additional cumulative consumption, $-\$28.43$ from lower investment gains, $-\$0.18$ from fees and $-\$0.82$ from sampled income differences (Figure Y.11). The mean-income term is zero analytically. The accounting and common-state saving results jointly support lower buffer demand rather than fees as the primary channel, without uniquely identifying pure prudence.

The duration calculations use the same 141- and 201-node grids and four paired batches of 1,024 paths per origin. All declared duration precision screens pass. A passing numerical screen does not

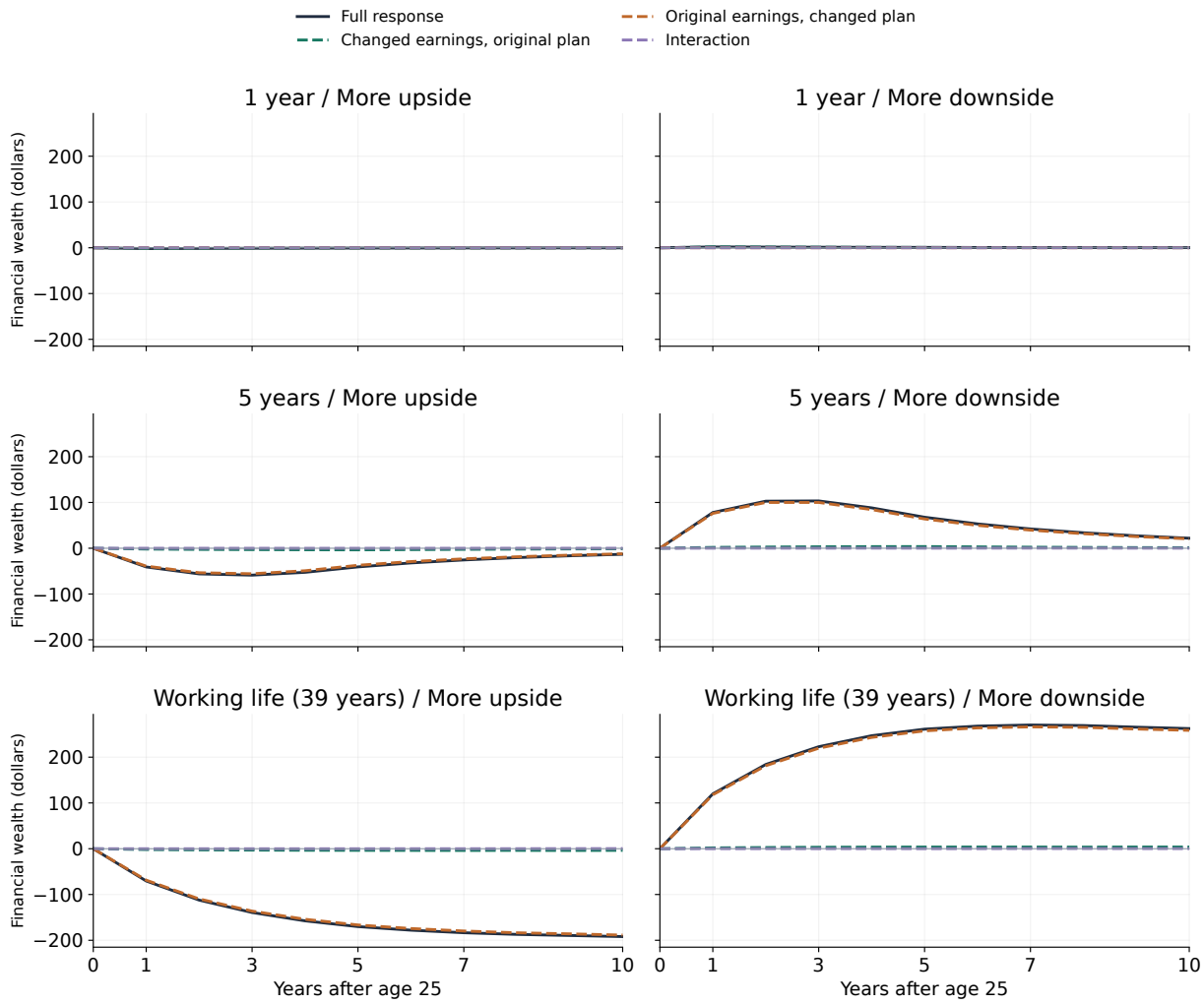


Figure Y.10. Duration and the resource–rule accounting for financial wealth

Notes: Mean wealth differences in real 2018 dollars for the same cohort and four-corner design as Figure Y.9. With persistent upside at year ten, resources-only contributes $-\$3.56$, rule-only $-\$188.64$, and their interaction $+\$0.25$, summing to the full $-\$191.95$ response before rounding. The rule contribution includes the changed saving and investment history, rather than isolating portfolio choice at fixed wealth.

make every tiny sign informative: persistent upside's year-ten cumulative-entry response is -0.0042 point with paired Monte Carlo standard error 0.0035 on the fine grid, and -0.0020 with standard error 0.0029 on the coarser grid. It is described as near zero. These simulation standard errors do not include parameter or empirical uncertainty. All financial parameters remain fixed, no earnings-risk coefficient is targeted, and the magnitudes remain illustrations.

Y.9 Welfare and the value of future resources

Within-model welfare comparisons hold preferences fixed and compare the optimized recursive values at specified current states. Preference changes and belief-only experiments are excluded from these welfare ratios. Resource-policy comparisons include the changed resource level and are not

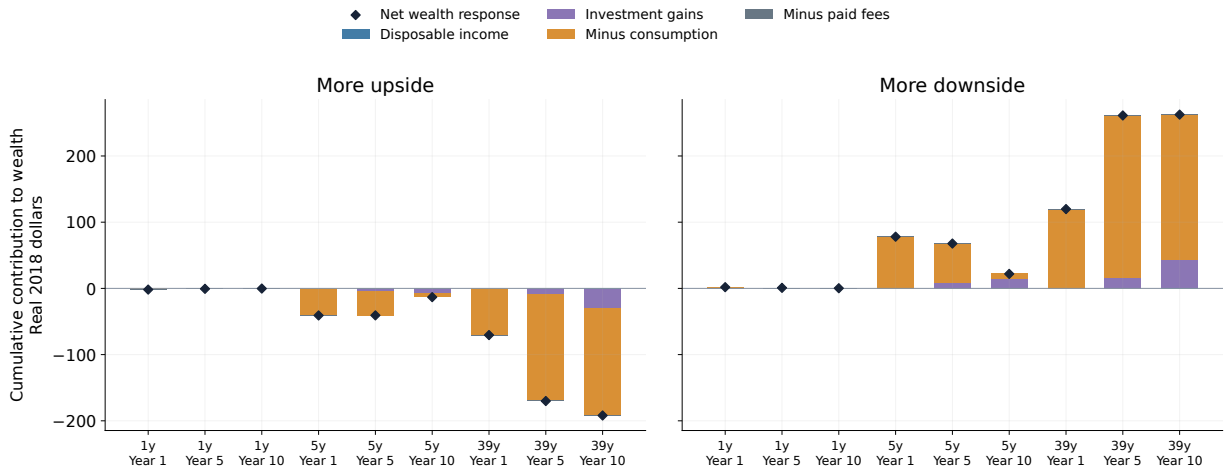


Figure Y.11. Duration and the consumption, return and fee contributions to wealth

Notes: Signed cumulative budget contributions to full mean-wealth differences, in real 2018 dollars, for the same 395 initial nonholders. Positive terms add wealth. Panels show each direction and duration at years one, five and ten. Diamonds mark net wealth, equal to income plus investment gains minus consumption and paid fees. Conditional expected disposable earnings are unchanged analytically; small sample income differences are simulation noise.

automatically compensated policy reforms. The static income illustrations instead evaluate one-period CRRA certainty equivalents at equal means and variances; they are distinct from household lifetime welfare.

A complementary valuation replaces all future nonfinancial resources, including retirement income, with current cash. If $V_t^B(m, z, d)$ is baseline value and $V_t^{Y=0}$ solves the same problem without future resources, the cash compensation H_t satisfies

$$V_t^{Y=0}(m + H_t, z, d) = V_t^B(m, z, d). \tag{Y.4}$$

Preferences, returns and access rules are unchanged. The calculation covers ages 35 and 45, cash of \$25,000, \$50,000 and \$100,000, and both selected industry–education groups. It is solved within the available cash grid, without extrapolation. Accounting present values discount expected resources at the safe rate, with both survivor-conditional and survival-weighted conventions reported. Neither accounting value is a market price. The gap between H_t and those present values reflects discounting, liquidity, mortality, access and preferences as well as income risk; it is not a pure risk premium.

Y.10 Entry rules under cell and pooled earnings laws

Figure Y.12 extends the same-resource comparison in Figure 8 to ages 30, 45 and 60 and three employed persistent earnings states in each of the two selected groups. Each curve changes future earnings laws while fixing the current household state. The response varies across cash positions, ages and earnings states. These comparisons therefore distinguish the effect of future earnings on current decision rules from the earnings and wealth paths in Figure 7; they do not imply that cell laws increase entry at every state.

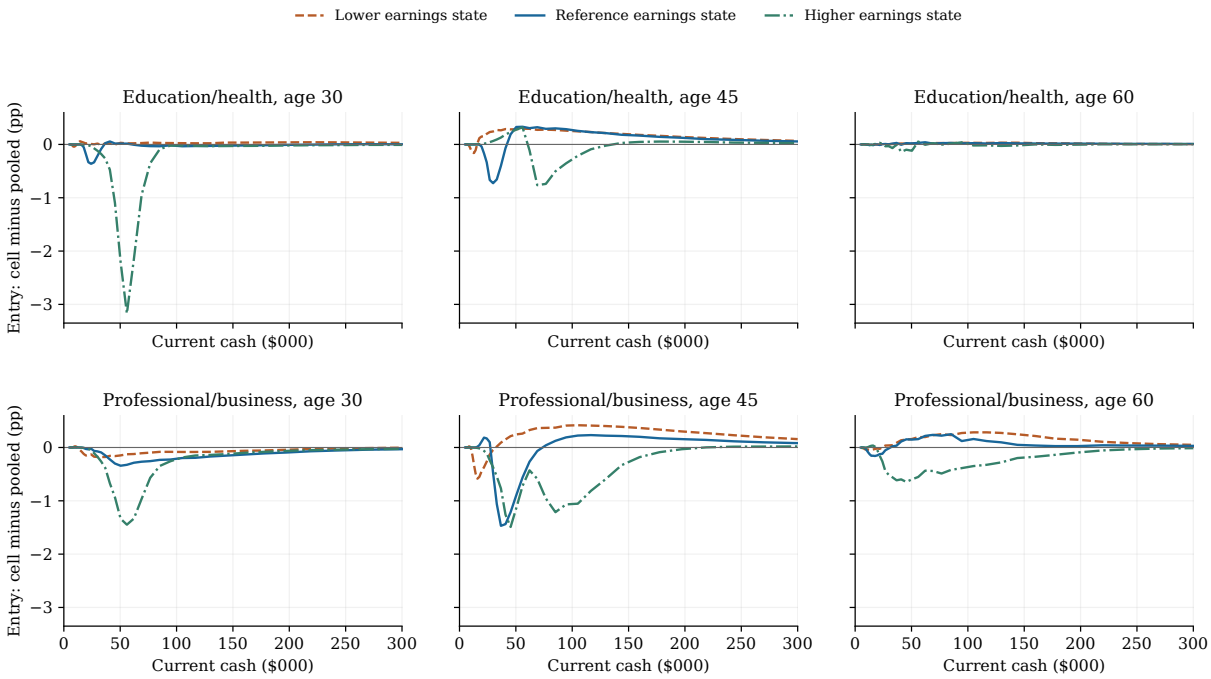


Figure Y.12. Entry-rule differences across ages and earnings states

Notes: Cell minus pooled entry probabilities for initially nonholding college workers in the two indicated groups. Columns fix age; curves distinguish employed persistent earnings states 4, 8 and 12. Lower, reference and higher states are model states, not empirical terciles. Each comparison holds cash, age, employment, the persistent earnings state, financial parameters, recurring costs and the opportunity probability fixed. Cash is in thousands of real 2018 dollars and differences are in percentage points. All declared ages, persistent states and cash positions for these two groups are displayed. Full future earnings laws change jointly.

Y.11 Numerical accuracy

The financial estimate uses 141 cash nodes and five ordinary-return nodes. The 201-node, seven-return evaluation checks all eight moments and the four net earnings-risk projections. The rate/share tolerance is the larger of 0.25 percentage point and half the empirical standard error; the wealth tolerance is \$250 and the projection tolerance is 0.05 point. These comparisons pass. Refined holder wealth growth is about \$498: it remains positive but does not retain the optimization's \$500 buffer. Model predictions do not receive sampling confidence intervals from these checks.

Accuracy is less uniform for some additional experiments. Across both specifications' life-cycle scenario family, 178 age-outcome comparisons exceed the declared grid tolerances, predominantly for consumption. The local state atlas has 920 flagged comparisons, concentrated in consumption and participation thresholds. Local marginal propensities should therefore be interpreted with the reported finer-grid comparisons. All cases remain in the results; the numerical check is not a rule for selecting favorable scenarios.

The controlled event paths receive an additional 301-node, nine-return-node evaluation and eight scrambled Sobol replications. Across the two models, 35 of 3,024 final grid comparisons remain outside tolerance. All asset-response comparisons satisfy their \$250 screen, while some ownership

Table Y.4. Opportunity access: numerical refinement

Outcome	Refined minus main	Financial tolerance
Entry: lower wealth tercile	0.014	0.250
Entry: middle wealth tercile	0.017	0.250
Entry: upper wealth tercile	0.022	0.250
Retention: lower initial risky-share bin	0.097	0.717
Retention: upper initial risky-share bin	0.077	0.614
Next risky share: all initial holders	0.057	0.421
Annual financial wealth growth: initial nonholders	-4.380	250.000
Annual financial wealth growth: initial holders	-12.910	250.000
Upside composition net projection	0.0049	–
Upper-tail derivative net projection	0.0059	–
Lower-tail derivative net projection	-0.0040	–
Dispersion at mean volatility net projection	0.0010	–

Notes: Compares 201 cash/seven ordinary-return nodes with 141/five. Rates and projections use percentage points; wealth changes use dollars. Both wealth-growth levels remain positive. These passing financial checks do not imply convergence of every subsequent local response or counterfactual. The online experiment discussion reports the retained finer-grid failures.

magnitudes remain sensitive. The largest ownership discrepancy is about 0.31 percentage point. Small pure-news effects should not be interpreted through their exact sign when grid differences are comparable. Resource cash compensation is more stable: all 144 valuation cases resolve within the grid, with a maximum finer-grid difference of about $231. Numerical precision and the economic fit discrepancies are separate considerations.

The working-life cell-minus-pooled ownership comparison in Figure 7 changes by at most 0.17 percentage point on the finer grid. The maximum change in the entry differences displayed in Figure 8 is 0.08 point. All entry-difference curves in Figure Y.12 satisfy the 0.5-point screen, although a small difference need not retain its sign. The complete cell-versus-pooled current-choice evaluation also retains continuation, financial balances, consumption and selected fees. That broader family contains 796 numerical-screen failures and 90 support-change rows; accuracy of the displayed entry comparisons does not establish uniform accuracy for those other outcomes.

Figures Y.13 and Y.14 compare the numerical grids for baseline participation rules and the life-cycle profiles. The first fixes the cell earnings law and contrasts entry with continued holding; it does not reproduce the cell-versus-pooled comparison in Figure 8. Their close overlap is an accuracy check for these displayed outcomes; it does not remove the limitations of the wider experiment family reported above.

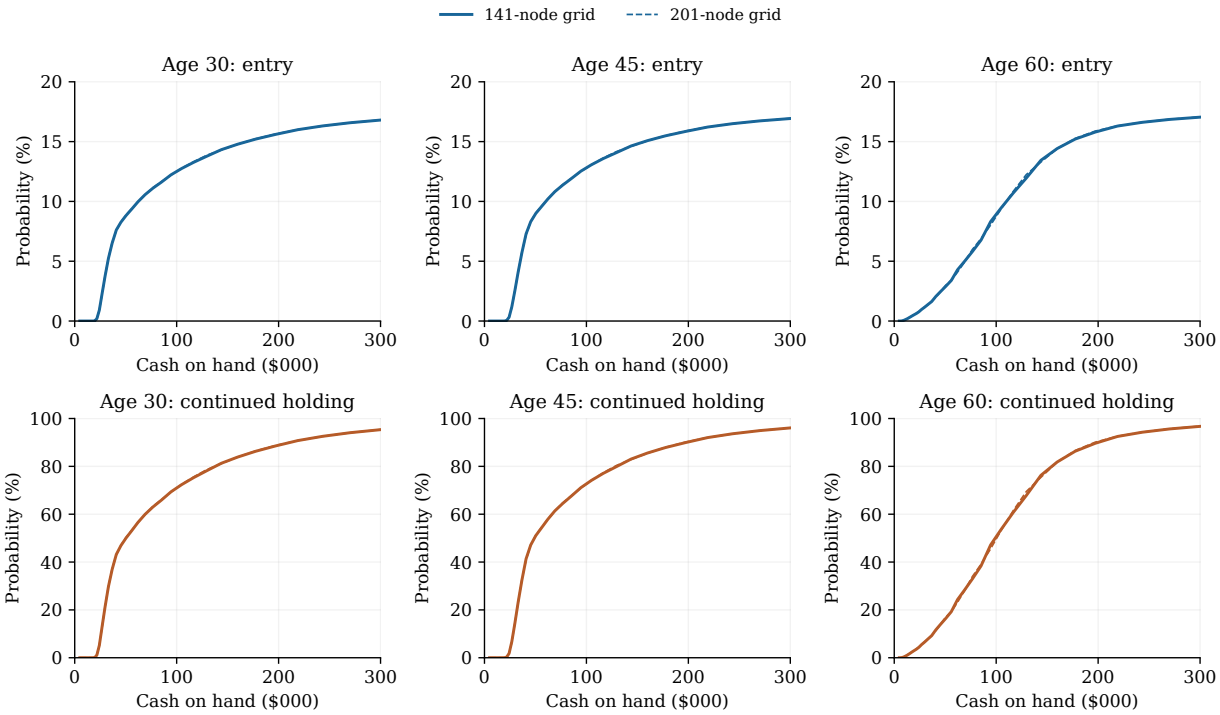


Figure Y.13. Numerical comparison of participation decision rules

Notes: College graduates in education and health at a common employed persistent earnings state. Columns fix age; the upper row shows entry after nonholding and the lower row continued holding. Note the different vertical scales. Probabilities integrate current opportunities and cost offers, at the same cash and earnings state within each comparison. Solid and dashed curves use 141 and 201 cash nodes. Cash is in thousands of real 2018 dollars. These are individual decision rules without the future-work and positive-asset conditioning imposed in the observed-transition comparison. The full two-group state exercise and numerical limitations are reported in Appendix Y.

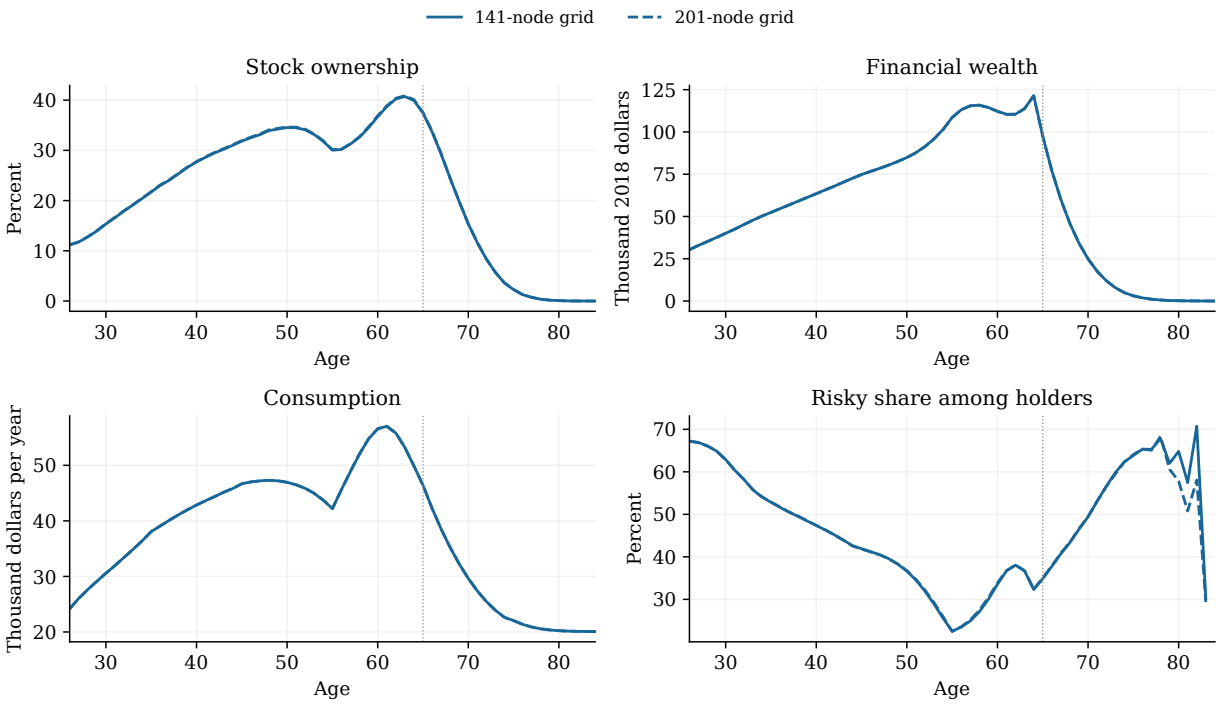


Figure Y.14. Numerical comparison of conditional life-cycle projections

Notes: Projections start from 452 observed age-25 origins in 29 cells, with survey-weighted initial wealth, ownership, shares and earnings-state posteriors as inputs. The first new decision is at age 26. Paths condition on survival without future employment or positive-asset selection. The dashed curves use the finer grid; the vertical line marks retirement at 65. These are conditional cohort projections, not a fitted unconditional distribution or an age–period–cohort comparison. There is no bequest motive and remaining resources are consumed at 84.

Y.12 Measured resource inputs

The PSID contribution and retirement schedules and dated portfolio-return inputs are common to all financial specifications. They are measured inputs to the conditional model, rather than fitted return forecasts or endogenous retirement-account balances.

Table Y.5. Measured contributions, retirement resources and calendar returns

A. Own annual retirement benefits (thousand 2018 dollars)

Education and age	Social security	Pension	Net total
Without a college degree, ages 65-74	14.94	6.58	15.07
Without a college degree, ages 75-84	14.50	6.90	14.98
College degree, ages 65-74	16.74	18.32	24.54
College degree, ages 75-84	18.15	18.94	25.96

B. Observed annual real gross return factors

Destination year	Market	Safe asset	CPI ratio
2020	1.2255	0.9923	0.9878
2021	1.1830	0.9552	0.9551
2022	0.7415	0.9389	0.9259
2023	1.2168	1.0079	0.9605
2024	1.2144	1.0224	0.9714

C. Own employee contributions relative to own wages (percent)

Education and age	Contribution rate	Coverage	Observations
Without a college degree, ages 25-34	2.01	95.8	5056
Without a college degree, ages 35-44	2.87	94.8	4804
Without a college degree, ages 45-54	3.75	94.6	3359
Without a college degree, ages 55-64	4.41	93.7	2705
College degree, ages 25-34	4.79	94.9	2223
College degree, ages 35-44	4.98	91.4	1689
College degree, ages 45-54	5.22	92.2	1051
College degree, ages 55-64	5.44	90.2	1051

Notes: Panel A uses own social-security and non-VA pension income of PSID reference persons aged 65–84 in income years 2012, 2014, 2016 and 2018, with zero benefits retained and equal total weight across waves. Spouse, IRA and annuity income are excluded. A calibrated 30 percent tax gives the net total; these are individual reference-person profiles, not a representative estimate for every adult. Panel B compounds the 12 monthly French-library broad-market returns (market excess return plus the risk-free rate) and risk-free returns separately, then applies the same annual CPI endpoint ratio as the SIPP balances. A factor of 1 means zero real return. These are common portfolio proxies; individual bond and mutual-fund exposures are not observed. Panel C takes the ratio of weighted own employee contributions to weighted own wages on the same observed contribution sample in each age/education bin. It includes required and voluntary employee contributions, including employee payments to defined-benefit plans; employer contributions are excluded. Known routed nonparticipation is zero, while unresolved routes remain missing; coverage is the weighted observed share. PSID waves are 2013, 2015, 2017 and 2019. These current-job annual proxies are not exact matched annual cash flows. Observed transitions apply calendar-year return proxies to inherited portfolios; the Bellman problem and hypothetical windfalls retain stationary stochastic returns. No displayed calibration was fitted to entry-risk coefficients.

Z Permanent Participation Types

This appendix asks whether persistent differences in access can account for the financial patterns organized by the opportunity model. Households have either permanent access to the nonretirement stock-and-mutual-fund market or no access to that market. Both types choose saving optimally. This alternative produces similar one-year financial fit but different responses to ownership history and persistent resources. A permanent nonparticipation fraction is also considered by Catherine (2022); the conditional treatment of initial ownership here reflects the observed-state design of this paper.

Z.1 Household types and the decision problem

Let $\tau_i \in \{A, E\}$ denote an active or excluded type, known to the household and constant throughout its simulated path. Active households always receive an investment opportunity and pay the recurring cost $K_t = \kappa_R \xi_t$ if they choose a funded risky position. They solve equation (X.3) with $q(d) = 1$ for both prior ownership states. An active nonholder can therefore enter whenever its optimal choices justify the fee. There is no additional cash cost after nonholding.

Excluded households cannot invest in the observed risky asset at any future date. Their safe-saving problem is

$$\begin{aligned} V_t^E(z,m) &= \max_{0 \le A < m} \left[(m-A)^\rho + \beta s_t \left\{ \mathbb{E}\left[V_{t+1}^E(z', AR_f + Y')^\omega \mid z \right] \right\}^{\rho/\omega} \right]^{1/\rho}, \\ V_{84}^E(z,m) &= m. \end{aligned} \tag{Z.1}$$

They pay no participation fee. Their saving policy incorporates the absence of a future stock option; it is not the active household's policy with stockholding mechanically set to zero. Earnings, taxes, own pension contributions, retirement resources, borrowing constraints, survival and returns otherwise follow Appendix X.

The researcher assigns each observed origin an active-type probability

$$p_i = \begin{cases} 1, & \text{if initially holding or affirmatively observed holding before the origin,} \\ p_A, & \text{otherwise.} \end{cases} \tag{Z.2}$$

The first group contains all 3,582 initial holders and 408 initial nonholders with quality-screened ownership in at most two preceding annual observations. The other 17,198 initial nonholders receive the common probability p_A. Missing or allocated histories do not establish exclusion, and no future outcome is used to assign type. The parameter is therefore a conditional initial mixture weight, not an identified population fraction of lifelong investors or a posterior inferred from a complete ownership history.

Households know their type, so p_i belongs in the researcher's aggregation rather than in either household's Bellman equation. For future eligibility $\mathcal{E}_{i1}$ and outcome X, define $e_i^\tau = \Pr(\mathcal{E}_{i1} \mid \mathcal{I}_i, \tau)$

and $n_i^\tau(X) = \mathbb{E}[X\mathbf{1}\{\mathcal{E}_{i1}\} \mid \mathcal{I}_i, \tau]$. The conditional prediction is

$$\widehat{X}_i = \frac{p_i n_i^A(X) + (1-p_i) n_i^E(X)}{p_i e_i^A + (1-p_i) e_i^E}. \tag{Z.3}$$

The type-specific outcome and eligibility masses are mixed before conditioning. Averaging conditional type means with the unadjusted p_i would be incorrect when future positive-asset eligibility differs by type. Survey weights then average these predictions within each observed target group, as in equation (X.8).

One type draw is retained throughout each origin's multiyear simulation and paired across counterfactuals. Separately conditioned observations of the same person are not assigned a jointly estimated lifetime type. Permanent exclusion concerns separately reported stocks and mutual funds outside retirement accounts; it does not imply that a household never owns equity through a pension, employer plan or retirement account.

Z.2 Estimation and financial fit

Table Z.1. Permanent access: parameters

Parameter	Value
Risk aversion γ	8.917
Annual discount factor β	0.810
Mean recurring offer (2018 dollars)	413
Log-cost standard deviation	1.013
Conditional active probability p_A	0.119
Elasticity of intertemporal substitution (fixed)	0.50

Notes: Five parameters are estimated from the eight financial moments in Table Z.2. The elasticity of intertemporal substitution is fixed. The recurring offer has a mean-one lognormal multiplier; the reported mean is not the selected payment. There is no additional entry payment. The access probability is a model restriction, not a measured attention or trust variable.

The alternative estimates $(\gamma, \beta, \kappa_R, \sigma_\kappa, p_A)$ on the same eight targets and criterion in equation (Y.1), replacing the opportunity probability with the initial active-type share. The parameter bounds and positive-growth search restriction are identical. Two coarse starts and a fine search precede one bounded continuation; the selected continuation converges after 53 objective calls. The initial fine search exhausted its evaluation budget, and the continuation was authorized before the untargeted comparisons. Both models retain their full search outcomes.

The fitted active share is about 12% among initial nonholders without affirmative holding history. It is a substantial participation restriction within that conditional group. The offered recurring mean is about $413. The criterion is close to that of the opportunity model, and the fitted entry rates and wealth changes are similar. Both specifications predict too much accumulation among initial nonholders and too little among initial holders. Refinement preserves positive group means, although holder growth of about $487 is slightly below the search buffer. Comparing the fitted specifications

Table Z.2. Permanent access: targeted financial outcomes

Financial outcome	Data	Model
Entry: lower wealth tercile (%)	2.93 (0.26)	3.21
Entry: middle wealth tercile (%)	5.84 (0.36)	5.49
Entry: upper wealth tercile (%)	10.36 (0.48)	10.39
Retention: lower initial risky-share bin (%)	66.19 (1.43)	68.01
Retention: upper initial risky-share bin (%)	76.31 (1.23)	73.83
Next risky share: all initial holders (%)	33.38 (0.84)	33.81
Annual financial wealth growth: initial nonholders (dollars)	3,939 (778)	10,005
Annual financial wealth growth: initial holders (dollars)	1,946 (5,177)	501

Notes: All eight outcomes are targets. Rates and portfolio shares are percentages; financial-wealth changes are annual real 2018 dollars and include portfolio gains. Data standard errors, in parentheses, use 299 household-lineage multiplier draws. Initial holders who exit have zero next-year shares. The conditional sample contains 17,606 initial nonholders and 3,582 holders. Observed initial states are inputs.

changes preferences and recurring costs as well as the access mechanism, so it is not a decomposition that changes only one parameter.

Z.3 Earnings-risk gradients and dynamic comparisons

Table Z.3. Permanent access: earnings-risk gradients

	Data	Data SE	Cell laws	Pooled laws	Difference
Upside composition	1.816^{***}	0.480	0.382	0.211	0.171
Upper-tail derivative	1.967^{***}	0.513	0.321	0.166	0.155
Lower-tail derivative	-1.665^{***}	0.470	-0.444	-0.256	-0.188
Dispersion at mean volatility	0.151	0.104	-0.061	-0.045	-0.016

Notes: All columns use the same 17,606 initial nonholders, weights, controls and source standardization. Units are percentage points per source standard deviation. Data use industry–education CR1 standard errors and two-sided t_{29} tests: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Model columns are deterministic conditional projections without sampling intervals or stars. No earnings-risk coefficient enters estimation. Tail rows are local partial-variance derivatives; dispersion is evaluated at common mean volatility. The difference is cell minus pooled laws at common parameters, not a causal share explained.

Both gross and pooled entry projections have positive upper-tail and negative lower-tail derivatives. The cell-minus-pooled composition increment is about 0.17 percentage point, compared with about 0.25 in the opportunity model and 1.82 in the matched data. The finer-grid increment is about 0.18. The empirical ordering therefore does not depend on allowing repeated opportunities, although its quantitative magnitude remains limited under either restriction. The same fixed regression operator and source standardization are used; no earnings-risk coefficient enters estimation.

Table Z.4. Permanent access: outcomes outside the financial objective

Outcome	Data	Model
Entry and holding at 2 annual follow-ups (%)	3.48	3.82–3.83
Entry and holding at 3 annual follow-ups (%)	2.27	2.22–2.26
Re-entry: reported prior holders (%)	27.16	57.68
Entry: no reported prior holding (%)	5.73	4.77
Windfall ownership response, year 0 (pp)	–	5.45
Windfall ownership response, year 1 (pp)	–	4.23
Windfall ownership response, year 2 (pp)	–	3.44
Entrants with a position at most $1,000 (%)	24.22	0.013–0.107

Notes: Sustained outcomes use the common 5,459- and 1,801-origin cohorts and show the range across two simulation seeds. Annual observations do not establish continuous ownership. Reported-history groups contain 408 prior holders and 17,198 other nonholders; the latter are not classified as lifetime nonparticipants. The hypothetical $150,000 windfall starts from 423 observed age-44 nonholders, arrives at age 45, and uses stationary stochastic returns; it differs from the Swedish lottery estimand. The small-position interval bounds one unresolved origin and is not a sampling interval. No row selects parameters.

The permanent model fits the level of three-follow-up sustained ownership more closely: approximately 2.2–2.3%, compared with 2.3% in the data and 3.3% under opportunity access. It overpredicts re-entry among known prior holders, however, at about 58% compared with 27% in the data. Those households are assigned active status and can re-enter whenever the recurring fee is worthwhile. Opportunity access has the opposite re-entry discrepancy. Ownership history therefore distinguishes the two restrictions even when their aggregate entry rates are close.

The $150,000 windfall increases permanent-model ownership by about 5.5 percentage points immediately but only 3.4 after two years. The opportunity model's corresponding response grows over that interval. Excluded households cannot use the windfall to acquire access, while among active households the untreated paths gradually catch up. The common U.S. conditional experiment is distinct from the Swedish lottery estimand. Both models also severely underproduce small entrant accounts; this distributional discrepancy is not resolved by permanent types.

Z.4 Projected life-cycle participation

At age 45, ownership is about 16% under permanent types and 32% under repeated opportunities, while mean financial wealth is approximately $74,000–$75,000 in both. Permanent types impose a ceiling on the households who can enter; repeated opportunities permit additional entry as resources accumulate. The two fitted models thus differ much more in participation than in aggregate wealth. Both display substantial retirement drawdown. Neither includes bequests, housing or a separately modeled retirement account.

Removing the recurring fee leaves a strong difference in eventual ownership. In the two-cell cohort at age 45, ownership is about 28% under permanent types and 98% under opportunity access. The excluded type remains excluded even when participation is free; active types and households receiving repeated opportunities can preserve access through holding. Removing both restrictions

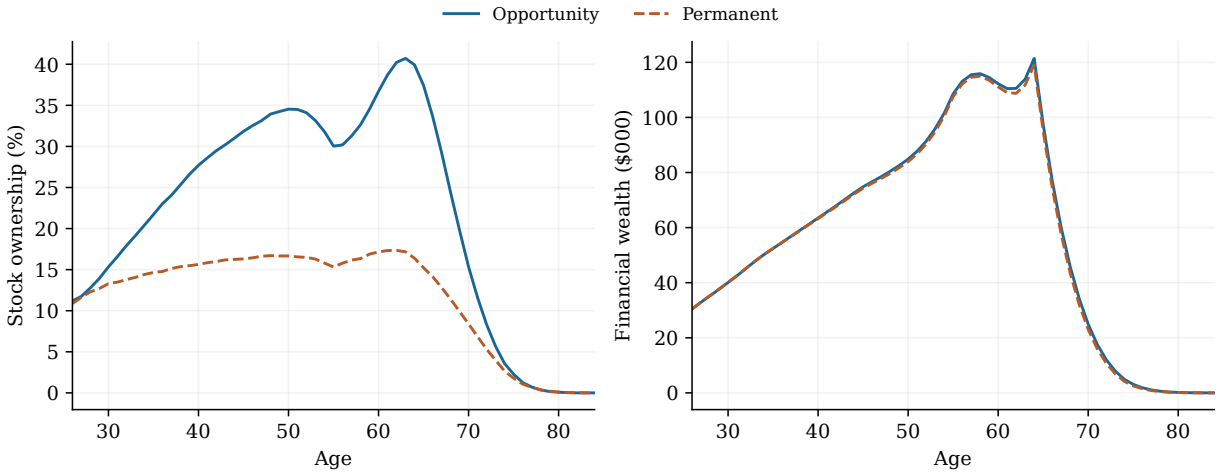


Figure Z.1. Repeated opportunities and permanent participation types over the life cycle

Notes: Both models start from the same 452 observed age-25 origins in 29 industry–education cells, with observed financial states and filtered earnings posteriors. The first new decision occurs at age 26. Paths condition on survival and impose no subsequent work or positive-asset selection. The permanent type is fixed along each path; its aggregate eligibility ceiling includes known-active histories as well as the conditional active fraction among other nonholders. Retirement begins at age 65 and remaining assets are consumed at age 84. These are conditional projections, not fitted unconditional age profiles.

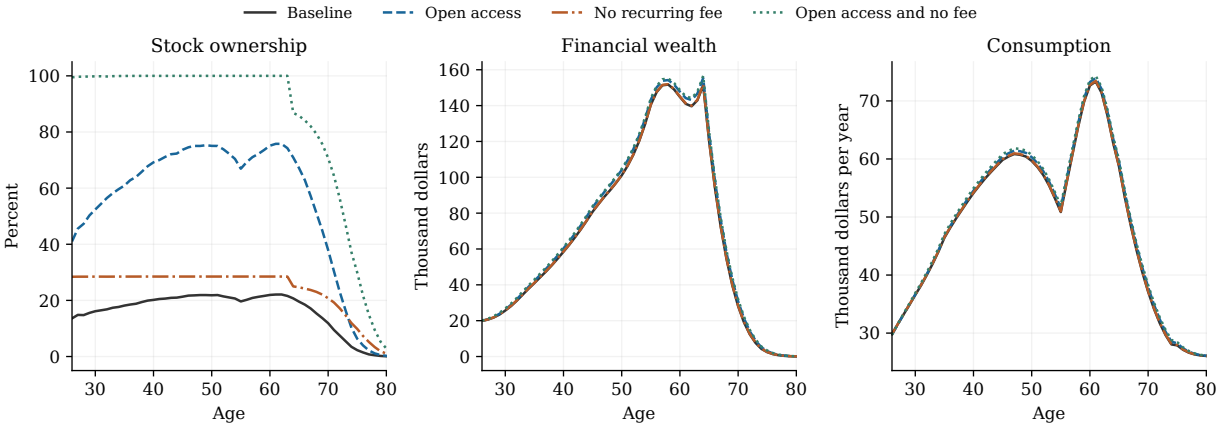


Figure Z.2. Access restrictions and recurring fees with permanent types

Notes: Permanent-type model: baseline, always-open access, zero recurring fee, and both restrictions removed. Counterfactuals hold the remaining permanent-model parameters fixed and reoptimize choices. The two-cell cohort contains college workers in education and health and in professional and business services, selected by initial survey weight. Paths begin from observed age-25 states and condition on survival. Always-open access removes permanent exclusion. None of the counterfactuals is refitted. Figure 9 presents the corresponding opportunity-model exercise.

makes ownership nearly universal in both models. These are state-dependent life-cycle outcomes, not alternative estimates of the empirical tail coefficients.

Z.5 Responses to realized earnings and distribution news

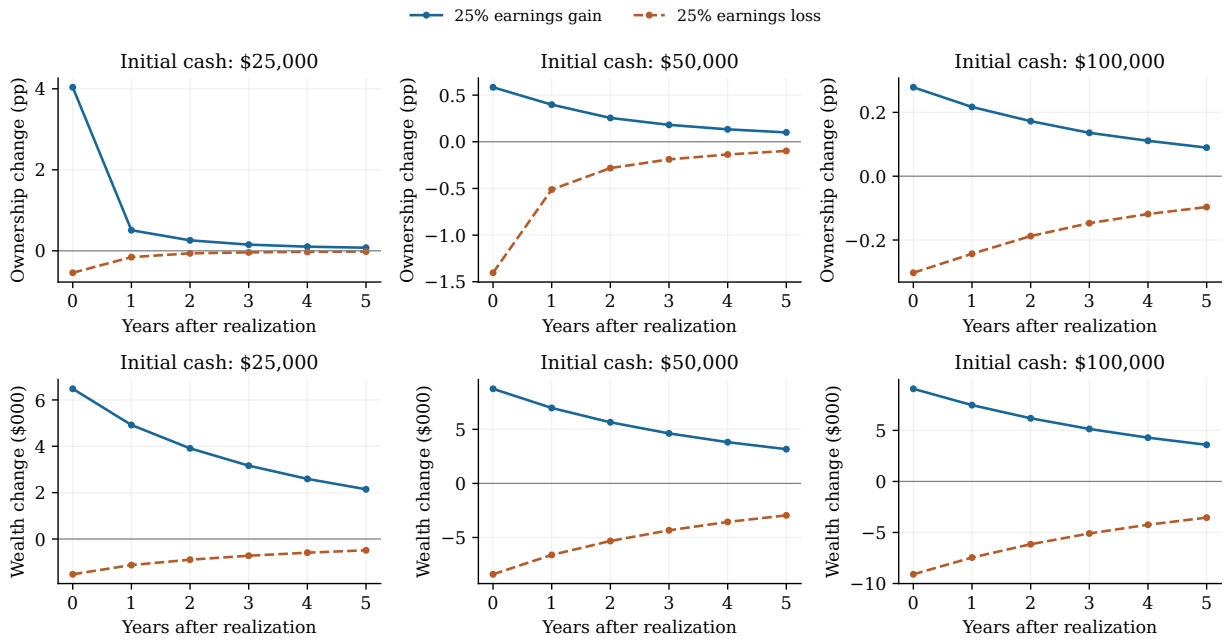


Figure Z.3. Earnings realizations with permanent participation types

Notes: Permanent-type model for initially nonholding college workers in education and health without known prior access, employed at age 45 with persistent state 8. Unexpected gains and losses equal 25% of the fixed current reference gross wage, translated into disposable cash with the common tax and contribution wedge. Initial cash before the imposed change is $25,000, $50,000 or $100,000. Future earnings laws and the persistent state are unchanged. Ownership, rather than a new-entry hazard, is followed after impact. The original cohort remains fixed without subsequent work or asset selection; stationary stochastic returns are paired across paths. Figure Y.1 gives the corresponding opportunity-model responses.

Both models turn a favorable realized wage change into higher ownership and a loss into lower ownership. At $50,000 of initial cash, the gain raises ownership among nonholders without known prior access by about 0.6 percentage point immediately and 0.1 after five years in the permanent model. The opportunity response is about 1.2 and 1.3 points. Yet the year-five asset increases are similar, at roughly $3,200. Permanent exclusion chiefly brings forward entry among active types, whereas repeated opportunities allow retained resources to support entry later.

Known prior-holder nonholders are active in the permanent model and have larger immediate responses than otherwise identical nonholders of unknown type. At $50,000, their gain response is about 4.9 percentage points, falling to 0.8 after five years. Current holders and known prior-holder nonholders have the same portfolio opportunities at a common state in that model. Opportunity access instead treats all nonholders alike after an exit. These differences reflect the modeled information about access history, rather than new evidence that the households' true lifetime types are observed.

The complete distribution-news family also remains available for both models. The four directions preserve expected employed wages at every persistent destination and leave current cash unchanged. In both models, the composition-up announcement response reverses sign at the low-cash state and the isolated upper-tail spread has almost no effect at the declared dose. Permanent exclusion therefore does not restore a universal positive response to upside news. The cash-dependent saving mechanism described in Appendix Y applies here as well.

Z.6 Additional experiments and accuracy

The accompanying results report the same 39 scenario definitions for both models: access and recurring fees; transitory and persistent earnings risk; income insurance and nonwork resources; retirement income; expected and realized stock returns; risk aversion, patience and intertemporal substitution; restricted portfolios; income shifts and initial wealth. All baseline and counterfactual paths, local decision rules, selected payments and numerical comparisons are retained. The welfare comparisons use each type's optimized value at unchanged preferences. For nonholders of unknown type, household certainty-equivalent changes are computed within type before averaging with initial type weights; these are not utility comparisons across the two fitted models.

Future-resource cash compensation in equation (Y.4) is likewise inverted separately for active and excluded types before averaging dollar amounts. Averaging values and then inverting would describe a different object when households know their type. All type-specific cases resolve within the numerical grid. Neither compensation nor its difference from expected resource present value is a market price or a pure measure of income risk.

The financial moments, net earnings-risk projections and four-corner accounting satisfy the stated 141/201-node refinement criteria for this alternative. The wider experimental family shares the numerical limits reported in Appendix Y: some local consumption, participation and event-response magnitudes remain sensitive to the grid. Both resolutions and the final event refinement are retained. The principal distinction between a fixed eligibility ceiling and repeated entry opportunities is much larger than those local differences, but small news effects should not be interpreted by their precise sign.